# Analytic Stacks

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autoscribed notes*

## Contents



*These notes were written primarily by an LLM (Claude Opus 5), with some manual retouching afterwards. Any further fixes will be posted at `https://github.com/dmanam/analyticstacksnotes`. There are undoubtedly still many errors here; use at your own risk.

# 1 Analytic Geometry, Condensed Mathematics, and Analytic Rings

This first lecture is an introduction to the first half of the course, whose goal is to set up foundations for analytic geometry: a general theory of *analytic stacks*. Clausen began with motivation, drawn from the existing classical theories of analytic geometry and their shortcomings; then introduced the condensed formalism as a replacement for topological spaces; and finally gave the notion of *analytic ring*, the affine local models of the theory, closing with the example of solid analytic rings. Throughout, this is an introduction: the definitions are stated in outline, and the details are deferred to the coming lectures.

## 1.1 Motivation: classical theories of analytic geometry

Classically there are several different theories of analytic geometry. Clausen listed the ones he is familiar with, with the caveat that the list is not exhaustive.

1. *Complex analytic spaces.* In the smooth case these are the complex manifolds: spaces obtained by gluing open subsets of $\mathbb{C}^n$ along biholomorphisms, i.e. maps that locally admit a power series expansion. In the non-smooth case one also allows, locally, the zero locus of a finite collection of analytic functions on such open subsets. This is perhaps the gold standard.

2. *Locally analytic manifolds.* This is a generalization of the smooth case, presented in Serre's book on Lie groups and Lie algebras [Ser92]. One starts with a complete normed field $(k, |\cdot|)$, which may be Archimedean or non-Archimedean, and glues open subsets of $k^d$ along locally analytic isomorphisms – maps admitting, locally around every point, a convergent power series expansion with coefficients in $k$. When $k = \mathbb{C}$ this recovers the smooth case of (1); when $k = \mathbb{R}$ it is a version of the theory of real manifolds; both are reasonable theories. But when $k$ is non-Archimedean the theory is not geometrically rich, absent some extra structure such as a group structure.

*Remark* 1.1 (Total disconnectedness in the non-Archimedean case). The failure of richness for, say, $k = \mathbb{Q}_p$ comes from the topology of $\mathbb{Q}_p^d$ being totally disconnected: every unit ball breaks up into $p$ smaller unit balls, each of which breaks up again, and so on. Serre's book [Ser92] contains the classification of compact $p$-adic manifolds; it is a simple combinatorial classification (by dimension and one further invariant), reflecting how little geometry is present.

The next two theories react to this by shifting attention from local topology to local rings of functions.

3. *Rigid analytic geometry*, introduced by Tate [Tat71], over a non-Archimedean field $k$ – motivated in part by the uniformization of elliptic curves. This is a geometrically rich theory that works in the non-Archimedean case. In contrast to (1)–(2) one does not think in terms of gluing open subsets of $k^d$, or even in topological terms at all, but rather algebraically. The shift is twofold:

   - one focuses on the *local rings of functions* rather than on local topology, letting the ring of functions dictate what the topology should be; and

   - the local models are rings of functions convergent not on an *open* disk but on a *closed* disk – which is what makes sense in the non-Archimedean setting.

Concretely the local rings of functions are (quotients of) rings of functions convergent on a closed polydisk; these are the Tate algebras. The manner of gluing these local models to global rigid analytic spaces is halfway between algebraic geometry and ordinary analytic geometry.

*Remark* 1.2 (Terminology). There was a brief exchange about whether "Tate algebra" should mean the rings of functions convergent on a polydisk, or the quotients of these; the usage varies between references. Clausen settled on calling the quotients *affinoid algebras*.

4. *Adic spaces*, Huber's generalization of rigid analytic geometry [Hub94]. Rigid analytic geometry takes place over a fixed base field; a loose analogy is that adic spaces are to rigid analytic spaces as general schemes are to varieties over a field. One again specifies local models algebraically, by a ring of functions, but now with a new twist: the datum is a pair

$$A^+ \subseteq A,$$

a topological ring $A$ ("local ring of functions", not local in the sense of commutative algebra) together with a subring $A^+$. To $A$ one attaches a space of valuations, and $A^+$ singles out those valuations for which the elements of $A^+$ are regarded as integral. A useful flexibility of Huber's theory is that one may consider different choices of $A^+$ for a fixed $A$. We will see later in the lecture what this choice is really doing.

5. *Berkovich's theory.* The rings of functions in the rigid and adic settings are typically rings complete with respect to a finitely generated ideal, given the inverse-limit topology (all quotients discrete), possibly with something inverted – provided the inversion inverts what one completed along. For example, functions on the closed unit disk in $\mathbb{A}^1$ over $\mathbb{Q}_p$ arise as

$$\left(\mathbb{Z}[T]_p^\wedge\right)\left[\frac{1}{p}\right],$$

the $p$-adic completion of $\mathbb{Z}[T]$ with $p$ inverted. These are $p$-adic Banach rings. Berkovich instead works with arbitrary Banach rings $(R, |\cdot|)$ as local models, and attaches to such a ring the space $\mathcal{M}(R, |\cdot|)$ of multiplicative seminorms, a compact Hausdorff space; the allowed gluings are organized by this Berkovich space. Since $\mathbb{R}$ and $\mathbb{C}$ are Banach rings, this theory – alone among those listed – allows both Archimedean and non-Archimedean phenomena. Complex analytic spaces, and in a certain sense rigid analytic geometry, appear as special cases. The global theory, however, does not function as smoothly as in the other settings; see [Ber12].

The relationships among these theories are more or less well understood: (1) is a special case of (2) and (of the smooth part of) others, rigid geometry is a special case of adic spaces and of Berkovich's theory, and so on. One can formulate the comparisons by hand, but there is no common framework in which all of these examples live and in which the comparisons take place.

### 1.2 Why introduce a new theory?

Clausen gave several reasons, theoretical and practical.

- *To accommodate all examples.* One would like a single general theory of analytic spaces that can be specialized to whichever context one is interested in, rather than a list of separate theories with their own flavors.

- *Archimedean and non-Archimedean geometry together, with good gluing.* Of the theories above, the only rich one allowing both Archimedean and non-Archimedean geometry is Berkovich's, but there the gluing is not so well worked out. Berkovich restricted essentially to the non-Archimedean case from the start; more general gluings were investigated by Poineau. In these approaches one fixes a Banach ring as a base, defines affine space of some dimension over it, and glues along certain subsets of affine space. The objects one glues along are often not themselves controlled by Banach rings, so the nature of the gluing is constrained to a finite-type situation and does not fit the mould of how one passes from affine schemes to general schemes by naive gluing of local models.

- *Flexibility, and issues of descent.* Even within their own contexts these theories are less flexible than the theory of schemes, largely for reasons of descent. For a scheme one has the category of quasi-coherent sheaves $\mathrm{QCoh}(-)$: over a ring one takes all modules, and this glues. In analytic geometry there is no such theory in any of the classical settings. One always has a local ring $R$ describing the local geometry, and one may assign to it the category of all $R$-modules, but this does not glue: modules on two pieces with gluing data on the overlap need not globalize. Only finitely presented modules glue, giving the theory of coherent sheaves $\mathrm{Coh}(-)$ – a beautiful and central tool, but constrained by finiteness. For instance, the pushforward of the structure sheaf along a map of analytic spaces is not coherent unless the map is finite, whereas such pushforwards are basic examples of quasi-coherent sheaves in algebraic geometry, where no finiteness is built in.

- *A speculative, practical reason from the Langlands program.* Fargues and Scholze geometrized the local Langlands correspondence [FS24], clarifying the local Langlands program. The geometrization is based on replacing $\mathbb{Q}_p$ by a more exotic object, the Fargues–Fontaine curve – or rather a family of such curves – produced in the language of adic spaces: an adic space over $\mathbb{Q}_p$, not at all of finite type, which the theory of adic spaces fortunately existed to accommodate. Speculatively, one might hope to geometrize not just local but global Langlands. (Clausen noted that Scholze is optimistic about this, and he himself less so.) This would involve replacing $\mathbb{Q}$ or $\mathbb{Z}$ by some family of exotic analytic spaces – which would have to have both Archimedean and non-Archimedean aspects (there should also be a version over $\mathbb{R}$, which Clausen believes Scholze is working out) and would not be of finite type in any sense. No existing theory could give the language to describe such an object, if it exists; and it is valuable to have a precise theory to guide the exploration of such possibilities.

The goal of the course is thus to introduce a new theory of analytic geometry and to explain its relation to the previous theories. Today's lecture treats only the affine situation – the analytic rings.

## 1.3 Condensed mathematics

The descent problems above can be traced to the fact that the local rings in all these theories are not abstract rings but *topological* rings, and it is essential to remember the topology.

**Example 1.3** (Completed vs. algebraic tensor products)**.** In algebraic geometry the polynomial ring in two variables, functions on $\mathbb{A}^2$, is the tensor product $\mathbb{C}[x] \otimes_{\mathbb{C}} \mathbb{C}[y]$. In rigid analytic geometry, however, the algebraic tensor product of the one-variable Tate algebra with itself gives a wrong, "crazy" ring, because it forgets the $p$-adic topology; one must take the $p$-adically completed tensor product to obtain the correct ring of functions in two variables. Tensor products compute fiber products geometrically, so one must remember the topology when forming them.

This is the source both of the absence of a naive theory of quasi-coherent sheaves and of the gluing problems outside the finitely generated case. Yet topological modules over a topological ring are not suitable for a general theory: the category of topological modules over a topological ring is not abelian, however one dresses it up.

*Remark* 1.4 (Why topological modules fail to be abelian). The phenomenon is that a dense inclusion of modules – which occurs constantly in infinite-dimensional situations – is generally both an epimorphism and a monomorphism in reasonable such categories, without being an isomorphism. (One must impose a separatedness/Hausdorff condition for the literal statement; the underlying point is that a non-strict map has an image carrying two different topologies.)

The remedy is to go back to basics and define a replacement for the category of topological spaces, one on which it is easy to pile algebraic structure – analogues of topological rings and modules – so as to obtain an abelian category in the end. The basic idea is old (Clausen attributed frequent use of it to Grothendieck, and suspected it is older still).

The difficulty with a topological space is that it is second-order data: a set of points together with a set of subsets of it, which is what makes topology hard to mix with algebraic structure. One would rather record first-order data. The idea is to single out a collection of nice *test spaces* $S$, and to encode a space $X$ not as a set with a chosen family of open subsets, but by recording the data of the (would-be) continuous maps $S \to X$, axiomatizing the structure and properties visible in that situation. We care about $X$ only insofar as it is seen by maps from the test objects.

*Remark* 1.5 (Older incarnations). Traditionally, to do homotopy theory, one sometimes fixed a set $X$ together with a collection of maps of simplices into $X$ and defined homotopy and homology from that. Clausen tentatively named Spanier as a source; he was not certain of the attribution.

We now fix the test spaces to be profinite sets, and impose a countability restriction.

**Definition 1.6** (Light profinite set)**.** A *light profinite set* is a countable inverse limit of finite sets (each finite set discrete, the limit given the inverse-limit topology). Equivalently, it is a metrizable totally disconnected compact Hausdorff space; equivalently still, a compact Hausdorff space with a countable basis of clopen subsets. We write $\mathrm{Prof}^{\mathrm{light}}$ for the category of light profinite sets, morphisms being continuous maps (equivalently, maps of the associated countable pro-systems of finite sets).

*Remark* 1.7 (Why "light"). The first courses on condensed mathematics (Scholze's, following [CS19]) used arbitrary profinite sets, i.e. arbitrary totally disconnected compact Hausdorff spaces, as test spaces. That is a large category with no cardinal bound, so encoding the data below records more than a set's worth of information, causing technical trouble which was worked around. For this course we take instead the light variant; the reasons for this precise choice will be explained over the first few lectures.

**Definition 1.8** (Light condensed set)**.** A *light condensed set* is a sheaf of sets on $\mathrm{Prof}^{\mathrm{light}}$ for the Grothendieck topology in which a covering sieve is one containing a finite, jointly surjective collection of continuous maps. Thus finite disjoint unions are covers, and a surjection is a cover.

Unwinding the sheaf condition makes this elementary.

**Definition 1.9** (Light condensed set, explicitly)**.** A light condensed set is a functor $X\colon (\mathrm{Prof}^{\mathrm{light}})^{\mathrm{op}} \to \mathrm{Set}$ such that:

1. $X(\emptyset) = *$;
2. $X(S \amalg T) \xrightarrow{\sim} X(S) \times X(T)$; and

3. for every surjection $T \twoheadrightarrow S$,

$$X(S) \xrightarrow{\sim} \mathrm{eq}\big(X(T) \rightrightarrows X(T \times_S T)\big).$$

**Example 1.10** (Topological spaces as condensed sets)**.** Any topological space $X$ gives a light condensed set with functor of points

$$S \longmapsto X(S) := \mathrm{Cont}(S, X),$$

the set of continuous maps $S \to X$. Functoriality and the three properties of Definition 1.9 are immediate.

*Remark* 1.11 (Grothendieck topologies will be used in earnest). Although the definition can be unwound as above, the elementary appearance is somewhat illusory: the theory of Grothendieck topologies and sheaves is used seriously in this course, both here and later in gluing the affine (analytic ring) case to general analytic spaces. Clausen recommended that anyone unfamiliar with sheaves on sites read up on the subject before following the course.

Two remarks indicate why this precise definition is made.

*Remark* 1.12 (The two basic test objects). The two most important light profinite sets are the point $*$ and the one-point compactification $\mathbb{N} \cup \{\infty\}$ of the natural numbers. Evaluating at $*$ recovers the underlying set of a condensed set, and evaluating at $\mathbb{N} \cup \{\infty\}$ records its convergent sequences (for a topological space these are literally the underlying set and the set of convergent sequences; in general they are the abstract analogues).

*Remark* 1.13 (Effect of the two choices in the topology). Two features of the Grothendieck topology pull in complementary directions.

- Allowing *all* surjections to count as covers simplifies the structure of the category, and in particular yields good homological algebra when one passes to light condensed abelian groups: one has much flexibility to work locally.
- Requiring the topology to be *finitary* (covers generated by finite jointly surjective families) gives good categorical compactness properties for the light profinite sets, sitting by the Yoneda embedding inside all light condensed sets – and indeed for all metrizable compact Hausdorff spaces. For example, the unit interval has the surjection

  $$\{0, 1\}^{\mathbb{N}} \twoheadrightarrow [0, 1]$$

  from the Cantor set given by binary expansions; since $\{0, 1\}^{\mathbb{N}}$ is one of the basic test objects and the topology is finitary, the topological compactness of these compact Hausdorff spaces translates into a categorical compactness property inside the larger category. For this it is important to have the larger light profinite sets available, not merely the sets of convergent sequences.

*Notation* 1.14 (Dropping "light"). From now on *condensed set* means light condensed set and *profinite set* means light profinite set; the countability hypothesis is always present.

## 1.4 Analytic rings

Condensed sets give rise, immediately from the sheaf-theoretic viewpoint, to condensed rings and condensed modules: a *condensed ring* is a sheaf of (commutative) rings on the site of profinite sets, equivalently a commutative ring object in condensed sets, and a condensed module over it is a module object.

*Remark* 1.15*.* Concretely a condensed ring is a compatible family of abstract commutative rings $R(S)$, one for each profinite set $S$, satisfying the conditions of Definition 1.9 (so $R(S \amalg T) = R(S) \times R(T)$ is a *Cartesian* product of rings, etc.). "Ring" means commutative ring throughout.

One might hope to use condensed rings directly as the local models of analytic geometry, by analogy with schemes (which are built from discrete rings). This is not enough: condensed rings alone do not give a good geometry.

*Remark* 1.16 (Why condensed rings alone are insufficient)*.* The category of condensed rings has pushouts given by relative tensor products, just as for classical commutative rings, and these should geometrically compute fiber products – i.e. should be the completed tensor products of Example 1.3. But for condensed rings $A, B$ over $K$ one computes, from the nature of the Grothendieck topology,

$$\big(A \otimes_K B\big)(*) = A(*) \otimes_{K(*)} B(*),$$

so the underlying ring of the relative tensor product is the *abstract* (uncompleted) tensor product of underlying rings. The relative tensor product thus merely puts a (nontrivial) condensed structure on the abstract tensor product; it does not perform any completion, and completion does change the underlying set. So this is not yet doing the geometrically correct thing.

To fix this, one puts additional structure on a condensed ring, recording a class of condensed modules to be considered as *complete* in an appropriate sense. This is the notion of analytic ring: a condensed ring together with data telling which condensed modules are complete for the theory being described.

Before the definition, one must be precise about "ring".

*Remark* 1.17 (Which notion of ring: animated)*.* Experience in algebraic geometry shows that the correct fiber product of schemes is the *derived* fiber product, corresponding on affines to the derived relative tensor product of rings. Thanks to Lurie's work [Lur17] this is no longer a technical hassle, so the course works with derived rings. In practice almost all basic examples will carry no derived structure – they will be ordinary rings – so one can follow the course while imagining everything is an ordinary ring; but the general claims can fail if everything is taken to be ordinary.

Once one works derived there are inequivalent choices. The two basic options are $\mathbb{E}_\infty$-algebras (in the sense of spectra) and *animated* commutative rings, i.e. the objects presented by simplicial commutative rings; and in each case one may or may not allow negative homotopy. We do not allow negative homotopy (rings are connective), and we choose animated commutative rings, which are more directly tied to classical algebraic geometry: taking derived tensor products of ordinary rings always lands in this class. The whole theory works equally well in any of the variants – and is in fact slightly less technical to set up in this seemingly more complicated setting.

**Definition 1.18** (Condensed animated ring)**.** A *condensed animated ring* is a hypersheaf of animated rings on the site of (light) profinite sets. From now on "ring" means animated ring, and if we wish to stress that a ring lives in degree zero we call it *static* (or classical).

*Remark* 1.19 (Why hypersheaves)*.* Hypersheaves are used (rather than sheaves) because one wants, e.g., convergence of Postnikov towers, which can be proved for hypersheaves but is not known for sheaves; one cannot show sheaves and hypersheaves agree. Moreover everything can always be shown to be a hypersheaf, and in practice hypercovers cause no more trouble than ordinary covers, so nothing is lost.

**Definition 1.20** (Derived category of a condensed animated ring)**.** The basic invariant of a condensed animated ring $R$ is its full derived category $D(R)$: the (unbounded) $\infty$-category of hypersheaves of modules over the sheaf of rings $R$. If $R$ is static, $D(R)$ is the $\infty$-categorical enhancement of the usual unbounded derived category of the abelian category of condensed $R$-modules (again in the naive sense: a sheaf of modules over a sheaf of rings). For general $R$ one still has a $t$-structure on $D(R)$, but its heart is no longer the module category once $R$ is not static. We use homological grading conventions throughout.

We can now give the definition. Below, $R^{\triangleright}$ (Clausen's notation: the "triangle" denotes the underlying condensed ring) is a condensed animated ring.

**Definition 1.21** (Analytic ring)**.** An *analytic ring* is a pair $R = (R^{\triangleright}, D(R))$ consisting of a condensed (animated) ring $R^{\triangleright}$ and a full subcategory $D(R) \subseteq D(R^{\triangleright})$ – its "complete" objects, the derived category of the analytic ring – such that:

1. $D(R)$ is closed under all limits and colimits (formed in $D(R^{\triangleright})$);
2. if $N \in D(R)$ and $M \in D(R^{\triangleright})$, then the internal hom $\underline{R\mathrm{Hom}}_{R^{\triangleright}}(M, N)$ lies in $D(R)$;
3. writing $(-)^{\wedge}_{R}$ for the left adjoint to the inclusion $D(R) \hookrightarrow D(R^{\triangleright})$ (a completion functor), $(-)^{\wedge}_{R}$ preserves connectivity, i.e. sends $D(R^{\triangleright})_{\geq 0}$ into $D(R^{\triangleright})_{\geq 0}$; and
4. $R^{\triangleright} \in D(R)$ (the unit, the underlying ring itself, is complete).

*Remark* 1.22 (Existence of the completion functor)*.* The left adjoint $(-)^{\wedge}_{R}$ in axiom (3) exists automatically: given (1), the inclusion of the presentable subcategory $D(R)$ admits a left adjoint with no further set-theoretic hypothesis needed. Axiom (3) imposes that our analytic rings are, in a suitable sense, connective, mirroring the connectivity of the ring. The role of connectivity (axiom (3)) is to permit a reduction from a general animated ring to its static truncation $\pi_0$.

*Remark* 1.23 ($A^+$, revisited)*.* The analytic ring structure – the choice of $D(R)$ on top of $R^{\triangleright}$ – is the analogue, in this theory, of the choice of a subring $A^+ \subseteq A$ in Huber's theory (Definition 1.21 versus the discussion of adic spaces above).

**Definition 1.24** (Map of analytic rings)**.** A *map of analytic rings* $R \to S$ is a map of condensed rings $R^{\triangleright} \to S^{\triangleright}$ such that restriction of scalars along $R^{\triangleright} \to S^{\triangleright}$ carries $D(S)$ into $D(R)$: if $M \in D(S)$, then $M$ regarded as an $R^{\triangleright}$-module lies in $D(R)$.

*Remark* 1.25 (The $t$-structure on $D(R)$)*.* There is always a $t$-structure on $D(R)$, described naively by intersection with that of the enveloping category:

$$D(R)_{\geq 0} = D(R) \cap D(R^{\triangleright})_{\geq 0}, \qquad D(R)_{\leq 0} = D(R) \cap D(R^{\triangleright})_{\leq 0},$$

so everything can be checked in the potentially more familiar $D(R^{\triangleright})$. Its heart

$$D(R)^{\heartsuit} = D(R) \cap D(R^{\triangleright})^{\heartsuit}$$

is an abelian category. The modules are allowed to be $\mathbb{Z}$-graded, while the ring is connective (positively graded). This abelian category already determines the analytic ring: there is an equivalent axiomatics purely at the abelian level, specifying $R^{\triangleright}$ together with an abelian subcategory of $D(R^{\triangleright})^{\heartsuit}$ satisfying suitable axioms (one must, however, retain axiom (4), that the unit is complete).

A third, concrete perspective is in terms of free modules, i.e. spaces of measures.

**Definition 1.26** (Completed free modules)**.** For a profinite set $S$, the completed free module is

$$R[S] := \big(R^{\triangleright}[S]\big)^{\wedge}_{R},$$

the completion of the free $R^{\triangleright}$-module $R^{\triangleright}[S]$ on $S$. These objects generate $D(R)_{\geq 0}$ under colimits, and are the basic building blocks. There is a further equivalent axiomatics taking as second datum not all of $D(R)$ but just the collection of these free modules $R[S] \in D(R^{\triangleright})$.

*Remark* 1.27 (Measures, and the completeness/integration interpretation)*.* Intuitively $R^{\triangleright}[S]$ is the space of *finite* $R^{\triangleright}$-linear combinations of points of $S$ – of Dirac measures on $S$ – while its completion $R[S]$ is a larger space of measures. In general $R[S]$ need not lie in the heart (though in essentially all examples it does). An analytic ring structure may thus be thought of as a choice of a space of measures on each profinite set. Their role: for $M$ in the heart (say),

$$M \in D(R)^{\heartsuit} \iff \begin{array}{l}\text{every map } f\colon R^{\triangleright}[S] \to M \text{ of } R^{\triangleright}\text{-modules extends}\\ \text{uniquely along } R^{\triangleright}[S] \to R[S] \text{ to a map } R[S] \to M.\end{array} \tag{1.1}$$

Concretely, if $f$ is a function from $S$ to $M$ (a linear-algebra object with a topology) and $\mu \in R[S]$ is one of the allowed measures, one obtains a well-defined pairing $\int_S f\, d\mu \in M$. This is the sense in which $M$ is "complete": complete enough to integrate functions against the prescribed class of measures, which is precisely the data of the analytic ring. (The equivalence (1.1) is by unwinding the left-adjoint property.)

*Remark* 1.28 (Some monoidal structure not spelled out)*.* $D(R)$ is symmetric monoidal, the completion functor $(-)^{\wedge}_{R}$ is symmetric monoidal, and the base-change (pullback) left adjoints associated to maps of analytic rings are symmetric monoidal. The base-change functor – the left adjoint to the restriction-of-scalars in Definition 1.24 – is the most important functor of the theory; it is a completed tensor product, obtained by tensoring up and then completing (tensoring up on the level of the underlying condensed rings need not preserve $D(-)$). One of the theorems of the theory is that the expected linear-algebra operations (symmetric powers, etc.) carry over automatically; this is not obvious.

## 1.5 Colimits in analytic rings

**Proposition 1.29** (Filtered and sifted colimits)**.** *Filtered colimits (more generally, sifted colimits) of analytic rings are computed naively. For a filtered system $(R_i)_{i\in I}$,*

$$\Big(\varinjlim_{i} R_i\Big)^{\triangleright} = \varinjlim_{i\in I} R_i^{\triangleright}, \qquad \Big(\varinjlim_{i} R_i\Big)[S] = \varinjlim_{i} R_i[S]$$

*on underlying rings and on free modules.*

Pushouts are more interesting; they are the crucial construction, since they are meant to produce the completed tensor products computing geometrically good fiber products. Given maps of analytic rings from a base $K$ to $A$ and $B$, form the pushout, written as a relative tensor product $A \otimes_K B$:

$$\begin{tikzcd} K \arrow[r] \arrow[d] & A \arrow[d] \\ B \arrow[r] & A \otimes_K B. \end{tikzcd}$$

**Proposition 1.30** (Derived category of a pushout)**.** *Abstractly, $D(A \otimes_K B)$ is the full subcategory of $D\big(A^\triangleright \otimes_{K^\triangleright} B^\triangleright\big)$ consisting of those objects whose underlying $A^\triangleright$-module lies in $D(A)$ and whose underlying $B^\triangleright$-module lies in $D(B)$. (Equivalently, it is the relative tensor product $D(A) \otimes_{D(K)} D(B)$ of module categories.)*

*Remark* 1.31 (The underlying ring must be completed)*.* Caution: the pair

$$\big(A^\triangleright \otimes_{K^\triangleright} B^\triangleright,\ D(A \otimes_K B)\big) \tag{1.2}$$

satisfies axioms (1)–(3) of Definition 1.21 but not axiom (4): it is almost an analytic ring, the only failure being that the unit $A^\triangleright \otimes_{K^\triangleright} B^\triangleright$ is not complete. One repairs this by applying the completion, which changes the underlying ring but leaves the category $D(A \otimes_K B)$ unchanged; that the result is again a ring must be proved and is not obvious. In full generality the completion can be hard to understand: abstractly one iterates the completions for $A$ and for $B$, sandwiched, countably many times, and takes a colimit. In practice, however, it can be computed, and it produces the geometrically correct completed tensor products of analytic geometry.

*Remark* 1.32 (Trivial (maximal) analytic ring structure)*.* Every condensed ring $A^\triangleright$ carries the trivial analytic ring structure $D(A) := D(A^\triangleright)$ (everything is complete); write this analytic ring simply as $A^\triangleright$. With this convention the notation in (1.2) is consistent, and every condensed ring is an analytic ring with its maximal structure.

## 1.6 Example: solid analytic rings

We turn to an example, connected to adic spaces (Huber pairs): the solid analytic rings, which encode a non-Archimedean condition. Given an analytic ring $R$, it is illuminating to look at the free modules on profinite sets to see which spaces of measures occur. In the linear setting, the natural object built from the basic profinite set $\mathbb{N} \cup \{\infty\}$ is the following.

**Definition 1.33** (Measures on $\mathbb{N}$)**.** For an analytic ring $R$, set

$$\mathcal{M}_R(\mathbb{N}) := R[\mathbb{N} \cup \{\infty\}]\big/\, R[\{\infty\}],$$

the completed free module on the one-point compactification $\mathbb{N} \cup \{\infty\}$, modulo the free module on the point $\infty$ (which is just $R$). It is the space of measures on $\mathbb{N}$; dually it classifies null sequences in $R$-modules.

*Remark* 1.34 (The ring structure and the variable $T$)*.* Addition on $\mathbb{N}$ induces a ring structure on $\mathcal{M}_R(\mathbb{N})$; it is not hard to show this. Writing the resulting element as $T$, the ring $\mathcal{M}_R(\mathbb{N})$ sits between the two extremes

$$R[T] \longrightarrow \mathcal{M}_R(\mathbb{N}) \longrightarrow R[[T]], \tag{1.3}$$

the polynomial algebra and the power series algebra over $R$ – as one expects of a space of null sequences (really its dual, some summability condition). Geometrically one has the affine line, a version of the formal neighbourhood of the origin, and $\mathcal{M}_R(\mathbb{N})$ somewhere in between. (In the universal case $R = \mathbb{Z}_\square$ this ring lives in degree 0; in general it is a summand and need not lie in the heart, but by base change from the universal case it is a ring in whatever context one wants.)

**Definition 1.35** (Solid analytic ring)**.** An analytic ring $R$ is *solid* if

$$\mathcal{M}_R(\mathbb{N})\big/(T-1) = 0,$$

i.e. if $\mathcal{M}_R(\mathbb{N})$, viewed as sitting inside the affine line via $T$, avoids the point 1; equivalently, multiplication by $T-1$ on $\mathcal{M}_R(\mathbb{N})$ is an isomorphism. Morally, this places $\mathcal{M}_R(\mathbb{N})$ inside the open unit disk, a non-Archimedean condition.

*Remark* 1.36 (Solidity as summability of null sequences). Solidity is equivalent to the existence of a (necessarily unique) measure

$$\mu \in \mathcal{M}_R(\mathbb{N}) \quad \text{with} \quad (T-1)\mu = 1,$$

where $1 \in \mathcal{M}_R(\mathbb{N})$ is the unit, i.e. the sequence $(1, 0, 0, \dots)$. Under (1.3) the measure $\mu$ maps to $\sum_{n\geq 0} T^n$ in the power series ring. Since $\mathcal{M}_R(\mathbb{N})$ classifies null sequences, and $\mu$ pairs with a null sequence by taking it as the coefficients of $\sum_n c_n T^n$ and setting $T = 1$, the interpretation is that *every null sequence is summable* – the classical non-Archimedean condition.

Clausen then stated (without full detail) the fundamental example.

**Theorem 1.37** (The solid analytic ring $\mathbb{Z}_\square$)**.** *There exists a solid analytic ring*

$$\mathbb{Z}_\square = \big(\mathbb{Z},\, D(\mathbb{Z}_\square)\big),$$

*whose underlying condensed ring is the usual integers $\mathbb{Z}$ (with discrete topology), such that an analytic ring $R$ is solid if and only if there exists a (necessarily unique) map of analytic rings $\mathbb{Z}_\square \to R$.*

The category $D(\mathbb{Z}_\square)$ can be understood very explicitly; the following are the basic facts of linear algebra in it.

**Proposition 1.38** (Structure of $D(\mathbb{Z}_\square)$)**.** *Let $S = \varprojlim_n S_n$ be a profinite set.*

1. *The free module is*
$$\mathbb{Z}_\square[S] = \varprojlim_n \mathbb{Z}[S_n] = \varprojlim_n \mathbb{Z}^{S_n}, \tag{1.4}$$
*a countable inverse limit of finite free modules, abstractly isomorphic to a (countably infinite) product $\prod_I \mathbb{Z}$ unless $S$ is finite.*
2. *These products $\prod_I \mathbb{Z}$ are compact projective generators of the heart $D(\mathbb{Z}_\square)^\heartsuit$ and lie in degree 0; consequently $D(\mathbb{Z}_\square)$ is the usual derived category of its heart. They are flat for the completed tensor product $- \otimes_{\mathbb{Z}_\square} -$, and tensor products are computed by the infinite distributive law*
$$\Big(\prod_I \mathbb{Z}\Big) \otimes_{\mathbb{Z}_\square} \Big(\prod_J \mathbb{Z}\Big) = \prod_{I\times J} \mathbb{Z}.$$
3. *The finitely presented objects of $D(\mathbb{Z}_\square)^\heartsuit$ (which generate it under filtered colimits) form an abelian subcategory closed under extensions, and every finitely presented $M$ admits a resolution by products $\prod_I \mathbb{Z}$ of length $\leq 2$ (a complex with three nonzero terms and two nonzero maps).*

*Remark* 1.39 (Lightness is used). The flatness in Proposition 1.38(2) is a place where the light restriction matters: Efimov proved that it fails if one increases the cardinalities of the profinite sets.

*Remark* 1.40 ($\mathbb{Z}_\square$ as a regular ring of dimension 2). The length-$\leq 2$ resolutions of Proposition 1.38(3) say that $\mathbb{Z}_\square$ behaves like a regular ring of dimension 2: whereas $\mathbb{Z}$ is regular of dimension 1, one picks up an extra dimension, attributable to the non-Hausdorff phenomena occurring in solid abelian groups. All told, one gets a very good handle on this category.

*Remark* 1.41 (A single generator). In (1.4) the index set $I$ is countable, and in fact a single generator suffices: the Cantor set $\{0, 1\}^{\mathbb{N}}$ always generates, a general phenomenon.

### 1.7 Closing remarks

Clausen closed with two points raised in discussion. First, an application: the theory yields very general Riemann–Roch theorems in analytic geometry, using these derived categories and the flexibility of the formalism in an essential way. Second, on the choice of framework: one uses light condensed sets in preference to general condensed sets to avoid choosing a cardinality bound; with a strong-limit cardinal one does get compactly generated derived categories, which is psychologically comforting though not so important in practice. Both $D(\mathbb{Z}_\square)$ and its heart are $\aleph_1$-compactly generated, though not compactly generated – a consequence of the light restriction; Clausen thought, but was not certain, that $D(\mathbb{Z}_\square)$ is not dualizable. Finally, the reason for choosing $\mathbb{E}_\infty$ or animated commutative rings, rather than $\mathbb{E}_1$ or $\mathbb{E}_2$, is that for them coproducts coincide with relative tensor products; moving to settings where this fails is a real inconvenience.

## 2 Light Condensed Sets and Abelian Groups

This lecture, the first given by Scholze, develops the foundation on which the whole course rests: *light condensed sets* and, on top of them, light condensed abelian groups. After some motivating remarks on what one might ultimately want analytic geometry to do, Scholze recalls the category of profinite sets and Stone duality, isolates the *light* profinite sets by a countability condition, establishes the two special properties of light profinite sets that make the theory run, defines light condensed sets and compares the construction with topological spaces (and with earlier topos-theoretic proposals of Johnstone and of Escardó–Xu), and finally passes to light condensed abelian groups, whose good homological algebra is the point of the whole exercise.

### 2.1 Motivation: a global perfectoid space

Scholze framed the course by a personal goal of some ten years' standing. Much recent $p$-adic geometry – perfectoid spaces, prismatic cohomology, the Fargues–Fontaine curve and the geometrization of the local Langlands correspondence [FS24] – works beautifully with the $p$-adic numbers, but at the cost of quite fancy $p$-adic geometry. One would hope that similar techniques apply not only to $p$-adic local fields but also to real local fields – something Scholze thinks is now genuinely possible – and then also to the integers, globally. What one would like is some notion of a *global* object: an analytic space, modelled by rings carrying a nontrivial topology, incorporating both non-Archimedean and Archimedean parts, and extremely non-Noetherian, so that none of the usual finiteness properties can be imposed.

*Remark* 2.1 (A vague, possibly misguided idea). Scholze stressed repeatedly that the following is only a vague idea, quite possibly completely misguided. In the perfectoid world one adjoins to a perfect coefficient field a variable together with all of its $p$-power roots; the prototypical example is

$$\mathbb{C}_p\langle T^{\pm 1/p^\infty}\rangle = \mathbb{C}_p\langle T^{\mathbb{Z}[1/p]}\rangle.$$

To imitate this globally one wants to get rid of the fixed prime, so one is led to adjoin a variable with *all* of its rational powers, and – as the analogue at the infinite place – with all of its real powers:

$$K\langle T^{\mathbb{Q}}\rangle \subset K\langle T^{\mathbb{R}}\rangle. \tag{2.1}$$

It is unclear what kind of object $K\langle T^{\mathbb{R}}\rangle$ should be. One can of course treat $\mathbb{R}$ as discrete and simply adjoin the real powers, getting some algebra; but this feels artificial (it presents $\mathbb{R}$ as an uncountable-dimensional $\mathbb{Q}$-vector space and does not improve the situation). One really wants to remember the topology on $\mathbb{R}$; but then one is mixing the $p$-adic-type topology, where these objects sit on a Banach line, with the real topology – so this is definitely *not* a Banach algebra. It is unclear technically what it should be, and still less clear what geometric object should correspond to it. These may not be the right objects at all; the point is to have a language in which one can at least talk about them.

The reaction to these difficulties is the condensed formalism. When Clausen came to Bonn in 2018 he suggested replacing topological spaces by condensed sets, and since then the program has been to replace

$$\text{topological spaces / groups / rings} \quad \rightsquigarrow \quad \text{condensed sets / groups / rings.}$$

This switch resolves many of the foundational problems of analytic geometry – for instance it removes the Noetherian hypotheses pervasive in the theory of adic spaces and in the associated derived categories of complete modules – and it makes objects such as $K\langle T^{\mathbb{R}}\rangle$ at least *exist*, now as condensed rings. The attitude of the course is therefore to develop the most natural possible foundations for analytic geometry based on condensed rings, and to try them out on the known formalisms (allowing, along the way, such variants as overconvergent function theories), in the hope of eventually accommodating more exotic examples of the kind above.

## 2.2 Profinite sets and Stone duality

*Notation* 2.2. Throughout, "ring" means commutative ring, and (following Lecture 1) the word *light* is eventually dropped: from the point where light condensed sets are defined, "condensed set" means light condensed set and "profinite set" means light profinite set.

The starting point is the category ProFin of profinite sets, which can be described in three equivalent ways.

**Proposition 2.3** (Stone duality)**.** *The following categories are equivalent:*

1. Pro(Fin)*, the pro-category of finite sets. Its objects are formal cofiltered inverse limits "$\varprojlim_{i\in I} S_i$" with each $S_i$ finite and $I$ a cofiltered poset (nonempty, and for all $i, j \in I$ there is $k$ with $k \le i, j$); its morphisms are*

$$\mathrm{Hom}\Big(\text{“}\varprojlim_{i\in I} S_i\text{”},\ \text{“}\varprojlim_{j\in J} T_j\text{”}\Big) = \varprojlim_{j}\ \varinjlim_{i}\ \mathrm{Hom}(S_i, T_j).$$

2. *Totally disconnected compact Hausdorff spaces.*
3. *The opposite of the category of Boolean algebras, a Boolean algebra being a commutative ring $R$ with $x^2 = x$ for all $x \in R$.*

*The equivalences go*

$$\text{“}\varprojlim_{i} S_i\text{”} \ \longmapsto\ S = \varprojlim_{i} S_i \ \longmapsto\ \mathrm{Cont}(S, \mathbb{F}_2) = \varinjlim_{i} \mathbb{F}_2^{S_i},$$

*where $S$ is the actual inverse limit (a compact Hausdorff space) and* $\mathrm{Cont}(S, \mathbb{F}_2)$ *is its Boolean algebra of continuous $\mathbb{F}_2$-valued functions; conversely a Boolean algebra $A$ is sent to* $\mathrm{Hom}(A, \mathbb{F}_2) = \mathrm{Spec}\, A$*, its set of points, which is canonically the inverse limit* $\varprojlim_{A_j \subseteq A} \mathrm{Hom}(A_j, \mathbb{F}_2)$ *over the finite subalgebras $A_j$ (as $A = \varinjlim_j A_j$ is their filtered union, so* $\mathrm{Hom}(A, \mathbb{F}_2) = \varprojlim_j \mathrm{Hom}(A_j, \mathbb{F}_2)$ *exhibits* $\mathrm{Spec}\, A$ *as a profinite set).*

Note that in a Boolean algebra $(-1)^2 = -1$, whence $2 = 0$: these are $\mathbb{F}_2$-algebras. Scholze remarked that he will usually think of a profinite set through the presentation of Proposition 2.3(1), as a formal inverse limit of finite sets.

**Two ways to measure the size of a profinite set**

**Definition 2.4** (Size and weight)**.** Let $S = \varprojlim_i S_i$ be a profinite set.

1. The *size* of $S$ is the cardinality of its underlying set, $\kappa := |S|$.
2. The *weight* of $S$ is $\lambda := \big|\mathrm{Cont}(S, \mathbb{F}_2)\big|$, the cardinality of the associated Boolean algebra.
3. $S$ is *light* if $\lambda \leq \omega$, i.e. if its weight is at most countable.

*Remark* 2.5. If $\lambda$ is infinite it equals $|I|$ for the smallest possible index poset $I$ in a presentation $S = \varprojlim_{i \in I} S_i$ (a point, say, can be presented with an arbitrarily large index poset, but there is always a minimal choice). In particular a profinite set is light if and only if it can be written as a *countable* (sequential) inverse limit of finite sets.

**Example 2.6** (Size vs. weight)**.**

1. The one-point compactification $S = \mathbb{N} \cup \{\infty\} = \varprojlim_n \{0, 1, \dots, n, \infty\}$ (transition maps collapsing everything from $n$ onward to $\infty$): $\kappa = \omega$, $\lambda = \omega$. This is the smallest infinite profinite set.
2. The Cantor set $S = \{0,1\}^{\mathbb{N}} = \varprojlim_n \{0,1\}^n$: $\kappa = 2^\omega$, $\lambda = \omega$. It is much bigger than $\mathbb{N} \cup \{\infty\}$ in cardinality of points, but still light.
3. The Stone–Čech compactification $S = \beta\mathbb{N}$: here
$$\mathrm{Cont}(\beta\mathbb{N}, \mathbb{F}_2) = \mathrm{Cont}(\mathbb{N}, \mathbb{F}_2) = \{\text{subsets of } \mathbb{N}\},$$
by the universal property of $\beta\mathbb{N}$ and discreteness of $\mathbb{F}_2$, so $\lambda = 2^\omega$ and $\kappa = 2^{2^\omega}$. This is not light.

Locally compact spaces have two compactifications: the one-point compactification, which is small, and the Stone–Čech compactification, which is huge.

**Proposition 2.7** (Comparison of size and weight)**.** *For any profinite set,*
$$\lambda \leq 2^\kappa, \qquad \kappa \leq 2^\lambda.$$
*Moreover, if $\kappa = \omega$ then $\lambda = \omega$.*

*Proof.* For the first inequality, $\lambda = |\,\mathrm{Cont}(S, \mathbb{F}_2)| \leq |\mathrm{Map}(S, \mathbb{F}_2)| = 2^\kappa$. For the second, writing $A$ for the Boolean algebra (so $|A| = \lambda$), $\kappa = |\,\mathrm{Hom}(A, \mathbb{F}_2)| \leq |\mathrm{Map}(A, \mathbb{F}_2)| = 2^\lambda$. For the last statement, enumerate $S = \{s_0, s_1, s_2, \dots\}$. Inductively choose, for each $n$, a finite quotient $S \twoheadrightarrow S_n$ (compatibly with $S_n \to S_{n-1}$) into which $\{s_0, \dots, s_n\}$ injects. Then $S \to \varprojlim_n S_n$ is injective (any two points are already separated in some finite quotient) and surjective (being a sequential limit of surjections), hence a bijection, exhibiting $S = \varprojlim_n S_n$ with countable weight. ☐

*Remark* 2.8 (Weight versus size). Scholze noted that the same proof in fact shows more: if $\kappa$ is infinite then $\lambda \leq \kappa$. Indeed, since $\kappa^2 = \kappa$, one can choose for each of the $\kappa$ pairs of distinct points of $S$ a finite quotient separating them; these generate a point-separating subalgebra of $\mathrm{Cont}(S, \mathbb{F}_2)$ of size at most $\kappa$, which must then be all of it. Thus in the infinite case the weight never exceeds the size. The bound $\kappa \leq 2^\lambda$, on the other hand, is attained (e.g. for the Cantor set and $\beta\mathbb{N}$, where

$\kappa = 2^\lambda$). Scholze had at first suspected that $\lambda > \kappa$ might be achievable by some complicated example, but this improved bound rules that out – whatever that example computed, it was something else. For us only the elementary facts of Proposition 2.7 are needed.

Specializing Stone duality to the light case gives:

**Proposition 2.9** (Stone duality, light version)**.** *The following categories are equivalent:*

1. $\mathrm{Pro}_{\mathbb{N}}(\mathrm{Fin})$*, the sequential pro-category of finite sets, whose objects are formal inverse limits "*$\varprojlim_{n\in\mathbb{N}} S_n$*" along the integers with each* $S_n$ *finite, and whose morphisms are*

$$\mathrm{Hom}\Big(\text{“}\varprojlim_n S_n\text{”},\ \text{“}\varprojlim_m T_m\text{”}\Big) = \varprojlim_m \varinjlim_n \mathrm{Hom}(S_n, T_m) = \operatorname*{colim}_{f:\mathbb{N}\to\mathbb{N}\text{ str. incr.}} \varprojlim_m \mathrm{Hom}\big(S_{f(m)}, T_m\big). \tag{2.2}$$

2. *Metrizable totally disconnected compact Hausdorff spaces.*

3. *The opposite of the category of countable Boolean algebras.*

The second description of morphisms in (2.2) uses that in a pro-object one may always pass to a cofinal subsequence: a morphism is a strictly increasing reindexing $f$ of the source together with a compatible family of maps $S_{f(m)} \to T_m$. The metrizability in (2) is exactly the reformulation, for compact Hausdorff spaces, of the sequential/compatibility condition in (1).

**Proposition 2.10** (Countable limits)**.** *The category of light profinite sets has all countable limits. Moreover a sequential limit of surjections is surjective.*

Here surjectivity is meant on underlying point sets (equivalently on the underlying compact Hausdorff spaces). In the sequential case this is elementary – one successively lifts – whereas the corresponding statement for all profinite sets requires a rather high-powered compactness argument.

## 2.3 Two properties special to light profinite sets

**Proposition 2.11** (Cantor surjections are universal)**.** *Let $S$ be a light profinite set. Then there exists a surjection from the Cantor set,*

$$\{0,1\}^{\mathbb{N}} \twoheadrightarrow S.$$

*Proof.* A simple induction: write $S = \varprojlim_n S_n$ and at each stage pick a large enough finite power $\{0,1\}^{k_n}$ surjecting onto $S_n$ compatibly. (Exercise.) ☐

The next two properties genuinely single out the light profinite sets; they may hold for some other profinite sets, but that would be hard to characterize, and they are among the key technical reasons for making the switch to the light setting.

**Proposition 2.12** (Open subsets are tame)**.** *Let $S$ be a light profinite set and $U \subseteq S$ open. Then $U$ is a* countable disjoint union *of light profinite sets.*

*Proof.* Write $S = \varprojlim_n S_n$, let $Z = S \setminus U$ be the closed complement, $Z = \varprojlim_n Z_n$ with $Z_n = \mathrm{im}(Z \to S_n) \subseteq S_n$, and $\pi_n : S \to S_n$ the projections. Then

$$U = \bigcup_n \pi_n^{-1}(S_n \setminus Z_n),$$

each $\pi_n^{-1}(S_n \setminus Z_n) \subseteq S$ being clopen (hence itself light profinite), so $U$ is an increasing union of clopens. Writing it as the associated disjoint union,

$$U = \coprod_n \Big( \pi_{n+1}^{-1}(S_{n+1} \setminus Z_{n+1}) \setminus \pi_n^{-1}(S_n \setminus Z_n) \Big),$$

a countable disjoint union of light profinite sets. $\square$

*Remark* 2.13 (Wildness in the general case)*.* This is to be contrasted with the general (non-light) case: there exists a profinite set $S$ with an open subset $U \subseteq S$ that is *not* a disjoint union of profinite sets, and indeed with $H^i(U, \mathbb{Z}) \neq 0$ for some $i > 0$. Disjoint unions take cohomology to products, and profinite sets are totally split (no nontrivial covers, so higher cohomology vanishes), so such higher cohomology on $U$ witnesses the failure. In the light case, by contrast, Proposition 2.12 gives complete control. In discussion an audience member gave concrete non-light counterexamples to both this proposition and Proposition 2.14, built from the ordinal $[0, \omega_1]$ (a profinite set): its complement of the top point $\omega_1$ is an open subset that is not a disjoint union of clopens, and a suitable closed subset of $[0, \omega_1]^2$ is not a retract of the product. Scholze added that one can also produce opens with nonvanishing higher cohomology, e.g. from $\beta\mathbb{N}$ by removing boundary points; he thought the first infinite profinite set with a removed limit point also works but was not certain in that case.

**Proposition 2.14** (Injectivity)**.** *Let $S$ be a light profinite set. Then $S$ is an injective object in* Pro(Fin)*: for every injection $Z \hookrightarrow X$ of profinite sets, every map $Z \to S$ extends to $X$.*

That is, one can always fill in the dashed arrow in

$$\begin{tikzcd} Z \arrow[r] \arrow[d, hook] & S \\ X \arrow[ur, dashed, "\exists"'] & \end{tikzcd}$$

*Proof.* First take $S = \{0, 1\} = \mathbb{F}_2$. Then the claim is that $\mathrm{Cont}(X, \mathbb{F}_2) \to \mathrm{Cont}(Z, \mathbb{F}_2)$ is surjective, i.e. any clopen subset of $Z$ extends to a clopen subset of $X$. In general write $S = \varprojlim_n S_n$ with all $S_{n+1} \to S_n$ surjective and induct on $n$: having extended the map to $S_n$, decompose the situation into the disjoint union over the fibres of $S_{n+1} \to S_n$, reducing to the case $S_n = *$ and hence to $S_{n+1}$ finite, which follows from the $\mathbb{F}_2$-case. $\square$

*Remark* 2.15*.* These light profinite sets do not exhaust the injective objects of Pro(Fin): injective objects are closed under products, so any product of light profinite sets is again injective, and there are much larger ones. It is an instructive exercise to see exactly where the inductive argument above breaks down for a general (non-sequential) profinite set. Scholze also raised the edge case: if $Z$ is empty but $X$ is not, one must know $S \neq \emptyset$ for the extension to exist.

## 2.4 Light condensed sets

**Definition 2.16** (Light condensed set)**.** A *light condensed set* is a sheaf on the category of light profinite sets for the Grothendieck topology generated by

1. finite disjoint unions, and
2. all surjective maps.

Equivalently, it is a functor $X\colon \mathrm{Pro}_{\mathbb{N}}(\mathrm{Fin})^{\mathrm{op}} \to \mathrm{Sets}$, $S \mapsto X(S)$, such that

(0) $X(\emptyset) = *$;

(1) $X(S_1 \amalg S_2) \xrightarrow{\sim} X(S_1) \times X(S_2)$;

(2) for every surjection $T \xrightarrow{f} S$,

$$X(S) \xrightarrow{\sim} \mathrm{eq}\Big(X(T) \overset{p_1^*}{\underset{p_2^*}{\rightrightarrows}} X(T \times_S T)\Big),$$

where $p_1, p_2 \colon T \times_S T \to T$ are the two projections.

Condition (2) is exactly what the sheaf condition for the surjective cover $f$ unwinds to: a map $S \to X$ is determined by its restriction to $T$, and the maps $T \to X$ that descend to $S$ are those whose two pullbacks to $T \times_S T$ agree (they "agree on fibres" of $f$).

**Example 2.17** (Topological spaces)**.** Any topological space $A$ gives a light condensed set

$$\underline{A}\colon S \longmapsto \mathrm{Cont}(S, A),$$

the continuous maps from the compact Hausdorff space $S$ into $A$, with all its functoriality. That condition (2) holds is not a complete tautology: it says that a set-theoretic map $S \to A$ which becomes continuous after restriction along a surjection $T \twoheadrightarrow S$ was already continuous – equivalently, that $S$ carries the quotient topology from $T$, which holds because surjections of compact Hausdorff spaces are quotient maps. Evaluating,

$$\underline{A}(*) = A \text{ as a set}, \qquad \underline{A}(\mathbb{N} \cup \{\infty\}) = \{\text{convergent sequences in } A, \text{ with a choice of limit point}\},$$

and $\underline{A}(\{0,1\}^{\mathbb{N}})$ (functions on the Cantor set) carries an action of $\mathrm{End}(\{0,1\}^{\mathbb{N}})$.

For a general light condensed set $X$ one thinks of $X(*)$ as the underlying set and of $X(\mathbb{N} \cup \{\infty\})$ as the "convergent sequences in $X$", where giving such a convergent sequence includes giving a witness for its limit (and there may be several limit points).

*Remark* 2.18 (A light condensed set is almost a monoid action). A light condensed set $X$ is completely determined by the single set $X(\{0,1\}^{\mathbb{N}})$ together with the action on it of the abstract monoid $\mathrm{End}(\{0,1\}^{\mathbb{N}})$:

$$\underbrace{X(\{0,1\}^{\mathbb{N}})}_{\text{abstract set}} \circlearrowleft \underbrace{\mathrm{End}(\{0,1\}^{\mathbb{N}})}_{\text{abstract monoid}}.$$

Indeed, every profinite set is covered by a Cantor set (Proposition 2.11), and the relevant fibre products are again so covered, so all values $X(S)$ are recovered from $X(\{0,1\}^{\mathbb{N}})$ via the sheaf condition, the transition maps being induced by endomorphisms of the Cantor set. One could therefore describe light condensed sets as certain sets with an action of this (very large) abstract monoid, but Scholze regards this as a strictly worse way to think about them: the functor from light condensed sets to $\mathrm{End}(\{0,1\}^{\mathbb{N}})$-sets is fully faithful but not essentially surjective – the covering (sheaf) condition cuts out which actions arise, and that condition is not visible from the monoid alone. The point of the remark is that a light condensed set is nevertheless a very concrete, almost combinatorial object.

### Topological realization

**Proposition 2.19** (Left adjoint to $A \mapsto \underline{A}$)**.** *The functor* Top $\to$ CondSet$^{\text{light}}$*,* $A \mapsto \underline{A}$*, has a left adjoint* $X \mapsto X(*)_{\text{top}}$*, where* $X(*)_{\text{top}}$ *is the underlying set* $X(*)$ *equipped with the quotient topology from*

$$\coprod_{\substack{S \text{ light profinite} \\ \alpha \in X(S)}} S \longrightarrow X(*), \tag{2.3}$$

*the map sending a point of the copy of* $S$ *indexed by* $\alpha \in X(S)$ *to the corresponding point of* $X(*)$*. The resulting space* $X(*)_{\text{top}}$ *is metrizably compactly generated.*

Here a topological space is *metrizably compactly generated* if continuity of a map out of it may be tested on maps into it from metrizable compact Hausdorff spaces; by construction of the quotient topology one only has to test continuity on the (metrizable) Cantor sets appearing in (2.3).

**Corollary 2.20** (Topological spaces as condensed sets)**.** *The functor* $A \mapsto \underline{A}$ *restricts to a fully faithful embedding*

$$\{\textit{metrizably compactly generated topological spaces}\} \hookrightarrow \text{CondSet}^{\text{light}},$$

*and for any metrizably compactly generated space* $A$ *the counit is an isomorphism,* $\underline{A}(*)_{\text{top}} \xrightarrow{\sim} A$*.*

This is a very weak condition: virtually all topological spaces arising in practice – in particular all the topological modules, Banach spaces and the like used in condensed functional analysis – are metrizably compactly generated. The condition is even implied by being metrizable, or more generally sequential.

*Remark* 2.21 (Earlier topos-theoretic proposals). Two related constructions have appeared before.

1. Johnstone's "topological topos" [Joh79] is built on just the single test object $\mathbb{N} \cup \{\infty\}$ (the sequence space), with the *canonical* topology (the finest for which all representable presheaves are sheaves). This topology is not finitary: it admits genuinely infinite covers with no finite subcover. Much of what Johnstone does goes through, but the non-finitary topology leads to bad algebraic behaviour.
2. Escardó–Xu [EX16] use light profinite sets but with a finitary topology, taking only finite disjoint unions as covers – *not* all surjective maps. Allowing all surjective maps as covers is, however, important for the good algebraic properties of the theory (as will be seen for abelian groups below), and this is precisely the point at which the present setup differs from theirs.

## 2.5 Light condensed abelian groups

Piling abelian-group structure on top gives the category CondAb$^{\text{light}}$ of light condensed abelian groups: sheaves of abelian groups on the site of light profinite sets.

**Proposition 2.22** (Grothendieck abelian category)**.** *For sheaves on any site, sheaves of abelian groups form a Grothendieck abelian category: all limits and colimits exist and filtered colimits are exact. In particular* CondAb$^{\text{light}}$ *is a Grothendieck abelian category.*

The whole point of passing to condensed abelian groups, as recalled in Lecture 1, is that dense inclusions – which make categories of topological abelian groups fail to be abelian – are now handled correctly. Scholze illustrated this with the two maps

$$\underline{\mathbb{Q}} \hookrightarrow \underline{\mathbb{R}}, \qquad \underline{\mathbb{R}_{\text{disc}}} \to \underline{\mathbb{R}},$$

where $\mathbb{Q}$ carries its usual topology and $\mathbb{R}_{\text{disc}}$ is $\mathbb{R}$ with the discrete topology. Both are perfectly nice maps of topological abelian groups whose cokernels are problematic topologically; condensed mathematics computes them cleanly.

**Example 2.23** (Two cokernels)**.** For the first map, the cokernel $\underline{\mathbb{R}}/\underline{\mathbb{Q}}$ has

$$\big(\underline{\mathbb{R}}/\underline{\mathbb{Q}}\big)(*) = \mathbb{R}/\mathbb{Q}, \qquad \big(\underline{\mathbb{R}}/\underline{\mathbb{Q}}\big)(S) = \mathrm{Cont}(S,\mathbb{R})\big/\,\mathrm{Cont}(S,\mathbb{Q}),$$

and one can prove that this naive quotient presheaf is already a sheaf, so no sheafification is needed: the value at $*$ is the plain quotient, while the values at general $S$ remember the topology (continuous $\mathbb{R}$-valued functions modulo continuous $\mathbb{Q}$-valued ones).

For the second map the phenomenon is more dramatic. The underlying set is

$$\big(\underline{\mathbb{R}}/\underline{\mathbb{R}_{\text{disc}}}\big)(*) = \mathbb{R}/\mathbb{R} = 0,$$

yet the cokernel is not zero: on a general profinite set

$$\big(\underline{\mathbb{R}}/\underline{\mathbb{R}_{\text{disc}}}\big)(S) = \mathrm{Cont}(S,\mathbb{R})\big/\,\mathrm{Cont}(S,\mathbb{R}_{\text{disc}}), \tag{2.4}$$

where $\mathrm{Cont}(S, \mathbb{R}_{\text{disc}})$ is the *locally constant* real functions. For $S$ the Cantor set, embedded in $\mathbb{R}$ in the standard way, the embedding is continuous but not locally constant, so (2.4) is very much nonzero. The cokernel is controlled by exactly these "non-locally-constant" real-valued functions on general light profinite sets.

Scholze closed by stating (with proofs deferred to the next lecture) the good categorical properties that motivate working with *light* condensed abelian groups specifically.

**Theorem 2.24** (Properties of $\mathrm{CondAb}^{\text{light}}$)**.** *The category* $\mathrm{CondAb}^{\text{light}}$ *is a convenient homological setting. In particular:*

1. *It has all countable limits, and countable products are exact; a form of Grothendieck's axiom* AB6 *holds for countable products (products commuting suitably with filtered colimits). This is weaker than for all condensed abelian groups, where* all *products are exact; here only the countable ones are.*
2. *The free light condensed abelian group on the convergent sequence, $\mathbb{Z}[\mathbb{N} \cup \{\infty\}]$, is* internally projective*.*

The forgetful functor $\mathrm{CondAb}^{\text{light}} \to \mathrm{CondSet}^{\text{light}}$ has a left adjoint $X \mapsto \mathbb{Z}[X]$, the free light condensed abelian group; applied to a light profinite set $S$ it gives the free object $\mathbb{Z}[S]$.

*Remark* 2.25 (Internal projectivity is special to the light setting)*.* Scholze emphasized Theorem 2.24(2) as one of the main reasons for the switch to the light setting. The category $\mathrm{CondAb}^{\text{light}}$ no longer has enough compact projective objects. Nevertheless $\mathbb{Z}[\mathbb{N} \cup \{\infty\}]$ – which would *not* be projective among all condensed abelian groups – happens to be projective, and even internally projective, in $\mathrm{CondAb}^{\text{light}}$. This is the only variant of condensed abelian groups in which Scholze is aware of any nontrivial internally projective object: in all condensed abelian groups there are plenty of projectives, but (apart from trivial cases such as $\mathbb{Z}$) none are internally projective, and those that exist come from extremely disconnected sets and are impractically large. Since the free group on a convergent sequence is a very basic building block of the theory, it is particularly valuable that in the light setting it enjoys these good categorical properties. Scholze added that the failure of *all* products to be exact was one reason he was at first hesitant to make the switch.

# 3 Compactness, Free Condensed Abelian Groups, and Cohomology

Scholze's second lecture continues the foundations. It opens by recalling the site of light profinite sets and the crucial feature of its Grothendieck topology – that sequential limits of covers are again covers – and by revisiting the topological realization functor, improving the statement of Section 2 to the identification of *metrizably compactly generated* with *sequential*. It then isolates the topos-theoretic notions of *quasicompact* and *quasiseparated* object, checks that they recover the expected classes of spaces, and uses them to explain why the formalism is forced to use totally disconnected test objects and a finitary topology with the Cantor set available. The bulk of the lecture develops light condensed abelian groups: the tensor product and the free-abelian-group functor (with the running example $\mathbb{Z}[\mathbb{R}]$), and the three good properties of the light setting – exactness of countable products, exactness of sequential limits of surjections, and internal projectivity of the free group on a convergent sequence – the last proved in detail. The lecture closes with a comparison to all condensed abelian groups, the AB axioms, and the definition of cohomology in the condensed topos.

## 3.1 Recollections: the site of light profinite sets

Recall from Section 2 the category of light profinite sets, described equivalently as

$$\begin{aligned}\mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin}) &\simeq \{\text{metrizable totally disconnected compact Hausdorff spaces}\}\\ &= \{\text{countable Boolean algebras}\}^{\mathrm{op}},\end{aligned}$$

i.e. as sequential (countable) pro-systems of finite sets, as metrizable totally disconnected compact Hausdorff spaces, or dually as countable Boolean algebras. We equip this category with the Grothendieck topology whose covers are generated by two families:

- finite disjoint unions, and
- *all* surjective maps.

As stressed in Section 2, this has the important consequence, which Scholze underlined again, that

$$\text{sequential limits of covers are still covers} \tag{3.1}$$

(cf. Proposition 2.10: a sequential limit of surjections of light profinite sets is surjective). This property is what will make the resulting homological algebra of light condensed abelian groups behave well, and Scholze returns to it repeatedly below.

Light condensed sets are the sheaves for this topology,

$$\mathrm{CondSet}^{\mathrm{light}} = \mathrm{Shv}\big(\mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})\big).$$

Whenever one has sheaves on a subcanonical site there is a Yoneda embedding of the generating category, so the light profinite sets sit inside $\mathrm{CondSet}^{\mathrm{light}}$ via

$$S \longmapsto \big(T \mapsto \mathrm{Hom}_{\mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})}(T,S) = \mathrm{Cont}(T,S)\big),$$

sending a profinite set $S$ to the sheaf of (continuous) maps into it. A general feature is that the image of the Yoneda embedding generates $\mathrm{CondSet}^{\mathrm{light}}$ under colimits: every light condensed set is built from light profinite sets by a gluing procedure.

There is also the comparison to topological spaces of Section 2: a topological space $A$ gives the light condensed set

$$\underline{A}\colon S \longmapsto \mathrm{Cont}(S,A),$$

and this functor $A \mapsto \underline{A}$ has a left adjoint $X \mapsto X(*)_{\mathrm{top}}$, the underlying set $X(*)$ equipped with the quotient topology from the surjection $\coprod_{\alpha \in X(\mathrm{Cantor})}(\text{Cantor set}) \to X(*)$ (Proposition 2.19).

### 3.2 Topological realization: sequential spaces

Scholze improved on one point from Section 2. In forming the quotient topology on $X(*)_{\mathrm{top}}$ one may use, in place of Cantor sets, the convergent sequences: crediting a remark of Morita after the previous lecture, the coproduct

$$\coprod_{\beta \in X(\mathbb{N} \cup \{\infty\})} (\mathbb{N} \cup \{\infty\}) \longrightarrow X(*)_{\mathrm{top}}$$

is also a quotient map of topological spaces. The reason is that the convergence of the Cantor set can be detected by sequences; more precisely, mapping all convergent sequences to the Cantor set is still a quotient map, so that in Top one may write

$$\text{Cantor set} \simeq \operatorname{colim}\,(\text{countable closed subsets}) \tag{3.2}$$

as a colimit over its countable closed subsets, e.g. finite unions of convergent sequences. Scholze emphasized that (3.2) holds *in* Top *but not in condensed sets* – a point he returns to below.

Consequently the corresponding topological space is determined by its convergent sequences alone, and the notion of being *metrizably compactly generated* from Section 2 is the same as being *sequential*: continuity may be checked on convergent sequences (a map is continuous iff it sends convergent sequences to convergent sequences).

**Proposition 3.1** (Sequential spaces embed)**.** *The functor $A \mapsto \underline{A}$ restricts to a fully faithful embedding*

$$\{\textit{sequential topological spaces}\} \hookrightarrow \mathrm{CondSet}^{\mathrm{light}}.$$

*Remark* 3.2 (Sequential versus first-countable). Scholze was unsure off-hand how "sequential" compares with "first countable"; in fact every first-countable space is sequential and the containment is strict. (The large discrete space floated in the discussion does not separate the two, since discrete spaces are always first countable.) He noted that in an earlier version of the theory the relevant condition was first-countability with all points closed; here one no longer needs all points closed, having dropped the set-theoretic bounds.

### 3.3 Quasicompact and quasiseparated objects

Scholze now recalled why the Cantor set is allowed as a test object, through two general topos-theoretic notions. In any topos – sheaves on any site – there is an intrinsic notion of an object being "compact" and of being "Hausdorff".

**Definition 3.3** (Quasicompact, quasiseparated)**.** Let $X$ be a light condensed set.

1. $X$ is *quasicompact* (qc) if any cover $\coprod_i X_i \twoheadrightarrow X$ admits a finite subcover. For light condensed sets this amounts to: there exists a surjection from the Cantor set,

$$\{0,1\}^{\mathbb{N}} \twoheadrightarrow X,$$

or $X = \emptyset$.

2. $X$ is *quasiseparated* (qs) if for all quasicompact $Y, Z \to X$ the fiber product $Y \times_X Z$ is quasicompact. For light condensed sets this amounts to: for all pairs of maps $\{0,1\}^{\mathbb{N}} \to X$ from the Cantor set,
$$\{0,1\}^{\mathbb{N}} \times_X \{0,1\}^{\mathbb{N}} \quad \text{is quasicompact.} \tag{3.3}$$

If both hold, $X$ is called *qcqs*.

*Remark* 3.4 (Terminology and provenance). The condition of being quasiseparated is Hausdorffness at the level of the topos. It was introduced by Grothendieck in SGA 4; using the algebraic-geometry convention where "separated" means Hausdorff, and this being a general topos-theoretic notion rather than a specific separatedness in algebraic geometry, he called it *quasiseparated*. The intuition for (3.3): a map $\{0,1\}^{\mathbb{N}} \to X$ need not be injective, so it factors over a quotient of the Cantor set; quasiseparatedness declares that this is the quotient by a *closed* equivalence relation on the Cantor set, one way to express Hausdorffness.

For a finitary topology the quasicompact objects are exactly those covered by a finite union of generating objects, such as $\{0,1\}^{\mathbb{N}}$. Scholze used this to explain the design of the site:

*Remark* 3.5 (Why the Cantor set must be a test object). If one allowed only the convergent sequence $\mathbb{N} \cup \{\infty\}$ in the test category, then the quasicompact objects would all be countable, and in particular the Cantor set would *not* be quasicompact – one could only access it as a huge colimit of smaller objects, as in (3.2). Since the Cantor set is morally a basic compact object, this would destroy the intuition attached to quasicompactness. So the abstract topos-theoretic notion mirrors closely what a compact space should be, and this forces the topology to be finitary and forces the Cantor set into the formalism.

The abstract notions recover exactly the expected classes of spaces.

**Proposition 3.6** (qcqs objects are metrizable compact Hausdorff spaces)**.** *Under $A \mapsto \underline{A}$ there is an equivalence*

$$\{\textit{qcqs light condensed sets}\} \xrightarrow{\sim} \{\textit{metrizable compact Hausdorff spaces}\}.$$

*If one removes "light" on the left, one removes "metrizable" on the right.*

**Example 3.7** (The interval and its Cantor covering)**.** The unit interval $[0,1]$ is a metrizable compact Hausdorff space, hence quasicompact as a condensed set, so there must be a surjection from the Cantor set onto it – and indeed a canonical one. A sequence $(a_0, a_1, a_2, \dots)$ of 0's and 1's is sent to the real number with binary expansion $0.a_0a_1a_2\cdots$, giving

$$\{0,1\}^{\mathbb{N}} \twoheadrightarrow [0,1],$$

which is surjective since every point of the interval admits such a representation. One recovers $[0,1]$ as the quotient by the closed equivalence relation identifying the two binary expansions of a dyadic rational, e.g. $0.0111\cdots = 0.1000\cdots$. One genuinely needs something as large as the Cantor set to surject onto $[0,1]$.

Dropping the compactness but keeping the Hausdorff condition, the quasiseparated objects also have a standard description.

**Proposition 3.8** (Quasiseparated light condensed sets)**.** *There is an equivalence*

$$\{\textit{qs light condensed sets}\} \simeq \mathrm{Ind}_{\mathrm{inj}}\big(\{\textit{metrizable compact Hausdorff spaces}\}\big),$$

*the ind-category of metrizable compact Hausdorff spaces along injective transition maps. Concretely the objects are filtered systems $\{X_i\}_{i\in I}$ (I a filtered poset) of metrizable compact Hausdorff spaces in which all transition maps are injective – hence automatically closed immersions – and one should think of such an object as a rising union of its quasicompact subspaces $X_i$.*

*Remark* 3.9 (Comparison with compactly generated weak Hausdorff spaces). Inside the quasiseparated light condensed sets one finds the *compactly generated weak Hausdorff spaces* ("CGWH") of algebraic topology, which by the discussion above are exactly the sequential ones. The two categories are nevertheless genuinely different. In Top (or CGWH) the Cantor set is the colimit (3.2) of its countable closed subsets, e.g. finite unions of convergent sequences; but this colimit identity *fails* in quasiseparated light condensed sets. There the Cantor set is a single quasicompact object, whereas the colimit of countable closed subsets is a genuine (formal) filtered colimit – so the two are very different as condensed sets, the latter not even being quasicompact. Scholze noted that the discrepancy only appears once the indexing colimit is large: restricting to countable systems, the two notions agree. Removing "light" throughout removes "metrizable".

### 3.4 Light condensed abelian groups

We now put algebra on top of the formalism. Light condensed abelian groups admit the two equivalent descriptions

$$\mathrm{CondAb}^{\mathrm{light}} = \{\text{abelian group objects in } \mathrm{CondSet}^{\mathrm{light}}\} = \mathrm{Shv}\big(\mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin}),\, \mathrm{Ab}\big),$$

either abelian group objects in light condensed sets, or sheaves of abelian groups on the site; the same remark applies to any algebraic structure (e.g. condensed rings are ring objects in $\mathrm{CondSet}^{\mathrm{light}}$, equivalently sheaves of rings).

From the general theory of sheaves of abelian groups on a site (cf. Proposition 2.22), $\mathrm{CondAb}^{\mathrm{light}}$ is a Grothendieck abelian category: all limits and colimits exist and filtered colimits are exact. It also carries a symmetric monoidal tensor product, with the following features.

*Notation* 3.10 (Unit and tensor product). The unit object is $\underline{\mathbb{Z}}$, the (discrete) condensed abelian group associated to $\mathbb{Z}$, with

$$\underline{\mathbb{Z}}\colon S \longmapsto \mathrm{Cont}(S, \mathbb{Z}) = \{\text{locally constant maps } S \to \mathbb{Z}\}.$$

The tensor product $M \otimes N$ is the sheafification of the presheaf

$$S \longmapsto M(S) \otimes N(S).$$

The forgetful functor $\mathrm{CondAb}^{\mathrm{light}} \to \mathrm{CondSet}^{\mathrm{light}}$ has a left adjoint, the free condensed abelian group functor.

**Definition 3.11** (Free condensed abelian group)**.** The free condensed abelian group on a light condensed set $X$ is $\mathbb{Z}[X]$, the sheafification of the presheaf

$$S \longmapsto \mathbb{Z}[X(S)],$$

the levelwise free abelian group. Its underlying group is the ordinary free abelian group $\mathbb{Z}[X(*)]$ on the underlying set; the construction puts a condensed ("topological") structure on it.

This is a structure one rarely contemplates for topological spaces – one does not usually form the free abelian group on a topological space – but here it exists and is completely fundamental.

**Example 3.12** (The free group on the reals)**.** Take $X = \mathbb{R}$ (from now on the underline is usually dropped, everything having implicitly become a condensed set) and form $\mathbb{Z}[\mathbb{R}]$. Its elements are finite formal sums

$$\sum_{x\in\mathbb{R}} n_x[x], \qquad n_x \in \mathbb{Z} \text{ almost all } 0,$$

thought of as finite integer combinations of Dirac measures, but now carrying a topology recording that a point $x$ may move continuously: $[x]-[y]$ is a nonzero element in general, yet it collapses to 0 as $y \to x$. It is a little awkward to describe the open subsets of $\mathbb{Z}[\mathbb{R}]$ as a topological group, but as a condensed set it is transparent. Since the free construction commutes with colimits,

$$\mathbb{Z}[\mathbb{R}] = \operatorname*{colim}_c \ \mathbb{Z}\big[[-c,c]\big],$$

the colimit over intervals $[-c,c]$, now with compact pieces, and each $\mathbb{Z}[I]$ ($I$ a compact interval) is the rising union

$$\mathbb{Z}[I] = \bigcup_{n\in\mathbb{N}} \mathbb{Z}[I]_{\le n}, \qquad \mathbb{Z}[I]_{\le n} = \Big\{ \textstyle\sum_x n_x[x] \ : \ \sum_x |n_x| \le n \Big\},$$

each $\mathbb{Z}[I]_{\le n}$ being compact Hausdorff. Indeed, whenever $K$ is a compact Hausdorff space, the integer combinations of points of $K$ with $\sum |n_x| \le n$ carry a canonical compact Hausdorff topology; the free construction produces it automatically. Passing from a profinite set into $\mathbb{Z}[\mathbb{R}]$, one always lands in one of these compact subsets.

*Remark* 3.13 (Nontrivial identifications require surjective covers)*.* The subtle content of Example 3.12 is precisely the identification collapsing $[x]-[y]$ to 0 as $y \to x$. Making this come out correctly uses that the Grothendieck topology allows not only finite disjoint unions but also surjective maps as covers; with the smaller topology the free construction would give the wrong answer.

*Remark* 3.14 (Free groups on group objects are rings; real powers)*.* If $X$ already carries a group structure, then $\mathbb{Z}[X]$ acquires a ring structure, the multiplication coming from the group law on $X$; so $\mathbb{Z}[\mathbb{R}]$ is naturally a condensed ring. Regarding $\mathbb{R}$ multiplicatively as the group of exponents of a formal variable $T$, this ring is $\mathbb{Z}[T^{\mathbb{R}}]$: it adjoins to $\mathbb{Z}$ a variable together with all of its real powers. This is a first incarnation of the object $K\langle T^{\mathbb{R}}\rangle$ that motivated Section 2 (see (2.1)), where its very nature was deliberately left open – only the wish that the condensed formalism should make such objects at least *exist* as condensed rings; the free construction now produces one for free. Editorially: this is not yet the completed or topologized version one ultimately wants.

### 3.5 Three properties special to the light setting

We turn to features specific to *light* condensed abelian groups.

**Theorem 3.15** (Good properties of $\mathrm{CondAb}^{\mathrm{light}}$)**.** *In* $\mathrm{CondAb}^{\mathrm{light}}$*:*

1. *countable products are exact;*
2. *sequential limits of surjective maps are surjective; and*
3. *the free group* $\mathbb{Z}[\mathbb{N}\cup\{\infty\}]$ *on the convergent sequence is* internally projective*.*

*Properties* (1) *and* (2) *in fact hold in all condensed abelian groups (there* all *products are exact); property* (3) *is special to the light setting.*

Property (2) is what one repeatedly needs in functional analysis: sequential limits of surjections – e.g. a Fréchet space as a sequential limit of Banach spaces – should behave well.

**Definition 3.16** (Internal projectivity)**.** Recall that in any Grothendieck abelian category $P$ is *projective* if $\mathrm{Ext}^i(P, -) = 0$ for $i > 0$. When the category is symmetric monoidal one has an *internal* Ext: for $M, N \in \mathrm{CondAb}^{\mathrm{light}}$,

$$\underline{\mathrm{Ext}}^i(M, N) := \text{sheafification of} \quad S \longmapsto \mathrm{Ext}^i\big(M \otimes \mathbb{Z}[S],\, N\big).$$

An object $P$ is *internally projective* if $\underline{\mathrm{Ext}}^i(P, -) = 0$ for $i > 0$.

*Proof of* (1) *and* (2)*. Reduction of* (1) *to* (2)*.* For a countable product of maps, products of injections are injections and preserve exactness on the left; the only point needing proof is that a countable product of surjections $f_n\colon M_n \twoheadrightarrow N_n$ is surjective. Write the map $\prod_n M_n \to \prod_n N_n$ as the sequential limit

$$\prod_n M_n \to \prod_n N_n \;=\; \varprojlim_m \Big( \prod_{n \le m} M_n \times \prod_{n > m} N_n \;\twoheadrightarrow\; \prod_n N_n \Big).$$

Each map in the limit is surjective: finite products are finite direct sums, which preserve surjections and are exact, so one can lift the first $m$ coordinates while leaving the rest. Thus $\prod f_n$ is a sequential limit of surjections, and (2) finishes the argument.

*Proof of* (2)*.* Let $M_\infty = \varprojlim(\cdots \to M_2 \to M_1 \twoheadrightarrow M_0)$ be a sequential limit of surjections of light condensed abelian groups; we show $M_\infty \to M_0$ is surjective in the sheaf sense. Surjectivity means: for every light profinite set $S$ with a map $S \to M_0$, there is a surjection $S_\infty \twoheadrightarrow S$ from a light profinite set and a lift $S_\infty \to M_\infty$. Since $M_1 \twoheadrightarrow M_0$ is surjective, there is a light profinite set $S_1 \twoheadrightarrow S$ with a lift $S_1 \to M_1$; inductively there is $S_{i+1} \twoheadrightarrow S_i$ with a lift to $M_{i+1}$. Put

$$S_\infty = \varprojlim_i S_i \;\in\; \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin}).$$

This is a countable limit of surjections, hence still surjective onto $S$ (Proposition 2.10), so $S_\infty \twoheadrightarrow S$ is a cover; and the compatible lifts assemble to a map $S_\infty \to M_\infty$. This is exactly the required lift, and (3.1) – that sequential limits of covers are covers – is the crux. □

*Remark* 3.17 (Why totally disconnected building blocks)*.* Property (2), i.e. that countable limits of covers are covers, is exactly what forces the test objects to be totally disconnected. Suppose one tried instead a "smooth" site of manifolds with the usual topology of open covers. Cover a small interval by two subintervals, cover each of those by two smaller ones, and continue; in the limit the intervals shrink to a cloud of dust – a Cantor set – with no reasonable smooth structure. So iterating covers drives one to totally disconnected compact Hausdorff spaces as building blocks, and among these Scholze takes the metrizable ones with all surjections as covers.

*Remark* 3.18 (Provenance: the pro-étale topology)*.* Scholze noted, as a historical aside, that this prototype is not new: the condensed formalism grows out of the pro-étale topology for schemes [BS14], where the very wish was that limits of surjective maps remain surjective, so that certain sheaves arising as inverse limits would be well behaved. Allowing countable (or all) limits of covers to remain covers is the origin of the whole theory. He added that any surjection of light profinite sets can be written as a sequential limit of open covers – for instance $\mathbb{N} \cup \{\infty\}$ is covered by two copies of $\mathbb{N} \cup \{\infty\}$ meeting at $\infty$ – so if one wants open covers and their sequential limits, one must allow all surjective maps as covers.

## 3.6 Internal projectivity of the free group on a convergent sequence

Before the proof, a warning: the phenomenon appears only after passing to abelian groups.

*Remark* 3.19 ($\mathbb{N} \cup \{\infty\}$ is not projective as a condensed set). The convergent sequence $\mathbb{N} \cup \{\infty\}$ is *not* projective in $\mathrm{CondSet}^{\mathrm{light}}$. Take the surjection of light profinite sets

$$S' = (2\mathbb{N} \cup \{\infty\}) \sqcup \big((2\mathbb{N}+1) \cup \{\infty\}\big) \twoheadrightarrow S = \mathbb{N} \cup \{\infty\},$$

a disjoint union of two convergent sequences (the evens, the odds), each with its own limit point, both mapping to $\infty$. The identity map $\mathbb{N} \cup \{\infty\} \to S$ admits no lift to $S'$: an integer has a unique preimage, so a lift would send the even and odd subsequences to the two *different* limit points of $S'$, and the resulting map does not converge.

$$\begin{tikzcd} & S' = (2\mathbb{N} \cup \{\infty\}) \sqcup \big((2\mathbb{N}+1) \cup \{\infty\}\big) \arrow[d, two heads] \\ \mathbb{N} \cup \{\infty\} \arrow[ur, dashed, "\nexists"] \arrow[r] & S = \mathbb{N} \cup \{\infty\} \end{tikzcd}$$

Miraculously, once one passes to abelian groups this obstruction disappears: any null sequence *can* be lifted, which is the content of Theorem 3.15(3).

It is convenient to prove the statement for a direct factor of $\mathbb{Z}[\mathbb{N} \cup \{\infty\}]$.

**Definition 3.20** (The free group on a null sequence)**.** Set

$$M := \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/\mathbb{Z} \cdot [\infty],$$

the free condensed abelian group on the pointed convergent sequence (the limit point sent to 0). It is the free group on a *null sequence* and classifies null sequences: maps out of $M$ are null sequences in the target. Since $\{\infty\} \hookrightarrow \mathbb{N} \cup \{\infty\}$ splits (project to a point), $M$ is a direct factor of $\mathbb{Z}[\mathbb{N} \cup \{\infty\}]$, the complementary factor $\mathbb{Z}$ being projective; so it suffices to prove $M$ internally projective.

*Proof of Theorem 3.15(3).* We prove ordinary projectivity of $M$ and remark afterward on the internal statement. Let $\widetilde{N} \twoheadrightarrow N$ be a surjection of light condensed abelian groups and $M \to N$ a map – equivalently a null sequence in $N$, i.e. a map $\mathbb{N} \cup \{\infty\} \to N$ sending $\infty \mapsto 0$. We must lift it to $\widetilde{N}$.

$$\begin{tikzcd} & \widetilde{N} \arrow[d, two heads] \\ M \arrow[ur, dashed, "\exists?"] \arrow[r] & N \end{tikzcd}$$

Since $\widetilde{N} \twoheadrightarrow N$ is a surjection of condensed abelian groups, the null sequence lifts after a cover: there is a surjection $S \twoheadrightarrow \mathbb{N} \cup \{\infty\}$ from a light profinite set together with a map $g\colon S \to \widetilde{N}$ lifting it. Shrinking $S$, we may assume that over each integer the fibre is a single point, i.e.

$$S \times_{\mathbb{N} \cup \{\infty\}} \mathbb{N} \xrightarrow{\sim} \mathbb{N},$$

so that $S$ is merely some compactification of $\mathbb{N}$. Let

$$S_\infty := S \times_{\mathbb{N} \cup \{\infty\}} \{\infty\} \subseteq S,$$

the fibre over $\infty$, a closed subspace.

Here lightness enters decisively: $S_\infty$ is light, hence injective in ProFin(Fin) (Proposition 2.14), so the inclusion $S_\infty \hookrightarrow S$ admits a retraction $r\colon S \to S_\infty$. Consider the map

$$g - g \circ r\colon\ S \longrightarrow \widetilde{N}.$$

On $S_\infty$ one has $r|_{S_\infty} = \mathrm{id}$, so $g - g\circ r$ vanishes on $S_\infty$.

Now the square

$$\begin{tikzcd} S_\infty \arrow[r, hook] \arrow[d] & S \arrow[d, two heads] \\ \{\infty\} \arrow[r, hook] & \mathbb{N}\cup\{\infty\} \end{tikzcd}$$

is a pushout in light condensed sets – a form of the gluing condition, valid because $S \twoheadrightarrow \mathbb{N}\cup\{\infty\}$ is surjective. Descending a map out of $S$ to $\mathbb{N}\cup\{\infty\}$ requires only that it be constant on fibres; over the integers the fibres are points, so the sole condition is at the fibre $S_\infty$ over $\infty$, where indeed $g - g\circ r = 0$. Hence $g - g \circ r$ factors through a map

$$\widetilde{f}\colon \mathbb{N}\cup\{\infty\} \longrightarrow \widetilde{N}, \qquad \widetilde{f}(\infty) = 0,$$

i.e. a null sequence in $\widetilde{N}$, equivalently a map $M \to \widetilde{N}$.

It remains to check that $\widetilde{f}$ lifts the original map, i.e. projects to it in $N$. Since $\mathbb{Z}[S] \twoheadrightarrow M$ is surjective, it suffices to check after precomposing with $S$. Projecting $g - g\circ r$ to $N$: the term $g$ projects to the original lift, while the correction $g \circ r$ projects to the composite $S \to N$ that factors through $r$ and hence through $S_\infty$; because the original null sequence sends $\infty \mapsto 0$, this composite vanishes on $S_\infty$ and so projects to 0 in $N$. Therefore $\widetilde{f}$ projects to the original null sequence, and $M$ is projective.

For internal projectivity the argument is essentially the same, carrying along an auxiliary profinite set that one also covers; again the decisive point is that sequential limits of surjections are surjections. □

### 3.7 Comparison with all condensed abelian groups

It is worth contrasting the light setting with the category CondAb of *all* condensed abelian groups.

*Remark* 3.21 (Projectives in CondAb). In CondAb all products are exact, and there are projective generators $\mathbb{Z}[S]$ with $S$ extremally disconnected, e.g. $S = \beta I$ the Stone–Čech compactification of a discrete set $I$. These are huge and inexplicit – their structure depends on the model of set theory – but they exist by the axiom of choice. The catch is that *none* of them are internally projective: this involves the badly-behaved products

$$\beta I \times \beta J \qquad (I, J \text{ infinite})$$

of extremally disconnected sets. The product $\beta I \times \beta J$ is itself *not* extremally disconnected, and Scholze suggested this may be what obstructs internal projectivity. It is a rather severe technical nuisance – many arguments really want the projectives to be internally projective.

*Remark* 3.22 (Projectives and injectives in $\mathrm{CondAb}^{\mathrm{light}}$). By contrast $\mathrm{CondAb}^{\mathrm{light}}$ has *no* projective generators: the free group $\mathbb{Z}[\{0,1\}^{\mathbb{N}}]$ on the Cantor set cannot be covered by a projective object, so the category does not have enough projectives – one of the reasons Scholze was initially hesitant to make the switch, though in practice it turns out not to matter much. What one does have is the single very explicit internally projective object $\mathbb{Z}[\mathbb{N}\cup\{\infty\}]$ (equivalently $M$ of Definition 3.20), which is worth a great deal. Dually, $\mathrm{CondAb}^{\mathrm{light}}$ has *enough injectives* – they exist for general set-theoretic reasons but cannot be written down explicitly – whereas CondAb has *no* nonzero injective objects at all.

*Remark* 3.23 ($\mathbb{Z}[\mathbb{N}\cup\{\infty\}]$ is not projective in CondAb). The free group on the convergent sequence, projective in $\mathrm{CondAb}^{\mathrm{light}}$, is *not* projective in CondAb. The universal compactification of $\mathbb{N}$ is the Stone–Čech compactification $\beta\mathbb{N}$, and the surjection $\mathbb{Z}[\beta\mathbb{N}] \to \mathbb{Z}[\mathbb{N}\cup\{\infty\}]$ admits no splitting – the Stone–Čech obstruction. So light-projectivity here is genuinely a light phenomenon, and $\mathbb{Z}[\mathbb{N}\cup\{\infty\}]$ does not even remain projective under the embedding $\mathrm{CondAb}^{\mathrm{light}} \hookrightarrow \mathrm{CondAb}$.

*Remark* 3.24 (Relation to injective Banach spaces). These phenomena are shadows of facts about Banach spaces. Under the duality whereby a projective object of the condensed world becomes an injective Banach space, the relevant objects are the spaces $\mathrm{Cont}(S,\mathbb{R})$ with $S$ extremally disconnected (a Stone–Čech compactification, or a retract of one); it is a classical theorem that every injective Banach space is a retract of such a $\mathrm{Cont}(S,\mathbb{R})$. The object dual to the free group on a null sequence is the Banach space $c_0(\mathbb{N})$ of null sequences, which is *not* injective but is *separably injective* – injectivity tested only against separable Banach spaces [ACCGM16]. Scholze remarked that he arrived at the light-projectivity of $\mathbb{Z}[\mathbb{N}\cup\{\infty\}]$ after reading the proof that $c_0$ is separably injective. He added that whether $\mathrm{Cont}(\beta\mathbb{N},\mathbb{R})$ is separably injective was an open question in the Banach-space literature, and that the analytic/complex-geometry framework of the course can show it is *not* even separably injective – so this space does not behave like a projective object even when tested only against light condensed abelian groups.

*Remark* 3.25 (AB axioms). Grothendieck's Tôhoku axioms for an abelian category are relevant here. The axiom that all products be exact (AB5*, the dual product form) holds in CondAb but fails in $\mathrm{CondAb}^{\mathrm{light}}$, where only countable products are exact. The next axiom, AB6 – a certain commutation of infinite products with filtered colimits – holds whenever there are enough projective generators, in particular in CondAb, and, restricted to countable products, also in $\mathrm{CondAb}^{\mathrm{light}}$:

$$\text{AB6 holds for countable products in } \mathrm{CondAb}^{\mathrm{light}}.$$

One way to see this: the fully faithful embedding $\mathrm{CondAb}^{\mathrm{light}} \hookrightarrow \mathrm{CondAb}$ commutes with all colimits and with countable limits, so countable products in $\mathrm{CondAb}^{\mathrm{light}}$ agree with those in CondAb, where AB6 holds; it can also be checked by hand. (This sharpens Theorem 2.24(1).)

### 3.8 Cohomology in the condensed topos

The lecture closed with cohomology. Working in any topos, one automatically inherits a notion of cohomology, and the point is that in the condensed topos it recovers the expected invariants of topological spaces even though no geometry was built into the site.

**Definition 3.26** (Condensed cohomology)**.** Let $X$ be any condensed set and $M$ an abelian group (for the moment a discrete one; more generally a condensed abelian group). Define

$$H^*(X,M) := \mathrm{Ext}^*_{\mathrm{CondAb}^{\mathrm{light}}}\big(\mathbb{Z}[X],\, M\big),$$

the Ext computed in light condensed abelian groups (the same as in all condensed abelian groups).

**Theorem 3.27** (Recovery of singular cohomology)**.** *If $X$ is a CW complex, then*

$$H^i(\underline{X},M) \;\cong\; H^i_{\mathrm{sing}}(X,M),$$

*the singular cohomology of $X$.*

*Remark* 3.28 (Geometry from totally disconnected data). This is striking: the site used only totally disconnected test objects, yet the resulting cohomology detects, for instance, that the circle is not contractible. The interval, as a condensed set, is a genuine interval and not a point; one passes to homotopy precisely by forming these Ext-groups, since the interval admits no nonzero map to a discrete abelian group.

*Remark* 3.29 (Dold–Thom: $\mathbb{Z}[X]$ models homology). The heuristic behind Theorem 3.27 is the Dold–Thom theorem [DT58]: the free abelian group $\mathbb{Z}[X]$ – finite integer combinations of points of $X$ – is, once one passes to homotopy, a model for the homology of $X$. Dualizing (via Ext into $M$) then gives cohomology, fitting the picture.

*Remark* 3.30 (Internal version and continuous group cohomology). Using the internal $\underline{\mathrm{Ext}}$ instead unravels, by adjunction, to replacing the coefficient $M$ by the continuous functions $\underline{\mathrm{Cont}}(S, M)$ out of a test object $S$ – still a condensed abelian group – so it does not carry much more information. A more substantial payoff is for continuous group cohomology: for a topological group, regarded as a condensed group, it is clear what an action on a condensed abelian group is and what its cohomology should be, via the general topos-theoretic answer. This always gives the expected answer, which need not coincide with the one computed by explicit continuous cochains – and where it differs, the condensed answer is the better one. For a CW complex with a local system one gets a sheaf on condensed sets over $\underline{X}$, and the usual cohomology is singular sheaf cohomology; the comparison rests on cohomological descent for surjections of compact Hausdorff spaces.

*Remark* 3.31 (Why the finitary site is the right one for cohomology). Had one worked instead in the sequential topos generated by $\mathbb{N} \cup \{\infty\}$ alone, computing the cohomology of the interval would first require expressing the interval as the huge colimit (3.2) of countable closed subsets; the cohomology would then return an enormous derived limit, and it is very unlikely to produce the expected answer. That the finitary, Cantor-set-based site does give the right answer is a strong indication that it is the good framework.

*Remark* 3.32 (Enough projectives, again). Asked whether $\mathrm{CondAb}^{\mathrm{light}}$ has enough projective objects, Scholze reiterated that it does not – one does not expect a projective cover of the free group on the Cantor set. In practice one embeds $\mathrm{CondAb}^{\mathrm{light}} \hookrightarrow \mathrm{CondAb}$, a functor preserving all colimits and all countable limits, hence compatible with many operations, and carries out computations in CondAb, which does have enough projectives.

# 4 Ext-Computations in Condensed Abelian Groups

Scholze's third lecture is devoted entirely to *computing* Ext-groups in light condensed abelian groups. The point of departure is that $\mathrm{CondAb}^{\mathrm{light}}$ is a very convenient framework for homological algebra: it is a nice abelian category, one can form its derived category without any fuss, and – since topological spaces embed fully faithfully into condensed sets, and (under mild hypotheses) topological abelian groups embed fully faithfully into condensed abelian groups – all the usual objects of functional analysis now live in a setting with a well-behaved Ext. For the theory to be useful these Ext-groups must have interpretable answers, and the bulk of the lecture is about establishing that they do. There are two main computations: cohomology of a topological space against a discrete coefficient group recovers sheaf (hence, for CW complexes, singular) cohomology; and the Ext-groups between locally compact abelian groups recover the classical Yoneda-Ext of topological abelian groups. The lecture closes with the key "solidity" computation $\mathrm{Ext}^*(\prod_{\mathbb{N}} \mathbb{Z}, \mathbb{Z})$ and a striking set-theoretic theorem about when products turn into direct sums.

Much of this and the following lectures follows Scholze's first course on condensed mathematics; the relevant reference is the *Condensed Mathematics* lecture notes [CS19].

### 4.1 Condensed cohomology recovers singular cohomology

Recall from Section 3 the definition of condensed cohomology: for a condensed set $X$ and an abelian group $M$ (for now discrete, more generally a condensed abelian group), one sets

$$H^*(X, M) := \mathrm{Ext}^*_{\mathrm{CondAb}^{\mathrm{light}}}\big(\mathbb{Z}[\underline{X}],\, M\big),$$

the Ext computed in light condensed abelian groups between the free condensed abelian group on $X$ and $M$. In any topos there is such an internal notion of cohomology – Ext in the category of abelian sheaves – and the content of the theorem stated at the end of Section 3 (Theorem 3.27) is that in the condensed world it recovers singular cohomology.

**Theorem 4.1** (Singular cohomology, recovered)**.** *Let $X$ be a CW complex and $M$ an abelian group. Then*

$$\mathrm{Ext}^i_{\mathrm{CondAb}^{\mathrm{light}}}\big(\mathbb{Z}[\underline{X}],\, M\big) \;\cong\; H^i_{\mathrm{sing}}(X, M). \tag{4.1}$$

The rest of the subsection sketches the proof, which proceeds through sheaf cohomology. Two reductions get us from a general CW complex to a compact space.

*Proof sketch, first reductions.* A CW complex is an increasing union $X = \bigcup_i X_i$ of compact (and metrizable) Hausdorff subspaces $X_i$. Both sides of (4.1) take filtered colimits in $X$ to derived limits, so – with a slightly sharper statement comparing the full complexes rather than the individual cohomology groups – one reduces to the case where $X$ is a compact (CW) space. The point is then that there is a more general statement, valid for compact Hausdorff spaces that need not be CW complexes at all. □

**Theorem 4.2** (Compact Hausdorff case: sheaf cohomology)**.** *Let $X$ be a metrizable compact Hausdorff space and $M$ an abelian group. Then*

$$\mathrm{Ext}^i_{\mathrm{CondAb}^{\mathrm{light}}}\big(\mathbb{Z}[\underline{X}],\, M\big) \;=\; H^i_{\mathrm{sheaf}}(X, M),$$

*the sheaf cohomology of $X$ with coefficients in the constant sheaf $M$.*

Here $H^i_{\mathrm{sheaf}}$ is ordinary sheaf cohomology on the topological space $X$: one has the global-sections functor

$$\Gamma(X, -)\colon \mathrm{Shv}(X, \mathrm{Ab}) \longrightarrow \mathrm{Ab}$$

on the category of sheaves of abelian groups on $X$, and its right derived functors $H^i_{\mathrm{sheaf}}(X, -)$, applied to the constant sheaf $M$.

It is classical that for a CW complex sheaf and singular cohomology agree – $H^i_{\mathrm{sheaf}}$ restricted to CW complexes satisfies the Eilenberg–Steenrod axioms, hence coincides with $H^i_{\mathrm{sing}}$ – so Theorem 4.2 implies Theorem 4.1. In general, however, the two differ, and it is sheaf cohomology (equivalently, the Ext-cohomology) that is the correct invariant.

**Example 4.3** (Totally disconnected spaces: sheaf vs. singular)**.** Let $X$ be a totally disconnected compact Hausdorff space – i.e. a profinite set. Then $\Gamma(X, -)$ is already exact, so

$$H^i_{\mathrm{sheaf}}(X, M) = \begin{cases} \{\text{locally constant maps } X \to M\} & i = 0, \\ 0 & i > 0. \end{cases}$$

By contrast the singular cohomology is

$$H^i_{\mathrm{sing}}(X, M) = \begin{cases} \{\underline{\text{all}} \text{ maps } X \to M\} & i = 0, \\ 0 & i > 0. \end{cases}$$

Indeed the singular cochain complex is built from maps of simplices into $X$, and from the point of view of simplices there is no way to tell $X$ apart from a discrete set of points: any simplex is connected, so it factors through a single point of $X$. Thus singular cohomology sees nothing of the topology of $X$, whereas sheaf cohomology – returning the locally constant maps in degree 0 – does. This is exactly why in general one should use sheaf cohomology (the same as the Ext-cohomology), not singular cohomology.

*Remark* 4.4 (Locally contractible and paracompact hypotheses). Scholze indicated the range of validity somewhat tentatively. He expects that for the Ext-cohomology, crossing $X$ with an interval does not change the answer, though he was not completely sure. For a *locally contractible* space one can compare with sheaf cohomology, and if in addition the space is *paracompact* one can compare sheaf cohomology with singular cohomology (by the usual results). The relevant notion of locally contractible is the "funny" one: not that every point has a basis of contractible open neighborhoods, but that for any point and any neighborhood there is a smaller neighborhood that can be contracted *inside the larger one*. With this hypothesis everything one knows for CW complexes extends. Scholze stressed that these better compactness assumptions only matter at the level of the intermediate sheaf cohomology: if one goes all the way back to the Ext-groups themselves, the paracompactness issue probably disappears again. He also emphasized that he does *not* want to work with topological spaces up to weak homotopy equivalence – otherwise a totally disconnected space would become a discrete set of points and could not be treated at all; the interest is genuinely in the actual topological space.

### 4.2 The sheaf-cohomology upgrade via a geometric morphism

Theorem 4.2 can be upgraded from constant coefficients to arbitrary abelian sheaves, and the cleanest way to see it is through a geometric morphism of topoi. Fix a metrizable compact Hausdorff space $X$. There are two sites over $X$:

$$\mathrm{CondSet}^{\mathrm{light}}_{/\underline{X}} = \big(\mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})_{/\underline{X}}\big)^{\sim} \xrightarrow{\ \lambda\ } \mathrm{Op}(X)^{\sim} = \mathrm{Shv}(X),$$

on the left the topos of sheaves on light profinite sets equipped with a map to $\underline{X}$ (light condensed sets over $X$), on the right the topos of sheaves on the poset $\mathrm{Op}(X)$ of open subsets of $X$ – ordinary sheaves on the topological space. This $\lambda$ is a geometric morphism: any sheaf on $X$ pulls back to a light condensed set over $X$, and to define the pullback it suffices to define it on generators, by

$$\lambda^*\underline{U} = \underline{U} \quad (\to \underline{X}), \qquad U \subseteq X \text{ open}.$$

Passing to abelian sheaves and derived categories gives

$$\lambda^* \colon D\big(\mathrm{Shv}(X, \mathrm{Ab})\big) \longrightarrow D\big(\mathrm{Ab}(\mathrm{CondSet}^{\mathrm{light}}_{/\underline{X}})\big).$$

**Theorem 4.5** (Full faithfulness of $\lambda^*$)**.** *Let $X$ be metrizable compact Hausdorff. On bounded-below objects, $\lambda^*$ is fully faithful:*

$$\lambda^* \colon D^+\big(\mathrm{Shv}(X, \mathrm{Ab})\big) \hookrightarrow D^+\big(\mathrm{Ab}(\mathrm{CondSet}^{\mathrm{light}}_{/\underline{X}})\big).$$

*Consequently, for every $\mathcal{F} \in \mathrm{Shv}(X, \mathrm{Ab})$,*

$$H^i\big(\mathrm{CondSet}^{\mathrm{light}}_{/\underline{X}},\, \lambda^*\mathcal{F}\big) \;\cong\; H^i_{\mathrm{sheaf}}(X, \mathcal{F}).$$

*Applying this to $\mathcal{F}$ the constant sheaf $M$ recovers Theorem 4.2.*

*Remark* 4.6 (Robustness). This is the "fancy" statement Scholze alluded to: it is not merely that a single very specific sheaf (the constant one) has the same cohomology on both sides, but that the whole bounded-below derived category of sheaves on $X$ embeds fully faithfully into the condensed side, and cohomology of *any* abelian sheaf is computed either way. In particular the argument works for non-constant sheaves.

*Proof sketch.* By adjunction, full faithfulness of $\lambda^*$ is equivalent to the unit map being an isomorphism: for all $A \in D^+(\mathrm{Shv}(X, \mathrm{Ab}))$,

$$A \xrightarrow{\sim} R\lambda_*\, \lambda^* A.$$

This is a map in $D^+(\mathrm{Shv}(X, \mathrm{Ab}))$, and whether it is an isomorphism can be checked on stalks at points $x \in X$:

$$A_x \xrightarrow{\sim} \big(R\lambda_*\, \lambda^* A\big)_x.$$

The key point is a *base-change property*: taking the stalk at $x$ commutes with $R\lambda_*$. This is where the topos-theoretic nature of $\underline{X}$ is used. One invokes the general results that cohomology commutes with filtered colimits for *coherent* topoi, coherence meaning quasicompact-plus-quasiseparated; the key geometric input is precisely that $\underline{X}$ is qcqs (Proposition 3.6) – the same fact, recalled from Section 3, that to compute the cohomology of e.g. the interval one needs a surjection onto it from one of the generating objects of the site, a Cantor set. Pulling the stalk functor inside $R\lambda_*$ then reduces the whole statement to the case where $X = \{x\}$ is a point. Over a point the assertion is clear: condensed abelian groups sit fully faithfully, and $\mathbb{Z}$ (as a condensed abelian group) has the two required cohomology objects, so the unit is an isomorphism. □

*Remark* 4.7 (Boundedness below and closed neighborhoods). The restriction to $D^+$ is genuine. Without a bound to the left there are issues with how cohomology interacts with the passage to limits; there are several versions of the unbounded derived category (the naive derived category of sheaves, and its left completion), and to retain full faithfulness one should use the left completion. For a CW complex, or in finite cohomological dimension, this makes no difference. When executing the stalk argument one should moreover regard a point $x$ as the limit of its *closed* neighborhoods rather than its open ones: in a compact Hausdorff space the closed and open neighborhoods are both cofinal, but pulling back along a closed neighborhood keeps one within profinite (qcqs) sets, which is what the coherence argument needs.

### 4.3 A concrete proof by hypercover resolutions

Scholze also indicated the more hands-on proof of Theorem 4.2, which is instructive and breaks into two steps. Concretely, for $X$ metrizable compact Hausdorff one wants to compute $\mathrm{Ext}^i(\mathbb{Z}[\underline{X}], \mathbb{Z})$ by finding a projective – or at least $\mathrm{Ext}^i(-, \mathbb{Z})$-acyclic – resolution of $\mathbb{Z}[\underline{X}]$.

**Step 1: totally disconnected $X$**

**Proposition 4.8** (Acyclicity for profinite sets)**.** *Let $S \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$ be a light profinite set. Then*

$$\mathrm{Ext}^i\big(\mathbb{Z}[S],\, \mathbb{Z}\big) = \begin{cases} \mathrm{Cont}(S, \mathbb{Z}) & i = 0, \\ 0 & i > 0. \end{cases}$$

For $S = \mathbb{N} \cup \{\infty\}$ this is immediate from Theorem 3.15(3): $\mathbb{Z}[S]$ is (internally) projective. For general $S$ one must genuinely resolve, and the point is the following. Any higher Ext-class against $\mathbb{Z}[S]$ can be split after passing to a suitable cover of $S$; iterating produces a hypercover. Recall that for a hypercover

$$\cdots \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} S_2 \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} S_1 \rightrightarrows S_0 \longrightarrow S$$

of light profinite sets – meaning $S_0 \twoheadrightarrow S$, then $S_1 \twoheadrightarrow S_0 \times_S S_0$, and $S_{n+1}$ surjecting onto the appropriate matching object at each stage – the associated augmented chain complex of free groups

$$\cdots \to \mathbb{Z}[S_2] \to \mathbb{Z}[S_1] \to \mathbb{Z}[S_0] \to \mathbb{Z}[S] \to 0 \tag{4.2}$$

is exact – this is a completely general fact about hypercovers in any site.

*Proof of Proposition 4.8.* Given the exact resolution (4.2), one must show that applying $\mathrm{Cont}(-, \mathbb{Z}) = \underline{\mathrm{Hom}}(\mathbb{Z}[-], \mathbb{Z})$ keeps it exact, i.e. that

$$0 \to \mathrm{Cont}(S, \mathbb{Z}) \to \mathrm{Cont}(S_0, \mathbb{Z}) \to \mathrm{Cont}(S_1, \mathbb{Z}) \to \cdots$$

is exact. There are two ways. *Either* treat every term as the global sections of a sheaf on $S$: push everything forward to $S$, so that it suffices to prove exactness of sheaves on $S$, which is checked on stalks, and over a point a cover is split so the statement is elementary (this is the homological descent of SGA 4, via the proper base change theorem for "proper" = universally closed separated maps of compact Hausdorff spaces). *Or* – the argument Scholze used in the lecture notes – write the hypercover of profinite sets as a cofiltered limit of hypercovers of *finite* sets by finite sets; for finite sets every hypercover is split and the exactness is obvious, and the exactness passes to the filtered colimit. It takes a little unravelling to see the hypercover can always be so written, but it reduces to the finite case where the claim is clear. □

**Step 2: general $X$**

Now let $X$ be a general metrizable compact Hausdorff space. We resolve $\mathbb{Z}[\underline{X}]$ by $\mathbb{Z}[S]$'s with $S$ profinite. Using that every such $X$ receives a surjection from a Cantor set (indeed from a light profinite set), pick $S_0 \twoheadrightarrow X$ and form the Čech nerve, the simplicial light profinite set with

$$S_i = \underbrace{S_0 \times_X \cdots \times_X S_0}_{i+1}, \qquad S_1 = S_0 \times_X S_0, \quad S_2 = S_0 \times_X S_0 \times_X S_0, \ \ldots$$

Because $S_0$ is already totally disconnected, each fibre product $S_i$ is a closed subspace of a product of totally disconnected spaces, hence again totally disconnected – so no further resolution is needed. The Čech nerve $S_\bullet \to \underline{X}$ is a hypercover, and by the same general principle its augmented chain complex is a resolution

$$\cdots \to \mathbb{Z}[S_2] \to \mathbb{Z}[S_1] \to \mathbb{Z}[S_0] \to \mathbb{Z}[\underline{X}] \to 0$$

whose terms $\mathbb{Z}[S_i]$ are $\mathrm{Ext}^i(-,\mathbb{Z})$-acyclic by Proposition 4.8. Hence $\mathrm{Ext}^i(\mathbb{Z}[\underline{X}],\mathbb{Z})$ is computed by the complex

$$0 \to \mathrm{Cont}(S_0,\mathbb{Z}) \to \mathrm{Cont}(S_1,\mathbb{Z}) \to \mathrm{Cont}(S_2,\mathbb{Z}) \to \cdots, \tag{4.3}$$

an admittedly awkward-looking formula for something like singular cohomology: one covers $X$ by a Cantor set, forms all the fibre products, takes locally constant $\mathbb{Z}$-valued functions everywhere, and the claim is that the resulting complex computes the right thing. That it does is proved exactly as before – treat the terms as global sections of sheaves on $X$ and check on stalks that (4.3) resolves the constant sheaf $\underline{\mathbb{Z}}$ on $X$.

## 4.4 Locally compact abelian groups

The second main computation concerns the classical world of locally compact abelian groups, which embeds fully faithfully into condensed abelian groups and whose Ext-groups one can therefore ask about.

*Notation* 4.9 (Metrizable LCA groups). Write $\mathrm{LCA}_m$ for the category of metrizable locally compact abelian groups. Examples include discrete abelian groups, $\mathbb{R}$, the circle $\mathbb{R}/\mathbb{Z}$, the $p$-adic integers $\mathbb{Z}_p$, and the adeles

$$\mathbb{A} = \widehat{\mathbb{Z}} \otimes \mathbb{Q} \times \mathbb{R}.$$

The adeles illustrate the general structure: $\mathbb{Q}$ sits inside $\mathbb{A}$ as a discrete subgroup with compact quotient,

$$0 \to \mathbb{Q} \to \mathbb{A} \to \mathbb{A}/\mathbb{Q} \to 0,$$

with $\mathbb{Q}$ discrete and $\mathbb{A}/\mathbb{Q}$ compact. In general every $A \in \mathrm{LCA}_m$ admits a filtration with three pieces:

$$\text{a discrete group,} \quad \text{a finite-dimensional real vector space,} \quad \text{a compact metrizable abelian group.} \tag{4.4}$$

**Theorem 4.10** (Condensed Ext of LCA groups)**.** *For $A, B \in \mathrm{LCA}_m$,*

$$\mathrm{Ext}^i_{\mathrm{CondAb}^{\mathrm{light}}}(\underline{A},\underline{B}) \cong \begin{cases} \mathrm{Hom}_{\mathrm{LCA}}(A,B) & i=0, \\ \mathrm{Ext}^1_{\mathrm{LCA}}(A,B) & i=1, \\ 0 & i \geq 2, \end{cases}$$

*where $\mathrm{Ext}^1_{\mathrm{LCA}}$ is the classical Yoneda* Ext *of extensions of $A$ by $B$ within locally compact abelian groups (up to the appropriate notion of equivalence).*

Full faithfulness already forces the $i=0$ term to be the ordinary (topological) homomorphisms. What is not a priori clear is that no exotic higher Ext-classes appear: the condensed $\mathrm{Ext}^1$ turns out to be exactly the classical Yoneda $\mathrm{Ext}^1$, and everything above degree 1 vanishes – matching the classical fact that LCA is an exact category with $\mathrm{Ext}^i_{\mathrm{LCA}} = 0$ for $i \geq 2$. (Scholze noted this is special: there *are* pairs of condensed abelian groups with surprising higher Ext, and the solidity computation below is one such.)

**Example 4.11** (Pontryagin duality)**.** For any $A \in \mathrm{LCA}_m$,

$$\mathrm{Ext}^i(\underline{A},\underline{\mathbb{R}/\mathbb{Z}}) = \begin{cases} A^\vee & i=0, \\ 0 & i>0, \end{cases}$$

where $A^\vee = \mathrm{Hom}(A, \mathbb{R}/\mathbb{Z})$ is the Pontryagin dual. Thus $A \mapsto A^\vee$ is exact on locally compact abelian groups, sitting in degree 0 with nothing in higher degrees – the condensed reflection of Pontryagin duality.

**Example 4.12** (No maps from the reals to the integers)**.**

$$\mathrm{Ext}^i\big(\underline{\mathbb{R}}, \mathbb{Z}\big) = 0 \qquad \text{for all } i \geq 0.$$

The intuition is that $\mathbb{R}$ is connected, so it can never map to the discrete group $\mathbb{Z}$ (giving $\mathrm{Ext}^0 = 0$), and all higher groups vanish as well. This is a computation one really has to do; the vanishing is proved below.

## 4.5 The Breen–Deligne resolution

To compute anything one needs something close to a projective resolution. After a dévissage reducing $A$ and $B$ to the three basic pieces of (4.4) (mapping to a discrete group is easy; the finite-dimensional real vector spaces are handled separately), one may assume $A$ is a compact metrizable group. The engine is a functorial resolution valid for *all* abelian groups.

**Theorem 4.13** (Breen–Deligne resolution)**.** *There is a resolution, functorial in the abelian group $M$,*

$$\cdots \to \mathbb{Z}[M^{n_i}] \to \cdots \to \mathbb{Z}[M^2] \to \mathbb{Z}[M] \to M \to 0, \tag{4.5}$$

*in which every term is a single free abelian group $\mathbb{Z}[M^{n_i}]$ on a power of $M$ and all differentials are given by* universal formulas*. The augmentation is $[m] \mapsto m$ and the first differential is*

$$[(a, b)] \longmapsto [a] + [b] - [a + b].$$

*Remark* 4.14 (Provenance and (in)explicitness)*.* This resolution was used substantially by Breen in an algebraic-geometry setting, but the statement in the form needed here was never put in the literature by him; there is an unpublished letter of Deligne to Breen proving the result. Two features are striking. First, no *explicit* such resolution is known: the differentials exist and are natural, but one cannot write them down, and Scholze noted (with mild surprise) that the proof uses a little stable homotopy theory, in some incarnations the finiteness of the stable homotopy groups of spheres – one has to kill something like the stable stems, and there is no explicit way to do so. Second, one can arrange a *single* free group $\mathbb{Z}[M^{n_i}]$ in each degree rather than a finite direct sum: any such direct sum can itself be covered by one free group of the same kind, so one chooses one term per degree.

Because (4.5) is given by universal formulas, functoriality makes it work for abelian sheaves on any site. Applied to a condensed abelian group $A$ it gives

$$\cdots \to \mathbb{Z}[\underline{A}^{n_i}] \to \cdots \to \mathbb{Z}[\underline{A}] \to \underline{A} \to 0, \tag{4.6}$$

which reduces the computation of $\mathrm{Ext}^i(\underline{A}, -)$ to that of $\mathrm{Ext}^i(\mathbb{Z}[\underline{A}^n], -)$ – and these are exactly the cohomology computations of the previous subsections, since $\underline{A}^n$ is a nice space. One more input is needed for real coefficients.

**Proposition 4.15** (Cohomology with real coefficients)**.** *For $X$ metrizable compact Hausdorff,*

$$\mathrm{Ext}^i_{\mathrm{CondAb}^{\mathrm{light}}}\big(\mathbb{Z}[\underline{X}], \underline{\mathbb{R}}\big) = \begin{cases} \mathrm{Cont}(X, \mathbb{R}) & i = 0, \\ 0 & i > 0. \end{cases} \tag{4.7}$$

*The same holds with $\mathbb{R}$ replaced by any Banach space $V$, but this uses the* local convexity *of $V$: the proof runs through partition-of-unity arguments, which require the target to have a basis of convex neighborhoods.*

*Remark* 4.16 (Non-locally-convex coefficients)*.* As a preview: when real vector spaces are treated inside the analytic theory later, one is forced to consider *non-locally-convex* vector spaces, for which Proposition 4.15 can fail – suspension need not behave well on all compact Hausdorff spaces. There one really has to resolve by totally disconnected objects instead.

**Example 4.17** ($\mathrm{Ext}^i(\mathbb{R},\mathbb{Z})=0$, worked out)**.** Resolve $\underline{\mathbb{R}}$ by the Breen–Deligne resolution (4.6), with terms $\mathbb{Z}[\underline{\mathbb{R}}^n]$. Since $\mathbb{R}^n$ is contractible,

$$\mathrm{Ext}^i\big(\mathbb{Z}[\underline{\mathbb{R}}^n],\mathbb{Z}\big) = H^i_{\mathrm{sing}}(\mathbb{R}^n,\mathbb{Z}) = \begin{cases}\mathbb{Z} & i=0,\\ 0 & i>0,\end{cases} \tag{4.8}$$

so each term contributes exactly one copy of $\mathbb{Z}$, and one must see that these assemble to 0. The trick is to run the *same* resolution on the zero group, which resolves 0; comparing the two augmented complexes column by column,

$$\begin{tikzcd}
\cdots \arrow[r] & \mathbb{Z}[0] \arrow[r] \arrow[d] & \cdots \arrow[r] & \mathbb{Z}[0] \arrow[r] \arrow[d] & \mathbb{Z}[0] \arrow[r] \arrow[d] & 0 \arrow[d] \\
\cdots \arrow[r] & \mathbb{Z}[\underline{\mathbb{R}}^n] \arrow[r] & \cdots \arrow[r] & \mathbb{Z}[\underline{\mathbb{R}}^2] \arrow[r] & \mathbb{Z}[\underline{\mathbb{R}}] \arrow[r] & \underline{\mathbb{R}}
\end{tikzcd}$$

the map $\mathbb{Z}[0^n]=\mathbb{Z}[0]\to\mathbb{Z}[\underline{\mathbb{R}}^n]$ induces an isomorphism on each $\mathrm{Ext}^*(-,\mathbb{Z})$ by (4.8). Hence $\mathrm{Ext}^*(\underline{\mathbb{R}},\mathbb{Z})$ agrees with $\mathrm{Ext}^*$ computed from the top row, which resolves 0; so it vanishes.

### 4.6 The cubical $Q$-construction

One might be uneasy about running explicit computations off an inexplicit resolution. In fact the inexplicitness never causes trouble – only the existence and functoriality of Theorem 4.13 are used – but there is also an *explicit* complex that can be used instead: MacLane's $Q$-construction, also called the cubical construction, worked out by Commelin in the course of the formalization effort.

**Definition 4.18** (The cubical construction)**.** For an abelian group $M$, let $Q(M)$ be the explicit complex

$$Q(M): \qquad \cdots\to\mathbb{Z}[M^8]\to\mathbb{Z}[M^4]\to\cdots\to\mathbb{Z}[M^2]\to\mathbb{Z}[M],$$

with $i$-th term $\mathbb{Z}[M^{2^i}]$. The first differential is $[(a,b)]\mapsto[a]+[b]-[a+b]$, and the next is the "cube" differential: viewing $(a,b,c,d)$ as the four corners of a square,

$$[(a,b,c,d)]\longmapsto[(a,b)]+[(c,d)]-[(a+c,b+d)]-[(a,c)]-[(b,d)]+[(a+b,c+d)], \tag{4.9}$$

subtracting the opposite faces in each of the two directions; higher differentials arrange the $2^i$ arguments on the vertices of an $i$-cube and take the alternating sum of the pairs of opposite faces.

*Remark* 4.19*.* Scholze cautioned that he might have the signs in (4.9) slightly wrong; if $d^2=0$ fails as written there is an easy variant that works. Editorially: for the two differentials as transcribed here the composite does vanish – expanding $d_1\circ d_2[(a,b,c,d)]$ via $[(x,y)]\mapsto[x]+[y]-[x+y]$, all nine terms cancel in pairs, so $d^2=0$ holds at least for this first composite.

The key property is that $Q$ is, up to torsion corrections, *linear* in $M$.

**Theorem 4.20** (Linearity of the cubical construction)**.** *There is a natural equivalence*

$$Q(M) \;\simeq\; Q(\mathbb{Z}) \otimes^{\mathbb{L}}_{\mathbb{Z}} M,$$

*a derived base change; if $M$ is torsion-free this reduces to $H_i(Q(M)) \cong H_i(Q(\mathbb{Z})) \otimes_{\mathbb{Z}} M$, with an additional* Tor *term in general. Moreover*

$$Q(\mathbb{Z}) \;\simeq\; \bigoplus_{i\geq 0} \big(\mathbb{Z} \otimes^{\mathbb{L}}_{\mathbb{S}} \mathbb{Z}\big)^{\oplus 2^i}[i],$$

*the tensor product being over the sphere spectrum $\mathbb{S}$; computing it uses a little stable homotopy theory. Consequently, for $A \in \mathrm{CondAb}^{\mathrm{light}}$ and any $B$,*

$$\mathrm{Ext}^i(\underline{A}, B) = 0 \;\; \forall i > 0 \quad\iff\quad \mathrm{Ext}^i(Q(\underline{A}), B) = 0 \;\; \forall i > 0.$$

This linearity in $M$ turns out to be exactly what is needed: any of the vanishing statements one wants to prove can be put in the form "$\mathrm{Ext}^i = 0$ for all $i \geq i_0$", and one may then use the explicit complex $Q(A)$ in place of a resolution of $A$ (it is not literally a resolution, but for this purpose it is just as good). The whole point is comfort: nothing in the arguments changes, but one knows the objects involved explicitly, and the shape of the answer "should have something to do with" the stable stems whenever one writes down something explicit.

## 4.7 The solidity computation

The final computation is the one that seeds the theory of solid abelian groups. It concerns an object that is no longer locally compact but genuinely large: the countable product $\prod_{\mathbb{N}} \mathbb{Z}$.

**Corollary 4.21** (Ext out of $\prod_{\mathbb{N}} \mathbb{Z}$)**.** *For every discrete abelian group $M$,*

$$\mathrm{Ext}^i_{\mathrm{CondAb}^{\mathrm{light}}}\Big(\prod_{\mathbb{N}} \mathbb{Z},\, M\Big) = \begin{cases} \bigoplus_{\mathbb{N}} M & i = 0, \\ 0 & i > 0. \end{cases}$$

The degree-0 statement is the classical fact that a homomorphism $\prod_{\mathbb{N}} \mathbb{Z} \to M$ factors through finitely many coordinates; the critical new content is that there is *nothing in higher degrees.* This is very different from the naive classical guess and is exactly what makes $\prod_{\mathbb{N}} \mathbb{Z}$ a compact projective generator of the full subcategory

$$\mathrm{Solid} \subset \mathrm{CondAb}^{\mathrm{light}}$$

of solid abelian groups (cf. Proposition 1.38, where the products $\prod_I \mathbb{Z}$ appeared as the compact projective generators of the heart of $D(\mathbb{Z}_{\square})$): if one wants $\prod_{\mathbb{N}} \mathbb{Z}$ to be projective, one had certainly better have all the higher Ext-groups vanish.

*Remark* 4.22 (A warning: $\prod_{\mathbb{N}} \mathbb{Z}$ is a huge colimit)*.* As a condensed set, $\prod_{\mathbb{N}} \mathbb{Z}$ is the filtered union

$$\prod_{\mathbb{N}} \mathbb{Z} = \bigcup_{f\colon \mathbb{N}\to\mathbb{N}} \; \prod_{n\in\mathbb{N}} [-f(n), f(n)],$$

over all functions $f\colon \mathbb{N} \to \mathbb{N}$, where each $\prod_n [-f(n), f(n)]$ (a product of finite integer intervals) is a profinite set. This is a *huge* colimit – of size the continuum. So the naive approach, resolving by free groups $\mathbb{Z}[S]$ on profinite sets, is essentially impossible to execute: mapping out of such a huge colimit turns into an extremely hard-to-control huge limit.

*Proof of Corollary 4.21.* The trick is to resolve in the "other" direction, embedding $\prod_{\mathbb{N}} \mathbb{Z}$ into a product of copies of $\mathbb{R}$. There is a short exact sequence

$$0 \to \prod_{\mathbb{N}} \mathbb{Z} \to \prod_{\mathbb{N}} \underline{\mathbb{R}} \to \prod_{\mathbb{N}} \underline{\mathbb{R}/\mathbb{Z}} \to 0, \tag{4.10}$$

in which $\prod_{\mathbb{N}} \mathbb{R}/\mathbb{Z}$ is a metrizable compact abelian group (a countable product of circles). By the compact/LCA computation (Theorem 4.10 applied to this compact group),

$$\operatorname{Ext}^i\Big(\prod_{\mathbb{N}} \underline{\mathbb{R}/\mathbb{Z}},\, M\Big) = \begin{cases} 0 & i = 0, \\ \bigoplus_{\mathbb{N}} M & i = 1, \\ 0 & i > 1. \end{cases} \tag{4.11}$$

It remains to show

$$\operatorname{Ext}^i\Big(\prod_{\mathbb{N}} \underline{\mathbb{R}},\, M\Big) = 0 \qquad \forall i \geq 0. \tag{4.12}$$

Granting (4.12), the long exact sequence obtained by applying $\operatorname{Ext}^*(-, M)$ to (4.10) gives $\operatorname{Ext}^i(\prod_{\mathbb{N}} \mathbb{Z}, M) \cong \operatorname{Ext}^{i+1}(\prod_{\mathbb{N}} \mathbb{R}/\mathbb{Z}, M)$, which by (4.11) is $\bigoplus_{\mathbb{N}} M$ for $i = 0$ and 0 for $i > 0$, as claimed.

For (4.12) one uses that $\prod_{\mathbb{N}} \mathbb{R}$ carries a module structure over the condensed ring $\underline{\mathbb{R}}$. Hence, by adjunction,

$$\operatorname{RHom}\Big(\prod_{\mathbb{N}} \underline{\mathbb{R}},\, M\Big) = \operatorname{RHom}_{\underline{\mathbb{R}}}\Big(\prod_{\mathbb{N}} \underline{\mathbb{R}},\, \underbrace{\operatorname{RHom}(\underline{\mathbb{R}}, M)}_{=0}\Big) = 0,$$

the internal $\operatorname{RHom}(\underline{\mathbb{R}}, M)$ vanishing because $\mathbb{R}$ is connected and $M$ discrete (the generalization of Example 4.17 to any discrete $M$). One does not even need to know what $\operatorname{RHom}(\underline{\mathbb{R}}, M)$ is – only that $\prod_{\mathbb{N}} \mathbb{R}$ is an $\mathbb{R}$-module and that $\mathbb{R}$ admits no maps to a discrete module. $\square$

*Remark* 4.23 (Hilbert-cube alternative)*.* There is a second route to (4.12). Running the analogous operation on $\prod_{\mathbb{N}} \mathbb{R}$ turns each term of the resolution into a product of intervals – a Hilbert cube. The comparison (4.7) between condensed and singular cohomology holds for the Hilbert cube as well (no higher cohomology), so the argument given for $\mathbb{R}$ applies verbatim to the Hilbert cube variant.

## 4.8 A set-theoretic theorem

Scholze closed with a "fun theorem with a set-theoretic proof". It asks in what generality the phenomenon of Corollary 4.21 – pulling a product out as a direct sum – persists, not just for products but for sequential limits.

**Definition 4.24** (The assertion $(*)$)**.** Let $(*)$ be the following assertion. For all sequential limits

$$\cdots \to M_1 \to M_0$$

of countable discrete abelian groups with surjective transition maps, and all discrete abelian groups $N$,

$$\operatorname{Ext}^i\Big(\varprojlim_n M_n,\, N\Big) \;\cong\; \operatorname*{colim}_n \operatorname{Ext}^i(M_n, N) \qquad (= 0 \text{ for } i \geq 2).$$

The right-hand vanishing for $i \geq 2$ is automatic, since $\operatorname{Ext}^i(M_n, N) = 0$ for $i \geq 2$ between discrete groups; the content is in degrees 0 and 1. Computing such a limit can be resolved by a two-term complex involving products of the $M_n$, which reduces the limit to a product; resolving the $M_n$ by countable free abelian groups then reduces $(*)$ to the vanishing

$$\mathrm{Ext}^i\Big(\prod_{\mathbb{N}}\bigoplus_{\mathbb{N}}\mathbb{Z},\ \bigoplus_{I}\mathbb{Z}\Big)=0 \qquad \text{for } i>0.$$

*Remark* 4.25 ($(*)$ fails under CH). Clausen and Scholze quickly realized that $(*)$ *fails* under the continuum hypothesis $2^{\aleph_0}=\aleph_1$: there one finds $\mathrm{Ext}^1\neq 0$. They thought it might be too much to hope for at all – but it turned out that set theorists had independently studied precisely the same question in a different language. Unravelling, the relevant Ext-group is a huge filtered colimit, over functions $f\colon \mathbb{N}\to\mathbb{N}$, of products of finite direct sums $\prod_n\bigoplus_{m\le f(n)}\mathbb{Z}$, and computing it amounts to a huge derived limit – exactly the higher derived limits studied in that literature [BL21; BHL21].

**Theorem 4.26** (Bergfalk–Lambie-Hanson–Hrušák–Bannister)**.**

1. *$(*) \implies 2^{\aleph_0}>\aleph_\omega$.*
2. *It is consistent that $(*)$ holds and $2^{\aleph_0}=\aleph_{\omega+1}$; in fact $(*)$ holds in any forcing extension obtained by adjoining $\beth_\omega$ many Cohen reals.*

So $(*)$ forces the continuum to be large; it cannot equal $\aleph_\omega$ (by König's theorem), and the first possibility above it, $2^{\aleph_0}=\aleph_{\omega+1}$, is already consistent with $(*)$. The consistency is realized by Cohen's most basic forcing – the one he invented to make the continuum hypothesis false, adjoining new real numbers to a ground model – here adjoining $\beth_\omega$ many Cohen reals, $\beth_\omega$ being in some sense the minimal number of reals one can adjoin and still have a chance.

*Remark* 4.27 (Why it is mentioned). The principle $(*)$ is never actually used in the course. It is nonetheless useful to know it can be arranged: when computing certain things it is easy to guess what the answer would be if $(*)$ held, and the results one genuinely needs can usually be proved without invoking it. But in some situations one might want it – for instance to control Ext-groups out of Fréchet-type spaces, where the expected answer holds under $(*)$; in general these Ext-groups are genuinely set-theory-dependent, their value tied to the order type of the poset of functions $\mathbb{N}\to\mathbb{N}$ under eventual dominance, which depends heavily on the model of set theory.

# 5 Solid Abelian Groups

Scholze's fourth lecture isolates, inside $\mathrm{CondAb}^{\mathrm{light}}$, a good class of "complete" objects: the *solid* abelian groups. The motivation is that naive tensor products of condensed abelian groups are badly behaved – their underlying object is the algebraic (uncompleted) tensor product – so one wants a reflective subcategory of complete objects together with a completion functor, so that completing a naive tensor product yields something reasonable. The lecture gives a presentation of the theory of solid abelian groups that is *specific to the light setting* and quite different from the one in the first *Condensed Mathematics* notes [CS19]: solidity is defined by the single condition that every null sequence be uniquely summable, phrased through the internally projective object $P=\mathbb{Z}[\mathbb{N}\cup\{\infty\}]/\mathbb{Z}[\infty]$ of Section 3. From this one gets, essentially for free, that $\mathrm{Solid}\subset\mathrm{CondAb}^{\mathrm{light}}$ is an abelian subcategory closed under all the usual operations; that the reals are annihilated by solidification; and, after a computation, that Solid has a single compact projective generator $\prod_{\mathbb{N}}\mathbb{Z}$, recovering the structural facts stated in Sections 1 and 4. The lecture closes with the reading of solidity as the ability to integrate against $\mathbb{Z}$-valued measures.

## 5.1 Motivation: a class of complete objects

*Remark* 5.1 (Why one wants completeness). Working inside $\mathrm{CondAb}^{\mathrm{light}}$, computations that start with reasonable examples and only take limits and colimits stay reasonable. The trouble begins with constructions like the tensor product. If one forms the tensor product of two power-series algebras, the answer one wants is a *completed* tensor product – morally a power-series algebra in both variables. But the naive tensor product in condensed abelian groups has underlying object the ordinary algebraic tensor product,

$$\big(M \otimes_{\mathrm{CondAb}^{\mathrm{light}}} N\big)(*) = M(*) \otimes_{\mathrm{Ab}} N(*),$$

a nasty, essentially indescribable object. So tensor products of condensed abelian groups are not nice, and the remedy is to single out a subclass of *complete* objects and to complete the naive tensor product into that subclass, hoping for a reasonable answer.

We want the notion of completeness to cut out an abelian category, closed enough that cokernels of complete objects stay complete. Among the objects that should count as complete are $\mathbb{Z}$, the $p$-adics $\mathbb{Z}_p$, and $\mathbb{Q}_p$; and if the class is to be closed under extensions it must also contain non-Hausdorff objects such as the quotient in

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}_p \longrightarrow \mathbb{Z}_p/\mathbb{Z} \longrightarrow 0.$$

The quotient $\mathbb{Z}_p/\mathbb{Z}$ is a very non-Hausdorff thing, so one *cannot* phrase completeness by asking that convergent sequences have unique limits. In the condensed setting there is no good abstract notion of a convergent sequence other than one carrying a specified limit point, and even granting such a notion one could not demand that limits exist, precisely because one wants to allow these non-Hausdorff examples.

*Remark* 5.2 (Nonarchimedean first, reals later). Scholze recalled that when he and Clausen first thought about this, finding the right notion of completeness was one of the key questions. They originally wanted a single notion for which the real numbers, and ideally all locally compact abelian groups, would be complete, while still allowing cokernels and staying complete; they could not directly make that work. Forgetting the reals for the moment and aiming only for a nonarchimedean theory, there is something that works very cleanly. So it is difficult to find a notion under which $\mathbb{R}$ is complete, but there is a theory that works well for all nonarchimedean examples; the real numbers are recovered later, by a different story (the liquid theory).

*Remark* 5.3 (A presentation special to the light setting). The presentation below is very different from the one in the first *Condensed Mathematics* lecture notes [CS19], and it really only works in the light setting. There, showing that solid abelian groups form an abelian category was among the *last* things one could prove; here it will be essentially the *first*.

The guiding idea is a basic fact of nonarchimedean analysis: a sequence is summable if and only if it is a null sequence (tends to 0). We turn this into a definition of completeness.

$$\boxed{\text{$M \in \mathrm{CondAb}^{\mathrm{light}}$ is “nonarchimedean complete” if every null sequence in $M$ is (uniquely) summable.}} \tag{5.1}$$

## 5.2 The definition via the free group on a null sequence

To formalize (5.1) we use the object that classifies null sequences.

*Notation* 5.4 (The free group on a null sequence). Write

$$P := \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/\mathbb{Z}[\infty] \ \in \ \mathrm{CondAb}^{\mathrm{light}},$$

the free condensed abelian group on a null sequence (the object $M$ of Definition 3.20). Maps out of $P$ are exactly null sequences in the target. Recall from Theorem 3.15(3) that $P$ is *internally projective.*

Internal projectivity says precisely that the internal Hom out of $P$ is exact. Recall the internal Hom functor

$$\underline{\mathrm{Hom}}(P, -) \colon \mathrm{CondAb}^{\mathrm{light}} \longrightarrow \mathrm{CondAb}^{\mathrm{light}}, \qquad M \longmapsto \Big(S \mapsto \mathrm{Hom}\big(P \otimes \mathbb{Z}[S],\, M\big)\Big),$$

whose value on the point is the group of null sequences in $M$. By internal projectivity, $\underline{\mathrm{Hom}}(P, -)$ is exact; being a right adjoint (to $P \otimes -$) it preserves all limits, and it preserves all colimits as well. This exactness is the engine of everything below.

Consider the endomorphism "identity minus shift"

$$f = \mathrm{id} - \mathrm{shift} \colon P \longrightarrow P, \qquad [n] \longmapsto [n] - [n+1]. \tag{5.2}$$

**Definition 5.5** (Solid abelian group)**.** An object $M \in \mathrm{CondAb}^{\mathrm{light}}$ is *solid* if the induced map

$$f^* \colon \underline{\mathrm{Hom}}(P, M) \xrightarrow{\sim} \underline{\mathrm{Hom}}(P, M) \tag{5.3}$$

is an isomorphism. Write $\mathrm{Solid} \subset \mathrm{CondAb}^{\mathrm{light}}$ for the full subcategory of solid objects.

*Remark* 5.6 (The condition is unique summability). Evaluating at a point, $\underline{\mathrm{Hom}}(P, M)$ is the space of null sequences $(m_0, m_1, \dots)$ in $M$, and $f^*$ is the difference operator

$$(m_0, m_1, m_2, \dots) \longmapsto (m_0 - m_1,\ m_1 - m_2,\ \dots).$$

Its inverse must be the telescoping/summation operator

$$(m_0, m_1, m_2, \dots) \longmapsto \big(m_0 + m_1 + m_2 + \cdots,\ m_1 + m_2 + \cdots,\ \dots\big),$$

recovering $m_0 = (m_0 - m_1) + (m_1 - m_2) + \cdots$ by telescoping. Thus (5.3) says exactly that every null sequence in $M$ has a unique sum, with the tails themselves forming a null sequence. Imposing the condition on the *internal* Hom, rather than merely on the naive Hom of null sequences, is a strengthening that turns out to be much better behaved.

## 5.3 Solid abelian groups form an abelian category

**Proposition 5.7** (Solid is a well-behaved abelian subcategory)**.** *The subcategory* $\mathrm{Solid} \subset \mathrm{CondAb}^{\mathrm{light}}$ *is an abelian subcategory, stable under kernels, cokernels, extensions, all limits and all colimits, internal Hom* $\underline{\mathrm{Hom}}$*, and internal Ext* $\underline{\mathrm{Ext}}^i$*; and it contains* $\mathbb{Z}$*. In the terminology to be introduced later, this proposition exhibits an* analytic ring structure on the integers.

*Proof.* Because $P$ is internally projective, $\underline{\mathrm{Hom}}(P, -)$ is exact and preserves all limits and colimits; the endomorphism $f$ of (5.2) induces a natural endomorphism $f^*$ of $\underline{\mathrm{Hom}}(P, -)$, and solidity is the assertion that $f^*$ is an isomorphism on the object. Every closure property is now inherited from a corresponding property of the functor $\underline{\mathrm{Hom}}(P, -)$.

*Kernels, cokernels, extensions, limits, colimits.* Given a map of solid groups $M \to M'$, apply the exact functor $\underline{\mathrm{Hom}}(P, -)$: it carries the kernel to the kernel and the cokernel to the cokernel, and $f^*$ acts compatibly, so it is an isomorphism on both – the same condition holds for the (co)kernel. For an extension $0 \to M' \to M \to M'' \to 0$ with $M', M''$ solid, exactness produces a short exact sequence of the $\underline{\mathrm{Hom}}(P, -)$'s on which $f^*$ is an isomorphism on the outer terms, hence on the middle by the five lemma. Stability under all limits and colimits is likewise immediate, since $\underline{\mathrm{Hom}}(P, -)$ preserves them and being solid is a pointwise-isomorphism condition.

*Internal Hom and internal Ext.* Let $M$ be solid and $N \in \mathrm{CondAb}^{\mathrm{light}}$ arbitrary; we claim $\underline{\mathrm{Hom}}(N, M)$ is solid. By tensor–Hom adjunction,

$$\underline{\mathrm{Hom}}\big(P, \underline{\mathrm{Hom}}(N, M)\big) = \underline{\mathrm{Hom}}\big(P \otimes N,\ M\big) = \underline{\mathrm{Hom}}\big(N, \underline{\mathrm{Hom}}(P, M)\big),$$

and $f^*$ becomes the map induced by $f$ on the last object, which is an isomorphism because $M$ is solid; so $f^*$ is an isomorphism on $\underline{\mathrm{Hom}}(P, \underline{\mathrm{Hom}}(N, M))$, i.e. $\underline{\mathrm{Hom}}(N, M)$ is solid. Since $\underline{\mathrm{Hom}}(P, -)$ is exact, the identical manipulation with $\underline{\mathrm{Ext}}^i$ in place of $\underline{\mathrm{Hom}}$ gives

$$\underline{\mathrm{Hom}}\big(P, \underline{\mathrm{Ext}}^i(N, M)\big) = \underline{\mathrm{Ext}}^i\big(P \otimes N,\ M\big) = \underline{\mathrm{Ext}}^i\big(N, \underline{\mathrm{Hom}}(P, M)\big),$$

so $\underline{\mathrm{Ext}}^i(N, M)$ is solid as well.

*$\mathbb{Z}$ is solid.* A null sequence in $\mathbb{Z}$ is eventually zero, so

$$\underline{\mathrm{Hom}}(P, \mathbb{Z}) = \bigoplus_{\mathbb{N}} \mathbb{Z}, \tag{5.4}$$

and $f = \mathrm{id} - \mathrm{shift}$ is an isomorphism on $\bigoplus_{\mathbb{N}} \mathbb{Z}$: an eventually-zero sequence is trivially summable. (A little care is needed to see this internally, again by adjunction.) Hence $\mathbb{Z}$ is solid, and by the closure properties so is everything built from it. $\square$

*Remark* 5.8 (The point of the argument)*.* Previously, constructing an analytic ring structure was something of an art: there were many conditions to check and none easy to guarantee, so essentially only two examples were built by hand – the solid structure on $\mathbb{Z}$ and the liquid structure on $\mathbb{R}$ and related rings – by hard, handcrafted proofs. Because $P$ is internally projective, imposing the isomorphism condition (5.3) for the single endomorphism $f$ makes all of the stability properties come for free; for any endomorphism of $P$ one could ask such a condition. What is *not* for free is the existence of a nonzero object satisfying it, which is exactly the content of (5.4).

The inclusion of a subcategory closed under all limits admits a left adjoint.

**Corollary 5.9** (Solidification)**.** *The inclusion* Solid $\subset$ $\mathrm{CondAb}^{\mathrm{light}}$ *admits a left adjoint, the* solidification

$$(-)^{\square} \colon \mathrm{CondAb}^{\mathrm{light}} \longrightarrow \mathrm{Solid}, \qquad M \longmapsto M^{\square},$$

*characterized by*

$$\mathrm{Hom}(M, N) = \mathrm{Hom}(M^{\square}, N) \qquad \textit{for all } N \in \mathrm{Solid}.$$

*It is colimit-preserving, and* Solid *acquires a unique symmetric monoidal tensor product* $\otimes^{\square}$ *making* $(-)^{\square}$ *symmetric monoidal, given concretely by*

$$M \otimes^{\square} N := (M \otimes N)^{\square}.$$

*Sketch.* Existence of $(-)^{\square}$ is the adjoint functor theorem: the inclusion preserves all limits, and under mild set-theoretic hypotheses (presentability) the left adjoint then exists. For the monoidal structure one must check

$$(M \otimes N)^{\square} \xrightarrow{\sim} \big(M^{\square} \otimes N^{\square}\big)^{\square},$$

which one verifies by mapping into an arbitrary solid $S$: by adjunction and tensor–Hom,

$$\mathrm{Hom}(M \otimes N, S) = \mathrm{Hom}\big(N, \underline{\mathrm{Hom}}(M, S)\big),$$

and since $\underline{\mathrm{Hom}}(M, S)$ is solid (Proposition 5.7) one may replace $N$ by $N^{\square}$; replacing $M$ by $M^{\square}$ symmetrically gives $\mathrm{Hom}(M^{\square} \otimes N^{\square}, S)$. The key input is exactly stability of Solid under internal Hom. $\square$

*Remark* 5.10 (A possible right adjoint). Since Solid is also stable under all colimits, the inclusion preserves colimits and so, formally, often admits a *right* adjoint as well; Scholze remarked he had never had occasion to consider it.

## 5.4 The reals are annihilated

We now record how solidity interacts with the real numbers: not only is $\mathbb{R}$ not solid, but it admits no nonzero map to any solid group.

**Lemma 5.11** ($\mathbb{R}^{\square} = 0$)**.** *The solidification of the reals vanishes:*

$$\underline{\mathbb{R}}^{\square} = 0.$$

*Proof.* Solidification of a ring is a ring (the solid tensor product is symmetric monoidal), so $\underline{\mathbb{R}}^{\square}$ is a ring and it suffices to show $1 = 0$ in it. Consider the null sequence in $\mathbb{R}$

$$1,\ \tfrac{1}{2}, \tfrac{1}{2},\ \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4},\ \ldots \qquad \big(\tfrac{1}{2^n} \text{ repeated } 2^n \text{ times}\big),$$

i.e. a map $P \to \underline{\mathbb{R}}$. Composing to $\underline{\mathbb{R}} \to \underline{\mathbb{R}}^{\square}$ and using that the target is solid, the null sequence becomes uniquely summable: there is a unique map from the summed sequence, intuitively the sequence of tails. Restricting along the inclusion $\mathbb{Z} \xrightarrow{[0]} P$ of the first term picks out the total sum

$$x := 1 + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4} + \cdots \ \in\ \underline{\mathbb{R}}^{\square}.$$

Now

$$x = 1 + \Big(\underbrace{\tfrac{1}{2} + \tfrac{1}{2}} + \underbrace{\tfrac{1}{4} + \tfrac{1}{4}} + \underbrace{\tfrac{1}{4} + \tfrac{1}{4}} + \cdots\Big) = 1 + \big(1 + \tfrac{1}{2} + \tfrac{1}{2} + \cdots\big) = 1 + x,$$

where in the second step one is allowed to first sum adjacent pairs – a finite-character operation which cannot change the value – reproducing the same series. Hence $x = 1 + x$, so $1 = 0$ and $\underline{\mathbb{R}}^{\square} = 0$. $\square$

*Remark* 5.12 (Provenance of the argument). Scholze noted that finding an argument for Lemma 5.11 took him and Clausen considerable effort; the divergent-null-sequence argument above he had worked out only the day before the lecture.

**Corollary 5.13** ($\underline{\mathbb{R}}$-modules are killed)**.** *If* $M \in \mathrm{CondAb}^{\mathrm{light}}$ *admits an* $\underline{\mathbb{R}}$*-module structure, then* $M^{\square} = 0$*; in fact*

$$\underline{\mathrm{Ext}}^i(M, \textit{solid}) = 0 \qquad \textit{for all } i \geq 0.$$

*Proof.* Resolve the $\underline{\mathbb{R}}$-module $M$ by the bar resolution of free $\underline{\mathbb{R}}$-modules,

$$\cdots \longrightarrow M\otimes\underline{\mathbb{R}}\otimes\underline{\mathbb{R}} \longrightarrow M\otimes\underline{\mathbb{R}} \longrightarrow M \longrightarrow 0 \qquad \text{(exact)}.$$

Solidifying (which is right exact), each term $M\otimes\underline{\mathbb{R}}$ becomes

$$(M\otimes\underline{\mathbb{R}})^{\square} = M^{\square}\otimes^{\square}\underline{\mathbb{R}}^{\square} = 0$$

by Lemma 5.11, so $M^{\square}$ is a colimit of zeros, i.e. $M^{\square}=0$. Equivalently, the groups $\underline{\mathrm{Ext}}^i(M,\mathrm{solid})$ are modules over $\underline{\mathbb{R}}^{\square}=0$, hence vanish. (For the $\underline{\mathrm{Ext}}$ statement one uses the derived solidification, or one observes directly that the internal Ext against a solid object is an $\underline{\mathbb{R}}^{\square}$-module.) ☐

*Remark* 5.14 (It is about the topology, not the field)*.* The obstruction is topological, not algebraic. A characteristic-zero field with its *discrete* topology is perfectly solid; the rationals $\mathbb{Q}$ with the *subspace* topology from $\mathbb{R}$ are killed, just as $\mathbb{R}$ is. Likewise anything pro-discrete – an inverse limit of discrete abelian groups, such as $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n$ – is solid.

## 5.5 Computing the solidification of $P$

The goal is now to compute $P^{\square}$. Solidifying $P$ forces in many new elements: there is a genuine null sequence in $P$, so its sum $1+1+1+\cdots$ (of the constant sequence) must become defined, and so on; one expects $P^{\square}$ to be much larger than $P$. The precise statement passes through a bounded product.

*Notation* 5.15 (Bounded product)*.* Set

$$\prod_{\mathbb{N}}^{\mathrm{bdd}}\mathbb{Z} := \bigcup_{n\in\mathbb{N}}\prod_{\mathbb{N}}\bigl(\mathbb{Z}\cap[-n,n]\bigr) \ \subseteq\ \prod_{\mathbb{N}}\mathbb{Z},$$

the subspace of bounded integer sequences (an increasing union of profinite subspaces). The standard basis vectors $e_n=(0,\dots,0,1,0,\dots)$ (1 in the $n$-th coordinate) form a null sequence in $\prod_{\mathbb{N}}^{\mathrm{bdd}}\mathbb{Z}$, because its topology is that of pointwise (coordinatewise) convergence; this gives a map $P\to\prod_{\mathbb{N}}^{\mathrm{bdd}}\mathbb{Z}$, $[n]\mapsto e_n$.

**Lemma 5.16** ($P^{\square}$ via the bounded product)**.** *The map $[n]\mapsto e_n$ induces an isomorphism*

$$P^{\square} \xrightarrow{\ \sim\ } \Bigl(\prod_{\mathbb{N}}^{\mathrm{bdd}}\mathbb{Z}\Bigr)^{\square},$$

*and more precisely, for every solid $M$ and all $i\geq 0$,*

$$\underline{\mathrm{Ext}}^i\Bigl(\prod_{\mathbb{N}}^{\mathrm{bdd}}\mathbb{Z},\, M\Bigr) \xrightarrow{\ \sim\ } \underline{\mathrm{Ext}}^i(P,M).$$

*Idea.* Both the identity of $\prod_{\mathbb{N}}^{\mathrm{bdd}}\mathbb{Z}$ and the composite through $f\otimes\mathrm{id}$ are expressed as sums of null sequences of endomorphisms. Writing $T=\prod_{\mathbb{N}}^{\mathrm{bdd}}\mathbb{Z}$, a map $P\otimes T\to X$ is the same as a null sequence of maps $T\to X$ (a map $P\to\underline{\mathrm{Hom}}(T,X)$). The evaluation/sum map $P\otimes T\to T$ corresponds to the null sequence of endomorphisms $p_n\colon T\to T$, "projection to the $n$-th coordinate", whose formal sum is the identity; the composite obtained after applying $f\otimes\mathrm{id}$ corresponds instead to the tails $q_n$, "projection to the coordinates $\geq n$", with $q_n-q_{n+1}=p_n$. Each $p_n$ factors through a single copy of

$\mathbb{Z}$, hence through the image of $P \to T$; this is exactly where the *bounded* product is needed, since for the full product the coordinates would be unbounded and the factorization would fail.

Solidifying the resulting square, $f \otimes \mathrm{id}$ becomes an isomorphism $(f \otimes \mathrm{id})^{\square}$ (as $f$ solidifies to an isomorphism, by the very definition of solidity), and the sum map becomes an isomorphism as well (solidification is monoidal). One reads off that the induced map $P^{\square} \to T^{\square}$ is a split surjection whose splitting composes to the identity; a short diagram chase upgrades it to an isomorphism (the resulting idempotent of $P^{\square}$ is the identity). The identical argument applies to $\underline{\mathrm{Ext}}^i(-, \mathrm{solid})$. $\square$

Next we replace the bounded product by the full product.

**Lemma 5.17** (Bounded product versus full product)**.** *The inclusion induces an isomorphism*

$$\Big(\prod_{\mathbb{N}}^{\mathrm{bdd}} \mathbb{Z}\Big)^{\square} \xrightarrow{\sim} \Big(\prod_{\mathbb{N}} \mathbb{Z}\Big)^{\square} = \prod_{\mathbb{N}} \mathbb{Z}, \tag{5.5}$$

*the full product being already solid, and similarly on* $\underline{\mathrm{Ext}}^i(-, \mathit{solid})$*.*

*Proof.* Apply $\underline{\mathrm{Ext}}^*(-, M)$, $M$ solid, to the short exact sequence

$$0 \longrightarrow \prod_{\mathbb{N}}^{\mathrm{bdd}} \mathbb{Z} \longrightarrow \prod_{\mathbb{N}} \mathbb{Z} \longrightarrow \prod_{\mathbb{N}} \mathbb{Z} \Big/ \prod_{\mathbb{N}}^{\mathrm{bdd}} \mathbb{Z} \longrightarrow 0.$$

It suffices to show the quotient has vanishing $\underline{\mathrm{Ext}}^i(-, \mathrm{solid})$ for all $i \geq 0$. The point is:

$$\prod_{\mathbb{N}} \mathbb{Z} \Big/ \prod_{\mathbb{N}}^{\mathrm{bdd}} \mathbb{Z} \;\cong\; \prod_{\mathbb{N}} \underline{\mathbb{R}} \Big/ \prod_{\mathbb{N}}^{\mathrm{bdd}} \underline{\mathbb{R}} \tag{5.6}$$

is an $\underline{\mathbb{R}}$-module. Scholze verified this by comparing the integer and real versions of the short exact sequence $0 \to \prod_{\mathbb{N}}^{\mathrm{bdd}}(-) \to \prod_{\mathbb{N}}(-) \to \prod_{\mathbb{N}}(-)/\prod_{\mathbb{N}}^{\mathrm{bdd}}(-) \to 0$ along the inclusion $\mathbb{Z} \hookrightarrow \underline{\mathbb{R}}$, laid out as a $3 \times 3$ diagram with exact rows and columns:

$$\begin{tikzcd}
0 \arrow[r] & \prod_{\mathbb{N}}^{\mathrm{bdd}} \mathbb{Z} \arrow[r] \arrow[d] & \prod_{\mathbb{N}}^{\mathrm{bdd}} \underline{\mathbb{R}} \arrow[r] \arrow[d] & \prod_{\mathbb{N}}(\mathbb{R}/\mathbb{Z}) \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \prod_{\mathbb{N}} \mathbb{Z} \arrow[r] \arrow[d] & \prod_{\mathbb{N}} \underline{\mathbb{R}} \arrow[r] \arrow[d] & \prod_{\mathbb{N}}(\mathbb{R}/\mathbb{Z}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \prod_{\mathbb{N}} \mathbb{Z}/\prod_{\mathbb{N}}^{\mathrm{bdd}} \mathbb{Z} \arrow[r, "\sim"] & \prod_{\mathbb{N}} \underline{\mathbb{R}}/\prod_{\mathbb{N}}^{\mathrm{bdd}} \underline{\mathbb{R}} \arrow[r] & 0 \arrow[r] & 0
\end{tikzcd}$$

Because $\mathbb{R}/\mathbb{Z}$ is bounded, the cokernel of $\mathbb{Z} \hookrightarrow \underline{\mathbb{R}}$ is the *same* object $\prod_{\mathbb{N}}(\mathbb{R}/\mathbb{Z})$ at both the bounded and the full level (the "difference" Scholze marked on either side), and the comparison map between these two cokernels is the identity, so the right-hand column is exact with vanishing bottom term. The nine lemma then forces the bottom row to be exact, i.e. the map (5.6) to be an isomorphism. Being an $\underline{\mathbb{R}}$-module, it is annihilated by Corollary 5.13, and the long exact sequence gives (5.5). $\square$

Combining the two lemmas yields the fundamental computation.

**Corollary 5.18** ($P^\square$ is a product)**.** *There is a canonical isomorphism*

$$P^\square \xrightarrow{\sim} \prod_{\mathbb{N}} \mathbb{Z},$$

*and for all solid $M$, $\underline{\mathrm{Ext}}^i(P, M) \xleftarrow{\sim} \underline{\mathrm{Ext}}^i_{\mathrm{CondAb}}\big(\prod_{\mathbb{N}} \mathbb{Z}, M\big)$. In particular, for discrete $M$,*

$$\mathrm{Ext}^i\Big(\prod_{\mathbb{N}} \mathbb{Z},\, M\Big) = \begin{cases} \bigoplus_{\mathbb{N}} M & i = 0, \\ 0 & i > 0, \end{cases} \tag{5.7}$$

*so that $\prod_{\mathbb{N}} \mathbb{Z}$ is a compact projective object of* Solid*.*

*Remark* 5.19 (An independent route to Corollary 4.21). Equation (5.7) is precisely Corollary 4.21, the vanishing of the higher Ext-groups out of $\prod_{\mathbb{N}} \mathbb{Z}$. In Section 4 it was a hard input, proved by embedding into products of $\underline{\mathbb{R}}$; here it comes out as an *output* of the light presentation of solidity, an independent argument. (The proof of Lemma 5.17 does use, through Corollary 5.13 and Lemma 5.11, that products of $\underline{\mathbb{R}}$ are annihilated.)

The completed tensor product now does what was wanted in Remark 5.1.

**Corollary 5.20** (The completed tensor square)**.** *There is a canonical isomorphism*

$$\prod_{\mathbb{N}} \mathbb{Z} \;\otimes^\square\; \prod_{\mathbb{N}} \mathbb{Z} \;\cong\; \prod_{\mathbb{N}\times\mathbb{N}} \mathbb{Z},$$

*coming from $\prod_{\mathbb{N}} \mathbb{Z} \otimes^\square \prod_{\mathbb{N}} \mathbb{Z} = (P \otimes P)^\square$ (monoidality of solidification) and the two-variable analogue of Corollary 5.18. Identifying $\prod_{\mathbb{N}} \mathbb{Z}$ with the power-series ring in one variable (measures, cf. Remark 1.34), this reads*

$$\mathbb{Z}[\![T]\!] \;\otimes^\square\; \mathbb{Z}[\![U]\!] \;\cong\; \mathbb{Z}[\![T, U]\!],$$

*so the completed tensor product of two power-series rings is the power-series ring in both variables – exactly the reasonable answer sought at the outset.*

## 5.6 The solidification of a general free group

The computation extends from the convergent sequence to an arbitrary profinite set, and the resulting formula is the one that *defined* solidification in the original approach.

**Proposition 5.21** (Solidification of $\mathbb{Z}[S]$)**.** *Let $S$ be any infinite light profinite set, written $S = \varprojlim_n S_n$ with the $S_n$ finite and surjective transition maps. Then there is a map $g\colon P \to \mathbb{Z}[S]$ inducing an isomorphism*

$$\mathbb{Z}[S]^\square \;\cong\; \prod_{\mathbb{N}} \mathbb{Z},$$

*and canonically*

$$\mathbb{Z}[S]^\square \;\xrightarrow{\sim}\; \varprojlim_n \mathbb{Z}[S_n], \tag{5.8}$$

*compatibly with $\underline{\mathrm{Ext}}^i(-, solid)$.*

*Sketch.* Let $\pi_n\colon S \to S_n$ be the projection. Choose sections inductively: a section $i_0\colon S_0 \hookrightarrow S$ of $\pi_0$; then, on the elements of $S_1$ not in the image of $i_0$, a further section $S_1 \setminus S_0 \to S$; gluing gives

$i_1 \colon S_1 \to S$ (equal to $i_0\pi_0$-lift on old elements, new on the rest); and so on.

$$\begin{tikzcd}
& S \arrow[dl, "\pi_n"'] & & \\
S_n \arrow[ur, dashed, bend right, "i_n"'] \arrow[r, two heads] & S_{n-1} \arrow[r, two heads] & \cdots \arrow[r, two heads] & S_0
\end{tikzcd}$$

The union of the images is a countable subset of $S$, and one enumerates

$$\mathbb{N} = S_0 \sqcup (S_1 \setminus S_0) \sqcup (S_2 \setminus S_1) \sqcup \cdots .$$

Define $g \colon P \to \mathbb{Z}[S]$ on this enumeration: on $S_0$ by $i_0$, and on $S_n \setminus S_{n-1}$ by the difference of the maps induced by $i_n$ and by $S_n \setminus S_{n-1} \to S_{n-1} \xrightarrow{i_{n-1}} S$. As in Lemma 5.16, the identity endomorphism of $\mathbb{Z}[S]$ is the sum of the null sequence of endomorphisms

$$\mathrm{id},\ \mathrm{id} - i_0\pi_0,\ \mathrm{id} - i_1\pi_1,\ \ldots,\ \mathrm{id} - i_n\pi_n,\ \ldots,$$

equivalently of the differences $i_0\pi_0,\ i_1\pi_1 - i_0\pi_0, \ldots$, each of which factors through the image of $g$ (this uses that $S$ is a *light* profinite set, so that it is exhausted through countably many finite quotients). Solidifying as before produces the isomorphism. $\square$

*Remark* 5.22 (The formula that used to be the definition). In the original set-up of the solid theory [CS19], the formula $\mathbb{Z}[S]^{\square} = \varprojlim_n \mathbb{Z}[S_n]$ of (5.8) was the *starting point*: one declared by hand what the solidification of each generator $\mathbb{Z}[S]$ should be, taking the naive inverse limit, and then checked by hand that this yields a valid theory – a much harder route. In the present light presentation, solidity is defined first (Definition 5.5) and (5.8) becomes a computation.

## 5.7 Structure theorem and integration

**Theorem 5.23** (Structure of Solid)**.** *The subcategory* Solid $\subset \mathrm{CondAb}^{\mathrm{light}}$ *is abelian and stable under all limits and colimits, and:*

1. *it has a single compact projective generator $\prod_{\mathbb{N}} \mathbb{Z}$, which is internally projective and satisfies*

$$\prod_{\mathbb{N}} \mathbb{Z} \ \otimes^{\square} \ \prod_{\mathbb{N}} \mathbb{Z} \ \cong \ \prod_{\mathbb{N}\times\mathbb{N}} \mathbb{Z};$$

2. *an object $M \in \mathrm{CondAb}^{\mathrm{light}}$ is solid if and only if, for every $S \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$ (with $S = \varprojlim_n S_n$) and every map $g \colon S \to M$, there is a unique extension of $g$ along $S \to \mathbb{Z}[S]^{\square} = \varprojlim_n \mathbb{Z}[S_n]$,*

$$\mathbb{Z}[S]^{\square} = \varprojlim_n \mathbb{Z}[S_n] \dashrightarrow M. \tag{5.9}$$

The characterization (5.9) was the *original* definition of solidity in [CS19]; note it involves only ordinary maps out of $S$, not internal Hom, in contrast to Definition 5.5.

*Proof.* Only the "$\Leftarrow$" direction of (2) is new. Present $M$ in $\mathrm{CondAb}^{\mathrm{light}}$ as a cokernel of free groups,

$$\bigoplus_j \mathbb{Z}[S_j] \longrightarrow \bigoplus_i \mathbb{Z}[S_i] \longrightarrow M \longrightarrow 0.$$

The hypothesis of (2) says that every map $\mathbb{Z}[S_i] \to M$ extends uniquely to $\mathbb{Z}[S_i]^\square$, and likewise for the $S_j$, so the presentation solidifies to

$$\bigoplus_j \mathbb{Z}[S_j]^\square \longrightarrow \bigoplus_i \mathbb{Z}[S_i]^\square \dashrightarrow M,$$

exhibiting $M$ as a cokernel of solid objects; since Solid is closed under cokernels (Proposition 5.7), $M$ is solid. Equivalently, the universal property makes $M$ a *retract* of its solidification $M^\square$ (the composite $M \to M^\square \to M$ being the identity, and a comparison map going back), and retracts of solid objects are solid. The forward direction is clear, since a map $S \to M$ is a map $\mathbb{Z}[S] \to M$, which factors uniquely through the solidification when $M$ is solid. $\square$

*Remark* 5.24 (Solidity as integration against measures). The philosophical content of the condition is integration. One has

$$\mathbb{Z}[S]^\square = \underline{\mathrm{Hom}}\big(\mathrm{Cont}(S, \mathbb{Z}),\, \mathbb{Z}\big),$$

because $\mathrm{Cont}(S, \mathbb{Z})$ is the filtered colimit of the functions on the finite quotients $S_n$, and dualizing gives the inverse limit (5.8); so $\mathbb{Z}[S]^\square$ is the group of *$\mathbb{Z}$-valued measures* on $S$. Thus, by (5.9), $M$ is solid precisely when, given a map $g\colon S \to M$ from a profinite set and a $\mathbb{Z}$-valued measure $\mu$ on $S$, one can form the integral

$$\int_S g \,\mathrm{d}\mu \;\in\; M.$$

Note this final characterization speaks only of ordinary Hom, whereas the null-sequence definition Definition 5.5 was phrased on the internal Hom.

*Remark* 5.25 (Lightness is essential). The characterization by null sequences is genuinely a light phenomenon. Were one to attempt the same definition for *all* condensed abelian groups, the null-sequence condition would be too weak: it only constrains countable data, whereas a general profinite set is not exhausted by countably many finite quotients. It is exactly the fact that a light profinite set *is* exhausted through countably many finite quotients that makes Proposition 5.21 work.

*Remark* 5.26 (A synthetic description; and the liquid question). Having a single compact projective generator $\prod_{\mathbb{N}} \mathbb{Z}$ means Solid is equivalent to modules over its endomorphism ring, so one can describe Solid purely algebraically, with no mention of profinite sets. The morphisms of the generator are

$$\mathrm{Hom}\Big(\prod_{\mathbb{N}} \mathbb{Z}, \prod_{\mathbb{N}} \mathbb{Z}\Big) = \prod_{\mathbb{N}} \bigoplus_{\mathbb{N}} \mathbb{Z},$$

i.e. infinite matrices each of whose rows is eventually zero; Scholze deferred this synthetic description to the next lecture. Asked whether the liquid (archimedean) theory of vector spaces admits an analogous null-sequence-style characterization, he said he expected it should be possible and that he and Clausen were currently working out the details.

# 6 Complements on Solid Abelian Groups

Scholze's last lecture before Clausen resumes is a collection of complements to the light theory of solid abelian groups built in Section 5. It has four movements. First, the whole picture is upgraded to the derived category: one defines solidity of a complex, gets a fully faithful $D(\mathrm{Solid}) \hookrightarrow D(\mathrm{CondAb}^{\mathrm{light}})$ with a derived solidification left adjoint, and observes that *derived* solidification is where genuine homotopy theory enters – for a CW complex $X$ the derived solidification of $\mathbb{Z}[\underline{X}]$ is the singular chain complex. Second, the abelian category Solid is analyzed: its finitely presented objects form an abelian category and are classified, the crux being a set-theoretic "freeness" lemma about finitely generated submodules of $\prod_{\mathbb{N}} \mathbb{Z}$, and $\prod_{\mathbb{N}} \mathbb{Z}$ turns out to be flat – a fact special to the light setting. Third, a family of solid tensor computations is proved, showing that $\otimes^{\square}$ reproduces the completed tensor products of formal geometry. Fourth, the same machinery is run on $p$-adic functional analysis (Smith spaces, Banach spaces, Fréchet spaces), where the solid tensor product recovers the classical completed tensor products – with one derived-duality question that, like the phenomenon of Section 4, depends on the model of set theory.

Throughout, the presentation continues to follow Scholze's first course [CS19], though the light-specific arguments are new.

## 6.1 Recollection

Recall from Section 5 the object

$$P := \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/\mathbb{Z}[\infty] \in \mathrm{CondAb}^{\mathrm{light}},$$

the free light condensed abelian group on a null sequence (Notation 5.4), which is *internally projective* (Theorem 3.15(3)), together with the endomorphism $f = \mathrm{id} - \mathrm{shift}\colon P \to P$, $[n] \mapsto [n] - [n+1]$ (Equation (5.2)). An object $M \in \mathrm{CondAb}^{\mathrm{light}}$ is solid (Definition 5.5) iff $f^*\colon \underline{\mathrm{Hom}}(P, M) \to \underline{\mathrm{Hom}}(P, M)$ is an isomorphism, i.e. iff every null sequence in $M$ is uniquely summable. The main theorem of Section 5 (Theorem 5.23, Corollary 5.9) was:

**Theorem 6.1** (Recalled from Section 5)**.** Solid $\subset \mathrm{CondAb}^{\mathrm{light}}$ *is an abelian subcategory, stable under all limits and colimits; the inclusion acquires a colimit-preserving left adjoint $M \mapsto M^{\square}$ (*solidification*) and a unique symmetric monoidal $\otimes^{\square}$ making solidification symmetric monoidal. It is generated by a single compact projective object $\prod_{\mathbb{N}} \mathbb{Z}$, with*

$$\prod_{\mathbb{N}} \mathbb{Z} \otimes^{\square} \prod_{\mathbb{N}} \mathbb{Z} \cong \prod_{\mathbb{N}\times\mathbb{N}} \mathbb{Z}, \tag{6.1}$$

*and for $S = \varprojlim_n S_n \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$,*

$$\mathbb{Z}[S]^{\square} \xrightarrow{\sim} \varprojlim_n \mathbb{Z}[S_n].$$

## 6.2 The derived enhancement

The definition of solidity makes sense verbatim in the derived category. Write $D(\mathrm{CondAb}^{\mathrm{light}})$ for the (unbounded) derived category of light condensed abelian groups.

**Definition 6.2** (Solid complexes)**.** An object $A \in D(\mathrm{CondAb}^{\mathrm{light}})$ is *solid* if

$$f^*\colon R\underline{\mathrm{Hom}}(P, A) \xrightarrow{\sim} R\underline{\mathrm{Hom}}(P, A) \tag{6.2}$$

is an isomorphism.

Because $P$ is internally projective, $R\underline{\mathrm{Hom}}(P, -)$ is exact, so $H^i\big(R\underline{\mathrm{Hom}}(P, A)\big) = \underline{\mathrm{Hom}}\big(P, H^i(A)\big)$ and (6.2) is equivalent to asking that $f^*$ be an isomorphism on $\underline{\mathrm{Hom}}(P, H^i(A))$ for every $i$, that is:

$$A \text{ solid} \iff H^i(A) \in \mathrm{Solid} \ \text{ for all } i \in \mathbb{Z}.$$

So solidity of a complex may be tested on cohomology groups. This is again immediate from the internal projectivity of $P$; that completeness for an analytic ring structure can be checked on cohomology is in general a subtle fact.

**Corollary 6.3.** *The class of solid objects of* $D(\mathrm{CondAb}^{\mathrm{light}})$ *is a triangulated subcategory, stable under all direct sums, all products and all limits and colimits, and stable under internal* $R\underline{\mathrm{Hom}}$.

The proofs are the same as in Section 5 (via the exactness of $R\underline{\mathrm{Hom}}(P,-)$). Passing to derived categories functorially, one gets:

**Proposition 6.4** (Derived solidification)**.** *The functor*

$$D(\mathrm{Solid}) \longrightarrow D(\mathrm{CondAb}^{\mathrm{light}}) \tag{6.3}$$

*is fully faithful with essential image the solid objects. The inclusion admits a colimit-preserving left adjoint, the* derived solidification

$$(-)^{\mathbb{L}\square}\colon D(\mathrm{CondAb}^{\mathrm{light}}) \longrightarrow D(\mathrm{Solid}), \qquad A \longmapsto A^{\mathbb{L}\square},$$

*and* $D(\mathrm{Solid})$ *carries a unique symmetric monoidal* $\otimes^{\mathbb{L}\square}$ *making* $A \mapsto A^{\mathbb{L}\square}$ *symmetric monoidal; it is the left derived functor of* $\otimes^{\square}$ *(and the derived tensor product is likewise obtained by deriving the one already present on* $\mathrm{CondAb}^{\mathrm{light}}$*).*

*Remark* 6.5 (On the existence of the adjoint)*.* Full faithfulness of (6.3) is not automatic: it uses that the compact projective generators stay Ext-vanishing after passage to $\mathrm{CondAb}^{\mathrm{light}}$, i.e. that derived solidification of a complex whose terms are direct sums of the generator $\prod_{\mathbb{N}} \mathbb{Z}$ is computed term by term. Existence of the left adjoint is the adjoint functor theorem for presentable $\infty$-categories (Lurie [Lur17]); Scholze noted that theorems of Neeman and others would also suffice, though they often assume genuine compact generation, which does not quite hold here, whereas Lurie's version is definitely sufficient. In any case the adjoint can be constructed by hand, since it is already identified on the generators. The abstract theorem applies in both variances: a colimit-preserving functor admits a right adjoint, while a functor preserving all limits and all sufficiently filtered colimits (i.e. an accessible one) admits a left adjoint.

On the generators, derived solidification agrees with the underived one – there are no higher homotopies:

$$\mathbb{Z}[S]^{\mathbb{L}\square} \xrightarrow{\sim} \varprojlim_n \mathbb{Z}[S_n] = \mathbb{Z}[S]^{\square}, \qquad P^{\mathbb{L}\square} \xrightarrow{\sim} \prod_{\mathbb{N}} \mathbb{Z},$$

and correspondingly $\prod_{\mathbb{N}} \mathbb{Z} \otimes^{\mathbb{L}\square} \prod_{\mathbb{N}} \mathbb{Z} \xrightarrow{\sim} \prod_{\mathbb{N}\times\mathbb{N}} \mathbb{Z}$ is the underived product.

### 6.3 Derived solidification and singular homology

One might think nothing derived ever appears in practice. It does.

**Proposition 6.6** (Solidification computes singular homology)**.** *Let $X$ be a CW complex. Then there is a quasi-isomorphism*

$$C_\bullet^{\mathrm{sing}}(X,\mathbb{Z}) \xrightarrow{\ \sim\ } \mathbb{Z}[\underline{X}]^{\mathbb{L}\square},$$

*in particular* $H_i^{\mathrm{sing}}(X,\mathbb{Z}) \cong H_i\big(\mathbb{Z}[\underline{X}]^{\mathbb{L}\square}\big)$*, and the derived solidification is a complex of* discrete *abelian groups.*

This is the homological counterpart of the cohomological statement $\mathrm{Ext}^i(\mathbb{Z}[\underline{X}],\mathbb{Z}) \cong H^i_{\mathrm{sing}}(X,\mathbb{Z})$ of Theorem 4.1, and is again a form of the Dold–Thom philosophy: $\mathbb{Z}[\underline{X}]$ models the homology of $X$.

**Example 6.7** (Contractible vs. non-contractible)**.** The free light condensed abelian group $\mathbb{Z}[\underline{[0,1]}]$ on the interval is a large object (its underlying group is finite sums of points), but solidifying kills the interval:

$$\mathbb{Z}[\underline{[0,1]}]^{\mathbb{L}\square} \simeq \mathbb{Z}.$$

For the circle,

$$\mathbb{Z}[\underline{S^1}]^{\mathbb{L}\square} \simeq \mathbb{Z} \oplus \mathbb{Z}[1].$$

For a space with homology in unboundedly many degrees – e.g. an infinite wedge of spheres of all dimensions – the object $\mathbb{Z}[\underline{X}]$ sits in degree 0, yet its derived solidification has homology in arbitrarily high degrees, i.e. is genuinely unbounded.

*Remark* 6.8 (Solidification as a partial passage to homotopy types). Thus the passage $\mathrm{CondAb}^{\mathrm{light}} \to$ Solid is, at least on CW complexes, like passing to homotopy types: an interval is contracted, and homotopy equivalent CW complexes are identified. On totally disconnected spaces it retains much finer information. One inverts homotopy equivalences but *not* weak homotopy equivalences. The same holds, Scholze noted, for locally contractible spaces (in the "funny" sense of Remark 4.4); and Mohan Ramachandran pointed out to him that the paracompactness hypotheses in comparing singular and sheaf cohomology are actually not needed – there are recent papers to that effect.

*Proof sketch of Proposition 6.6.* A formal argument reduces to $X$ a finite CW complex (a filtered colimit in $X$ becomes a limit, and one keeps the finite generation of the homology in each degree). Assume then $X$ finite CW, hence compact. Choose a surjection $S \twoheadrightarrow X$ from a light profinite set $S \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$ and form the Čech nerve; as in Section 4 its fibre products $S \times_X S, \ldots$ are again totally disconnected, giving a resolution

$$\cdots \to \mathbb{Z}[S \times_X S] \to \mathbb{Z}[S] \to \mathbb{Z}[\underline{X}] \to 0.$$

Applying derived solidification term by term, each $\mathbb{Z}[S_i]^{\mathbb{L}\square}$ is concentrated in degree 0, equal to

$$\mathbb{Z}[S_i]^{\square} = \underline{\mathrm{Hom}}\big(\mathrm{Cont}(S_i,\mathbb{Z}),\mathbb{Z}\big) = \Big(\mathrm{Cont}(S_i,\mathbb{Z})\Big)^{\vee},$$

the $\mathbb{Z}$-valued measures on $S_i$ (Remark 5.24). Each such term is its own double dual, so $\mathbb{Z}[\underline{X}]^{\mathbb{L}\square}$ is computed by the complex of measures, which one rewrites as

$$\mathbb{Z}[\underline{X}]^{\mathbb{L}\square} \xrightarrow{\ \sim\ } R\underline{\mathrm{Hom}}\Big(R\underline{\mathrm{Hom}}\big(\mathbb{Z}[\underline{X}],\mathbb{Z}\big),\ \mathbb{Z}\Big).$$

The inner $R\underline{\mathrm{Hom}}(\mathbb{Z}[\underline{X}],\mathbb{Z})$ is the singular cochain complex $C^{\bullet}_{\mathrm{sing}}(X,\mathbb{Z})$ by Theorem 4.1; since $X$ is finite CW, each $H^i_{\mathrm{sing}}(X,\mathbb{Z})$ is finitely generated, so dualizing back turns cohomology into homology and yields $C_{\bullet}^{\mathrm{sing}}(X,\mathbb{Z})$. $\square$

### 6.4 The structure of Solid

We now describe the abelian category Solid more precisely. Since $\prod_{\mathbb{N}}\mathbb{Z}$ is a compact projective generator, there are the usual finiteness notions.

**Definition 6.9** (Finitely generated and finitely presented solid groups)**.** An object $M \in$ Solid is *finitely generated* if it is a quotient of a single $\prod_{\mathbb{N}}\mathbb{Z}$, and *finitely presented* if it is the cokernel of a map

$$\prod_{\mathbb{N}}\mathbb{Z} \longrightarrow \prod_{\mathbb{N}}\mathbb{Z}. \tag{6.4}$$

Solid $\mathbb{Z}$-modules turn out to behave like modules over a coherent ring.

**Theorem 6.10** (Finitely presented solid groups form an abelian category)**.** *The finitely presented objects of* Solid *form an abelian category, stable under kernels, cokernels and extensions, and*

$$\mathrm{Solid} \;\simeq\; \mathrm{Ind}\big(\mathrm{Solid}^{\mathrm{fp}}\big).$$

*The only nontrivial closure property is stability under kernels; cokernels and extensions then follow formally.*

*Remark* 6.11 (Resolutions of length one)*.* This is slightly stronger than what was announced in Section 1: Clausen stated that finitely presented objects admit resolutions of length two, but in fact length one suffices. Every finitely presented $M \in \mathrm{Solid}$ has a resolution

$$0 \longrightarrow \prod_{\mathbb{N}} \mathbb{Z} \longrightarrow \prod_{\mathbb{N}} \mathbb{Z} \longrightarrow M \longrightarrow 0, \tag{6.5}$$

with the first map *injective*; equivalently, finitely presented objects are exactly the cokernels of injective maps of products of copies of $\mathbb{Z}$.

The engine of both statements is the following.

**Lemma 6.12** (Key Lemma)**.** *Every finitely generated submodule of $\prod_{\mathbb{N}} \mathbb{Z}$ is isomorphic to a product of copies of $\mathbb{Z}$ (an infinite product generically, possibly zero or finite).*

Granting Lemma 6.12, Theorem 6.10 follows: to see that the finitely presented objects are stable under kernels, given a map with target $M$ one takes the kernel, which is a finitely generated submodule of a product, hence by the Lemma itself a product $\prod_{\mathbb{N}} \mathbb{Z}$; the resolution (6.5) results, and the rest is formal. The proof of Lemma 6.12 uses countability and is genuinely a *light* phenomenon (the analogous statement fails for all condensed abelian groups). Its combinatorial core is a classical fact of abelian group theory.

**Lemma 6.13** (Freeness fact)**.** *Let $N$ be a countable abelian group that embeds into a product of copies of $\mathbb{Z}$. Then $N$ is free.*

*Remark* 6.14*.* Scholze remarked he was not sure whether this is stated in the literature, and noted that the countability hypothesis is essential: an uncountable product $\prod_I \mathbb{Z}$ is not free as an abstract abelian group. *Editorially:* this is the $\aleph_1$-freeness of the Baer–Specker group $\prod_{\mathbb{N}} \mathbb{Z}$ – every countable subgroup is free – a classical result of Specker; we record it only as a pointer.

*Proof of Lemma 6.12.* Let $M \subseteq \prod_{\mathbb{N}} \mathbb{Z}$ be finitely generated, so that there is a surjection $\prod_{\mathbb{N}} \mathbb{Z} \twoheadrightarrow M$ and an injection $M \hookrightarrow \prod_{\mathbb{N}} \mathbb{Z}$; the composite $g\colon \prod_{\mathbb{N}} \mathbb{Z} \to \prod_{\mathbb{N}} \mathbb{Z}$ is dual to a map of direct sums

$$h\colon \bigoplus_{\mathbb{N}} \mathbb{Z} \longrightarrow \bigoplus_{\mathbb{N}} \mathbb{Z},$$

and the task is to show that $\mathrm{im}(g)$ is a product of copies of $\mathbb{Z}$.

First reduce to $h$ injective. By Lemma 6.13 the image $\mathrm{im}(h)$, being countable and embedded in a direct sum (hence in a product) of copies of $\mathbb{Z}$, is free; therefore $\ker(h) \hookrightarrow \bigoplus_{\mathbb{N}} \mathbb{Z}$ splits off as a summand, and after discarding it we may assume $h$ is injective, sitting in a short exact sequence

$$0 \longrightarrow \bigoplus_{\mathbb{N}} \mathbb{Z} \xrightarrow{h} \bigoplus_{\mathbb{N}} \mathbb{Z} \longrightarrow Q \longrightarrow 0. \tag{6.6}$$

Next simplify $Q$. Let $\overline{Q}$ be the image of $Q$ under the natural map into $\prod_{\varphi\in\mathrm{Hom}(Q,\mathbb{Z})}\mathbb{Z}$ (evaluate against all maps $Q\to\mathbb{Z}$), so that there is a short exact sequence

$$0\longrightarrow Q'\longrightarrow Q\longrightarrow\overline{Q}\longrightarrow 0,\qquad \mathrm{Hom}(Q',\mathbb{Z})=0.$$

By Lemma 6.13 $\overline{Q}$ is free, hence projective, so this sequence splits and one may split $\overline{Q}$ off from $\bigoplus_{\mathbb{N}}\mathbb{Z}$ as well; without loss of generality

$$\mathrm{Hom}(Q,\mathbb{Z})=0.$$

(Concretely, $\overline{Q}$ is the cokernel of $Q$ in the category of countably generated free abelian groups.)

Now dualize (6.6). It gives an exact sequence

$$0\to\mathrm{Hom}(Q,\mathbb{Z})\to\prod_{\mathbb{N}}\mathbb{Z}\xrightarrow{g=h^\vee}\prod_{\mathbb{N}}\mathbb{Z}\to\mathrm{Ext}^1(Q,\mathbb{Z})\to 0.$$

The point is that $\mathrm{Hom}(Q,\mathbb{Z})$ vanishes as a *condensed* abelian group, not merely on points: its $S$-valued points are

$$\mathrm{Hom}\big(Q\otimes\mathbb{Z}[S],\mathbb{Z}\big)=\mathrm{Hom}\big(Q,\mathrm{Cont}(S,\mathbb{Z})\big)=0,$$

using tensor–Hom adjunction and the fact that $\mathrm{Cont}(S,\mathbb{Z})$ is a discrete free abelian group, into which $Q$ (with no maps to $\mathbb{Z}$, hence to any free abelian group) has no maps. Therefore $g$ is injective, and $\mathrm{im}(g)\cong\prod_{\mathbb{N}}\mathbb{Z}$. The reductions made at the start amounted precisely to arranging that $g$ became injective. $\square$

**Corollary 6.15** (Classification of finitely presented solid groups)**.** *Every finitely presented $M\in\mathrm{Solid}$ is a direct sum of a product of copies of $\mathbb{Z}$ and a group of the form*

$$\mathrm{Ext}^1(Q,\mathbb{Z}),$$

*for some countable abelian group $Q$ with $\mathrm{Hom}(Q,\mathbb{Z})=0$. If $Q$ is torsion then $\mathrm{Ext}^1(Q,\mathbb{Z})$ is profinite; in general one gets certain (non-quasiseparated) solid groups.*

**Corollary 6.16** (Flatness of $\prod_{\mathbb{N}}\mathbb{Z}$)**.** *$\prod_{\mathbb{N}}\mathbb{Z}$ is flat for $\otimes^{\square}$; and for solid $M$,*

$$M\otimes^{\square}\prod_{\mathbb{N}}\mathbb{Z}\;\cong\;\prod_{\mathbb{N}}M.$$

*Proof.* One must show $M\otimes^{\mathbb{L}\square}\prod_{\mathbb{N}}\mathbb{Z}$ sits in degree 0 for every solid $M$; reducing to $M$ finitely presented, tensor the resolution (6.5) with $\prod_{\mathbb{N}}\mathbb{Z}$. The injective map stays injective, so the derived tensor product is concentrated in degree 0 and equals $\prod_{\mathbb{N}}M$. $\square$

*Remark* 6.17 (Flatness is special to the light setting). Abstractly a compact projective object need not be flat. Corollary 6.16 in fact *fails* for all (non-light) solid abelian groups, but becomes true again for light condensed abelian groups – which is why no obstruction to it was ever seen in mathematical practice. Scholze recalled that after passing to $p$-adic coefficients something like it was known, and that he and Clausen had been puzzled about the solid case before observing this light/heavy discrepancy.

### 6.5 Some $\otimes^{\square}$-computations

A striking feature of the solid tensor product is that many computations "come out right", often in non-obvious ways, just as the earlier Ext-computations did. The relevant setting is $p$-adic formal geometry, where one takes derived $p$-completions.

*Notation* 6.18 (Derived $p$-completion). For an abelian group (or complex) $M$, write

$$M_p^\wedge := R\varprojlim_n M/^{\mathbb{L}}p^n, \qquad M/^{\mathbb{L}}p^n = \left[\, M \xrightarrow{p^n} M \,\right] \text{ (in degrees } 1,0),$$

the *derived* $p$-completion. Being derived $p$-complete can be checked on cohomology groups. In the classical situation one often uses the completed tensor product $\widehat{\otimes}$, the $p$-adic completion of the usual tensor product. Almost always $M_p^\wedge$ agrees with the naive $p$-adic completion (it is in degree 0), but in full generality the derived version is the better operation.

**Proposition 6.19** ($\otimes^{\mathbb{L}\square}$ preserves derived $p$-completeness)**.** *If $N, M \in D_{\geq 0}(\text{Solid})$ are derived $p$-complete, then $N \otimes^{\mathbb{L}\square} M$ is derived $p$-complete.*

**Corollary 6.20.** *For the completed free modules,*

$$\Big(\bigoplus_{\mathbb{N}} \mathbb{Z}\Big)_p^\wedge \otimes^{\mathbb{L}\square} \Big(\bigoplus_{\mathbb{N}} \mathbb{Z}\Big)_p^\wedge \;\cong\; \Big(\bigoplus_{\mathbb{N}\times\mathbb{N}} \mathbb{Z}\Big)_p^\wedge.$$

*So in situations of formal geometry, where one completes, the tensor product does what one expects.*

*Remark* 6.21 (Nothing special about $p$). There is nothing special about the number $p$ or about $\mathbb{Z}$. For any ring $A$ and any element $x \in A$, let $\text{Solid}_A$ denote the $A$-modules inside solid abelian groups. Then for $M, N \in D_{\geq 0}(\text{Solid}_A)$ derived $x$-complete, $M \otimes_A^{\mathbb{L}\square} N$ is again derived $x$-complete. (Here $\text{Solid}_A$ means, for now, simply $A$-modules in solid abelian groups.)

The proof of Proposition 6.19, in the model case, exhibits the combinatorial mechanism behind all such statements.

*Proof sketch of Corollary 6.20.* First, one may work with $\mathbb{Z}_p$ in place of $\mathbb{Z}$, because $\mathbb{Z}_p \otimes^{\mathbb{L}\square} \mathbb{Z}_p = \mathbb{Z}_p$: indeed

$$\mathbb{Z}[\![T]\!] \otimes^{\mathbb{L}\square} \mathbb{Z}[\![U]\!] = \mathbb{Z}[\![U, T]\!]$$

by Corollary 5.20, and quotienting by $(T - p,\, U - p)$ gives $\mathbb{Z}_p$ on each factor while quotienting the two-variable power series ring by $T = p$ yields $\mathbb{Z}_p[\![U]\!]$, then further by $U = p$ gives $\mathbb{Z}_p$. Consequently

$$D(\text{Solid}_{\mathbb{Z}_p}) \;\subset\; D(\text{Solid})$$

is a full subcategory: being a $\mathbb{Z}_p$-module is a *condition*, not a structure, on a solid abelian group, and anything derived $p$-complete acquires it. Take $M, N$ in this subcategory.

Consider the case $N = M = \left(\bigoplus_{\mathbb{N}} \mathbb{Z}_p\right)_p^\wedge$. Writing it in terms of the compact projective generators is a small exercise: one has

$$\Big(\bigoplus_{\mathbb{N}} \mathbb{Z}_p\Big)_p^\wedge \;\cong\; \operatorname*{colim}_{\substack{f\colon \mathbb{N}\to\mathbb{N} \\ f\to\infty}} \prod_{n\in\mathbb{N}} p^{f(n)}\mathbb{Z}_p,$$

the colimit over functions $f$ tending to $\infty$ (each fixed at value 0 or 1 or…only finitely often). There is an obvious map from each $\prod_n p^{f(n)}\mathbb{Z}_p$ into the completed direct sum $\left(\bigoplus_{\mathbb{N}} \mathbb{Z}_p\right)_p^\wedge = \varprojlim_n \left(\bigoplus_{\mathbb{N}} \mathbb{Z}/p^n\right)$, since $f \to \infty$ forces almost all coordinates to vanish modulo any fixed power of $p$. It is injective;

surjectivity is the point. Given $S \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$ and a map $g\colon S \to \big(\bigoplus_{\mathbb{N}} \mathbb{Z}_p\big)^{\wedge}_p = \varprojlim_n \big(\bigoplus_{\mathbb{N}} \mathbb{Z}/p^n\big)$ with components $g_n$, each $g_n$ factors – $S$ being compact and the target a direct sum – through $\bigoplus_{n \le a(n)} \mathbb{Z}/p^n$ for some $a(n)$; choosing a sufficiently slowly increasing $f$ makes $g$ factor through $\prod_n p^{f(n)}\mathbb{Z}_p$.

Now compute the tensor product, pulling the colimits out (which $\otimes^{\square}$ commutes with):

$$\begin{aligned} M \otimes^{\mathbb{L}\square} N &= \Big(\operatorname*{colim}_{f \to \infty} \prod_n p^{f(n)}\mathbb{Z}_p\Big) \otimes^{\mathbb{L}\square} \Big(\operatorname*{colim}_{g \to \infty} \prod_m p^{g(m)}\mathbb{Z}_p\Big) \\ &= \operatorname*{colim}_{f,g \to \infty} \prod_{n,m \in \mathbb{N}} p^{f(n)+g(m)}\mathbb{Z}_p \xrightarrow{\ \sim\ } \operatorname*{colim}_{\substack{h\colon \mathbb{N}\times\mathbb{N}\to\mathbb{N} \\ h \to \infty}} \prod_{n,m} p^{h(n,m)}\mathbb{Z}_p = \Big(\bigoplus_{\mathbb{N}\times\mathbb{N}} \mathbb{Z}_p\Big)^{\wedge}_p, \end{aligned} \tag{6.7}$$

using that products of $\mathbb{Z}_p$'s tensor to products (the two-variable form of (6.1)). The final isomorphism is a combinatorial lemma. ☐

**Lemma 6.22** (Combinatorial domination)**.** *Any function $h\colon \mathbb{N}\times\mathbb{N} \to \mathbb{N}$ tending to $\infty$ dominates an (even more slowly increasing) function, still tending to $\infty$, that is a sum $f(n)+g(m)$ of functions of the individual coordinates. (In the dual situation of Proposition 6.26 one uses the same fact with the inequality reversed.)*

At first sight the colimit over sums $f(n)+g(m)$ looks far smaller than the colimit over all two-variable functions $h$; the lemma says they are cofinal. Scholze left the (not hard) proof as an exercise. The general case of Proposition 6.19 reduces $M, N$ to completed direct sums of generators and runs the same argument, with an extra step reducing an uncountable index set to the countable case (derived completions commute with the relevant colimits).

### 6.6 Solid functional analysis

Further computations come out right when one does $p$-adic functional analysis inside the solid framework. Work over $\mathbb{Q}_p$; most of what follows works over any complete nonarchimedean field. Inverting $p$ and passing to $\mathbb{Z}_p$-modules gives a chain of full subcategories

$$D(\mathrm{Solid}_{\mathbb{Q}_p}) \ \subset\ D(\mathrm{Solid}_{\mathbb{Z}_p}) \ \subset\ D(\mathrm{Solid}).$$

Tensor products stay within each subcategory. The category $\mathrm{Solid}_{\mathbb{Q}_p}$ of solid $\mathbb{Q}_p$-vector spaces has a compact projective generator

$$\Big(\prod_{\mathbb{N}} \mathbb{Z}_p\Big)\big[\tfrac{1}{p}\big], \tag{6.8}$$

a $p$-adic *Smith space*, while the more usual objects are the $p$-adic Banach spaces

$$\Big(\bigoplus_{\mathbb{N}} \mathbb{Z}_p\Big)^{\wedge}_p\big[\tfrac{1}{p}\big].$$

*Remark* 6.23 (Smith spaces are not normed)*.* The generator (6.8) is a curious condensed vector space that does come from a topological vector space, but is not a normed space. It looks like one with a distinguished "unit ball" $\prod_{\mathbb{N}} \mathbb{Z}_p$ that is *compact*; but no infinite-dimensional Banach (or normed) space has compact unit ball, so if one tried to endow it with a norm making that set the unit ball, the norm would fail to be continuous. Such objects have appeared in classical functional analysis under the name *Smith spaces*: one considers topological vector spaces that are increasing unions obtained by scaling out a fixed compact convex set, dually to Banach spaces. Scholze attributed the notion to Marianne F. Smith, who studied the analogue over $\mathbb{R}$.

The Smith spaces and (separable) Banach spaces are exchanged by duality.

**Proposition 6.24** (Smith–Banach duality). *There is an anti-equivalence*

$$\big(\{\textit{light Smith spaces}\}\big)^{\mathrm{op}} \;\simeq\; \{\textit{separable Banach spaces}\},$$

*given in both directions by $\mathbb{Q}_p$-linear duality $V \mapsto \underline{\mathrm{Hom}}(V, \mathbb{Q}_p)$, $W \mapsto \underline{\mathrm{Hom}}(W, \mathbb{Q}_p)$.*

Concretely, the dual of the Smith space $(\prod_{\mathbb{N}} \mathbb{Z}_p)[\frac{1}{p}]$ is the Banach space $(\bigoplus_{\mathbb{N}} \mathbb{Z}_p)^{\wedge}_p[\frac{1}{p}]$ and vice versa; "separable/light" on one side matches a countable direct sum, on the other a countable product.

*Remark* 6.25 (Derived duality and set theory). One may ask what happens to duality *derived*. Going from a Smith space $V$,

$$R\underline{\mathrm{Hom}}(V, \mathbb{Q}_p) = \underline{\mathrm{Hom}}(V, \mathbb{Q}_p),$$

with no higher terms – true and essentially obvious, since $V$ is internally projective. Going the other way, from a Banach space $W$, the question whether

$$\underline{\mathrm{Hom}}(W, \mathbb{Q}_p) \;\overset{?}{=}\; R\underline{\mathrm{Hom}}(W, \mathbb{Q}_p)$$

*depends on the model of set theory*, exactly as with the phenomenon of Section 4: it fails under the continuum hypothesis, but holds under the principle $(*)$ of Definition 4.24, which is consistent (Theorem 4.26). The reason is that a Banach space is a special pro-$p$-type object, a limit of direct sums $\varprojlim_n \big(\bigoplus_{\mathbb{N}} \mathbb{Q}_p/p^n\mathbb{Z}_p\big)$, so the higher Ext against $\mathbb{Q}_p$ is precisely controlled by $(*)$. Scholze said he was increasingly tempted to just work in a model where $(*)$ holds, so that these higher Ext's are 0 rather than meaningless junk; in the computations of solid functional analysis (many duals and tensor products) one does at some point take an $R\underline{\mathrm{Hom}}$ of a Banach space against $\mathbb{Q}_p$, and then one may choose the well-behaved model. He and Clausen never found the question genuinely relevant, but it does come up.

Finally, the solid tensor product recovers the classical completed tensor products of functional analysis – even though it was defined to be compatible with *colimits*, whereas those tensor products are usually designed to be compatible with *limits*.

**Proposition 6.26** (Fréchet spaces). *A* Fréchet space *is a countable limit of Banach spaces along dense transition maps; these carry a standard completed tensor product $\widehat{\otimes}$, compatible with such limits. For Fréchet $\mathbb{Q}_p$-vector spaces $V, W$,*

$$V \otimes^{\mathbb{L}\square} W \;\cong\; V \mathbin{\widehat{\otimes}} W,$$

*concentrated in degree 0. For example*

$$\prod_{\mathbb{N}} \mathbb{Q}_p \;\otimes^{\mathbb{L}\square}\; \prod_{\mathbb{N}} \mathbb{Q}_p \;\cong\; \prod_{\mathbb{N}\times\mathbb{N}} \mathbb{Q}_p. \tag{6.9}$$

*Proof in the example.* Write the Fréchet space $\prod_{\mathbb{N}} \mathbb{Q}_p$ as a colimit of Smith spaces with ever larger denominators,

$$\prod_{\mathbb{N}} \mathbb{Q}_p \;=\; \underset{\substack{f\colon \mathbb{N}\to\mathbb{N} \\ f\to\infty \text{ fast}}}{\mathrm{colim}} \Big(\prod_n p^{-f(n)}\mathbb{Z}_p\Big)\big[\tfrac{1}{p}\big],$$

the colimit now over *very fast* increasing $f$ (any profinite set mapping in factors through some $\prod_n p^{-f(n)}\mathbb{Z}_p$). Then

$$\prod_{\mathbb{N}} \mathbb{Q}_p \otimes^{\mathbb{L}\square} \prod_{\mathbb{N}} \mathbb{Q}_p = \underset{f,g\to\infty}{\mathrm{colim}} \Big(\prod_{\mathbb{N}\times\mathbb{N}} p^{-f(n)-g(m)}\mathbb{Z}_p\Big)\big[\tfrac{1}{p}\big] = \prod_{\mathbb{N}\times\mathbb{N}} \mathbb{Q}_p,$$

again by the combinatorial domination Lemma 6.22. □

*Remark* 6.27 (A huge space, not a compact projective one). The answer $\prod_{\mathbb{N}\times\mathbb{N}}\mathbb{Q}_p$ in (6.9) might look like a formal statement about compact projective generators, but it is not: the spaces $\prod_{\mathbb{N}}\mathbb{Q}_p$ are very far from compact projective (only the unit balls $(\prod_{\mathbb{N}}\mathbb{Z}_p)[\frac{1}{p}]$ are). The general proof combines the argument for the compact case with the present one; Scholze noted Efimov has a slicker way to combine them, used in his categorical work.

### 6.7 Questions

*Remark* 6.28 (Light vs. all condensed sets for Banach spaces). For functional analysis in large Banach spaces it makes no difference whether one works in light or all condensed sets: all Banach spaces – and all Fréchet spaces – are light, separable or not, since even an uncountable-dimensional Banach space is a naive filtered colimit of countable-dimensional (separable) Banach subspaces. It is only the predual Smith spaces that see the distinction.

*Remark* 6.29 (Solidification of $K$-theory spectra). The relation between the singular complex and solidification (Proposition 6.6) is no accident: one can replace $\mathrm{CondAb}^{\mathrm{light}}$ everywhere by condensed spectra and solidify there, since Solid sits inside all condensed spectra just as solid abelian groups sit inside condensed abelian groups. A telling example is connective complex $K$-theory $ku$. The algebraic $K$-theory spectrum $K(\mathbb{C})$ is naturally a *condensed* spectrum: $\mathbb{C}$ is a condensed ring, and one sets $S \mapsto K\big(\mathrm{Cont}(S,\mathbb{C})\big)$. This is not yet the topological object – e.g. $\pi_1$ is $\mathbb{C}^\times$ as a condensed abelian group – but its solidification contracts the groups $\mathrm{GL}_n(\mathbb{C})$ down to their homotopy types (by the mechanism of Proposition 6.6), and one recovers topological connective $ku$. More generally, for any $\mathbb{C}$-algebra $A$ (with any topology, even discrete) one may form the condensed spectrum $K(A)$ and solidify, obtaining the *semi-topological* $K$-theory of $A$; the analogous constructions over $\mathbb{R}$ give $KO$, etc. The output is connective (not entirely obviously); recovering the periodic $KU$ – inverting the Bott element – is more subtle, and is one motivation for the *analytic* $K$-theory Clausen and Scholze developed, which allows negative degrees. The underlying mechanism is that solidification factors through passage to homotopy types on suitable objects: in the ambient category of condensed anima, the "homotopy type" functor is a left adjoint defined on a class of objects (e.g. on classifying spaces $B\mathrm{GL}_n(\mathbb{C})$), and solidification applies it.

## 7 The Solid Affine Line

Clausen resumes the course and begins to do geometry. The object of study is the *solid affine line*: the polynomial ring $\mathbb{Z}[T]$, viewed through the lens of solid modules, and the various subspaces of it (open and closed unit disks) that the solid formalism cuts out. The lecture has two halves. First, a relative version of solidity over $\mathbb{Z}[T]$ is set up, its structure theory is proved by transporting Scholze's arguments through an explicit formula for the relative solidification, and the resulting categories are shown to fit into a *recollement* – from which Clausen reads off a definite verdict on what is "open" and what is "closed" in nonarchimedean geometry. Second, for a general solid ring $R$ the theory is used to define the notions of *power-bounded* and *topologically nilpotent* elements, to build the *rational localizations* $R\langle f_1,\ldots,f_n/g\rangle$, and to compare the result with Huber's theory – of which it is a derived, non-quasiseparated refinement that repairs the failure of Huber's structure presheaf to be a sheaf.

Throughout, "ring" means commutative ring, and everything takes place inside light condensed abelian groups. We freely use the theory of solid abelian groups $\mathrm{Solid} \subset \mathrm{CondAb}^{\mathrm{light}}$ from Sections 5 and 6.

## 7.1 Recap

Recall from Section 5 the abelian category

$$\mathrm{Solid} \;\subset\; \mathrm{CondAb}^{\mathrm{light}}$$

of solid abelian groups: a full subcategory, closed under limits, colimits, extensions and much else, which one thinks of as an algebraically well-behaved replacement for complete nonarchimedean topological abelian groups (Theorem 6.1). The inclusion has a colimit-preserving left adjoint, *solidification* $M \mapsto M^{\square}$, and a symmetric monoidal structure $\otimes^{\square}$ (a completed tensor product) for which solidification is symmetric monoidal (Corollary 5.9). The compact projective generator is $\prod_{\mathbb{N}} \mathbb{Z}$.

Solidity was defined (Definition 5.5) through the free light condensed abelian group on a null sequence

$$P = \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/\mathbb{Z}[\infty],$$

which is internally projective (Theorem 3.15(3)): an object $M$ is solid iff the endomorphism $\mathrm{id} - \mathrm{shift}$ of the shift on $\mathbb{N}$ induces an isomorphism

$$\underline{\mathrm{Hom}}(P, M) \xrightarrow{1-\sigma} \underline{\mathrm{Hom}}(P, M), \tag{7.1}$$

i.e. every null sequence in $M$ is uniquely summable.

## 7.2 The ring $P$ and its solidification

The new observation is that $P$ is not merely an abelian group but a *ring*, and that there is a ring map

$$\mathbb{Z}[T] \longrightarrow P, \qquad T \longmapsto \sigma, \tag{7.2}$$

under which the variable $T$ acts as the (injective) shift $\sigma$ that enters (7.1).

*Remark* 7.1 (The ring structure on $P$, via measures). The ring structure is cleanest when one attaches the measure construction to $\mathbb{N}$ rather than to $\mathbb{N} \cup \{\infty\}$. For any countable set $S$ one defines the measures on $S$ as the free module on any compactification modulo the free module on the boundary; the result is independent of the compactification, is functorial for proper maps, and is symmetric monoidal, in that measures on $S_1$ tensored with measures on $S_2$ are measures on $S_1 \times S_2$ (seen using independence of the compactification). Since addition $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ is a proper map with finite fibres, it induces a multiplication on $P$, making $P$ a ring, and (7.2) is the induced $\mathbb{Z}[T]$-algebra structure.

Solidifying $P$ is very clean. As an abelian group $P^{\square} = \prod_{\mathbb{N}} \mathbb{Z}$ (Corollary 5.18); taking the ring structure into account, this is the one-variable power series ring:

$$P^{\square} \;\cong\; \mathbb{Z}[\![T]\!].$$

Thus, in the solid world, the universal carrier of a null sequence is the power series ring $\mathbb{Z}[\![T]\!]$.

The following idempotence is the technical heart of the geometric picture.

**Lemma 7.2** ($\mathbb{Z}[\![T]\!]$ is idempotent over $\mathbb{Z}[T]$)**.** *The multiplication map is an isomorphism*

$$\mathbb{Z}[\![T]\!] \ \otimes^{\mathbb{L}\square}_{\mathbb{Z}[T]} \ \mathbb{Z}[\![T]\!] \ \xrightarrow{\ \sim\ } \ \mathbb{Z}[\![T]\!], \tag{7.3}$$

*already at the level of the derived solid tensor product.*

*Proof.* By the basic computation of the solid tensor product (Corollary 5.20),

$$\mathbb{Z}[\![T_1]\!] \ \otimes^{\mathbb{L}\square}_{\mathbb{Z}} \ \mathbb{Z}[\![T_2]\!] \ \cong \ \mathbb{Z}[\![T_1, T_2]\!]. \tag{7.4}$$

Forming (7.3) amounts to imposing $T_1 = T_2$, i.e. quotienting by $T_1 - T_2$, and $\mathbb{Z}[\![T_1, T_2]\!]/(T_1 - T_2) = \mathbb{Z}[\![T]\!]$. Alternatively, both sides are derived $T$-complete (Proposition 6.19), so the isomorphism may be checked modulo $T$, where it is a triviality. $\square$

*Remark* 7.3 (Derived solid tensor product)*.* Here $\otimes^{\mathbb{L}\square}$ is either the left derived functor of $\otimes^{\square}$, or equivalently the derived tensor product followed by derived solidification. One need not know that there are enough flat objects to form it, though in fact all the basic objects (products of copies of $\mathbb{Z}[T]$) are flat (Corollary 6.16), so a resolution of one variable suffices.

### 7.3 First interpretation: the open unit disk

Because $\mathbb{Z}[\![T]\!]$ is idempotent over $\mathbb{Z}[T]$ (Lemma 7.2), inverting it inside $\mathbb{Z}[T]$-algebras does not change it; such idempotent algebras correspond to *subspaces* of $\operatorname{Spec}\mathbb{Z}[T]$, i.e. of the affine line, rather than to arbitrary spaces mapping to it. So $\mathbb{Z}[\![T]\!]$ should be thought of as some subspace of $\mathbb{A}^1$.

The naive guess – that $\mathbb{Z}[\![T]\!]$ is the *formal completion of* $\mathbb{A}^1$ *at the origin* – is *wrong*. The reason is that the formal-neighbourhood interpretation is not stable under base change, whereas $\mathbb{Z}[\![T]\!]^{\square}$ is. One sees the difference after base change to a nonarchimedean field.

**Example 7.4** (Base change to $\mathbb{Q}_p$: bounded functions on the open disk)**.** Base change $\mathbb{Z}[\![T]\!]$ to $\mathbb{Q}_p$. Since $\mathbb{Q}_p = \mathbb{Z}_p[\frac{1}{p}]$ and the solid tensor product commutes with colimits in each variable,

$$\mathbb{Q}_p \ \otimes^{\mathbb{L}\square}_{\mathbb{Z}} \ \mathbb{Z}[\![T]\!] \ = \ \Big(\mathbb{Z}_p \ \otimes^{\mathbb{L}\square}_{\mathbb{Z}} \ \mathbb{Z}[\![T]\!]\Big)\big[\tfrac{1}{p}\big].$$

Now $\mathbb{Z}_p = \mathbb{Z}[\![U]\!]/(U - p)$ has a two-term resolution by copies of the power series ring, so, using (7.4) to compute $\mathbb{Z}[\![U]\!] \otimes^{\mathbb{L}\square} \mathbb{Z}[\![T]\!] = \mathbb{Z}[\![U, T]\!]$ and pulling the (naive) limit through,

$$\mathbb{Q}_p \ \otimes^{\mathbb{L}\square}_{\mathbb{Z}} \ \mathbb{Z}[\![T]\!] \ \cong \ \mathbb{Z}_p[\![T]\!]\big[\tfrac{1}{p}\big].$$

This is *not* the formal completion $\mathbb{Q}_p[\![T]\!]$; it is the subring of $\mathbb{Q}_p[\![T]\!]$ of power series whose coefficients are *bounded* in $p$-adic norm. Its interpretation: it is the ring of bounded rigid-analytic functions on the *open* unit disk. For any nonarchimedean extension of $\mathbb{Q}_p$, such a series converges exactly at points of absolute value strictly less than 1; all these functions converge on the open unit disk and no further, but boundedness of the coefficients cuts out the bounded functions rather than all holomorphic ones. Over a discrete ring such as $\mathbb{Z}$ the open unit disk and the formal neighbourhood cannot be told apart; the distinction emerges only after base change to a ring with more room to move.

## 7.4 The closed unit disk and $\mathbb{Z}[T]$-solidity

In rigid geometry the fundamental object is the *closed* unit disk, not the open one. To reach it, put the open unit disk "at infinity": working with the variable $T^{-1}$, the ring $\mathbb{Z}[\![T^{-1}]\!]$ is the open unit disk centred at $\infty$, and puncturing at $\infty$ to land in the affine line gives the $\mathbb{Z}[T]$-algebra

$$\mathbb{Z}[\![T^{-1}]\!] \ \otimes^{\square}_{\mathbb{Z}[T^{-1}]} \ \mathbb{Z}[T, T^{-1}] \ = \ \mathbb{Z}(\!(T^{-1})\!), \tag{7.5}$$

the Laurent series in $T^{-1}$. This is the ring of the *complement* of the closed unit disk. To pass to the closed unit disk itself one must localize away from this complement – equivalently, *kill* the object $\mathbb{Z}(\!(T^{-1})\!)$.

As a $\mathbb{Z}[T]$-module, $\mathbb{Z}(\!(T^{-1})\!)$ has a simple two-term resolution. Writing $U = T^{-1}$, one inverts $U$ inside $\mathbb{Z}[\![U]\!]$ by adjoining a formal inverse and imposing the relation, giving

$$\Big[\ \mathbb{Z}[\![U]\!][T] \xrightarrow{UT-1} \mathbb{Z}[\![U]\!][T]\ \Big] \ \xrightarrow{\sim} \ \mathbb{Z}(\!(T^{-1})\!). \tag{7.6}$$

The two terms are the base change of $P$ to $\mathbb{Z}[T]$-modules in solid abelian groups: $\mathbb{Z}[\![U]\!][T] = P^{\square} \otimes_{\mathbb{Z}} \mathbb{Z}[T]$ is the compact projective generator of $\mathbb{Z}[T]$-modules in solid abelian groups. Thus $\mathbb{Z}(\!(T^{-1})\!)$ is resolved by compact projective generators, and killing it should mean requiring the map induced by $UT - 1$ to become an isomorphism. By analogy with (7.1) we make the following definition.

**Definition 7.5** ($\mathbb{Z}[T]$-solid modules)**.** Let $M$ be a module over $\mathbb{Z}[T]$ in $\mathrm{CondAb}^{\mathrm{light}}$, with $t\colon M \to M$ the action of $T$. Then $M$ is $\mathbb{Z}[T]$-*solid* if

$$\underline{\mathrm{Hom}}(P, M) \ \xrightarrow{\sigma t-1} \ \underline{\mathrm{Hom}}(P, M) \tag{7.7}$$

is an isomorphism, where $\sigma$ is the shift. Write $\mathrm{Solid}_{\mathbb{Z}[T]}$ for the full subcategory of $\mathbb{Z}[T]$-solid objects.

By (7.6), passing to $R\underline{\mathrm{Hom}}$, the condition (7.7) is equivalent to

$$R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}\big(\mathbb{Z}(\!(T^{-1})\!),\ M\big) \ = \ 0,$$

i.e. to the demand that $M$ "see nothing" of the complement of the closed unit disk.

**Theorem 7.6** (Structure of the $\mathbb{Z}[T]$-solid theory)**.** *Inside* $\mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{CondAb}^{\mathrm{light}})$*, the category* $\mathrm{Solid}_{\mathbb{Z}[T]}$ *satisfies:*

1. *it is an abelian subcategory, closed under limits, colimits and extensions, and* $\mathrm{Solid}_{\mathbb{Z}[T]} \subseteq \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid}_{\mathbb{Z}}) \subseteq \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{CondAb}^{\mathrm{light}})$*;*
2. *if* $M \in \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{CondAb}^{\mathrm{light}})$ *is arbitrary and* $N \in \mathrm{Solid}_{\mathbb{Z}[T]}$*, then* $\underline{\mathrm{Ext}}^i_{\mathbb{Z}[T]}(M, N) \in \mathrm{Solid}_{\mathbb{Z}[T]}$ *for all* $i$*;*
3. *the inclusion has a colimit-preserving left adjoint, the* $\mathbb{Z}[T]$-solidification $(-)^{T\text{-}\square}$*, and a unique symmetric monoidal structure making it symmetric monoidal;*
4. *the derived analogue of* (1)–(3) *holds, and the derived theory is "naive": a complex is derived* $\mathbb{Z}[T]$*-solid iff each of its homology groups is* $\mathbb{Z}[T]$*-solid, so that* $D(\mathrm{Solid}_{\mathbb{Z}[T]})$ *is the derived category of the abelian category* $\mathrm{Solid}_{\mathbb{Z}[T]}$*.*

The name is well chosen: the $\mathbb{Z}[T]$-solid theory over $\mathbb{Z}[T]$ is as robust as the solid theory over $\mathbb{Z}$.

*Remark* 7.7 ($\mathbb{Z}[T]$ and $\mathbb{Z}[\![T]\!]$ are $\mathbb{Z}[T]$-solid). Both $\mathbb{Z}[T]$ and $\mathbb{Z}[\![T]\!]$ are $\mathbb{Z}[T]$-solid. Indeed $\mathbb{Z}[T]$ is a retract of the compact projective generator (below), and it is even solid as an abelian group, since every discrete abelian group is solid (being generated under colimits by $\mathbb{Z}$). And $\mathbb{Z}[\![T]\!]$ is built from $\mathbb{Z}[T]$ by limits and colimits, hence is $\mathbb{Z}[T]$-solid too.

*Remark* 7.8 (Nonarchimedean interpretation). Just as $\mathrm{Solid}_{\mathbb{Z}}$ is an algebraic surrogate for complete nonarchimedean topological abelian groups – those with a basis of neighbourhoods of 0 consisting of subgroups – the category $\mathrm{Solid}_{\mathbb{Z}[T]}$ is a surrogate for complete nonarchimedean objects admitting a basis of neighbourhoods of 0 consisting of $\mathbb{Z}[T]$-*submodules*. The ring $\mathbb{Z}(\!(T^{-1})\!)$ is the prototypical nonarchimedean $\mathbb{Z}[T]$-module for which no such basis exists: its neighbourhood basis is stable under multiplication by $T^{-1}$ rather than by $T$. Killing this single object accounts for the whole difference between the two notions.

## 7.5 An explicit formula for the $\mathbb{Z}[T]$-solidification

The formal properties (1)–(3) of Theorem 7.6 are proved exactly as Scholze proved their $\mathrm{Solid}_{\mathbb{Z}}$-analogues in Section 5: everything followed from $P$ being internally projective and from asking that an endomorphism of $\underline{\mathrm{Hom}}(P,-)$ become invertible, which is exactly the shape of (7.7). What must be redone is the identification of the free objects; and here, thanks to the reading of the theory as killing $\mathbb{Z}(\!(T^{-1})\!)$, matters are actually easier than over $\mathbb{Z}$, because $\mathrm{Solid}_{\mathbb{Z}}$ is already available.

The key is an explicit formula for the (derived) $\mathbb{Z}[T]$-solidification.

**Proposition 7.9** (Formula for $(-)^{\mathbb{L}T\text{-}\square}$). *For $M \in D(\mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{CondAb}^{\mathrm{light}}))$,*

$$M^{\mathbb{L}T\text{-}\square} \;=\; R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}\Big(\; \mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T]\;[-1],\;\; M\;\Big), \tag{7.8}$$

*and the natural map $M \to M^{\mathbb{L}T\text{-}\square}$ is induced by $R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(\mathbb{Z}[T], M) \simeq M$ under the boundary map of the fibre sequence $\mathbb{Z}[T] \to \mathbb{Z}(\!(T^{-1})\!) \to \mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T]$.*

*Sketch.* This is formal from the idempotence of $\mathbb{Z}(\!(T^{-1})\!)$ over $\mathbb{Z}[T]$,

$$\mathbb{Z}(\!(T^{-1})\!)\;\otimes^{\mathbb{L}\square}_{\mathbb{Z}[T]}\;\mathbb{Z}(\!(T^{-1})\!)\;\xrightarrow{\;\sim\;}\;\mathbb{Z}(\!(T^{-1})\!), \tag{7.9}$$

which follows from (7.5) and Lemma 7.2 by transporting to $\infty$. The object $\mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T]$ is the homotopy fibre of $\mathbb{Z}[T] \hookrightarrow \mathbb{Z}(\!(T^{-1})\!)$ shifted, and (7.9) says $R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(\mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T][-1], -)$ is an idempotent operation whose local objects are exactly the $M$ with $R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(\mathbb{Z}(\!(T^{-1})\!), M) = 0$, i.e. the $\mathbb{Z}[T]$-solid ones. This is precisely the situation of a Bousfield localization, and (7.8) is the localization functor. $\square$

Something better happens when $M$ is base-changed from a solid $\mathbb{Z}$-module. This is the computation that identifies the free objects.

**Proposition 7.10** (Pushforward from $\mathrm{Solid}_{\mathbb{Z}}$). *The functor*

$$D(\mathrm{Solid}_{\mathbb{Z}}) \longrightarrow D(\mathrm{Solid}_{\mathbb{Z}[T]}), \qquad M \longmapsto \big(M \otimes_{\mathbb{Z}} \mathbb{Z}[T]\big)^{\mathbb{L}T\text{-}\square}, \tag{7.10}$$

*is t-exact, preserves all limits and colimits, and sends $\mathbb{Z}$ to $\mathbb{Z}[T]$. Concretely, there is a natural isomorphism*

$$\big(M \otimes_{\mathbb{Z}} \mathbb{Z}[T]\big)^{\mathbb{L}T\text{-}\square} \;\cong\; R\underline{\mathrm{Hom}}_{\mathbb{Z}}\big(U\,\mathbb{Z}[\![U]\!]\,[-1],\;M\,\big), \tag{7.11}$$

*where the residual $\mathbb{Z}[T]$-module structure is induced by the endomorphism of $U\,\mathbb{Z}[\![U]\!]$ sending $U \mapsto 0$, $U^2 \mapsto U$, $U^3 \mapsto U^2$, and so on.*

*Sketch.* Apply (7.8) to $M[T] := M \otimes_{\mathbb{Z}} \mathbb{Z}[T]$ and compute the $R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}$ by disambiguating the two occurrences of $T$ into $U$ and $T$ and then equalizing. Using the two-term presentation of $\mathbb{Z}((T^{-1}))$ from (7.6), one is led to

$$R\underline{\mathrm{Hom}}_{\mathbb{Z}}\big(U\mathbb{Z}[\![U]\!][-1],\ M\big)[T] \xrightarrow{T-\sigma} R\underline{\mathrm{Hom}}_{\mathbb{Z}}\big(U\mathbb{Z}[\![U]\!][-1],\ M[T]\big),$$

after which one equalizes $T$ against the shift $\sigma$ induced by multiplication by $T$. Now $U\mathbb{Z}[\![U]\!]$ is a compact projective object of $\mathrm{Solid}_{\mathbb{Z}}$, so $R\underline{\mathrm{Hom}}$ out of it commutes with the colimit $M[T] = \bigoplus_n M \cdot T^n$; the resulting fibre (from "$T$ minus a shift") together with the extra shift cancel, and one is left with the clean formula (7.11). Since $U\mathbb{Z}[\![U]\!]$ is internally projective in $\mathrm{Solid}_{\mathbb{Z}}$, the functor is $t$-exact and preserves limits and colimits; and limits and colimits in $\mathrm{Solid}_{\mathbb{Z}[T]}$ are computed on underlying objects, being a module category. Finally, plugging $M = \mathbb{Z}$ into (7.11), $R\underline{\mathrm{Hom}}_{\mathbb{Z}}$ out of the product $U\mathbb{Z}[\![U]\!] = \prod_{\mathbb{N}} \mathbb{Z}$ into $\mathbb{Z}$ turns into a direct sum, and a natural comparison exhibits the result as $\mathbb{Z}[T]$ with its evident $\mathbb{Z}[T]$-module structure. □

**Corollary 7.11** (The compact projective generator)**.** *Solidifying the sequence space produces the free module of the theory: for a product of copies of $\mathbb{Z}$,*

$$\Big(\big(\textstyle\prod_{\mathbb{N}} \mathbb{Z}\big) \otimes_{\mathbb{Z}} \mathbb{Z}[T]\Big)^{T\text{-}\square} \ \cong\ \textstyle\prod_{\mathbb{N}} \mathbb{Z}[T],$$

*a product of copies of $\mathbb{Z}[T]$, and this is the compact projective generator of* $\mathrm{Solid}_{\mathbb{Z}[T]}$.

*Proof.* The functor of Proposition 7.10 commutes with products and sends $\mathbb{Z}$ to $\mathbb{Z}[T]$; apply it to $\prod_{\mathbb{N}} \mathbb{Z}$. □

*Remark* 7.12 (Where the derived solidification fails to be $t$-exact)*.* The pushforward (7.10) is $t$-exact, but the $\mathbb{Z}[T]$-solidification $(-)^{\mathbb{L}T\text{-}\square}$ on all of $D(\mathrm{Mod}_{\mathbb{Z}[T]})$ is *not*. By (7.8) it is $R\underline{\mathrm{Hom}}$ against an object with a two-term compact resolution, so it is off from $t$-exactness by at most one: connective objects stay connective, and coconnective objects go to at most a one-fold shift of coconnective. It is thus very controlled, but genuinely not exact. The pushforward (7.10), by contrast, is $t$-exact even though it factors through $(-)^{\mathbb{L}T\text{-}\square}$: it is the composite of the $t$-exact base change $M \mapsto M \otimes_{\mathbb{Z}} \mathbb{Z}[T]$ with the symmetric monoidal quotient $\mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid}_{\mathbb{Z}}) \to \mathrm{Solid}_{\mathbb{Z}[T]}$, and although this second, quotient step is not $t$-exact in general, on base-changed objects the composite is.

**Example 7.13** (Base change to $\mathbb{Q}_p$: the Tate algebra)**.** Take $\mathbb{Q}_p[T]$, the functions on $\mathbb{A}^1_{\mathbb{Q}_p}$, and restrict to the closed unit disk by applying $(-)^{T\text{-}\square}$, an instance of Proposition 7.10. Pulling $\frac{1}{p}$ and the limit through,

$$\mathbb{Q}_p[T]^{T\text{-}\square} \ =\ \varprojlim_n \Big((\mathbb{Z}/p^n)[T]\Big)^{T\text{-}\square}\big[\tfrac{1}{p}\big] \ =\ \varprojlim_n (\mathbb{Z}/p^n)[T]\big[\tfrac{1}{p}\big] \ =\ \mathbb{Z}[T]^{\wedge}_p\big[\tfrac{1}{p}\big] \ =\ \mathbb{Q}_p\langle T\rangle,$$

using that $(\mathbb{Z}/p^n)[T]$ is base-changed from a discrete ring, so its $\mathbb{Z}[T]$-solidification is itself. This is the Tate algebra: power series in $T$ with coefficients in $\mathbb{Q}_p$ tending to 0. The smaller ring of functions converges on a larger region – one is now allowed to sit on the boundary $|T| = 1$ – exactly as the closed unit disk should.

## 7.6 Open versus closed: a recollement

We can now settle a question of terminology that nonarchimedean geometry leaves ambiguous. In rigid geometry the closed unit disk is treated as an open (it is affinoid, quasicompact), while the open unit disk is a non-quasicompact increasing union of closed disks; from the Berkovich viewpoint the roles can seem to switch again. Clausen's claim is that the categories attached to these objects give a definite answer.

Consider the three symmetric monoidal derived categories: for the affine line $\mathbb{A}^1$ the modules $D(\mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid}_{\mathbb{Z}}))$; for the closed unit disk the quotient $D(\mathrm{Solid}_{\mathbb{Z}[T]})$; and for the complement of the closed unit disk (the open unit disk moved to $\infty$) the modules $D(\mathrm{Mod}_{\mathbb{Z}(\!(T^{-1})\!)}(\mathrm{Solid}_{\mathbb{Z}}))$. They sit in a diagram

$$D\big(\mathrm{Mod}_{\mathbb{Z}(\!(T^{-1})\!)}(\mathrm{Solid}_{\mathbb{Z}})\big) \xrightarrow{\;i_*\;} D\big(\mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid}_{\mathbb{Z}})\big) \xrightarrow{j^*=(-)^{\mathbb{L}T\text{-}\square}} D\big(\mathrm{Solid}_{\mathbb{Z}[T]}\big) \tag{7.12}$$

which is formally a *recollement* (localization sequence), realized by the idempotent algebra $\mathbb{Z}(\!(T^{-1})\!)$ inside the symmetric monoidal category of $\mathbb{Z}[T]$-modules. The two sides carry the standard adjunctions.

- *The closed unit disk, as an open $j$.* The symmetric monoidal functor $j^* = (-)^{\mathbb{L}T\text{-}\square}$ has a right adjoint $j_*$, namely the inclusion $D(\mathrm{Solid}_{\mathbb{Z}[T]}) \hookrightarrow D(\mathrm{Mod}_{\mathbb{Z}[T]})$; it is fully faithful because the idempotence (7.9) forces at most one $\mathbb{Z}(\!(T^{-1})\!)$-module structure on a $\mathbb{Z}[T]$-module, so $D(\mathrm{Solid}_{\mathbb{Z}[T]})$ is a symmetric monoidal quotient. It also has a left adjoint $j_!$, given by

  $$j_! \colon M \longmapsto \mathrm{fib}\Big(M \to M \otimes_{\mathbb{Z}[T]} \mathbb{Z}(\!(T^{-1})\!)\Big),$$

  i.e. "base change to $\infty$ and take the fibre". This $j_!$ is better behaved than $j_*$: it satisfies the projection formula over the two symmetric monoidal categories, and is $j^*$-linear. As at infinity this is extension by zero – one keeps the sections and kills those living near the boundary.

- *The complement, as a closed $i$.* The inclusion $i_* \colon D(\mathrm{Mod}_{\mathbb{Z}(\!(T^{-1})\!)}) \hookrightarrow D(\mathrm{Mod}_{\mathbb{Z}[T]})$ is fully faithful; its left adjoint is the symmetric monoidal base change $i^* = (-) \otimes_{\mathbb{Z}[T]} \mathbb{Z}(\!(T^{-1})\!)$, and its right adjoint $i^!$ is a certain $R\underline{\mathrm{Hom}}$, which is less well-behaved. Here $i_*$ satisfies the projection formula over $i^*$.

This is formally identical to the six-functor package attached to a topological space $X$ with a closed subset $Z$ and open complement $U$:

$$D(\mathrm{Shv}(Z)) \xrightarrow{\;i_*\;} D(\mathrm{Shv}(X)) \xrightarrow{\;j^*\;} D(\mathrm{Shv}(U)), \tag{7.13}$$

with $i^* \dashv i_* \dashv i^!$ and $j_! \dashv j^* \dashv j_*$, the well-behaved (projection-formula) functors being $i_*$ and $j_!$. Both (7.12) and (7.13) are instances of the same abstract input: a symmetric monoidal category together with an idempotent algebra – here $\mathbb{Z}(\!(T^{-1})\!)$, there the pushforward of the constant sheaf from $Z$.

*Remark* 7.14 (The verdict: closed disks are open). Comparing (7.12) with (7.13), the closed unit disk $D(\mathrm{Solid}_{\mathbb{Z}[T]})$ occupies the slot of the *open* $U$, and the complementary open unit disk occupies the slot of the *closed* $Z$. So from this perspective the closed unit disk should be thought of as open, and the open unit disk as closed. The same reasoning applies in usual algebraic geometry: a quasicompact (e.g. distinguished affine) Zariski open, being the locus where a function is inverted – an idempotent algebra – should be thought of as *closed*. Clausen offered this as a genuine reorientation, to be justified later, and noted that these recollements are exactly the sort of structure guiding the definition of analytic stack: one takes seriously the principle that the derived categories and the functors between them dictate what the geometry is, and it is in this condensed/solid framework that one obtains six-functor formalisms on derived categories of quasi-coherent sheaves in large generality, with proofs of e.g. Serre duality.

## 7.7 Vista: solid rings and the valuative spectrum

The plan for the rest of the theory is to run the above over a general *solid ring* – a commutative algebra object

$$R \in \mathrm{CAlg}\big(\mathrm{Solid}_{\mathbb{Z}}, \otimes^{\square}\big)$$

– and to organize the resulting subspaces (closed disks, open disks, and their many analogues coming from arbitrary functions) in a systematic way. The mechanism is to base-change the pictures found over $\mathbb{Z}[T]$ along all possible maps $\mathbb{Z}[T] \to R$, i.e. along all functions in $R$.

*Remark* 7.15 (Localization along the valuative spectrum). The outcome is that the derived category $D(\mathrm{Mod}_R(\mathrm{Solid}_{\mathbb{Z}}))$ – even as a symmetric monoidal category – localizes along the *valuative spectrum* $\mathrm{Spv}(R(*))$ of the underlying discrete ring $R(*)$: the space of all valuations of $R(*)$ (with no continuity or integrality restriction – an arbitrary valuation, given by a residue field and a valuation on it). This is a spectral space, with a basis of quasicompact opens; the particularly nice basis, analogous to distinguished affine opens, consists of the *rational opens*. A rational open is cut out by finitely many functions $f_1, \dots, f_n$ and a further function $g$,

$$X\Big(\frac{f_1, \dots, f_n}{g}\Big) \;=\; \big\{\, |f_i| \le |g| \;\forall i, \quad g \neq 0 \,\big\}, \tag{7.14}$$

i.e. one inverts $g$ (lands in the Zariski nonvanishing locus of $g$) and then shrinks to where each $|f_i| \le |g|$. One attaches to such an open a version of the solid theory: the $M \in D(\mathrm{Mod}_R(\mathrm{Solid}_{\mathbb{Z}}))$ such that multiplication by $g$ is an isomorphism on $M$, and such that for each $i$

$$\underline{\mathrm{Hom}}(P, M) \xrightarrow{(f_i/g)\,\sigma - 1} \underline{\mathrm{Hom}}(P, M)$$

is an isomorphism; in words, thinking of $g, f_i$ as maps $\mathrm{Spec}\, R \to \mathbb{A}^1$, one requires $g$ to land in the Zariski nonvanishing locus and each $f_i/g$ to land in the closed unit disk. This exhibits a sheaf of symmetric monoidal categories on $\mathrm{Spv}(R(*))$, and in particular a structure sheaf. The goal of the rest of the lecture is to make this structure sheaf explicit and to begin comparing it with Huber's theory; the full construction, by easy formal means, needs more language and is deferred.

## 7.8 Power-bounded and topologically nilpotent elements

Fix $R \in \mathrm{CAlg}(\mathrm{Solid}_{\mathbb{Z}})$ and an element $f \in R(*)$ of the underlying discrete ring. It determines a ring map $\mathbb{Z}[T] \to R$, $T \mapsto f$, i.e. the sequence $1, f, f^2, f^3, \dots$.

**Definition 7.16** (Topologically nilpotent, power-bounded)**.** With notation as above:

- $f$ is *topologically nilpotent* if the ring map $\mathbb{Z}[T] \to R$ factors (as a ring map) through $\mathbb{Z}[\![T]\!] = P^{\square}$;
- $f$ is *power-bounded* if $R$, viewed as a $\mathbb{Z}[T]$-module via $T \mapsto f$, is $\mathbb{Z}[T]$-solid, i.e. $\underline{\mathrm{Hom}}(P, R) \xrightarrow{f\sigma - 1} \underline{\mathrm{Hom}}(P, R)$ is an isomorphism.

Write $R^\circ \subseteq R(*)$ for the set of power-bounded elements and $R^{\circ\circ} \subseteq R(*)$ for the set of topologically nilpotent elements (Huber's notation).

*Remark* 7.17 (Comparison with the classical definitions). For topological nilpotence the agreement with the classical notion is immediate: since $T \mapsto f$ is a ring map, $T^n \mapsto f^n$, and factoring through $\mathbb{Z}[\![T]\!]$ says exactly that the sequence $1, f, f^2, \dots$ extends to a null sequence – topological nilpotence. The genuinely new feature in the solid setting is that such a factorization, *if it exists, is unique*, by the idempotence of $\mathbb{Z}[\![T]\!]$: even though solid abelian groups can be very non-Hausdorff, the limit built into solidity is imposed to exist uniquely.

*Remark* 7.18 (A subtlety: factorization as rings). The factorization defining topological nilpotence must be taken in the category of *rings*. A factorization merely as condensed abelian groups, or even as $\mathbb{Z}[T]$-modules, is weaker: one may have a null sequence $1, f, f^2, \dots$ (a factorization of abelian groups) without $f$ being topologically nilpotent in the ring sense. If, however, the target is quasiseparated – the condensed analogue of Hausdorff – then by density the factorization is automatically a ring map when it exists, and the two notions agree.

**Lemma 7.19** (Huber-type structure of $R^\circ$ and $R^{\circ\circ}$)**.** *$R^\circ \subseteq R(*)$ is an integrally closed subring, and $R^{\circ\circ} \subseteq R^\circ$ is a radical ideal.*

*Proof that $R^\circ$ is a subring.* That $1 \in R^\circ$ is the definition of solidity (put $f = 1$). For closure under the ring operations, suppose $f, g \in R^\circ$; we show $F(f,g) \in R^\circ$ for any $F \in \mathbb{Z}[X,Y]$. Consider the ring map $\mathbb{Z}[X,Y] \to R$, $X \mapsto f$, $Y \mapsto g$; the hypothesis is that $R$ is $\mathbb{Z}[X]$-solid and $\mathbb{Z}[Y]$-solid, and we want it to be $\mathbb{Z}[T]$-solid along $T \mapsto F(f,g)$.

$$\begin{array}{ccc} & & \mathbb{Z}[T] \\ & & \downarrow{\scriptstyle T\mapsto F} \\ \mathbb{Z}[X,Y] & \xrightarrow[X\mapsto f,\ Y\mapsto g]{} & R \end{array}$$

Resolve $R$ (which is solid as an abelian group) by direct sums of the compact projective generators $\prod_{\mathbb{N}} \mathbb{Z}$, base-changed to $\mathbb{Z}[X,Y]$-modules. Solidifying with respect to $X$ turns each term into $\bigoplus(\prod_{\mathbb{N}} \mathbb{Z}[\![X]\!])[Y]$ (by the properties of $\mathbb{Z}[X]$-solidification, Proposition 7.10) without changing $R$; solidifying then with respect to $Y$ gives $\bigoplus \prod_{\mathbb{N}} \mathbb{Z}[\![X,Y]\!]$. Each such term is $\mathbb{Z}[T]$-solid – a colimit of limits of things solid over $\mathbb{Z}$, discrete in $T$, hence solid along any $T \mapsto F(f,g)$ – and $\mathrm{Solid}_{\mathbb{Z}[T]}$ is closed under colimits, so $R$ is $\mathbb{Z}[T]$-solid. Thus $F(f,g) \in R^\circ$. □

*Proof of integral closedness.* Suppose $f \in R(*)$ satisfies a monic equation

$$f^n + c_{n-1} f^{n-1} + \cdots + c_0 = 0, \qquad c_i \in R^\circ.$$

Form the universal situation: over $\mathbb{Z}[X_0, \dots, X_{n-1}]$ adjoin a variable $T$ and impose $T^n + X_{n-1}T^{n-1} + \cdots + X_0 = 0$, with the map to $R$ sending $X_i \mapsto c_i$ and $T \mapsto f$. By hypothesis $R$ is solid over each $\mathbb{Z}[X_i]$; running the same resolution as above (solidify in each $X_i$), one finds $R$ resolved by direct sums of products of copies of the ring $\mathbb{Z}[X_0, \dots, X_{n-1}][T]/(T^n + \cdots + X_0)$, which is finite over $\mathbb{Z}[X_0, \dots, X_{n-1}]$; a finite (indeed Noetherian, so finitely presented) module can be pulled inside the products, and the resulting terms are solid over $\mathbb{Z}[T]$. Hence $R$ is $\mathbb{Z}[T]$-solid along $T \mapsto f$, i.e. $f \in R^\circ$. The proofs that $R^{\circ\circ}$ is an ideal and is radical are entirely analogous and are left as exercises in the same style.[1] □

[1] Board 33 writes the coefficients as $c_i \in R^{\circ\circ}$, but this is a slip: the statement being proved is integral closedness

## 7.9 Rational localizations and the structure sheaf

We can now build the rings that will be the values of the structure sheaf on the rational opens (7.14).

**Theorem 7.20** (Rational localization)**.** *Let $R \in \mathrm{CAlg}(\mathrm{Solid}_{\mathbb{Z}})$ and $g, f_1, \ldots, f_n \in R(*)$. Then there exists an initial solid ring*

$$R\Big\langle \frac{f_1, \ldots, f_n}{g} \Big\rangle^{\square}, \qquad R \longrightarrow R\Big\langle \frac{f_1, \ldots, f_n}{g} \Big\rangle^{\square},$$

*such that in the target*

1. *$g$ is invertible, and*
2. *$f_i/g$ is power-bounded for all $i$.*

*Explicitly, it is constructed as*

$$R\Big\langle \frac{f_1, \ldots, f_n}{g} \Big\rangle^{\square} \;=\; \Big(R[X_1, \ldots, X_n]^{X_1^{\square}, \ldots, X_n^{\square}} / (gX_1 - f_1, \; \ldots, \; gX_n - f_n)\Big)\big[\tfrac{1}{g}\big], \tag{7.15}$$

*where one first freely adjoins the solid variables $X_i$ (i.e. solidifies with respect to each), then imposes $gX_i = f_i$ (making $X_i = f_i/g$ power-bounded), then inverts $g$.*

*Proof.* The construction (7.15) manifestly has the correct universal property. Adjoining a solid variable $X_i$ and solidifying is what forces $X_i$ power-bounded (this "does something" when $R$ is nondiscrete: over $\mathbb{Q}_p$ with one variable it produces the Tate algebra, cf. Example 7.13); imposing $gX_i - f_i$ then sets $X_i = f_i/g$; and inverting $g$ makes $g$ a unit. The only point to check is that after these operations the result is still solid, which holds because they are all colimits. $\square$

*Remark* 7.21 (Two warnings: derived and non-quasiseparated). Two cautions about (7.15).

1. It is *not* exactly the value of the structure sheaf on $X(f_1, \ldots, f_n/g)$; it is $\pi_0$ of that value. The six-functor formalisms and the localization of Remark 7.15 live at the *derived* level, so what is produced a priori is a *derived* sheaf. In almost all practical cases the higher homotopy vanishes, $\pi_i = 0$ for $i > 0$, so this causes no trouble; but it must be kept in mind.
2. Even for very nice $R$ – a complete Huber ring – the quotient in (7.15) need not be quasiseparated. The solidified polynomial ring $R[X_1, \ldots, X_n]^{X_1^{\square}, \ldots, X_n^{\square}}$ is the usual Tate algebra in several variables (this follows from the arguments above for any complete Huber ring $R$), but the ideal generated by the $gX_i - f_i$ need not be closed, so the quotient can be non-Hausdorff. Again, in essentially all practical cases it is quasiseparated.

*Remark* 7.22 (Relation to Huber's theory). Let $R$ be a complete Huber ring and suppose the ideal $(f_1, \ldots, f_n) \subseteq R$ is open – the condition that describes rational opens in the space of *continuous* valuations, as opposed to all valuations. Then Huber [Hub94] defines a ring $R\langle f_1, \ldots, f_n/g\rangle^{\mathrm{Huber}}$. It is recovered from the solid construction as its *quasiseparation*:

$$R\Big\langle \frac{f_1, \ldots, f_n}{g} \Big\rangle^{\mathrm{Huber}} \;=\; \text{quasiseparation of } \pi_0\, R\Big\langle \frac{f_1, \ldots, f_n}{g} \Big\rangle^{\square}.$$

of $R^{\circ}$, which by definition concerns monic relations with $c_i \in R^{\circ}$, and the proof uses precisely that each $c_i$ is power-bounded (so that $R$ is $\mathbb{Z}[X_i]$-solid via $X_i \mapsto c_i$). Clausen indeed said "all $c_i$ are power-bounded" (01:16:55). Transcribed here as $R^{\circ}$ accordingly.

So Huber's ring is obtained functorially from the (more generally defined, possibly derived and non-quasiseparated) solid object, but the two need not be equal in general – though in all practical cases they are. This repairs a known defect of Huber's theory: in general his structure presheaf is only a presheaf, not a sheaf (though it is a sheaf in all practical cases). If one drops the insistence on quasiseparatedness and on being non-derived, the solid theory yields a genuine structure sheaf – indeed a derived category of quasi-coherent sheaves that localizes – and moreover allows one to define these objects even without the openness hypothesis on $(f_1, \ldots, f_n)$.

### 7.10 Closing remarks

*Remark* 7.23 (Non-continuous valuations and Colmez's ad hoc spaces). Asked whether non-continuous valuations are ever used, Clausen pointed to Colmez, who considered spaces (which he called "ad hoc spaces") designed to include objects such as the open unit disk and $\mathbb{Z}_p$ as honest affine objects, in the context of his work on the mod-$p$ Langlands correspondence for $\mathrm{GL}_2$. The open unit disk is the prototypical topological ring that is not a Huber ring, so in the usual rigid theory one is forced to present it as a non-quasicompact union of closed disks; Colmez advocated sometimes treating it directly as affine, which the present theory accommodates easily. Clausen noted this was done in an example rather than as a developed theory.

*Remark* 7.24 (Induced analytic ring structures; recovering Huber pairs). Every algebra $R$ in solid $\mathbb{Z}$-modules gives what will be called an *induced* analytic ring structure: its module category sits inside condensed $R$-modules and forms an analytic ring in the sense of Definition 1.21. For $R = \mathbb{Z}[T]$ the $\mathbb{Z}[T]$-solid theory is a further, more complete such structure; both exist and one wants both. This extra flexibility can be extended to any discrete ring, though for a general solid $\mathbb{Z}$-algebra Clausen knew of no completely canonical choice – one option is to force all power-bounded elements to be solid, giving the maximally complete structure obtainable with the present tools, though further completions may exist.

Feeding a *pair* $(R, R^+)$ into the analytic-ring formalism of Section 1, rather than a bare ring, recovers an analogue of Huber's theory for Huber pairs: one imposes solidity of the $f_i/g$ only for the chosen ring of integral elements $R^+$, not for all functions. This fits Huber's theory remarkably well, and topologically nilpotent elements are automatically solid in any case.

*Remark* 7.25 (Solidifications commute). A structural bonus of the present setup: whereas in the earlier theory forming the tensor product of analytic rings required solidifying in one direction and then the other (a colimit of iterated completions, cf. Remark 1.31), here the various solidifications all commute with one another, since each is given by an $R\underline{\mathrm{Hom}}$ out of a fixed object and any two such $R\underline{\mathrm{Hom}}$'s commute.

## 8 Huber Pairs and Analytic Rings

Scholze's last lecture (before the course turns to geometry proper) sets up the general notion of an *analytic ring* and compares it with Huber's theory of adic spaces. The material has two halves. First, the abstract definition: an analytic ring is an underlying light condensed ring $A^\triangleright$ together with a notion of "complete" modules – a full abelian subcategory $\mathrm{Mod}(A) \subseteq \mathrm{Cond}(A^\triangleright)$ closed under enough operations to behave like the solid or $\mathbb{Z}[T]$-solid categories of Sections 5 and 7. One derives from this a base-change left adjoint, a symmetric monoidal structure, a derived category $D(A)$ with a $t$-structure, and a derived symmetric monoidal tensor product. Second, the comparison with Huber: for a solid ring $A^\triangleright$ one recovers the topologically nilpotent and power-bounded elements, and one shows that Huber's *rings of integral elements* $A^+$ correspond exactly to the solid analytic ring structures on $A^\triangleright$ that are cut out by conditions of the form "$1 - f\sigma$ is invertible on $\underline{\mathrm{Hom}}(P, -)$". The match with Huber's classical theory is, in Scholze's words, striking.

Throughout, "ring" means commutative ring, everything happens in light condensed abelian groups, and we freely use solid abelian groups $\mathrm{Solid} \subseteq \mathrm{CondAb}^{\mathrm{light}}$ (Sections 5 and 6) and the relative solid theory over $\mathbb{Z}[T]$ (Section 7).

## 8.1 Motivation

By now we have seen several examples of the same package: a light condensed ring together with a notion of "complete" condensed module over it. In each case the complete modules form a full subcategory of all condensed modules, cut out by a completeness condition.

*Remark* 8.1 (The recurring examples). The examples encountered so far are

$$\begin{aligned}
&\big(\mathbb{Z},\ \mathrm{Solid}_{\mathbb{Z}} \subseteq \mathrm{Cond}(\mathrm{Ab})\big),\\
&\big(\mathbb{Z}[T],\ \mathrm{Solid}_{\mathbb{Z}[T]} \subseteq \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid}_{\mathbb{Z}}) \subseteq \mathrm{Cond}(\mathbb{Z}[T])\big),\\
&\big(\mathbb{Z}_p,\ \mathrm{Solid}_{\mathbb{Z}_p}\big), \qquad \big(\mathbb{Q}_p,\ \mathrm{Solid}_{\mathbb{Q}_p}\big),\ \ldots
\end{aligned} \tag{8.1}$$

The first is the solid theory (Section 5); the second is Clausen's affine line, where one may either take all $\mathbb{Z}[T]$-modules whose underlying group is solid, or isolate the stronger $\mathbb{Z}[T]$-solid condition (Definition 7.5), which geometrically distinguishes the affine line from the closed unit disk. In the last two one simply takes the modules whose underlying condensed abelian group is solid; over $\mathbb{Q}_p$ (or $\mathbb{C}_p$, or a $p$-adic Banach algebra) there is no meaningful way to strengthen this. The aim is to axiomatise this situation.

## 8.2 Huber's definitions

For comparison we recall Huber's basic objects [Hub94]; Huber uses different names, but the content is the following.

**Definition 8.2** (Huber)**.** A *Huber ring* is a topological ring $A$ that admits an open subring $A_0 \subseteq A$ (a *ring of definition*) and a finitely generated ideal $I \subseteq A_0$ (an *ideal of definition*) such that $A_0$ carries the $I$-adic topology.

**Definition 8.3** (Ring of integral elements)**.** A *ring of integral elements* in a Huber ring $A$ is an open, integrally closed subring

$$A^+ \ \subseteq\ A^\circ \ =\ \{\text{power-bounded elements of } A\}.$$

**Definition 8.4** (Huber pair)**.** A *Huber pair* is a pair $(A, A^+)$ of a Huber ring $A$ and a ring of integral elements $A^+$ of $A$.

**Example 8.5** (Huber rings)**.** 1. Any discrete ring $A$ is Huber: take $A_0 = A$, $I = (0)$.

2. Any ring $A$ with the $I$-adic topology for a finitely generated ideal $I$ is Huber: take $A_0 = A$, $I = I$.
3. $\mathbb{Q}_p$ is Huber: take $A_0 = \mathbb{Z}_p$, $I = (p)$ (which is an ideal in $\mathbb{Z}_p$, not in $\mathbb{Q}_p$). More generally any nonarchimedean field $K$ is Huber, with $A_0$ the valuation ring and $I$ the ideal generated by

a uniformizer; and any Banach algebra over $\mathbb{Q}_p$ (or another nonarchimedean local field) is Huber, with $A_0$ a unit ball and $I$ generated by a uniformizer.

Thus Huber rings are, roughly, localisations of adic rings of the form (f.g.-ideal)-adic rings, packaged with some extra technology.

*Remark* 8.6 (Completion; only complete rings). The completion $\widehat{A}$ of any Huber ring $A$ is again a Huber ring: if $A \supseteq A_0 \supseteq I$ as above, then $\widehat{A} \supseteq \widehat{A_0}$ = the $I$-adic completion of $A_0$, an open subring of $\widehat{A}$. Here "completion" is the classical one, allowing all convergent Cauchy sequences. Since the adic spaces attached to $A$ and to $\widehat{A}$ are definitionally the same, non-complete Huber rings play no essential role, and **henceforth all Huber rings are assumed complete**: whenever we write a Huber pair, the underlying ring is complete.

For a complete Huber ring one has the canonical subsets

$$A^{\circ\circ} \;=\; \{f \in A \mid f^n \to 0 \text{ as } n \to \infty\} \;\subseteq\; A^\circ \;=\; \{f \in A \mid (f^n)_{n\geq 0} \text{ is bounded}\} \;\subseteq\; A, \tag{8.2}$$

the *topologically nilpotent* and *power-bounded* elements. (Since $A$ is complete, $f^n \to 0$ is a condition, not extra data.) These are canonical, whereas $A_0$ and $I$ are choices; the two are related by

$$A^{\circ\circ} \;=\; \varinjlim_{I \subseteq A_0 \subseteq A} I, \qquad A^\circ \;=\; \varinjlim_{A_0 \subseteq A} A_0,$$

filtered colimits (unions) over all ideals, resp. rings, of definition: the $A_0$'s and $I$'s form filtered systems, and $A^\circ$ (resp. $A^{\circ\circ}$) is their union. For example $(p) \subseteq \mathbb{Z}_p \subseteq \mathbb{Q}_p$ realises $A^{\circ\circ} \subseteq A^\circ \subseteq A$; the elements of $I$ are topologically nilpotent, but $I = (p^2)$ is an equally valid ideal of definition, so the containment is not an equality for a single choice.

### 8.3 Analytic rings

*Notation* 8.7 (Single-letter convention). Huber denotes a Huber pair by the single letter $A$, its two constituents being a topological ring and a ring of integral elements. We follow this convention for analytic rings: an analytic ring is denoted by a single letter $A$, and its underlying light condensed ring is denoted $A^\triangleright$ (Clausen's triangle notation, Section 1).

**Definition 8.8** (Analytic ring structure)**.** Let $A^\triangleright$ be a light condensed ring. An *analytic ring structure* on $A^\triangleright$ is a full subcategory

$$\mathrm{Mod}(A) \;\subseteq\; \mathrm{Cond}(A^\triangleright)$$

of the light condensed $A^\triangleright$-modules, that is an abelian subcategory and is

- stable under all limits, all colimits, and all extensions;
- stable under all $\mathrm{Ext}^i_{A^\triangleright}(N, -)$, i.e. $\mathrm{Ext}^i_{A^\triangleright}(N, M) \in \mathrm{Mod}(A)$ for $M \in \mathrm{Mod}(A)$ and $N \in \mathrm{Cond}(A^\triangleright)$ (internal Ext);
- contains $A^\triangleright$;

and such that the inclusion $\mathrm{Mod}(A) \hookrightarrow \mathrm{Cond}(A^\triangleright)$ admits a left adjoint. We call the objects of $\mathrm{Mod}(A)$ the *complete* modules.

*Remark* 8.9 (Relation to Section 1). This is equivalent to the definition of analytic ring given by Clausen in Section 1 (Definition 1.21), but phrased from a slightly more elementary, abelian-categorical perspective: rather than list the long roster of good properties of $\mathrm{Solid}_{\mathbb{Z}}$ or $\mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid})$

every time, one takes closure under exactly those operations as the axioms. The clause that the inclusion admits a left adjoint was, in the lecture, initially omitted and then restored (see Remark 8.12).

*Remark* 8.10 (Grothendieck category; $D(A)$ need not be $D(\mathrm{Mod}(A))$). Gabber asked whether the axioms force $\mathrm{Mod}(A)$ to be a Grothendieck abelian category, i.e. to admit a set of generators; Scholze confirmed this is automatic. A second question – whether Ext-groups computed in $\mathrm{Mod}(A)$ agree with those in the ambient $\mathrm{Cond}(A^{\triangleright})$ – is more delicate. The point is that $D(A)$ will be defined *not* as the derived category of the abelian category $\mathrm{Mod}(A)$, but as a full subcategory of $D(\mathrm{Cond}(A^{\triangleright}))$ (Definition 8.14), of which $\mathrm{Mod}(A)$ is the *heart* of a $t$-structure (Proposition 8.16). In most practical cases the two derived categories coincide, but not in general.

## 8.4 Base change and the symmetric monoidal structure

**Proposition 8.11** (Base change)**.** *The inclusion* $\mathrm{Mod}(A) \hookrightarrow \mathrm{Cond}(A^{\triangleright})$ *has a left adjoint*

$$\mathrm{Cond}(A^{\triangleright}) \longrightarrow \mathrm{Mod}(A), \qquad M \longmapsto M \otimes_{A^{\triangleright}} A,$$

*the* base change. *Its kernel is a* $\otimes$*-ideal, and* $\mathrm{Mod}(A)$ *thereby acquires a unique symmetric monoidal structure making* $- \otimes_{A^{\triangleright}} A$ *symmetric monoidal; explicitly*

$$M \otimes_A N \;:=\; (M \otimes_{A^{\triangleright}} N) \otimes_{A^{\triangleright}} A. \tag{8.3}$$

*Proof sketch.* Existence of the left adjoint is formal (Remark 8.12). To see the kernel is a $\otimes$-ideal, let $M \in \mathrm{Cond}(A^{\triangleright})$ with $M \otimes_{A^{\triangleright}} A = 0$ and let $N \in \mathrm{Cond}(A^{\triangleright})$ be arbitrary; we must show $(N \otimes_{A^{\triangleright}} M) \otimes_{A^{\triangleright}} A = 0$, equivalently that $\mathrm{Hom}_{A^{\triangleright}}(N \otimes_{A^{\triangleright}} M, L) = 0$ for all $L \in \mathrm{Mod}(A)$. By tensor–Hom adjunction

$$\mathrm{Hom}_{A^{\triangleright}}(N \otimes_{A^{\triangleright}} M, L) \;=\; \mathrm{Hom}_{A^{\triangleright}}\big(M,\ \underline{\mathrm{Hom}}_{A^{\triangleright}}(N, L)\big),$$

and $\underline{\mathrm{Hom}}_{A^{\triangleright}}(N, L) \in \mathrm{Mod}(A)$ by the internal-Ext stability of Definition 8.8; since $M \otimes_{A^{\triangleright}} A = 0$, i.e. $M$ has no maps to any complete module, this vanishes. The tensor product (8.3) is then forced: one checks

$$(M \otimes_{A^{\triangleright}} N) \otimes_{A^{\triangleright}} A \;\xrightarrow{\sim}\; \big((M \otimes_{A^{\triangleright}} A) \otimes_{A^{\triangleright}} (N \otimes_{A^{\triangleright}} A)\big) \otimes_{A^{\triangleright}} A$$

for all $M, N \in \mathrm{Cond}(A^{\triangleright})$ by the same argument as in the solid case (Section 5). $\square$

*Remark* 8.12 (Why left-adjointness belongs in the definition). Scholze first hoped to *prove* the left adjoint exists: there is a general principle that a full subcategory of a presentable category, closed under all limits and all sufficiently filtered colimits, is reflective (cf. Remark 6.5). The non-trivial point is the existence of a generating set. However, a genuine gap appears one level up, at the derived stage: stability of $D(A)$ under *arbitrary* (uncountable) products cannot be had for free, because arbitrary products in $\mathrm{Cond}(A^{\triangleright})$ are not exact. The clean fix, already adopted by Clausen in Section 1, is to *require* the left adjoint to exist as part of Definition 8.8; arbitrary products are then stable because a category admitting a left adjoint to a product-preserving inclusion has well-behaved products.

*Remark* 8.13 (The analytic ring structure is an "induced" localisation). So an analytic ring structure is a localisation-type situation: $\mathrm{Mod}(A)$ is a reflective symmetric monoidal subcategory of the condensed modules over the underlying ring, and the reflector $- \otimes_{A^{\triangleright}} A$ is a symmetric monoidal localisation whose kernel is a $\otimes$-ideal. This is the abelian shadow of the induced analytic ring structures of Remark 7.24.

## 8.5 Derived categories of analytic rings

**Definition 8.14** (Derived category of an analytic ring)**.** Let $A$ be an analytic ring structure on $A^{\triangleright}$. The *derived category* $D(A) \subseteq D(\mathrm{Cond}(A^{\triangleright}))$ is the full subcategory

$$D(A) \;=\; \left\{ M \in D(\mathrm{Cond}(A^{\triangleright})) \;\middle|\; H_i(M) \in \mathrm{Mod}(A) \text{ for all } i \in \mathbb{Z} \right\}$$

(homological grading). There is a natural comparison functor $D(\mathrm{Mod}(A)) \to D(A)$, but it is *not* always an equivalence (though it is in essentially all cases arising in practice).

**Proposition 8.15** (Structure of $D(A)$)**.** *$D(A) \subseteq D(\mathrm{Cond}(A^{\triangleright}))$ is a triangulated subcategory, stable under all direct sums and all products. The inclusion admits a left adjoint, the* derived base change

$$- \otimes^{\mathbb{L}}_{A^{\triangleright}} A\colon\; D(\mathrm{Cond}(A^{\triangleright})) \longrightarrow D(A),$$

*whose kernel is a $\otimes$-ideal, and there is a unique symmetric monoidal tensor product $- \otimes^{\mathbb{L}}_{A} -$ on $D(A)$ making $- \otimes^{\mathbb{L}}_{A^{\triangleright}} A$ symmetric monoidal.*

*Proof sketch. Triangulated.* Given a triangle $M' \to M \to M'' \to M'[1]$ in $D(\mathrm{Cond}(A^{\triangleright}))$ with $M', M'' \in D(A)$, the long exact homology sequence

$$\cdots \to H_{i+1}(M'') \to H_i(M') \to H_i(M) \to H_i(M'') \to H_{i-1}(M') \to \cdots$$

exhibits each $H_i(M)$ as an extension of a subobject $K$ of $H_i(M'')$ by a quotient $Q$ of $H_i(M')$, both in $\mathrm{Mod}(A)$; stability under kernels, quotients and extensions gives $H_i(M) \in \mathrm{Mod}(A)$, so $M \in D(A)$.

*Direct sums and products.* Stability under $\bigoplus$ reduces to the $H_i$'s. Countable products $\prod_{\mathbb{N}}$ likewise reduce to homology; arbitrary (uncountable) products are fine only once one admits a left adjoint (Remark 8.12).

*Kernel is a $\otimes$-ideal.* Let $M \in \ker(D(\mathrm{Cond}(A^{\triangleright})) \to D(A))$ and $N \in D(\mathrm{Cond}(A^{\triangleright}))$; we must show $\mathrm{Hom}_{D(\mathrm{Cond}(A^{\triangleright}))}(M \otimes^{\mathbb{L}}_{A^{\triangleright}} N, L) = 0$ for all $L \in D(A)$, i.e. that $R\underline{\mathrm{Hom}}(N, L) \in D(A)$. Here light condensed sets are *replete* (countable limits of surjections are surjections, Proposition 2.10), so every $K \in D(\mathrm{Cond}(A^{\triangleright}))$ is its own Postnikov limit,

$$K \;\xrightarrow{\sim}\; R\varprojlim_{n} \tau_{\leq n} K. \tag{8.4}$$

This lets one reduce (writing $L \in D^+$, $N = \mathrm{colim}_n\, \tau_{\geq -n} N \in D^-$) via a spectral sequence to the abelian statement: for $L \in \mathrm{Mod}(A)$ and $N \in \mathrm{Cond}(A^{\triangleright})$, the condition $\mathrm{Ext}^i_{A^{\triangleright}}(N, L) \in \mathrm{Mod}(A)$ – which holds by Definition 8.8. The boundedness furnished by (8.4) is essential: the internal-Ext stability is available only after such a reduction, since the complexes are a priori unbounded in both directions.

*Tensor product.* Existence of $- \otimes^{\mathbb{L}}_{A} -$ is then formal, exactly as in Proposition 8.11. $\square$

**Proposition 8.16** ($t$-structure and heart)**.** *$D(A)$ carries a natural $t$-structure, for which the inclusion $D(A) \hookrightarrow D(\mathrm{Cond}(A^{\triangleright}))$ is $t$-exact and the derived base change $- \otimes^{\mathbb{L}}_{A^{\triangleright}} A$ preserves connective objects $D_{\geq 0}$. Its heart is* $\mathrm{Mod}(A)$*,*

$$D(A)^{\heartsuit} \;=\; \mathrm{Mod}(A),$$

*and taking hearts recovers the abelian-level functors: $(- \otimes^{\mathbb{L}}_{A^{\triangleright}} A)^{\heartsuit} = - \otimes_{A^{\triangleright}} A$ and $(- \otimes^{\mathbb{L}}_{A} -)^{\heartsuit} = - \otimes_A -$.*

*Proof.* $D(A)$ is stable under all $\tau_{\leq n}$ and $\tau_{\geq n}$ by Definition 8.14, giving the $t$-structure with $D(A)^{\heartsuit} = \mathrm{Mod}(A)$. A left adjoint to a $t$-exact functor preserves connective objects, so $- \otimes^{\mathbb{L}}_{A^{\triangleright}} A$ preserves $D_{\geq 0}$. The heart identities are immediate. $\square$

*Remark* 8.17 (Derived base change is not $t$-exact). The derived base change $- \otimes^{\mathbb{L}}_{A^{\triangleright}} A$ preserves connective objects but is not in general $t$-exact – as already seen for solidification, which can push a discrete object to the left (Proposition 6.6, Remark 7.12). The comparison $D(\mathrm{Mod}(A)) \to D(A)$ and the question of whether animating the abelian tensor product recovers $- \otimes^{\mathbb{L}}_{A} -$ both fail in general; when the product is recovered, its underlying abelian functor is automatically the right one, by adjunction.

## 8.6 Maps of analytic rings

**Definition 8.18** (Map of analytic rings)**.** A *map of analytic rings* $A = (A^{\triangleright}, \mathrm{Mod}(A)) \to B = (B^{\triangleright}, \mathrm{Mod}(B))$ is a map $A^{\triangleright} \to B^{\triangleright}$ of underlying condensed rings such that restriction of scalars sends complete modules to complete modules, i.e. the square

$$\begin{tikzcd}
\mathrm{Mod}(B) \arrow[r, hook] \arrow[d, "\mathrm{res}"'] & \mathrm{Cond}(B^{\triangleright}) \arrow[d, "\mathrm{res}"] \\
\mathrm{Mod}(A) \arrow[r, hook] & \mathrm{Cond}(A^{\triangleright})
\end{tikzcd}$$

commutes (the right vertical map being restriction along $A^{\triangleright} \to B^{\triangleright}$). Passing to left adjoints yields the *base change*

$$- \otimes_A B \colon\ \mathrm{Mod}(A) \longrightarrow \mathrm{Mod}(B),$$

a symmetric monoidal ($\otimes$-)functor, together with its derived version $- \otimes^{\mathbb{L}}_{A} B \colon D(A) \to D(B)$ (which one may compute by base changing as condensed modules and then completing). The passage to left adjoints is valid already on hearts, hence on derived categories, since it can be checked on underlying objects.

## 8.7 Comparison with Huber rings

Huber rings (assumed complete, Remark 8.6) define condensed rings, and this functor is fully faithful; in fact it lands in *solid* rings – those $A^{\triangleright}$ whose underlying condensed abelian group is solid. We now recover Huber's invariants inside the solid formalism.

**Definition 8.19** (Topologically nilpotent and power-bounded elements)**.** Let $A^{\triangleright}$ be a solid ring and $f \in A^{\triangleright}(*)$ an element of the underlying discrete ring, giving a ring map $\mathbb{Z}[T] \to A^{\triangleright}$, $T \mapsto f$ (the sequence $1, f, f^2, \dots$). Following Definition 7.16, define subsets $A^{\circ\circ} \subseteq A^{\circ} \subseteq A^{\triangleright}(*)$ by:

- $f \in A^{\circ\circ}$ (*topologically nilpotent*) if $T \mapsto f$ factors through $\mathbb{Z}[\![T]\!]$,

$$\begin{tikzcd}
\mathbb{Z}[T] \arrow[r, "T\mapsto f"] \arrow[d] & A^{\triangleright} \\
\mathbb{Z}[\![T]\!] \arrow[ur, dashed, "\exists"'] &
\end{tikzcd}$$

  i.e. the sequence $1, f, f^2, \dots$ extends to a null sequence (it is enough that $T \mapsto f$ factors as $\mathbb{Z}[T]$-modules, with the shift relation, cf. Remark 7.18);

- $f \in A^\circ$ (*power-bounded*) if $A^\triangleright$ is $\mathbb{Z}[T]$-solid via $T \mapsto f$, i.e. $1 - f\sigma$ is an isomorphism on $\underline{\mathrm{Hom}}(P, A^\triangleright)$, where $\sigma$ is the shift.

When $A^\triangleright$ arises from a Huber ring, these agree with the classical subsets (8.2).

Given a solid ring $A^\triangleright$ equipped with an analytic ring structure $A$, one extracts a ring of integral elements as follows.

**Definition 8.20** (Ring of integral elements of an analytic ring)**.** Let $A$ be a solid analytic ring structure on $A^\triangleright$. Set

$$\begin{aligned} A^+ = \big\{\, f \in A^\triangleright(*) \;\big|\; & (T \mapsto f)\colon \mathbb{Z}[T] \to A^\triangleright \\ & \text{induces a map of analytic rings } (\mathbb{Z}[T], \mathrm{Solid}_{\mathbb{Z}[T]}) \to A \,\big\}, \end{aligned}$$

where $(\mathbb{Z}[T], \mathrm{Solid}_{\mathbb{Z}[T]})$ is the closed-unit-disk analytic ring of Section 7. Equivalently, $f \in A^+$ iff $1 - f\sigma$ is an automorphism of $P \otimes_{\mathbb{Z}} A$ in $D(A)$ (via the derived base change $- \otimes^{\mathbb{L}}_{\mathbb{Z}} A$).

**Proposition 8.21** (Huber-type structure)**.** *With notation as above,*

$$A^{\circ\circ} \;\subseteq\; A^+ \;\subseteq\; A^\circ \;(\subseteq A^\triangleright(*)),$$

*$A^+$ is an integrally closed subring, and $A^{\circ\circ}$ is a radical ideal.*

*Idea.* This is the same argument Clausen gave in Lemma 7.19: everything reduces to statements of the form "being solid over one ring implies being solid over another". If $f \in A^{\circ\circ}$, then $A^\triangleright$ becomes a $\mathbb{Z}[\![T]\!]$-module, which is automatically $\mathbb{Z}[T]$-solid, giving $A^{\circ\circ} \subseteq A^+$; the ring/ideal and integral-closedness statements follow by resolving $A^\triangleright$ by direct sums of the compact projective generators $\prod_{\mathbb{N}} \mathbb{Z}$ base-changed appropriately, exactly as in Section 7. Indeed the argument there was already phrased for an arbitrary module, so it applies verbatim. $\square$

Conversely, a ring of integral elements produces an analytic ring structure.

**Definition 8.22** (Analytic ring of a Huber pair)**.** Let $(A^\triangleright, A^+)$ be a (solid) Huber pair. Define the analytic ring $(A^\triangleright, A^+)_\square$ with underlying ring $A^\triangleright$ and complete modules

$$\begin{aligned} \mathrm{Mod}\big((A^\triangleright, A^+)_\square\big) = \big\{\, M \in \mathrm{Cond}(A^\triangleright) \;\big|\; & \forall f \in A^+, \\ & 1 - f\sigma\colon \underline{\mathrm{Hom}}(P, M) \to \underline{\mathrm{Hom}}(P, M) \text{ is an isomorphism} \,\big\}. \end{aligned}$$

Declaring $1 - f\sigma$ invertible on $\underline{\mathrm{Hom}}(P, -)$ defines an analytic ring structure by the general mechanism of Definition 8.8; and if $S \subseteq A^+$ generates $A^+$ as a ring of integral elements (i.e. as an open integrally closed subring), it suffices to impose the condition for $f \in S$ – by integral closedness of the resulting power-bounded locus – so typically only one or two elements need be checked.

**Example 8.23** (Three structures on $\mathbb{Z}$ and $\mathbb{Z}[T]$)**.**

$$\begin{aligned} (\mathbb{Z}, \mathbb{Z})_\square &= (\mathbb{Z}, \mathrm{Solid}_{\mathbb{Z}}) = \mathbb{Z}_\square, \\ (\mathbb{Z}[T], \mathbb{Z})_\square &= \big(\mathbb{Z}[T], \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid}_{\mathbb{Z}})\big), \\ (\mathbb{Z}[T], \mathbb{Z}[T])_\square &= \big(\mathbb{Z}[T], \mathrm{Solid}_{\mathbb{Z}[T]}\big). \end{aligned} \tag{8.5}$$

The middle one imposes no condition on $T$ (only solidity over $\mathbb{Z}$), giving the affine line; the last imposes power-boundedness of $T$, giving the closed unit disk (Section 7).

**Proposition 8.24** (Huber pairs embed)**.** *For a Huber pair $(A^\triangleright, A^+)$, the associated analytic ring $A = (A^\triangleright, A^+)_\square$ recovers its ring of integral elements:*

$$\big((A^\triangleright, A^+)_\square\big)^+ \;=\; A^+.$$

*Consequently the assignment*

$$\{\textit{Huber pairs}\} \;\longrightarrow\; \{\textit{(solid) analytic rings}\}, \qquad (A^\triangleright, A^+) \longmapsto (A^\triangleright, A^+)_\square,$$

*is fully faithful.*

**Theorem 8.25** (Rings of integral elements $\leftrightarrow$ solid analytic structures)**.** *Fix a Huber ring $A^\triangleright$. The maps $A \mapsto A^+$ and $A^+ \mapsto (A^\triangleright, A^+)_\square$ set up an adjunction*

$$\{\textit{rings of integral elements } A^+ \subseteq A^\triangleright\} \;\rightleftarrows\; \{\textit{solid analytic ring structures on } A^\triangleright\},$$

*under which $A^+ \mapsto (A^\triangleright, A^+)_\square$ is a fully faithful left adjoint. Thus Huber's rings of integral elements are exactly the analytic ring structures of the special form "impose $1 - f\sigma$ invertible for $f$ in a subring".*

**Definition 8.26** (Solid analytic ring)**.** An analytic ring $A = (A^\triangleright, \mathrm{Mod}(A))$ is *solid* if it admits a (necessarily unique) map $\mathbb{Z}_\square \to A$, where $\mathbb{Z}_\square = (\mathbb{Z}, \mathrm{Solid}_\mathbb{Z})$. Equivalently:

$$A \text{ solid} \quad\Longleftrightarrow\quad \text{every } M \in \mathrm{Mod}(A) \text{ is solid} \quad\Longleftrightarrow\quad 1 - \sigma\colon P \otimes_\mathbb{Z} A \to P \otimes_\mathbb{Z} A \text{ is an isomorphism.}$$

Uniqueness of the map follows because a map of rings $\mathbb{Z} \to A^\triangleright$ is unique and solidity is then a condition, not extra structure.

*Remark* 8.27 (A posteriori motivation)*.* Scholze emphasised how closely the solid theory reproduces Huber's classical one: restricting attention to analytic ring structures cut out by conditions of the form "$1 - f\sigma$ acts invertibly on $P$" for $f$ ranging over a subring, one obtains *precisely* the analytic ring structures induced by Huber's rings of integral elements. That the ad hoc-looking axiom on $A^+$ falls out so exactly is, he suggested, a strong a posteriori justification for the whole definition of analytic ring. Feeding a Huber *pair* rather than a bare ring into the formalism thereby reproduces Huber's theory (cf. Remark 7.24).

# 9 Localization of Solid Analytic Rings

Clausen turns to *geometry*: how the derived category of an analytic ring glues over a space of "points". The lecture has two halves. First, some loose ends from Scholze's Section 8 are tied off – the existence of the reflector into $\mathrm{Mod}(R)$, and the stability of the derived category $D(R)$ under arbitrary products – and the class of *solid* analytic ring structures $(R^\triangleright, R^+)^\square$ cut out by a ring of integral elements is recalled. Second, and the heart of the lecture, is a *descent* theorem: for a discrete Huber pair $(R, R^+)$ the assignment

$$U \;\longmapsto\; D\big((\mathcal{O}(U), \mathcal{O}^+(U))^\square\big)$$

is a sheaf of $\infty$-categories on the *valuative spectrum* $\mathrm{Spv}(R, R^+)$, exactly parallel to the Zariski descent of $D(R)$ over $\operatorname{Spec} R$ – but only at the derived level, since the localization functors are the $\mathbb{Z}[T]$-solidifications of Section 7 and are not $t$-exact. The proof reduces, by a Tate-style cover-generation lemma, to a single elementary vanishing of a solid tensor product. The lecture closes by comparing the whole valuative spectrum with Huber's smaller adic spectrum $\mathrm{Spa}(R, R^+)$ of *continuous* valuations, which is recovered as a retract.

Throughout, "ring" means commutative ring, everything lives in light condensed abelian groups, and we use freely the solid theory $\mathrm{Solid} \subseteq \mathrm{CondAb}^{\mathrm{light}}$ (Sections 5 and 6), the relative solid theory over $\mathbb{Z}[T]$ (Section 7), and the abstract notion of analytic ring (Section 8).

## 9.1 Recap: analytic rings

Recall from Section 8 the notion of an analytic ring: a pair $R = (R^{\triangleright}, \mathrm{Mod}(R))$, denoted by the single letter $R$, where $R^{\triangleright}$ is a condensed ring (Clausen's triangle notation) and

$$\mathrm{Mod}(R) \ \subseteq \ \mathrm{Mod}_{R^{\triangleright}}$$

is a full subcategory of the condensed $R^{\triangleright}$-modules – the *complete* modules – satisfying the closure axioms:

1. $\mathrm{Mod}(R)$ is closed under all limits, colimits and extensions (in particular it is an abelian subcategory, but more, since infinite limits and colimits are allowed);
2. if $M \in \mathrm{Mod}_{R^{\triangleright}}$ and $N \in \mathrm{Mod}(R)$, then $\underline{\mathrm{Ext}}^i_{R^{\triangleright}}(M, N) \in \mathrm{Mod}(R)$ for all $i$ (internal Ext; this is what produces a good tensor product on $\mathrm{Mod}(R)$ and on its derived analogue);
3. the unit $R^{\triangleright}$ itself lies in $\mathrm{Mod}(R)$.

The inclusion admits a left adjoint (Theorem 9.1 below).

The derived category was defined (Definition 8.14) as the full subcategory

$$D(R) \ \subseteq \ D(\mathrm{Mod}_{R^{\triangleright}}) \ (= D(R^{\triangleright}))$$

of those $M$ with $H_i M \in \mathrm{Mod}(R)$ for all $i$, and one wants $D(R)$ to inherit the analogues of (1)–(3).

## 9.2 Two deferred technical points

### Existence of the left adjoint

The first point left open in Section 8 was whether the reflector $\mathrm{Mod}_{R^{\triangleright}} \to \mathrm{Mod}(R)$ exists automatically. It does, by a version of the adjoint functor theorem.

**Theorem 9.1** (Reflection principle; Adámek–Rosický [AR94])**.** *Let $\mathcal{C}$ be a presentable (*locally presentable*) category and $\mathcal{D} \subseteq \mathcal{C}$ a full subcategory closed under all limits. Suppose moreover that there exists a regular cardinal $\kappa$ such that $\mathcal{D}$ is closed under all $\kappa$-filtered colimits. Then $\mathcal{D}$ is presentable, and the inclusion $\mathcal{D} \hookrightarrow \mathcal{C}$ admits a left adjoint.*

Closure under all limits is the condition under which one would naively hope for a left adjoint; the extra clause on $\kappa$-filtered colimits (for $\kappa = \aleph_0$ this is ordinary filtered colimits, and larger $\kappa$ gives a weaker hypothesis) removes the set-theoretic obstructions. The $\infty$-categorical version – needed for $D(R)$ – is due to Ragimov–Schlank [RS26].

*Remark* 9.2 (An alternative for $D(R)$)*.* For the derived category one can avoid invoking the $\infty$-categorical reflection theorem. There is a general principle (attributed in discussion to Scholze) that presentable categories are closed under limits inside the category of (large) categories, provided the connecting functors preserve colimits; since the homology functors $H_i$ preserve colimits, $D(R)$ – cut out by conditions on the $H_i$ – is presentable, and then Lurie's more naive adjoint functor theorem [Lur17] supplies the left adjoint. One can also check presentability at the level of complexes directly. Clausen was content to believe the $\infty$-categorical input is avoidable.

### Stability of $D(R)$ under products

The second point was closure of $D(R)$ under limits, which reduces to closure under products. The subtlety (Section 3) is that in light condensed abelian groups *countable* products are exact but arbitrary products are not, so the product functor has higher right derived functors $R^i\prod$, and one must check these land in $\mathrm{Mod}(R)$.

**Proposition 9.3** (Derived products stay complete)**.** *Let $(M_\alpha)_{\alpha\in A}$ be a family of objects of $\mathrm{Mod}(R)$. Viewing each $M_\alpha$ in $D(R)$ in degree $0$, the derived product $\prod_\alpha M_\alpha$ lies in $D(R)$; equivalently $R^i\prod_{\alpha\in A} M_\alpha \in \mathrm{Mod}(R)$ for all $i$.*

*Proof.* Set $M = \bigoplus_{\alpha\in A} M_\alpha \in \mathrm{Mod}(R)$. Termwise, the system $(M_\alpha)$ is a retract of the constant-in-$M$ system, so $R^i\prod_\alpha M_\alpha$ is a retract of $R^i\prod_{\alpha\in A} M$. But the latter is internal Ext out of a free module,

$$R^i\prod_{\alpha\in A} M \;=\; \underline{\mathrm{Ext}}^i_{R^\triangleright}\Big(\bigoplus_{\alpha\in A} R^\triangleright,\; M\Big),$$

which lies in $\mathrm{Mod}(R)$ by axiom (2). Since $\mathrm{Mod}(R)$ is closed under retracts (being closed under kernels and cokernels), the claim follows. $\square$

The case of unbounded-below complexes was already handled in Section 8: products there are computed through the countable Postnikov limits, reducing to the countable-product case.

*Remark* 9.4 (Light vs. general condensed). Light condensed rings embed fully faithfully into all condensed rings, so the first coordinate transports without loss. The analytic ring *structure* is more subtle: a light analytic ring structure only decides what the free modules should be on *light* profinite sets, so passing to the general condensed setting is not a mere restriction of scalars.

## 9.3 Solid analytic ring structures

We specialise to the solid context. A *solid ring* $R^\triangleright$ is a condensed ring whose underlying condensed abelian group is solid, i.e. a commutative algebra object

$$R^\triangleright \in \mathrm{CAlg}(\mathrm{Solid}_{\mathbb{Z}}),$$

and it carries the subset $R^\circ \subseteq R^\triangleright(*)$ of power-bounded elements (Definition 7.16).

**Definition 9.5** (The structure $(R^\triangleright, R^+)^\square$)**.** For any subset $R^+ \subseteq R^\circ$ with $1 \in R^+$, one obtains an analytic ring structure on $R^\triangleright$, denoted $(R^\triangleright, R^+)^\square$, with the same underlying ring and complete modules

$$\mathrm{Mod}\big((R^\triangleright, R^+)^\square\big) \;=\; \Big\{ M \in \mathrm{Mod}_{R^\triangleright} \;\Big|\; \underline{\mathrm{Hom}}(P, M) \xrightarrow{1-\sigma f} \underline{\mathrm{Hom}}(P, M) \text{ is an iso } \forall f \in R^+ \Big\}, \quad (9.1)$$

where $P = \mathbb{Z}[\mathbb{N}\cup\{\infty\}]/\mathbb{Z}[\infty]$ is the free group on a null sequence and $\sigma$ its shift (Sections 5 and 7). In words, one requires that for each $f \in R^+$ the module be $\mathbb{Z}[T]$-solid along $T \mapsto f$.

*Remark* 9.6 (Reading the two conditions). The two boundary cases $f = 1$ and $R^+ \subseteq R^\circ$ have clean meanings.

- Taking $f = 1$ recovers the definition of $\mathrm{Solid}_{\mathbb{Z}}$ (Definition 5.5); so $1 \in R^+$ forces every $M \in \mathrm{Mod}(R)$ to be solid as a condensed abelian group.
- The condition $R^+ \subseteq R^\circ$ (power-bounded) is precisely what makes the ring $R^\triangleright$ itself complete, i.e. $R^\triangleright \in \mathrm{Mod}(R)$ (axiom (3)): the second coordinate specifies a completeness notion adapted to the situation, and one wants the ring to be already complete.

*Remark* 9.7 (No loss in taking $R^+$ integrally closed). There is no loss of generality in assuming $R^+ \subseteq R^\triangleright(*)$ is an integrally closed subring containing all topologically nilpotent elements $R^{\circ\circ}$. Indeed, given any solid analytic ring structure on $R^\triangleright$ (one for which every module is solid), the set of $f$ for which $1 - \sigma f$ acts invertibly on $\underline{\mathrm{Hom}}(P, M)$ for all such $M$ is already an integrally closed subring containing $R^{\circ\circ}$ (Lemma 7.19 and Proposition 8.21). So one may always replace $R^+$ by the integral closure of the subring it generates without changing the theory.

**Example 9.8** (Huber pairs)**.** If $R^\triangleright$ is (the condensed ring associated to) a complete Huber ring, then the integrally closed subrings $R^+$ with $R^{\circ\circ} \subseteq R^+ \subseteq R^\circ$ are the same as the *open* integrally closed subrings $R^+ \subseteq R^\circ$ – exactly the rings of integral elements of Huber's theory. So the general classification of admissible $R^+$ specialises, for a Huber ring, to Huber's own. Moreover $R^+$ is *recovered* from the analytic ring $(R^\triangleright, R^+)^\square$: the assignment

$$(R^\triangleright, R^+) \longmapsto (R^\triangleright, R^+)^\square$$

is fully faithful, so a Huber pair and its solid analytic ring carry the same information (Proposition 8.24 and Theorem 8.25; the converse – that no *other* $f$ can satisfy the defining condition – was proved for Huber rings in Section 8). "Huber ring" henceforth means *complete* Huber ring.

*Remark* 9.9 (Derived-complete rings). Asked about "derived complete" rings, Clausen confirmed these too give condensed/analytic objects and can be regarded as solid; there is "a fun story" there, deferred.

## 9.4 Localization: reduction to the discrete case

*Remark* 9.10 (All the data is already discrete). Here is a remark that is shocking at first glance but trivial. Fix a solid ring $R^\triangleright$ and $R^+ \subseteq R^\circ$ as above (say integrally closed, containing $R^{\circ\circ}$). The condition (9.1) defining the analytic ring structure is imposed only for $f \in R^+$, and $R^+$ is a subset of the *underlying discrete ring* $R^\triangleright(*)$. So all the data used to define the structure already appears at the discrete level: the same $R^+$ defines a structure on the *discrete* ring $R^\triangleright(*)$, where every element is power-bounded, giving the discrete Huber pair $(R^\triangleright(*), R^+)$.

Consequently, to understand $\mathrm{Mod}\big((R^\triangleright, R^+)^\square\big)$ it suffices to understand the theory over the discrete pair and then remember that $R^\triangleright$ is a commutative algebra object there:

$$\mathrm{Mod}\big((R^\triangleright, R^+)^\square\big) \;=\; \mathrm{Mod}_{R^\triangleright}\Big(\mathrm{Mod}\big((R^\triangleright(*), R^+)^\square\big)\Big),$$

i.e. the general theory is base-changed from the discrete case in a completely naive way. (Because we are doing *condensed* modules, a discrete ring already has a huge supply of non-discrete modules; in particular $R^\triangleright$ itself is an honest condensed ring over the discrete condensed ring $R^\triangleright(*)$.) Thus, to understand how these categories glue it is enough to treat the discrete case, and then put the $R^\triangleright$-module structure on top.

We describe one framework for gluing – not the most general, but quite general. It is organised by two analogies, one with the Zariski spectrum (classical) and one with the valuative spectrum (the solid analogue).

### The classical picture: $D(R)$ over $\operatorname{Spec} R$

*Remark* 9.11 (Zariski localization of $D(R)$). Let $R$ be an ordinary commutative ring and $D(R)$ its usual (unbounded) derived category of discrete $R$-modules. Then $D(R)$ *localizes over* $\operatorname{Spec} R = \{\text{prime ideals } \mathfrak{p} \subseteq R\}$. This space has a basis of quasicompact opens, closed under finite intersection, the *distinguished opens*

$$U(f) \;=\; \{\mathfrak{p} : \; f \notin \mathfrak{p}\} \qquad (f \in R),$$

carrying a structure sheaf with $\mathcal{O}(U(f)) = R[\frac{1}{f}]$. Neither $U(f)$ nor $R[\frac{1}{f}]$ determines $f$, but they determine each other: $U(f) \cong \operatorname{Spec} R[\frac{1}{f}]$, matching up distinguished opens. Assigning to a distinguished open $U$ the derived category $D(\mathcal{O}(U))$, with base-change functors for inclusions $V \subseteq U$, gives a presheaf of $\infty$-categories.

**Theorem 9.12** (Lurie [Lur17])**.** *The presheaf $U \mapsto D(\mathcal{O}(U))$ on distinguished opens of* $\operatorname{Spec} R$ *is a sheaf of $\infty$-categories. It is even a sheaf (indeed a hypersheaf) for the hypercover topology, and there is a sheaf on the level of abelian categories $U \mapsto \mathrm{Mod}_{\mathcal{O}(U)}$ as well.*

*Remark* 9.13 (The abelian sheaf fails in the solid setting). In the Zariski case one has a sheaf already on abelian categories because the localization maps $\mathcal{O}(U) \to \mathcal{O}(V)$ are *flat*, so derived statements descend to abelian ones. In the solid analogue below, the localization maps are *not* flat – one of them is the $\mathbb{Z}[T]$-solidification of Section 7, which is off from $t$-exactness by one (Remark 7.12) – and this obstructs the sheaf property at the abelian level. *We obtain a sheaf only on derived categories.*

### The solid picture: the valuative spectrum

Now let $(R, R^+)$ be a discrete Huber pair: $R$ an ordinary commutative ring and $R^+ \subseteq R$ an integrally closed subring. To it we attached the analytic ring $(R, R^+)^{\square}$ and the derived category $D\big((R, R^+)^{\square}\big)$, abbreviated $D(R, R^+)$. The claim is that $D(R, R^+)$ localizes over the *valuative spectrum* $\operatorname{Spv}(R, R^+)$.

**Definition 9.14** (Valuative spectrum)**.** Whereas a prime ideal records only whether a function vanishes, a valuation records the finer comparison of sizes of functions. Define

$$\operatorname{Spv}(R, R^+) \;=\; \Big\{ v \colon R \longrightarrow \Gamma \cup \{0\} \;\Big|\; \text{(axioms below)} \Big\} \big/ \sim,$$

where $\Gamma$ is an ordered abelian group written multiplicatively, and $v$ satisfies:

$$\begin{aligned} v(fg) &= v(f)v(g), & v(f+g) &\le \max\big(v(f), v(g)\big), \\ v(0) &= 0, \quad v(1) = 1, & v(f) &\le 1 \quad \forall f \in R^+. \end{aligned} \tag{9.2}$$

Two valuations are equivalent, $v \sim w$, iff they induce the same order relation on functions:

$$v \sim w \iff \big(\, \forall f, g \in R : \; v(f) \le v(g) \iff w(f) \le w(g) \,\big).$$

The genuinely important datum of a valuation is thus the binary relation "$f$ is smaller than $g$".

*Remark* 9.15 (Second description; higher rank). Equivalently, a point of $\operatorname{Spv}(R, R^+)$ is a pair of a prime ideal $\mathfrak{p} \subseteq R$ and a valuation subring (valuation domain) of the residue field $\kappa(\mathfrak{p})$ containing the image of $R^+$: the prime records vanishing, the valuation ring records which fractions of non-vanishing functions are $\le 1$. Since the continuity condition is vacuous for a discrete ring, this is the same as Huber's Spa in the discrete case. For $\mathbb{Q}$ (or $\mathbb{Z}$) one has the trivial valuations and the $p$-adic valuations, and the classification is manageable; but adding one variable makes valuations proliferate – in particular *higher-rank* valuations appear (many curves through a point of a surface), a genuine additional complication.

**Definition 9.16** (Rational opens and their structure sheaves)**.** The space $\mathrm{Spv}(R, R^+)$ has a basis of quasicompact opens, closed under finite intersection, the *rational opens.* For $f_1, \ldots, f_n, g \in R$,

$$U\Big(\tfrac{f_1,\ldots,f_n}{g}\Big) \;=\; \big\{\, v : \; v(g) \neq 0,\; v(f_i) \leq v(g)\ \forall i \,\big\} \;\subseteq\; \{v : \; v(g) \neq 0\},$$

i.e. one sits inside the Zariski open where $g$ is invertible and shrinks by the inequalities $|f_i| \leq |g|$. It carries *two* structure sheaves: the algebraic (Zariski) localization

$$\mathcal{O}\Big(U\big(\tfrac{f_1,\ldots,f_n}{g}\big)\Big) \;=\; R[\tfrac{1}{g}],$$

together with a ring of integral elements, the integral closure

$$\mathcal{O}^+\Big(U\big(\tfrac{f_1,\ldots,f_n}{g}\big)\Big) \;=\; \overline{R^+\big[\tfrac{f_1}{g}, \ldots, \tfrac{f_n}{g}\big]} \;\subseteq\; R[\tfrac{1}{g}],$$

obtained by adjoining the $f_i/g$ (forced $\leq 1$ on the open) to $R^+$ and taking the integral closure. Then

$$U\Big(\tfrac{f_1,\ldots,f_n}{g}\Big) \;\cong\; \mathrm{Spv}\big(\mathcal{O}(U), \mathcal{O}^+(U)\big), \tag{9.3}$$

matching up rational opens, exactly as $U(f) \cong \mathrm{Spec}\, R[\frac{1}{f}]$ in the Zariski case.

*Remark* 9.17 (Why $R^+$ is needed)*.* Equation (9.3) is where the necessity of carrying the datum $R^+$ becomes visible. Had one dropped the integral condition to get a bigger space, the rational opens would no longer be of the form Spv(ring pair): the abstract ring $R[\frac{1}{g}]$ does not remember that the $f_i/g$ were forced $\leq 1$. To keep rational opens of the same shape as the global object, one must include $R^+$ from the start.

We assign to a rational open $U$ the derived category $D\big((\mathcal{O}(U), \mathcal{O}^+(U))^{\square}\big)$, and to an inclusion $U \supseteq V$ the pullback (base-change) functor

$$D\big((\mathcal{O}(U), \mathcal{O}^+(U))^{\square}\big) \;\longrightarrow\; D\big((\mathcal{O}(V), \mathcal{O}^+(V))^{\square}\big), \tag{9.4}$$

which comes from a map of analytic rings $(\mathcal{O}(U), \mathcal{O}^+(U))^{\square} \to (\mathcal{O}(V), \mathcal{O}^+(V))^{\square}$ (Definition 8.18): restriction of scalars preserves completeness, and its left adjoint – tensor up the underlying module, then recomplete – is the base change.

**Theorem 9.18** (Descent for the solid affinoid theory)**.** *For a discrete Huber pair* $(R, R^+)$*, the presheaf on rational opens of* $\mathrm{Spv}(R, R^+)$*,*

$$U \;\longmapsto\; D\big((\mathcal{O}(U), \mathcal{O}^+(U))^{\square}\big),$$

*is a sheaf of* $\infty$*-categories.* Warning: *unlike the classical case, it is* not *a sheaf at the level of abelian categories, because the pullback functors* (9.4) *are not t-exact in general (Remark 9.13).*

Since the rational opens form a basis closed under finite intersection, a sheaf on them extends uniquely to a sheaf on all opens (an arbitrary open is covered by rationals and one takes the limit); so Theorem 9.18 determines a category for every open, rational or not.

*Remark* 9.19 (The discrepancy is bounded). The failure of $t$-exactness is controlled: the $\mathbb{Z}[T]$-solidification is given by $R\underline{\mathrm{Hom}}$ against an object with a two-term compact resolution, so it is off from $t$-exact by at most one – the homology of the solidification of an object concentrated in degree 0 vanishes above degree $n$ for some finite $n$ (indeed $n = 1$). The sheaf need not be hypercomplete in general, though hypercompleteness is automatic when the base space has finite Krull dimension.

## 9.5 Proof of the classical theorem

To motivate the proof of Theorem 9.18 we give a particular proof of Zariski descent (Theorem 9.12) that transports with little change. Work on the site of distinguished opens $U \subseteq \operatorname{Spec} R$ with the open-cover topology.

**Lemma 9.20** (Cover-generation)**.** *The Grothendieck topology on distinguished opens is generated by:*

- *the empty cover of the empty set; and*
- *for each distinguished open $U \subseteq \operatorname{Spec} R$ and each $f \in \mathcal{O}(U)$, the two-element cover*

$$\{\, U(f),\ U(1-f) \,\} \longrightarrow U. \tag{9.5}$$

*Proof.* A general cover of $U$ is given by $f_1, \ldots, f_n \in \mathcal{O}(U)$ generating the unit ideal – $\exists\, x_1, \ldots, x_n$ with $\sum_i x_i f_i = 1$ – and equals $\{U(f_i)\}_i$. (The empty cover of the empty set must be listed separately: it cannot be generated from nonempty data, and checking the sheaf condition there just says the value on $\emptyset$ is terminal.) Refining $f_i \rightsquigarrow f_i x_i$ we may assume $\sum_i f_i = 1$, and then induct on $n$. For instance with $f_1 + f_2 + f_3 = 1$, the basic cover $\{U(f_1 + f_2), U(f_3)\}$ (of the shape (9.5), since $f_3 = 1 - (f_1 + f_2)$) reduces the sheaf condition to its restrictions: over $U(f_1 + f_2)$ the element $f_1 + f_2$ becomes a unit and one divides through to reach the two-variable case; over $U(f_3)$ the cover splits (one of the covering opens is all of it), so the sheaf condition is automatic. This collection is closed under base change – an important technical point – and that makes the induction go through. This reduction is originally due to Quillen (in his proof of the Serre conjecture on projective modules). □

So it suffices to check the sheaf condition for the cover $\{U(f), U(1-f)\}$ of $U = \operatorname{Spec} R$ (no loss in taking $U = \operatorname{Spec} R$). Concretely this asks that the functor

$$D(R) \xrightarrow{\simeq} D\big(R[\tfrac{1}{f}]\big) \times_{D(R[\frac{1}{f(1-f)}])} D\big(R[\tfrac{1}{1-f}]\big) \tag{9.6}$$

be an equivalence of $\infty$-categories, where the right-hand side is the pullback of the cospan

$$\begin{tikzcd} & D\big(R[\tfrac{1}{f}]\big) \arrow[d] \\ D\big(R[\tfrac{1}{1-f}]\big) \arrow[r] & D\big(R[\tfrac{1}{f(1-f)}]\big). \end{tikzcd}$$

*Remark* 9.21 (What the pullback category is, concretely). An object of the pullback is a triple $(M, N, \alpha)$ of $M \in D(R[\frac{1}{f}])$, $N \in D(R[\frac{1}{1-f}])$, and an isomorphism $\alpha\colon M[\frac{1}{1-f}] \xrightarrow{\sim} N[\frac{1}{f}]$ of their common base changes – but an isomorphism in the $\infty$-*categorical* sense, e.g. (for bounded-above complexes of projectives) an explicit chain-homotopy equivalence, not merely an iso in the ordinary derived category. Essential surjectivity of (9.6) is the statement that such glueing data descends uniquely to a global object; full faithfulness is that global $R\operatorname{Hom}$'s can be computed locally, as the homotopy pullback of the three local $R\operatorname{Hom}$-complexes (a shifted mapping cone). So (9.6) both glues locally defined objects and computes global Ext by localizing.

*Proof of* (9.6). Each base change $D(R) \to D(R[\frac{1}{f}])$, etc., has a right adjoint (the forgetful functor), so the comparison functor (9.6) has a right adjoint too – an automatic candidate inverse. Explicitly it sends $(M,N,\alpha)$ to the fibre product $M \times_{M[\frac{1}{f}]=N[\frac{1}{1-f}]} N$ (apply the right adjoints and take the limit). One must check the unit and counit are isomorphisms; both are maps in one of the two categories. For the unit one needs, for $M \in D(R)$, that the square

$$\begin{tikzcd} M \arrow[r] \arrow[d] & M[\frac{1}{f}] \arrow[d] \\ M[\frac{1}{1-f}] \arrow[r] & M[\frac{1}{f(1-f)}] \end{tikzcd}$$

is a (homotopy) pullback. The difference between $M$ and the pullback is a mapping cone $N$ with $N[\frac{1}{f}] = N[\frac{1}{1-f}] = 0$, and the point is that

$$N \in D(R), \quad N[\tfrac{1}{f}] = N[\tfrac{1}{1-f}] = 0 \implies N = 0. \tag{9.7}$$

This reduces to the abelian level (an object vanishes iff its homology does) and, the localizations being flat, is elementary there. The counit reduces to the same statement (9.7). □

*Remark* 9.22 (The three formal ingredients). The proof uses only three facts about the localizations $M \mapsto M[\frac{1}{f}]$, $M \mapsto M[\frac{1}{1-f}]$:

1. each base change is a *localization* (its right adjoint is fully faithful);
2. the localizations *commute*: something on which $1-f$ acts invertibly still has $1-f$ invertible after inverting $f$, and the composite localization $M \mapsto M[\frac{1}{f(1-f)}]$ is the common further localization;
3. they are *jointly conservative*: if $M \mapsto 0$ under each element of the cover, then $M = 0$.

Given a square of localizations with these three properties, one automatically gets a pullback. This is what will be verified in the solid setting.

### 9.6 Proof of the solid analogue

Work on the site of rational opens $U \subseteq \mathrm{Spv}(R, R^+)$ with the open-cover topology. The cover-generation lemma now has two families, reflecting the extra inequalities.

**Lemma 9.23** (Cover-generation, cf. Tate [Tat71])**.** *The topology on rational opens of* $\mathrm{Spv}(R, R^+)$ *is generated by:*

- *the empty cover of the empty set; and*
- *for each rational open $U$ and each $f \in \mathcal{O}(U)$, the two covers*

$$\{\, U(\tfrac{f}{1}),\ U(\tfrac{1}{f}) \,\} \quad \textit{and} \quad \{\, U(\tfrac{1}{f}),\ U(\tfrac{1}{1-f}) \,\}. \tag{9.8}$$

*Here $U(\frac{f}{1}) = \{v(f) \leq 1\}$ is where $f$ is integral (the closed unit disk); $U(\frac{1}{f}) = \{v(f) \neq 0,\ v(f) \geq 1\}$ is where $f$ is well away from $0$ (the complement of the open unit disk). Both are refinements of the Zariski cover $\{U(f), U(1-f)\}$ that still cover.*

For the general reduction – Huber shows every cover is refined by one of the form: take $f_1, \ldots, f_n$ generating the unit ideal, and take the rational opens $U\big(\frac{f_1,\ldots,\widehat{f_i},\ldots,f_n}{f_i}\big)$ (a refinement of the Zariski cover $\{U(f_i)\}$ that still covers the valuative spectrum). Then, exactly as in the classical proof, one plays the same reduction until the theorem reduces to a single case.

**Proposition 9.24** (The two pullback squares)**.** *For $(R, R^+)$ a discrete Huber pair and $f \in R$, the theorem reduces to the statements that*

$$D(R, R^+) \ \xrightarrow{\simeq}\ D\big(R,\ \widetilde{R^+[f]}\big) \ \times_{D(R[\frac{1}{f}],\, \widetilde{R^+[\frac{1}{f}, f]})}\ D\big(R[\tfrac{1}{f}],\ \widetilde{R^+[\tfrac{1}{f}]}\big)$$

*and the analogous square for the second cover in* (9.8) *are pullbacks, where* $\widetilde{(-)}$ *denotes integral closure.*

The identification of the base-change functors is the crux, and here the theory of Section 7 enters.

**Lemma 9.25** (Base changes as $R\underline{\mathrm{Hom}}$'s)**.** *Let $f \in R$.*

1. *The closed-disk base change*
$$D(R, R^+) \longrightarrow D\big(R, \widetilde{R^+[f]}\big)$$
*is simply the $\mathbb{Z}[T]$-solidification for $\mathbb{Z}[T] \xrightarrow{T \mapsto f} R$, given (Proposition 7.9) by*
$$R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}\Big(\big(\mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T]\big)[-1],\ -\Big).$$

2. *The complementary base change*
$$D(R, R^+) \longrightarrow D\big(R[\tfrac{1}{f}], \widetilde{R^+[\tfrac{1}{f}]}\big)$$
*– first invert $f$, then solidify with respect to $1/f$ – is* also *an $R\underline{\mathrm{Hom}}$: for $\mathbb{Z}[T] \xrightarrow{T \mapsto f} R$,*
$$R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}\Big(\big(\mathbb{Z}[\![T]\!]/\mathbb{Z}[T]\big)[-1],\ -\Big), \tag{9.9}$$
*the localization killing the idempotent object $\mathbb{Z}[\![T]\!] \in D\big(\mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{Solid}_{\mathbb{Z}})\big)$.*

*Why* (9.9) *inverts $f$.* By Lemma 7.2 the object $\mathbb{Z}[\![T]\!]$ is idempotent over $\mathbb{Z}[T]$, so $R\underline{\mathrm{Hom}}$ against $(\mathbb{Z}[\![T]\!]/\mathbb{Z}[T])[-1]$ is the Bousfield localization killing $\mathbb{Z}[\![T]\!]$. Killing $\mathbb{Z}[\![T]\!]$ kills every module over it, and everything $T$-torsion is a $\mathbb{Z}[\![T]\!]$-module: a solid abelian group that is a filtered colimit of things killed by powers of $T$ is a $\mathbb{Z}[\![T]\!]$-module (reduce, using closure under limits and colimits, to something uniformly killed by a power of $T$, hence a module over the truncated power-series ring). So this localization automatically inverts $f$. After the change of variables inverting $T$, what remains is exactly the $1/f$-solidification, i.e. the composite "invert $f$, then solidify". ☐

*Proof of Theorem 9.18.* By Remark 9.22 it remains to verify the three ingredients for the two covers. Ingredients (1) and (2) are immediate: each base change is a localization by construction, and any two localizations given by $R\underline{\mathrm{Hom}}$ out of fixed objects commute (as $R\underline{\mathrm{Hom}}(A, R\underline{\mathrm{Hom}}(B, -)) = R\underline{\mathrm{Hom}}(A \otimes B, -)$ and the tensor product is symmetric). Only the joint conservativity (3) must be checked, and it translates into a single tensor-product vanishing per cover.

For the first cover $\{U(\frac{f}{1}), U(\frac{1}{f})\}$ – localize to the closed unit disk, resp. away from the open unit disk at 0 – ingredient (3) is:

$$\mathbb{Z}[\![T]\!] \otimes^{\mathbb{L}\square}_{\mathbb{Z}[T]} \mathbb{Z}(\!(T^{-1})\!) \;=\; \mathbb{Z}[\![T, U]\!]/(1 - UT) \;=\; 0 \qquad (U = T^{-1}). \tag{9.10}$$

For the second cover $\{U(\frac{1}{f}), U(\frac{1}{1-f})\}$ – away from the open unit disks at 0 and at 1 – it is:

$$\mathbb{Z}[\![T]\!] \otimes^{\mathbb{L}\square}_{\mathbb{Z}[T]} \mathbb{Z}[\![1 - T]\!] \;=\; \mathbb{Z}[\![T, U]\!]/\big(1 - (T + U)\big) \;=\; 0 \qquad (U = 1 - T). \tag{9.11}$$

In each case both variables are topologically nilpotent, so the displayed denominator is 1 minus a topologically nilpotent element, hence a unit (invert by geometric series), and the quotient is 0. □

*Remark* 9.26 (Geometric meaning of the vanishings). The two identities say that the two idempotent algebras being killed have disjoint support. Equation (9.10) says the open unit disk at 0 and the open unit disk at $\infty$ do not meet: their union of complements (closed disk at 0 together with closed disk at $\infty$) is the whole space. Equation (9.11) says the open unit disk at 0 and the open unit disk at 1 do not meet – strange until one recalls this is non-archimedean geometry, where it is exactly right.

## 9.7 Consequences and comparison with Huber

**Corollary 9.27** (Localization for a general solid ring)**.** *Let $R^{\triangleright}$ be any solid ring and $R^+ \subseteq R^\circ \subseteq R^{\triangleright}(*)$ with $1 \in R^+$. Then $D(R^{\triangleright}, R^+)$ localizes on the valuative spectrum* $\mathrm{Spv}(R^{\triangleright}(*), R^+)$ *of the underlying discrete ring: to a rational open $U$ one assigns*

$$U \;\longmapsto\; \mathrm{Mod}_{R^{\triangleright}}\Big(D\big((\mathcal{O}(U), \mathcal{O}^+(U))\big)\Big), \tag{9.12}$$

*the $R^{\triangleright}$-modules in the (discrete) affinoid category of the rational open. The unit object over $U(\frac{f_1,\ldots,f_n}{g})$ is $R^{\triangleright}[\frac{1}{g}]$ derived-solidified with respect to the $f_i/g$.*

This is the base-change of the discrete descent (Theorem 9.18) along $R^{\triangleright}$, by Remark 9.10. Note that for a nondiscrete $R^{\triangleright}$ this unit object generically carries genuine *derived* phenomena (higher homotopy): the structure sheaf is a priori a derived object. *Caveat:* the sheaf (9.12) uses the discrete rings $\mathcal{O}(U)$, $\mathcal{O}^+(U)$, so it does not by itself see the topology of a nondiscrete $R^{\triangleright}$.

*Remark* 9.28 (It lives over a closed subset). In fact $D(R^{\triangleright}, R^+)$ lies over a much smaller *closed* subset of the valuative spectrum,

$$\mathrm{Spv}(R^{\triangleright}(*), R^+, R^{\circ\circ}) \;=\; \big\{\, v \in \mathrm{Spv}(R^{\triangleright}(*), R^+) \;:\; v(f) < 1 \;\;\forall f \in R^{\circ\circ} \,\big\}, \tag{9.13}$$

where in addition to $v(f) \leq 1$ on $R^+$ one imposes $v(f) < 1$ strictly for every topologically nilpotent $f$. For a single $f$ this is a closed condition, and (9.13) is the (closed) intersection over all such $f$. The claim is that restricting the sheaf of categories to the open complement gives 0; so it genuinely lives over (9.13).

*Remark* 9.29 (Huber's adic spectrum as a retract). Huber [Hub94] instead works with the *adic spectrum*

$$\mathrm{Spa}(R, R^+) \;=\; \{\text{continuous valuations on } R,\ \leq 1 \text{ on } R^+\},$$

which is a subset of (9.13) cut out by a *stronger* condition: for $f \in R^{\circ\circ}$ and all $\gamma \in \Gamma$ there is $N \in \mathbb{N}$ with $v(f^N) < \gamma$. This subset is generally strictly smaller than (9.13), the distinction occurring only for higher-rank valuations – but in higher dimensions higher-rank valuations are everywhere, so the difference is large. Now

- $D(R, R^+)$ does *not* live over $\mathrm{Spa}(R, R^+) \subseteq \mathrm{Spv}(R, R^+)$ (unlike (9.13), restricting to the complement does not give 0);
- but there is a *retraction*

$$\mathrm{Spv}(R, R^+, R^{\circ\circ}) \xrightarrow{r} \mathrm{Spa}(R, R^+),$$

realizing $\mathrm{Spa}(R, R^+)$ as a *quotient* of the larger spectral space (it is a priori a subspace), and one *does* get a sheaf of categories on $\mathrm{Spa}(R, R^+)$ by pushing forward along $r$. This is the correct way to place a sheaf of categories on Huber's space.

The retraction $r$ is the well-behaved (quasicompact, spectral) map, while the inclusion $\mathrm{Spa} \hookrightarrow \mathrm{Spv}(\ldots, R^{\circ\circ})$ is not spectral; and $r$ is such that the direct image along it agrees with restriction to the subspace Spa (the pertinent direct images vanish, by Remark 9.28).

*Remark* 9.30 (More localizations than Huber; the structure sheaf). The valuative-spectrum picture allows strictly more localizations than Huber's. The rational opens on Spa correspond to data $f_1, \ldots, f_n, g$ generating an *open* ideal; pulling such a rational open back along $r$ recovers the corresponding rational open upstairs. But a general rational open of $\mathrm{Spv}(\ldots, R^{\circ\circ})$ – one *not* satisfying the openness condition – restricts to something new, not necessarily quasicompact and not a rational open, which must be written as a union of rational opens. This is like writing the open unit disk as an increasing union of closed disks; and objects such as the functions on the closed unit disk (Section 7) arise from the structure sheaf here but not in Huber's setting.

The structure sheaf itself is obtained by taking the globally defined $R$ and applying the localization functor to land in the category attached to a rational open; as a symmetric monoidal category the value is generated by its *unit*, which one can analyse. In good cases it corresponds to the Huber pair of the rational domain, but sometimes only up to a derived enhancement (one must remember not just the ordinary condensed ring in a full subcategory, but a derived enhancement of it; the abelian category of modules over the ordinary thing then suffices). Besides the derived subtlety there is a quasiseparatedness one: even in degree 0 the value can differ from Huber's if the relevant quotient is not by a closed ideal – though in all practical cases it does not, and even inverting $g$ can introduce non-quasiseparated behaviour in general. Iterating the construction shows the global analytic functions of such a rational domain genuinely carry more data than Huber's (they need not invert a fixed power of a uniformizer, only converge on each disk), so this theory sees rings of functions unavailable to the classical formalism.

# 10 Analytic (Tate) Adic Spaces

This lecture turns to the comparison with Huber's classical theory of *adic spaces*, in the special "analytic" – here called *Tate* – case. There are two halves. First, we recall Tate Huber rings (Banach algebras with a topologically nilpotent unit), the notion of a *sheafy* Huber pair, and the theorem of Kedlaya–Liu that finite projective modules glue to vector bundles when $(A, A^+)$ is sheafy. Second, and this is the point of the lecture, we re-prove and vastly generalise that theorem inside the solid formalism of Sections 7 and 9: for *any* Huber pair with $A$ Tate – with no sheafiness hypothesis – pseudocoherent complexes, perfect complexes and vector bundles all glue over the adic spectrum, because they are detected inside the (always well-behaved) sheaf of solid derived categories. The argument reduces the classical analysis to two abstract finiteness conditions, *pseudocoherence* and *nuclearity*, whose intersection is pinned down by a Fredholm-theoretic argument that is the one genuinely analytic step.

Throughout, "ring" means commutative ring, everything lives in light condensed abelian groups, and we use the solid theory $\mathrm{Solid} \subseteq \mathrm{CondAb}^{\mathrm{light}}$ (Sections 5 and 6), the relative theory over $\mathbb{Z}[T]$ (Section 7), the abstract notion of analytic ring (Section 8) and the localization of solid analytic rings over the valuative/adic spectrum (Section 9).

### 10.1 Tate adic spaces

*Remark* 10.1 (A conflict of terminology). Every adic space gives rise to an analytic space in our sense (Section 9). But within the theory of adic spaces there is already an established qualifier "analytic" for a certain subclass. To avoid overloading the word, in this lecture Scholze calls the members of that subclass *Tate* adic spaces.

**Definition 10.2** (Tate Huber ring; Tate adic space)**.** A Huber ring $A$ (Definition 8.2) is *Tate* if it has a topologically nilpotent unit $\pi \in A$, a so-called *pseudo-uniformizer*. An adic space $X$ is *Tate* (the property classically called *analytic*) if it is covered by affinoids $\mathrm{Spa}(A, A^+)$ with $A$ Tate. Equivalently,

$$X \text{ is Tate} \quad \Longleftrightarrow \quad \text{the completed residue fields at all points of } X \text{ are not discrete}$$

(so that they are honest nonarchimedean fields, rather than discrete rings).

**Example 10.3** (Tate rings)**.** $\mathbb{Q}_p$ is Tate, with pseudo-uniformizer $\pi = p$; more generally any nonarchimedean local field, and any Huber ring over such a field, is Tate – once a topological unit is present downstairs, it remains a topological unit after any base change up.

*Remark* 10.4 (Intuition: schemes, formal schemes, generic fibres). The heuristic is that adic spaces interpolate between schemes and formal schemes. A scheme sits inside a formal scheme; the associated adic space contains, at one extreme, the "scheme-like" locus, and at the other extreme the Tate locus, which is morally the generic fibre of a formal scheme with all scheme-like points removed. The basic picture is the formal scheme $\mathrm{Spf}\,\mathbb{Z}_p$, i.e. $\mathrm{Spa}(\mathbb{Z}_p, \mathbb{Z}_p)$, with its closed (scheme-like) point $\mathrm{Spec}\,\mathbb{F}_p$ and its open Tate generic fibre $\mathrm{Spa}(\mathbb{Q}_p, \mathbb{Z}_p)$:

$$\mathrm{Spec}\,\mathbb{F}_p \hookrightarrow \mathrm{Spa}(\mathbb{Z}_p, \mathbb{Z}_p) \hookleftarrow \mathrm{Spa}(\mathbb{Q}_p, \mathbb{Z}_p)$$

The closed point is scheme-like; the open complement is Tate.

### 10.2 Structure of Tate Huber rings

**Proposition 10.5** (Tate Huber rings are Banach algebras)**.** *Let $A$ be a Tate Huber ring, $\pi \in A$ a pseudo-uniformizer, and $A_0 \subseteq A$ a ring of definition. One may always arrange $\pi \in A_0$ (and that $A_0$ contains any prescribed power-bounded element); then $A_0$ carries the $\pi$-adic topology – a priori it could carry the $I$-adic topology for some finitely generated ideal $I$, but for a Tate ring the topology is always $\pi$-adic. This turns $A$ into a Banach algebra with unit ball $A_0$, via*

$$|f| \;=\; 2^{-n}, \qquad n \;=\; \sup\{\, m \in \mathbb{Z} \;\mid\; \pi^{-m} f \in A_0 \,\}, \tag{10.1}$$

*so that $|\pi| = \frac{1}{2}$ and $|\pi^{-1}| = 2$.*

*Remark* 10.6 (Equivalence with ultrametric Banach rings). The base 2 and the choice of $\pi$ in (10.1) are arbitrary; only the induced topology matters. Passing to that topology gives an equivalence of categories

$$\{\text{Tate Huber rings}\} \;\simeq\; \left\{ \begin{array}{c} \text{complete ultrametric Banach rings } B \text{ admitting } \pi \in B \\ \text{with } 0 < |\pi| < 1 \text{ and } |\pi|\,|\pi^{-1}| = 1 \end{array} \right\},$$

with continuous maps as morphisms. On the right one should not fix a norm but only require that *some* equivalent norm has the displayed multiplicativity on $\pi$. (The zero ring is a degenerate exception to the naive statements about pseudo-uniformizers, and is best set aside.)

### 10.3 Sheafiness and the theorem of Kedlaya–Liu

Recall (Section 9) that Huber equips $\mathrm{Spa}(A, A^+)$ with a structure presheaf $\mathcal{O}^{\mathrm{Huber}}$; its defect is that it is not always a sheaf.

**Definition 10.7** (Sheafy)**.** A Huber pair $(A, A^+)$ is *sheafy* if the Huber structure presheaf $\mathcal{O}^{\mathrm{Huber}}_{\mathrm{Spa}(A,A^+)}$ is a sheaf.

**Theorem 10.8** (Kedlaya–Liu [KL15])**.** *Let $A$ be Tate and $(A, A^+)$ sheafy. Then base change to the structure sheaf is an equivalence*

$$\mathrm{FProj}(A) \xrightarrow{\ \sim\ } \mathrm{Vect}\big(\mathrm{Spa}(A, A^+),\, \mathcal{O}^{\mathrm{Huber}}_{\mathrm{Spa}(A,A^+)}\big), \qquad M \longmapsto M \otimes_A \mathcal{O}^{\mathrm{Huber}}_{\mathrm{Spa}(A,A^+)}, \tag{10.2}$$

*between finite projective $A$-modules and vector bundles (locally free sheaves of finite rank) on the adic spectrum. In other words, finite projective modules glue. There is a version for (pseudo)coherent modules, whose precise statement is more subtle.*

*Remark* 10.9 (History; the plan for today). Scholze found this theorem striking when it appeared: one expected that in situations where sheafiness *could* be proved good things would follow, but not that sheafiness alone – the property most prone to fail – could be the sole hypothesis of a nontrivial theorem. The original Kedlaya–Liu proof is a delicate change-of-basis argument that "does some actual analysis". Kremnizer had suggested to Scholze already at the time that, working in derived categories with the right notion of modules, one can simply glue; the task today is to make that concrete and recover the vector-bundle statement. The reference for the solid treatment is Andreychev [And21]; the same argument proves the analogue of the corresponding statement in the complex-analytic setting of [CS20]. The lecture has two parts:

1. the relation between Huber's $\mathcal{O}^{\mathrm{Huber}}$ and the solid structure sheaf $\mathcal{O}$ of Section 9, culminating in the fact that *when* $(A, A^+)$ is sheafy the two agree;
2. the gluing of pseudocoherent and perfect complexes, hence of vector bundles, for an *arbitrary* Tate Huber pair.

### 10.4 Huber's presheaf versus the solid structure sheaf

Fix a Huber pair $(A, A^+)$ with $A$ Tate. Throughout we use the adic spectrum $\mathrm{Spa}(A, A^+)$ of *continuous* valuations, so that in forming a rational subset $U(f_1, \dots, f_n/g)$ we assume $(f_1, \dots, f_n)$ generates an open ideal – which, $A$ being Tate, is automatically the unit ideal, so $(f_1, \dots, f_n) = A$.

**Proposition 10.10** (Huber's structure presheaf)**.** *For a rational subset $U = U(f_1, \dots, f_n/g) \subseteq \mathrm{Spa}(A, A^+)$ (so $(f_1, \dots, f_n) = A$),*

$$\mathcal{O}^{\mathrm{Huber}}(U) \;=\; A\langle T_1, \dots, T_n\rangle \big/ \overline{(gT_1 - f_1, \dots, gT_n - f_n)}, \tag{10.3}$$

*the convergent power series algebra in $T_1, \dots, T_n$ modulo the* closure *of the ideal $(gT_i - f_i)$. Here $g$ is already invertible before quotienting (its ideal contains $f_1, \dots, f_n$, hence is the unit ideal), and $T_i$ is the new function $f_i/g$, bounded by $1$ on $U$.*

The point of Huber's construction is that a quotient of a Banach algebra by a *closed* ideal is again a Banach algebra – so one takes the closure and stays inside the world of Huber rings. The solid structure sheaf of Section 9 instead uses a *derived* quotient, and this is where the comparison lives.

*Notation* 10.11 (Derived quotient/Koszul complex). For a condensed ring $R$ and $x_1, \dots, x_n \in R$, write

$$R/^{\mathbb{L}}(x_1, \dots, x_n) \;=\; R \otimes^{\mathbb{L}}_{\mathbb{Z}[x_1,\dots,x_n]} \mathbb{Z}$$

for the derived quotient, an animated condensed ring, computed by the Koszul complex

$$\Big( \textstyle\bigwedge^n (\bigoplus_{i=1}^n R) \to \cdots \to \bigwedge^2 (\bigoplus_{i=1}^n R) \to \bigoplus_{i=1}^n R \xrightarrow{(x_i)_i} R \Big).$$

*Remark* 10.12 (The solid value of a rational open). The solid structure sheaf assigns to $U$ the endomorphisms of the unit of the analytic ring $(\mathcal{O}(U), \mathcal{O}^+(U))^{\square}$; concretely $\mathcal{O}(U)$ is $A[\frac{1}{g}]$ solidified so that each $f_i/g$ is power-bounded. Its value has the same shape as (10.3), but with a derived quotient in place of Huber's closure. In two steps (cf. Section 7):

- start with the derived localization $A[\frac{1}{g}] = A[T_1, \dots, T_n]/^{\mathbb{L}}(gT_i - f_i)$ (the sequence $gT_i - f_i$ is regular, so "derived" does no harm), and then $(T_1, \dots, T_n)$-solidify. Solidifying the polynomial variables produces the Tate algebra,

  $$A[T_1, \dots, T_n]^{\mathbb{L}(T_1,\dots,T_n)\text{-}\square} \;=\; A\langle T_1, \dots, T_n \rangle,$$

  because joining a variable and solidifying commute with the presentation $A = A_0[\frac{1}{\pi}]$, $A_0 = \varprojlim_n A_0/\pi^n$, as limits and colimits (Section 7).

**Corollary 10.13** (Huber = quasiseparation of $\pi_0$ of the solid value)**.** *With $\mathcal{O}(U)$ the solid value as above,*

$$\mathcal{O}^{\mathrm{Huber}}(U) \;=\; \big(\pi_0 \mathcal{O}(U)\big)^{\mathrm{qs}},$$

*Huber's ring is the* quasiseparation *of the underlying static condensed ring $\pi_0\mathcal{O}(U) = A\langle T_1, \dots, T_n\rangle/(gT_i - f_i)$.*

*Remark* 10.14 (The quasiseparation functor). The inclusion of quasiseparated condensed sets into all condensed sets, $\mathrm{CondSet}^{\mathrm{qs,light}} \subseteq \mathrm{CondSet}^{\mathrm{light}}$, has a left adjoint

$$X \;\longmapsto\; X^{\mathrm{qs}},$$

a "maximal Hausdorff quotient". For non-obvious reasons it commutes with finite products, hence preserves algebraic structures (being a group, ring, etc.), so $(\pi_0\mathcal{O}(U))^{\mathrm{qs}}$ is again a ring; on a light condensed set the quasiseparation is again light. Passing from $\pi_0\mathcal{O}(U)$ to its quasiseparation is exactly Huber's passage to the closure of the ideal $(gT_i - f_i)$.

**Lemma 10.15** (Quasiseparation via equivalence relations)**.** *Write a condensed set $X$ as a quotient $X = \widetilde{X}/R$ of a quasiseparated $\widetilde{X}$ (e.g. a profinite set surjecting onto it) by an equivalence relation $R \subseteq \widetilde{X} \times \widetilde{X}$. Then there is a minimal quasicompact injection $\overline{R} \subseteq \widetilde{X} \times \widetilde{X}$ that is again an equivalence relation and contains $R$, and*

$$X^{\mathrm{qs}} \;=\; \widetilde{X}/\overline{R}.$$

*One obtains $\overline{R}$ by intersecting all quasicompact-injective equivalence relations containing $R$ (a transfinite closure that may add more than the naive closure of $R$). If $X$ is a group one may take $\widetilde{X}$ a group and $R$ a subgroup, in which case $\overline{R}$ is the closure of the subgroup.*

**Lemma 10.16** (Closed subsets are quasicompact injections)**.** *Let $X$ be any sequential topological space (so that $\underline{X}$ is a light condensed set, Section 3). Then*

$$\{\textit{closed subsets of } X\} \xrightarrow{\sim} \{\textit{quasicompact injections into } \underline{X}\}.$$

*Indeed the pullback of a closed subset $Z \subseteq X$ along any map from a profinite set is a closed subset of a profinite set, hence profinite, hence quasicompact; the quotient topology on $X$ makes this reversible.*

We can now state the sharp comparison. It is essentially due to Kedlaya, phrased here in the present language.

**Theorem 10.17** (Sheafy ⇔ Huber = solid)**.** *For a Tate Huber pair $(A, A^+)$,*

$$\begin{aligned} (A, A^+) \textit{ sheafy} \quad &\iff \quad \mathcal{O}^{\mathrm{Huber}} = \mathcal{O} \\ &\iff \quad \forall\, U \textit{ rational, } \mathcal{O}(U) \textit{ sits in degree } 0 \textit{ and is quasiseparated.} \end{aligned}$$

*That is, whenever Huber's presheaf happens to be a sheaf, it agrees with the solid structure sheaf $\mathcal{O}$ – so the good properties of $\mathcal{O}$ transfer, which is why sheafiness alone forces good behaviour.*

*Sketch.* The implication "⇐" is clear: if $\mathcal{O}(U)$ sits in degree 0 and is quasiseparated for every rational $U$, then by Corollary 10.13 it equals $\mathcal{O}^{\mathrm{Huber}}(U)$, which is therefore the (sheafy) solid structure sheaf. For "⇒" it suffices to check the good properties of $\mathcal{O}(U)$ after possibly refining the cover. One always refines to *simple Laurent covers* $\{|f| \leq 1\} \cup \{|f| \geq 1\}$, and the problem reduces to Lemma 10.18 below. □

**Lemma 10.18** (Key Lemma)**.** *Let $A$ be a Tate Huber ring and $f \in A$. Assume the Laurent-cover sequence*

$$0 \;\to\; A \;\to\; A\langle \tfrac{f}{1}\rangle \oplus A\langle \tfrac{1}{f}\rangle \;\to\; A\langle \tfrac{f}{1}, \tfrac{1}{f}\rangle \;\to\; 0 \tag{10.4}$$

*is exact (as it is, e.g., when $A$ is sheafy) – here $A\langle \frac{f}{1}\rangle = \mathcal{O}^{\mathrm{Huber}}(\{|f| \leq 1\})$, $A\langle \frac{1}{f}\rangle = \mathcal{O}^{\mathrm{Huber}}(\{|f| \geq 1\})$ and $A\langle \frac{f}{1}, \frac{1}{f}\rangle = \mathcal{O}^{\mathrm{Huber}}(\{|f| = 1\})$. Then Huber's rings agree with the derived quotients,*

$$A\langle \tfrac{f}{1}\rangle = A\langle T\rangle/^{\mathbb{L}}(T - f), \qquad A\langle \tfrac{1}{f}\rangle = A\langle T\rangle/^{\mathbb{L}}(1 - Tf),$$

*i.e. multiplication by $T - f$ (resp. $1 - Tf$) on $A\langle T\rangle$ is injective with closed image.*

*Idea.* One must see that $A\langle T\rangle \xrightarrow{T-f} A\langle T\rangle$ (resp. $1 - Tf$) is injective with closed image. Using the exactness (10.4) the question may be analyzed over the two localizations $A\langle \frac{f}{1}\rangle$ and $A\langle \frac{1}{f}\rangle$ – in other words one may assume $|f| \leq 1$ or $|f| \geq 1$. In the case $|f| \leq 1$ the ideal $(T - f)$ is closed because one successively kills the highest coefficient; the case $|f| \geq 1$ amounts to inverting $f$ already at the level of $A_0$, after which the other localization becomes a tautology. Either way the statement can be checked by hand. This is the funny reduction, due to Kedlaya, by which sheafiness of a single Laurent cover controls everything. □

### 10.5 Gluing pseudocoherent and perfect complexes

We now drop the sheafiness hypothesis entirely. The classical finiteness notions are those of [BJGRKIB71] (SGA 6).

**Definition 10.19** (Pseudocoherent, perfect; SGA 6 [BJGRKIB71])**.** Let $R$ be any ring. A complex $C \in D(R)$ is *pseudocoherent* if it can be represented by a complex

$$\cdots \to C_n \to \cdots \to C_1 \to C_0 \to \cdots \to C_{-m} \to 0 \to 0 \to \cdots$$

that is bounded to the right (zero in low degrees) but possibly unbounded to the left, with each $C_i$ a finite free (equivalently, finite projective) $R$-module. It is *perfect* if such a representative can be chosen bounded on both sides. Over a noetherian ring pseudocoherence says each homology group is coherent; over a general ring it captures "finitely generated with successively finitely many relations".

**Theorem 10.20** (Gluing of finiteness conditions)**.** *Let $(A, A^+)$ be any Huber pair with $A$ Tate. Then each of the assignments sending a rational $U \subseteq \mathrm{Spa}(A, A^+)$ to*

$$D^{\mathrm{pcoh}}(\mathcal{O}(U)(*)), \quad D^{\mathrm{pcoh}}_{\geq n}(\mathcal{O}(U)(*)), \quad \mathrm{Perf}(\mathcal{O}(U)(*)), \quad \mathrm{Perf}^{[a,b]}(\mathcal{O}(U)(*)) \tag{10.5}$$

*– pseudocoherent complexes, pseudocoherent complexes in degrees $\geq n$, perfect complexes, and perfect complexes admitting a representative in degrees $[a, b]$, over the underlying static ring $\mathcal{O}(U)(*)$ – is a sheaf of $\infty$-categories. The special case* $\mathrm{Perf}^{[0,0]} = \mathrm{Vect}$ *recovers the Kedlaya–Liu theorem* (10.2)*, now without any sheafiness hypothesis.*

*Remark* 10.21 (Warning: discreteness does not globalize)*.* It is essential to work with the finiteness *subcategories* of the solid derived category rather than with the condition of being a discrete module. The condition $M \in D(A(*)) \subseteq D((A, A^+)_D)$ – being (a complex of) discrete $A(*)$-modules – does *not* globalize. Consider the Tate elliptic curve, the quotient of the adic $\mathbb{G}_m$ over $\mathbb{Q}_p$ by multiplication by $p$,

$$a\colon\ \mathbb{G}^{\mathrm{ad}}_{m,\mathbb{Q}_p} \longrightarrow \mathbb{G}^{\mathrm{ad}}_{m,\mathbb{Q}_p}/p^{\mathbb{Z}},$$

the nonarchimedean analogue of $\mathbb{C}^*/q^{\mathbb{Z}}$. Let $\mathrm{Spa}(A, A^+)$ be a large open subset of the quotient – remove a small disk around one point, but keep it large enough to carry a global monodromy, and so that it genuinely lies in the quotient rather than lifting to $\mathbb{G}_m$. Then the lower-shriek $a_! \mathcal{O}^{\mathrm{ad}}_{\mathbb{G}_{m,\mathbb{Q}_p}}$, restricted to $\mathrm{Spa}(A, A^+)$, defines an object of $D((A, A^+)_D)$ that is *locally* discrete – locally the covering map $a$ splits, so $a_!\mathcal{O}$ is a direct sum of copies of the base indexed by $\mathbb{Z}$ – but is *not* globally discrete: it does not lie in $D(A(*))$. The monodromy obstructs the global splitting.

*Remark* 10.22 (Outline of the argument)*.* By Section 9 we already know the assignment

$$U \longmapsto D\big((\mathcal{O}(U), \mathcal{O}^+(U))_D\big)$$

is a sheaf of $\infty$-categories, and all the finiteness categories of (10.5) are cut out by finiteness conditions inside it. So gluing reduces to a *membership* statement: if $M \in D((A, A^+)_D)$ is such that, over a cover, each base change $M \otimes^{\mathbb{L}}_{(A,A^+)_D} (\mathcal{O}(U), \mathcal{O}^+(U))_D$ lies in one of the subcategories, then so does $M$ globally. This is carried out in three steps: a pseudocoherent subcategory that globalizes formally, a *nuclear* subcategory that globalizes because internal Hom commutes with the localizations, and the identification of the discrete pseudocoherent complexes as the intersection of the two.

### Step 1: pseudocoherent complexes globalize

**Definition 10.23** (Solid pseudocoherent complexes)**.** Define $D^{\mathrm{pcoh}}((A, A^+)_D) \subseteq D((A, A^+)_D)$ to consist of the complexes representable by

$$\cdots \to C_n \to \cdots \to C_1 \to C_0 \to \cdots \to C_{-m} \to 0, \qquad \text{all } C_i = P,$$

bounded to the right, where $P$ is the compact projective generator (the free complete module on a null sequence), subject to the finiteness condition that for all $i \in \mathbb{Z}$ the functor $\mathrm{Ext}^i(C_i, -)\colon \mathrm{Mod}(A)_D \to \mathrm{Ab}$ commutes with filtered colimits.

That being in this subcategory globalizes is formal: it is a right-boundedness statement together with the finite-Tor-dimension of the localizations, so it descends.

### Step 2: nuclear modules

The eigenvector for detecting vector bundles requires a second class.

**Definition 10.24** (Nuclear modules)**.** Let $P$ be the compact projective generator. An object $C \in D((A, A^+)_D)$ is *nuclear* if the natural map

$$C \otimes^{\mathbb{L}}_{(A,A^+)_D} \underline{\mathrm{Hom}}(P, A) \xrightarrow{\sim} \underline{\mathrm{Hom}}(P, C) \tag{10.6}$$

is an isomorphism – "all maps from $P$ into $C$ are trace-class". Write $D^{\mathrm{nuc}}((A, A^+)_D)$ for the full subcategory of nuclear objects.

*Remark* 10.25 (Provenance and generation)*.* The name is borrowed, in spirit rather than to the letter, from Grothendieck's nuclear spaces in functional analysis. Nuclear modules are generated under colimits by the module $c_0(\mathbb{N}, A) = A\langle T\rangle$ of $A$-valued null sequences (and its shifts); in general by sequential colimits $\varinjlim(Q_0 \to Q_1 \to \cdots)$ in which each transition is trace-class, i.e. arises from an element of $Q_i^\vee \otimes Q_{i+1}$. Over $\mathbb{Q}_p$ they are generated by one-dimensional Banach spaces, still a large class. Nuclear modules form a dualizable category – a universal dualizable approximation to the whole category – which is one way to define analytic $K$-theory spaces (Scholze noted this is not over a field).

*Remark* 10.26 (Why nuclearity globalizes)*.* Both sides of (10.6) localize: the right by the same argument Clausen used in Section 9 – the different localizations commute because they are all given by internal Hom out of fixed objects, and $\underline{\mathrm{Hom}}(A, \underline{\mathrm{Hom}}(B, -)) = \underline{\mathrm{Hom}}(A \otimes B, -)$. The key point is that $\underline{\mathrm{Hom}}(P, -)$ commutes with the localizations of interest; for a rational (rather than a general algebraic) localization this holds because such a localization is itself given by an internal Hom. Hence nuclearity, being a local isomorphism condition, globalizes.

### Step 3: the intersection, and the Fredholm step

**Proposition 10.27** (Discrete pseudocoherent = solid pseudocoherent $\cap$ nuclear)**.** *Base change along $g\colon A(*) \to (A, A^+)_D$ identifies*

$$D^{\mathrm{pcoh}}(A(*)) \xrightarrow{\sim} D^{\mathrm{pcoh}}((A, A^+)_D) \cap D^{\mathrm{nuc}}((A, A^+)_D).$$

*Sketch.* Take $C$ in the intersection, represented as a pseudocoherent complex all of whose terms are the generator $P$, and (by shifting) starting in degree 0: $\cdots \to P \to P \to 0$. There is a tautological map $P \to C$, the identity in degree 0. Nuclearity makes this map trace-class, so it factors through an approximation of $C$; concretely, the map $P \to C$ factors through the two-term complex $[P \xrightarrow{1-g} P]$ for some trace-class endomorphism $g$ of $P$:

$$\begin{tikzcd} & {[P \xrightarrow{1-g} P]} \arrow[d] \\ P \arrow[ur] \arrow[r] & C \end{tikzcd}$$

It then suffices to show $[P \xrightarrow{1-g} P]$ is perfect over $A(*)$. This is classical Fredholm theory: with $g$ trace-class, $1-g$ is a Fredholm operator (the identity is Fredholm, and a compact – in particular trace-class – perturbation of a Fredholm operator is Fredholm), so its kernel and cokernel are finite-dimensional and the kernel is a perfect complex. Iterating, one truncates $C$ to a perfect complex of discrete modules. This is the single genuinely analytic step of the whole argument. $\square$

*Remark* 10.28 (Conclusion). Combining the three steps: the solid pseudocoherent and nuclear subcategories both glue, and their intersection is the discrete pseudocoherent (in particular perfect, and $\mathrm{Perf}^{[0,0]} = \mathrm{Vect}$) complexes, which therefore glue as well. The remaining superscripted variants of (10.5) glue by the same, easier, bookkeeping. Thus vector bundles glue over $\mathrm{Spa}(A, A^+)$ for every Tate Huber pair, recovering and generalising Theorem 10.8 – and up to the final Fredholm step, no analysis was done at all.

# 11 Solid Miscellany: Structure, Homological Dimension, and the Constructible Topology

This lecture collects several complements to the solid theory before the course moves on to a new topic. There are three parts. First, Clausen shows that for a solid analytic ring cut out by a ring of integral elements $(A^\triangleright, A^+)$ the derived category $D((A^\triangleright, A^+)^\square)$ really is the derived category of its heart – the derived $A^+$-solidification of the compact projective generator lives in degree 0 – with everything reduced, by a filtered-colimit argument, to the case of a ring finitely generated over $\mathbb{Z}$. Second, for such a ring $R$ he analyses the abelian category $\mathrm{Solid}_R$: it is Ind of its finitely presented objects, these form an abelian ("coherent") subcategory, the quasiseparated ones are exactly the countable pro-systems of finitely generated discrete modules, and the compact projective generator $\prod_{\mathbb{N}} R$ is flat. Third, when $R$ is moreover regular of dimension $d$ he proves the expected homological bound $\mathrm{Ext}^i = 0$ for $i > d$ on finitely presented modules – the one genuinely delicate point being a depth/Auslander–Buchsbaum argument. The lecture closes by pointing out a second, more radical way to localize $D((R, \mathbb{Z})^\square)$: not over the valuative spectrum of Section 9 but over the *constructible* topology of $\mathrm{Spec}\, R$, producing idempotent algebras like the higher local fields $\mathbb{Z}(\!(Y)\!)(\!(X)\!)$.

Throughout, "ring" means commutative ring, everything lives in light condensed abelian groups, and we use freely the solid theory $\mathrm{Solid} \subseteq \mathrm{CondAb}^{\mathrm{light}}$ (Sections 5 and 6), the relative solid theory over $\mathbb{Z}[T]$ (Section 7), the abstract notion of analytic ring (Section 8) and its localization over the valuative spectrum (Section 9).

## 11.1 The solid analytic ring of a pair $(A^\triangleright, A^+)$, revisited

Recall from Sections 8 and 9 the following construction. Let $A^\triangleright$ be a solid ring and $A^+ \subseteq A^\circ \subseteq A^\triangleright(*)$ a subring of power-bounded elements. To the pair one attaches a solid analytic ring $(A^\triangleright, A^+)^\square$, specified on the abelian level by the full subcategory

$$\mathrm{Mod}_{(A^\triangleright, A^+)} \;\subseteq\; \mathrm{Mod}_{A^\triangleright}(\mathrm{Solid}_{\mathbb{Z}}) \tag{11.1}$$

of those modules obtained by forcing every $f \in A^+$ to become a solidified variable (i.e. requiring $\mathbb{Z}[T]$-solidity along $T \mapsto f$ for all $f \in A^+$; Definition 9.5). The hypothesis $A^+ \subseteq A^\circ$ is what makes $A^\triangleright$ itself complete, so that $A^\triangleright \in \mathrm{Mod}_{(A^\triangleright, A^+)}$.

*Remark* 11.1 (When $A^+$ is not inside a ring of definition). If one drops the power-boundedness $A^+ \subseteq A^\circ$, then enforcing (11.1) may throw $A^\triangleright$ out of the category; one must then apply the localization (solidification) and obtain a new underlying ring – a completion of $A^{\triangleright}$ in the appropriate sense.

On the derived level (Definition 8.14) one has

$$D\big((A^{\triangleright}, A^+)^{\square}\big) \ \subseteq\ D\big(\mathrm{Mod}_{A^{\triangleright}}(\mathrm{Solid}_{\mathbb{Z}})\big), \tag{11.2}$$

the full subcategory of objects whose homology lies in $\mathrm{Mod}_{(A^{\triangleright},A^+)}$ in every degree. The inclusion has a left adjoint – the derived $A^+$-solidification – with the usual formal properties (symmetric monoidal, colimit-preserving). The subtlety, flagged in Section 8, is that in the generality of an arbitrary analytic ring (11.2) need *not* be the derived category of the abelian category (11.1): the obstruction is that applying derived solidification to a basic object may produce homology outside degree 0.

The abelian category $\mathrm{Mod}_{(A^{\triangleright},A^+)}$ has a compact projective generator, namely

$$\Big(\prod_{\mathbb{N}} \mathbb{Z}\Big) \otimes_{\mathbb{Z},\square} A^{\triangleright}, \tag{11.3}$$

the compact projective generator $\prod_{\mathbb{N}} \mathbb{Z}$ of $\mathrm{Solid}_{\mathbb{Z}}$ (Corollary 5.18) base-changed to $A^{\triangleright}$. Since $\prod_{\mathbb{N}} \mathbb{Z}$ is flat in $\mathrm{Solid}_{\mathbb{Z}}$ (Corollary 6.16), it is immaterial whether one uses the ordinary or derived solid tensor product here. A compact generator of the derived category (11.2) is obtained from (11.3) by applying the derived $A^+$-solidification.

**Theorem 11.2** (The derived category is the derived category of the heart)**.** *For $(A^{\triangleright}, A^+)$ as above, the derived $A^+$-solidification of $(\prod_{\mathbb{N}} \mathbb{Z}) \otimes_{\mathbb{Z},\square} A^{\triangleright}$ lives in degree* 0*. Consequently it is a compact projective generator of the abelian category* $\mathrm{Mod}_{(A^{\triangleright},A^+)^{\square}}$*, and*

$$D\big(\mathrm{Mod}_{(A^{\triangleright},A^+)^{\square}}\big) \ =\ D\big((A^{\triangleright}, A^+)^{\square}\big). \tag{11.4}$$

Thus, as long as the underlying solid ring $A^{\triangleright}$ is not itself derived, the whole theory is "flat", i.e. determined at the abelian level.

*Remark* 11.3 (This settles a point left open earlier)*.* In Section 8 it was not clear that (11.4) holds; that is exactly what Theorem 11.2 establishes.

## 11.2 Reduction to finite type over $\mathbb{Z}$

Being derived $A^+$-solid is an element-wise condition, so it can be tested on the finitely generated subrings of $A^+$.

**Proposition 11.4** (Solidification through finite-type subrings)**.** *Derived $A^+$-solidity is equivalent to derived $R$-solidity for all $R \subseteq A^+$ finite type over $\mathbb{Z}$, and for any $M$ the derived $A^+$-solidification is the filtered colimit*

$$M^{\mathbb{L}A^+\text{-}\square} \ =\ \varinjlim_{\substack{R\subseteq A^+\\ R\ \text{f.t.}/\mathbb{Z}}} M^{\mathbb{L}R\text{-}\square}, \tag{11.5}$$

*with structure map induced from $M$ (regarding $M$ as an $R$-module via $R \hookrightarrow A^+ \to A^{\triangleright}$).*

*Sketch.* The right-hand side of (11.5) is derived $A^+$-solid: for fixed $R$, every term past the $R$-term is even derived $R'$-solid for some $R' \supseteq R$ (hence $R$-solid), so a cofinal subsystem is $R$-solid, and a filtered colimit of $R$-solid objects is $R$-solid; thus the colimit is $R$-solid for every $R$, hence $A^+$-solid. And mapping out to any derived $A^+$-solid target does not distinguish $M$ from the colimit, by the same $R$-by-$R$ argument. □

By Proposition 11.4, Theorem 11.2 reduces to showing, for $R$ finite type over $\mathbb{Z}$, that the derived $R$-solidification of $(\prod_{\mathbb{N}} \mathbb{Z}) \otimes_{\mathbb{Z},\square} A^{\triangleright}$ lives in degree 0. Doing this in two steps – base change to $\mathrm{Solid}_R$, then tensor up to $A^{\triangleright}$ over $\mathrm{Solid}_R$ – reduces further to the two statements

$$\Big(\prod_{\mathbb{N}} \mathbb{Z} \otimes_{\mathbb{Z},\square} R\Big)^{\mathbb{L}R\text{-}\square} \text{ lives in degree 0} \quad \text{and is flat for the } \mathrm{Solid}_R\text{-tensor product.} \tag{11.6}$$

Before analysing (11.6), we record a general consequence of the filtered-colimit picture.

**Corollary 11.5** (All completion happens at finite type)**.** *For an arbitrary commutative ring $R$, write* $\mathrm{Solid}_R := \mathrm{Mod}_{(R,R)^\square}$ *(so that every element of $R$ is solidified). Then*

$$\mathrm{Solid}_R \;=\; \mathrm{Ind}\big(\mathrm{Solid}_R^{\mathrm{fp}}\big),$$

*where the finitely presented objects are the cokernels of maps between the compact projective generator $\prod_{\mathbb{N}} R$. Moreover the whole finitely presented theory reduces to the finite-type case: all of the completion is happening at the level of finite-type subrings of $R$, and the rest is an algebraic filtered colimit.*

### 11.3 The generator over a finite-type ring

**Proposition 11.6** (The generator is a product of copies of $R$)**.** *Let $R$ be finite type over $\mathbb{Z}$. Then*

$$\Big(\prod_{\mathbb{N}} \mathbb{Z} \otimes_{\mathbb{Z},\square} R\Big)^{\mathbb{L}R\text{-}\square} \;=\; \prod_{\mathbb{N}} R$$

(*an unlabelled index set is always countable*)*. For a general $R$, not of finite type, the answer is instead the smaller object*

$$\Big(\prod_{\mathbb{N}} \mathbb{Z} \otimes_{\mathbb{Z},\square} R\Big)^{\mathbb{L}R\text{-}\square} \;=\; \bigcup_{\substack{R' \subseteq R \\ R' \ \text{f.t.}}} \prod_{\mathbb{N}} R'.$$

*Proof.* First take $R = \mathbb{Z}[X_1, \dots, X_n]$. Recall (Proposition 7.10 and Corollary 7.11) that the derived $\mathbb{Z}[T]$-solidification of a module tensored with $\mathbb{Z}[T]$ commutes with all limits and colimits and sends $\mathbb{Z}$ to $\mathbb{Z}[T]$; the various solidifications commute, and once $M$ is $X_i$-solid it remains so after solidifying in the other variables. Solidifying one variable at a time therefore throws each $X_i$ into the product:

$$\Big(\prod_{\mathbb{N}} \mathbb{Z} \otimes_{\mathbb{Z},\square} \mathbb{Z}[X_1, \dots, X_n]\Big)^{X_1\text{-}\square} = \prod_{\mathbb{N}} \mathbb{Z}[X_1] \otimes_{\mathbb{Z}[X_1],\square} \mathbb{Z}[X_2, \dots, X_n] = \cdots = \prod_{\mathbb{N}} \mathbb{Z}[X_1, \dots, X_n].$$

In general, choose a surjection $\mathbb{Z}[X_1, \dots, X_n] \twoheadrightarrow R$. Solidifying in $\mathbb{Z}$ and then base-changing algebraically (in the derived sense) from $\mathbb{Z}[X_1, \dots, X_n]$ to $R$, one has already solidified with respect to a generating set of $R$, so the object is already $R$-solid and it remains to compute

$$\Big(\prod_{\mathbb{N}} \mathbb{Z}[X_1, \dots, X_n]\Big) \otimes^{\mathbb{L}}_{\mathbb{Z}[X_1,\dots,X_n]} R.$$

But $\mathbb{Z}[X_1, \dots, X_n]$ is noetherian, so $R$ admits a (possibly infinite) resolution by finite free $\mathbb{Z}[X_1, \dots, X_n]$-modules. For a finite free module the product manifestly commutes with the base change; passing to the class of $\mathbb{Z}[X_1, \dots, X_n]$-modules for which the claim holds – which contains the ring and is closed under the relevant colimits – gives

$$\Big(\prod_{\mathbb{N}} \mathbb{Z}[X_1, \dots, X_n]\Big) \otimes^{\mathbb{L}}_{\mathbb{Z}[X_1,\dots,X_n]} R \;\xrightarrow{\sim}\; \prod_{\mathbb{N}} R,$$

as an equivalence in the derived category, computed by resolving $R$ and applying the colimit-preserving tensor functor. (The finiteness of the ideal of relations is what turns the naive limit of quotients into an honest product of copies of $R$.) $\square$

### 11.4 Structure of $\mathrm{Solid}_R$ for $R$ finite type

**Theorem 11.7** (Coherence). *Let $R$ be finite type over $\mathbb{Z}$. Then:*

1. $\mathrm{Solid}_R = \mathrm{Ind}(\mathrm{Solid}_R^{\mathrm{fp}})$*, where the finitely presented objects* $\mathrm{Solid}_R^{\mathrm{fp}}$ *are the cokernels of maps between copies of the compact projective generator* $\prod_{\mathbb{N}} R$*;*

2. $\mathrm{Solid}_R^{\mathrm{fp}}$ *is an abelian subcategory of* $\mathrm{Solid}_R$*, closed under kernels, cokernels and extensions.*

Part (1) is formal from the existence of a compact projective generator. Part (2) is a coherence statement, and its proof mirrors classical commutative algebra. Recall that a discrete ring is *coherent* when its finitely presented modules form an abelian subcategory, and that this reduces – by manipulating short exact sequences – to the single point that every finitely generated ideal is finitely presented. The same reduction here shows it suffices to check that

$$\text{every finitely generated subobject of } \textstyle\prod_{\mathbb{N}} R \text{ is finitely presented.} \tag{11.7}$$

Any subobject of $\prod_{\mathbb{N}} R$ is quasiseparated (Hausdorff), being a subobject of a quasiseparated object, so (11.7) in turn follows from: every finitely generated *quasiseparated* solid $R$-module is finitely presented. This is the content of the next lemma, which identifies the quasiseparated finitely presented modules with the classical pro-systems.

**Lemma 11.8** (Quasiseparated finitely presented = pro-(f.g. modules)). *Let $M \in \mathrm{Solid}_R$ be quasiseparated. Then the following are equivalent:*

1. *$M$ is finitely presented;*

2. *$M$ is finitely generated;*

3. *$M = \varprojlim_n M_n$, where each $M_n$ is a finitely generated discrete $R$-module and the transition maps are surjective.*

*Moreover the category of such $M$ is the countable pro-category*

$$\mathrm{Pro}_{\mathbb{N}_0}\big(\{\textit{finitely generated } R\textit{-modules}\}\big),$$

*i.e. morphisms between such inverse limits are the maps of pro-objects.*

*Proof.* The nontrivial implications are (2)⇒(3) and (3)⇒(1).

For (2)⇒(3), a finite generation gives a surjection $\prod_{\mathbb{N}} R \twoheadrightarrow M$ with kernel $K$:

$$0 \longrightarrow K \hookrightarrow \textstyle\prod_{\mathbb{N}} R \twoheadrightarrow M \longrightarrow 0.$$

Since $M$ is quasiseparated, the inclusion $K \hookrightarrow \prod_{\mathbb{N}} R$ is a quasicompact map, so $K$ is quasiseparated: a filtered union of compact Hausdorff subspaces, on each of which the inclusion is a closed embedding. As $\prod_{\mathbb{N}} R$ is metrizable (sequential), being closed on each compact subset is the same as being closed, so $K$ is just a closed submodule of $\prod_{\mathbb{N}} R$ for the product topology. Projecting to the first $n$ coordinates and letting $K_n \subseteq \prod_{i\le n} R$ be the image, one gets $K = \varprojlim_n K_n$ with surjective transition maps, and analogously $M = \varprojlim_n M_n$ with $M_n = \prod_{i\le n} R/K_n$; both are of the form (3).

For (3)⇒(1) it suffices to prove (3)⇒(2): once $M$ is finitely generated it fits into a sequence as above whose kernel is again quasiseparated of the form (3), so it too is finitely generated, and $M$ is finitely presented. To see (3)⇒(2), resolve the tower: choose a finite free module surjecting onto $M_1$, a finite free $R^{\oplus d_2}$ onto the relevant kernel, and assemble the finite free modules $R^{\oplus d_1}$, $R^{\oplus d_1+d_2}, \ldots$ into a compatible system; in the inverse limit these give a surjection $\prod_{\mathbb{N}} R \twoheadrightarrow M$. (One can even arrange a topological splitting of the underlying map of spaces, by choosing compatible sections, if one does not insist on $R$-linearity.) $\square$

*Remark* 11.9 (Module theory over the endomorphism ring). Because $\mathrm{Solid}_R$ has a compact projective generator, it is equivalent to the category of modules over the (noncommutative) endomorphism ring of that generator, so the extension-closedness in Theorem 11.7(2) is just the usual statement from module theory over an associative ring, and one may quote it directly.

*Remark* 11.10 (Finitely presented $\neq$ quasiseparated). The lemma does not say finitely presented objects are quasiseparated. A cokernel of a map $\prod_{\mathbb{N}} R \to \prod_{\mathbb{N}} R$ need not be quasiseparated: over $\mathbb{Z}$ the module $\mathbb{Z}_p/\mathbb{Z}$ is finitely presented but not quasiseparated. A non-quasiseparated finitely presented object forces $\dim R \geq 1$; over a finite field every finitely presented solid module is quasiseparated.

*Remark* 11.11 (Where finite type is used). Finite type over $\mathbb{Z}$ enters *only* to identify $\prod_{\mathbb{N}} R$ as the compact projective generator (via the surjection from a polynomial ring). If one instead *defines* the category by declaring $\prod_{\mathbb{N}} R$ to be a compact projective generator with the prescribed endomorphisms, then the rest of Theorem 11.7 and Lemma 11.8 goes through for any noetherian ring.

## 11.5 Flatness of the generator

**Proposition 11.12** ($\prod_{\mathbb{N}} R$ is flat). *Let $R$ be finite type over $\mathbb{Z}$ and $M \in \mathrm{Solid}_R^{\mathrm{fp}}$. Then*

$$M \otimes_R^{\mathbb{L}\square} \textstyle\prod_{\mathbb{N}} R \;\cong\; \textstyle\prod_{\mathbb{N}} M, \tag{11.8}$$

*which in particular lives in degree 0. Consequently $\prod_{\mathbb{N}} R$ is flat for the solid tensor product $\otimes_R^{\square}$.*

*Proof.* Flatness follows from (11.8): the derived tensor product with any finitely presented object lives in degree 0, and every object is a filtered colimit of finitely presented ones, so the derived tensor product with $\prod_{\mathbb{N}} R$ lands in degree 0 throughout.

For (11.8), resolve $M$ by copies of $\prod_{\mathbb{N}} R$ (an infinite resolution: each successive kernel is quasiseparated, hence finitely presented by Lemma 11.8). This reduces the claim to $M = \prod_{\mathbb{N}} R$, and then, writing $\prod_{\mathbb{N}} R = (\prod_{\mathbb{N}} \mathbb{Z} \otimes_{\mathbb{Z},\square} R)$ solidified, to the analogous fact for $\prod_{\mathbb{N}} \mathbb{Z}$ over $\mathbb{Z}$: that $\prod_{\mathbb{N}} \mathbb{Z}$ distributes over infinite products. This was proved via the light condensed set $P = \mathbb{N} \cup \{\infty\}$, whose self product $P \times P$ is isomorphic to $P$ in the expected way (Corollaries 5.20 and 6.16). $\square$

*Remark* 11.13 (Defining the derived tensor product). Making sense of $\otimes^{\mathbb{L}}$ does not require producing flat objects by hand. On $D(\mathrm{CondAb}^{\mathrm{light}})$ one derived-sheafifies the section-wise tensor product of complexes of abelian groups; over a condensed ring one takes the relative tensor product (a bar construction of derived tensor products), and on a localization one has the symmetric monoidal structure supplied by the analytic ring structure. So the construction is available before any flatness is known. That said, in light condensed abelian groups there *are* enough flats – the free object on a light condensed set is flat – and, once the flatness and degree-0 statements above are in hand, $\otimes_R^{\mathbb{L}\square}$ is genuinely the left derived functor of $\otimes_R^{\square}$ (resolvable in either variable), and derived solidification is the left derived functor of solidification.

## 11.6 Homological dimension over a regular ring

Over $\mathbb{Z}$, Scholze showed that finitely presented solid abelian groups have a two-term resolution by products of copies of $\mathbb{Z}$ (Remark 6.11), so solid $\mathbb{Z}$ has homological dimension 1 on finitely presented objects, exactly as $\mathbb{Z}$ does. Here is the generalization.

**Theorem 11.14** (Ext-vanishing in the regular case)**.** *Let $R$ be finite type over $\mathbb{Z}$ and regular of dimension $d$. Then for all $M \in \mathrm{Solid}_R^{\mathrm{fp}}$ and all $N \in \mathrm{Solid}_R$ one has*

$$\mathrm{Ext}^i_R(M,N) = 0 \qquad \textit{and even} \qquad \underline{\mathrm{Ext}}^i_R(M,N) = 0, \qquad \forall\, i > d.$$

**Corollary 11.15** (Tor bound)**.** *For $M, N \in \mathrm{Solid}_R$ ($R$ finite type over $\mathbb{Z}$, regular of dimension $d$),*

$$H_i\big(M \otimes_R^{\mathbb{L}\square} N\big) = 0 \qquad \forall\, i > d.$$

Indeed Theorem 11.14 yields, for finitely presented $M$, a projective resolution of length $\leq d$ (or $d+1$, depending on the convention), and the projective objects are flat (Proposition 11.12); a filtered-colimit argument passes from finitely presented to arbitrary $M$.

*Remark* 11.16 (The hypothesis "finitely presented" is essential). Theorem 11.14 does *not* extend to arbitrary $M$. Unlike the classical situation over a regular noetherian ring of finite dimension – where the Ext-vanishing holds for all modules – the passage from finitely presented to general modules requires analysing a derived inverse limit along an arbitrary filtered system, and that step fails here. As a concrete (and cute) exercise over $R = \mathbb{Z}$, take two distinct primes $p \neq \ell$ and the non-finitely-presented module $\mathbb{Z}_p/\mathbb{Z}[1/\ell]$; then

$$\mathrm{Ext}^2\big(\mathbb{Z}_p/\mathbb{Z}[1/\ell],\ \mathbb{Z}\big) \ = \ \mathbb{Z}_\ell/\mathbb{Z}[1/p] \ \neq \ 0,$$

symmetric under interchanging the roles of $p$ and $\ell$. Clausen expects arbitrarily high non-vanishing Ext-groups to occur here even in ZFC (and they certainly do in some model of ZFC – the relevant Ext-group is one of those studied by set theorists, cf. Definition 4.24 and Theorem 4.26). If the continuum $2^{\aleph_0}$ is bounded by some $\aleph_n$, one does get cohomological-dimension bounds; if $2^{\aleph_0} > \aleph_n$ for all $n$, one should expect arbitrarily high Ext.

**Proof of Theorem 11.14**

*Step 1: $M$ quasiseparated, $N$ discrete.* Take $N$ a discrete finitely generated $R$-module and $M$ quasiseparated finitely presented, so $M = \varprojlim_n M_n$ with surjective transition maps (Lemma 11.8). We claim

$$\mathrm{Ext}^i_{R^\square}(M,N) \ = \ \varinjlim_n \mathrm{Ext}^i_R(M_n, N). \tag{11.9}$$

(The internal statement follows from the underlying one by replacing $N$ with $\mathrm{Cont}(S,N)$, another discrete module.) The content is that $R\,\mathrm{Hom}$ pulls the inverse limit out. The Mittag-Leffler resolution

$$\varprojlim_n M_n \ \longrightarrow \ \textstyle\prod_n M_n \ \xrightarrow{\ \mathrm{id}-\mathrm{shift}\ } \ \textstyle\prod_n M_n$$

turns the question into whether $R\,\mathrm{Hom}$ takes the product $\prod_n M_n$ to the direct sum; resolving $\prod_n M_n$ by copies of $\prod_{\mathbb{N}} R$ reduces this to

$$R\,\mathrm{Hom}\big(\textstyle\prod_{\mathbb{N}} R,\ N\big) \ = \ \textstyle\bigoplus_{\mathbb{N}} R\,\mathrm{Hom}_R(R,N),$$

which holds because $\prod_{\mathbb{N}} R$ is projective and by the basic computation of Hom. Finally, $R$ being regular of dimension $d$, the classical Ext-groups $\mathrm{Ext}^i_R(M_n, N)$ between discrete modules vanish for $i > d$, so the colimit (11.9) vanishes for $i > d$. Thus the quasiseparated question has been reduced to the classical one in discrete commutative algebra – and it is exactly here that regularity is used.

*Step 2: general finitely presented $M$, the naive bound.* For $M \in \mathrm{Solid}_R^{\mathrm{fp}}$ arbitrary, resolve

$$0 \to K \to \textstyle\prod_{\mathbb{N}} R \to M \to 0,$$

with $K$ quasiseparated and finitely presented. Chasing the long exact sequence and using Step 1 only gives

$$\mathrm{Ext}^i_R(M, N) = 0 \qquad \forall\, i > d + 1,$$

one degree worse than wanted. To recover the sharp bound, continue the resolution one further step (assume $d \geq 2$; the cases $d \leq 1$ are handled by variants of the same argument):

$$0 \to N \to \prod F_n \to \prod F_n' \to M \to 0, \qquad F_n, F_n' \text{ finite free,}$$

so that $N$ is a second syzygy. To prove $\mathrm{Ext}^i(M, X) = 0$ for $i > d$ it now suffices to prove

$$\mathrm{Ext}^i(N, X) = 0 \qquad \forall\, i > d - 2. \tag{11.10}$$

*Step 3: two presentations of the syzygy.* There are two ways to write the quasiseparated module $N$ as an inverse limit of finitely generated $R$-modules, each good in one respect and bad in the other:

1. $N_n := \ker(F_n \to F_n')$, so $N = \varprojlim_n N_n$. *Good:* each $N_n$ is a kernel of a map between finite frees, i.e. a second syzygy, so $\mathrm{Ext}^i(N_n, X) = 0$ for $i > d - 2$. *Bad:* no guarantee that $(N_n)$ is Mittag-Leffler – which is what (11.9) needs to compute $\mathrm{Ext}^i(N, X) = \varinjlim_n \mathrm{Ext}^i(N_n, X)$.
2. $N_n' := \mathrm{im}\big(N \to F_n\big) \subseteq N_n$. *Good:* Mittag-Leffler, indeed surjective transition maps by construction. *Bad:* no guarantee that $\mathrm{Ext}^{d-1}(N_n', X) = 0$.

*Step 4: interpolating.* One finds $N_n' \subseteq N_n'' \subseteq N_n$ with both good properties:

$$\mathrm{Ext}^i(N_n'', X) = 0 \;\; \forall\, i \geq d - 1, \qquad \text{and} \qquad (N_n'') \text{ is Mittag-Leffler.} \tag{11.11}$$

Then $\varprojlim_n N_n'' = N$ (it is sandwiched between two systems with the same inverse limit), and (11.11) yields (11.10).

The obstruction to $N_n'$ itself having $\mathrm{Ext}^{d-1}(N_n', X) = 0$ for all $X$ is a matter of *depth*. By the Auslander–Buchsbaum formula, for a regular noetherian local ring

$$\mathrm{pd}(M) + \mathrm{depth}(M) = \dim,$$

and because we already have the one-better estimate on projective dimension, the only obstruction is at the maximal ideals whose local ring has the full dimension $d$, where the issue is passing from depth 1 to depth 2: at such a maximal ideal $N_n'$ is only known to have depth 1.

To fix this, suppose (for simplicity) the problem occurs at a single closed point $x \in \mathrm{Spec}\, R$; in general one fixes finitely many, and by noetherianness the situation stabilizes. With $j \colon \mathrm{Spec}\, R \setminus \{x\} \hookrightarrow \mathrm{Spec}\, R$, replace $N_n'$ by $j_* j^* N_n'$. There is a map $N_n' \to j_* j^* N_n'$, which is an inclusion by the depth-1 hypothesis; and, since local cohomology at the maximal ideal computes depth, this operation improves the depth at $x$ to 2. Performed on $N_n$ (already of depth 2) it would change nothing, so the result $N''_n$ genuinely sits between $N'_n$ and $N_n$, and fixes the problem at $x$. Doing this compatibly at all the finitely many bad closed points produces the tower $(N''_n)$ with depth 2 everywhere – hence (11.11)'s Ext-vanishing.

For the Mittag-Leffler property of $(N''_n)$, use the short exact sequence of towers

$$0 \to (N'_n) \to (N''_n) \to (N''_n/N'_n) \to 0.$$

The subtower $(N'_n)$ is Mittag-Leffler (surjective transitions), and the quotient tower $(N''_n/N'_n)$ is supported at the finitely many closed points and finitely generated over $R$, hence *finite* as an abelian group (the residue fields are finite, $R$ being finite type over $\mathbb{Z}$); an inverse system of finite abelian groups is automatically Mittag-Leffler by a compactness argument. Thus $(N''_n)$ is Mittag-Leffler.

*Step 5: bootstrapping the coefficients.* With Steps 1–4 the theorem holds whenever $M$ is finitely presented and $N = X$ is a discrete module. Now let $X$ be quasiseparated finitely presented, $X = \varprojlim_n X_n$ with surjective transitions; then $R\operatorname{Hom}(M, X) = R\varprojlim_n R\operatorname{Hom}(M, X_n)$. A priori the $\varprojlim$ could push one degree higher, but the possible $\varprojlim^1$ lives on the $\operatorname{Ext}^d$-groups, and there the surjectivity of the transition maps makes the maps on $\operatorname{Ext}^d$ surjective (the obstruction is an $\operatorname{Ext}^{d+1}$ into a kernel, which vanishes), so no $\varprojlim^1$ term contributes in degree $d+1$. Hence the bound $i > d$ persists for such $X$. For $X$ arbitrary finitely presented one resolves by quasiseparated modules; for $X$ a filtered colimit $\varinjlim_j X_j$ of finitely presented modules one uses pseudocoherence of $M$ (finitely presented objects have infinite resolutions by copies of $\prod_{\mathbb{N}} R$, which are compact projective) to commute $\operatorname{Ext}^i(M, -)$ with the colimit. Finally, the internal $\underline{\operatorname{Ext}}^i$ statements follow from the underlying ones by passing to $S$-valued points, and the $\varprojlim^1$ arguments go through at the internal level because in condensed abelian groups the condensed $\varprojlim^1$ is the sheafification of the section-wise $\varprojlim^1$, which can be computed termwise since countable products are exact. □

### 11.7 A second localization: the constructible topology

We turn from structure theory to geometry. Recall from Section 9 that for a discrete commutative ring $R$ the derived category $D((R, \mathbb{Z})^{\square})$ localizes over the valuative spectrum $\operatorname{Spv}(R)$. That localization is convenient because its unit, restricted to a basic quasicompact open, stays discrete: over a rational open $U(f_1, \ldots, f_n/g)$ the structure ring is simply the discrete localization $R[\frac{1}{g}]$. It keeps one close to Huber's world. But it is by no means the only localization of this category, and we now describe a genuinely different one, departing more radically from Huber's formalism.

**Definition 11.17** (Idempotent algebra of a basic constructible subset)**.** Let $C \subseteq \operatorname{Spec} R$ be a constructible subset, that is, a finite union of locally closed subsets, where a basic locally closed subset is an intersection

$$D(g) \cap Z(f_1, \ldots, f_n) \;=\; \{g \neq 0\} \cap \{f_1 = \cdots = f_n = 0\}$$

of a distinguished quasicompact open with the common zero locus of finitely many functions. To such a basic locally closed subset one assigns the object

$$A \;=\; R\big[\tfrac{1}{g}\big]^{\wedge}_{(f_1,\ldots,f_n)} \;\in\; D(\mathrm{Solid}_{\mathbb{Z}}),$$

the $(f_1, \ldots, f_n)$-*derived* completion of $R[\frac{1}{g}]$, formed in the derived category of solid abelian groups – i.e. one inverts $g$ and then puts the inverse-limit (adic) topology on the derived inverse limit. This object depends only on the constructible subset. When $R$ is noetherian the derived completion agrees with the ordinary completion.

*Remark* 11.18 (Only locally closed pieces are idempotent). One should attach the idempotent algebra only to a basic locally closed subset, not to an arbitrary constructible one. For a general constructible $C$, written as a union of basic locally closed pieces, one takes the corresponding limit of the pieces (and their intersections, which are tensor products, Remark 11.19); the resulting object $R\Gamma$ can have higher cohomological degree, depending on the number of pieces used to cover $C$. The only derived behaviour arises from a union of principal opens.

*Remark* 11.19 (Idempotence). Recall (Proposition 6.19 and its generalization) that the solid tensor product of *connective* derived complete objects is derived complete. Writing $A = R[\frac{1}{g}]^{\wedge}_{(f_1,\dots,f_n)}$, idempotence

$$A \otimes^{\mathbb{L}\square}_{(R,\mathbb{Z})^\square} A \xrightarrow{\sim} A$$

is checked modulo $(f_1, \dots, f_n)$: since both sides are derived complete one may reduce modulo the sequence, whereupon the statement becomes the obvious fact that a localization of a ring, tensored over the ring with itself, is that localization again. Intersections of basic constructible subsets correspond to solid tensor products of the associated algebras.

**Proposition 11.20** (Constructible covers give a descent datum)**.** *Let $C_1, \dots, C_n \subseteq \operatorname{Spec} R$ be locally closed constructible subsets with $\bigcup_i C_i = \operatorname{Spec} R$, and $A_1, \dots, A_n$ the associated idempotent algebras. Then the $A_i$* cover *$D((R, \mathbb{Z})^\square)$:*

$$M \in D\big((R,\mathbb{Z})^\square\big), \quad M \otimes A_i = 0 \ \forall i \implies M = 0, \tag{11.12}$$

*and consequently $D((R, \mathbb{Z})^\square)$ is recovered as the totalization of the associated Čech diagram*

$$D\big((R,\mathbb{Z})^\square\big) \ = \ \varprojlim\Big( \textstyle\prod_i \operatorname{Mod}_{A_i} \ \rightrightarrows \ \prod_{i<j} \operatorname{Mod}_{A_i \otimes A_j} \ \substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow} \ \cdots \Big).$$

*Idea.* It suffices to show that $R$ is generated, under the triangulated operations, by $A_i$-modules for varying $i$. Consider the simplest cover $\operatorname{Spec} R = D(f) \cup Z(f)$. Here $R[\frac{1}{f}]$ is already an $A_{D(f)}$-module, while the fibre of $R \to R[\frac{1}{f}]$ is built from $R/f^n$'s and is an $R^\wedge_f$-module, hence lies over $Z(f)$. So $R$ is generated by an $A_{D(f)}$-module and an $A_{Z(f)}$-module; tensoring, one hits every object, and (11.12) follows. ☐

**Example 11.21** (Higher local fields)**.** Take $R = \mathbb{Z}[X, Y]$ and cover $\operatorname{Spec} R$ by three locally closed pieces – successively imposing $X \neq 0$; then $X = 0, \ Y \neq 0$; then $X = Y = 0$:

$$\begin{aligned} C_1 &= \{X \neq 0\}, & A_1 &= \mathbb{Z}[X^{\pm 1}, Y], \\ C_2 &= \{X = 0, \ Y \neq 0\}, & A_2 &= \mathbb{Z}[Y^{\pm 1}][\![X]\!], \\ C_3 &= \{X = Y = 0\}, & A_3 &= \mathbb{Z}[\![X, Y]\!]. \end{aligned} \tag{11.13}$$

The interesting object is the triple intersection. Tensoring $A_2$ with $A_1$ inverts $X$, giving $\mathbb{Z}[Y^{\pm 1}](\!(X)\!)$; tensoring further over $\mathbb{Z}[X, Y]$ (in $\mathrm{Solid}_{\mathbb{Z}}$) with $A_3 = \mathbb{Z}[\![X, Y]\!]$ inverts $Y$ as well, and one computes

$$A_{C_1 \cap C_2 \cap C_3} \ = \ \mathbb{Z}(\!(Y)\!)(\!(X)\!),$$

each such value obtained by iterating "complete, then invert". The order in which one completes and inverts famously matters, yet here it all arises from a manifestly commutative construction. These objects are the higher local fields that have been studied classically; as topological rings or fields they are notoriously ill-behaved, but in the condensed world they are perfectly well-behaved objects, arising from this natural, idempotent (even derived) and functorial procedure.[2]

[2] Clausen attributes to A. Mathew a paper documenting how badly higher local fields behave as topological rings; the precise reference was not given.

*Remark* 11.22 (Where the solid $p$-adic functional analysis is written up). The solid functional analysis underlying the "derived complete" facts used above (Remark 11.19) – Scholze's treatment from earlier in the course – is written down by Bosco, in an appendix on $p$-adic functional analysis from the solid perspective [Bos23] (the $p$-complete case), and in general by Mann in his thesis [Man22].

# 12 Liquid and Gaseous Analytic Rings, and the Tate Elliptic Curve

This lecture is a change of direction. So far the course has developed one notion of completeness – the solid theory (Sections 5 and 6) and its relative and Huber-pair variants (Sections 7 to 9) – which is well suited to nonarchimedean geometry and stays extremely close to Huber's theory of adic spaces. Scholze now takes the broader view: completeness is not to be fixed once and for all but is part of the datum of an *analytic ring* (Section 8), and one would like many more examples, in particular ones that accommodate archimedean geometry over $\mathbb{R}$ and $\mathbb{C}$, and ones that combine the archimedean and nonarchimedean worlds. The organising example is the *Tate elliptic curve*. Following its coefficients leads to a Banach ring $\mathbb{Z}(\!(q)\!)$ (with several natural norms), whose Berkovich spectrum glues nonarchimedean and archimedean pieces; the question of putting an analytic ring structure on it is answered by the *liquid* theory, giving $\alpha$-liquid structures on $\mathbb{R}$ and $\mathbb{Q}_p$; and the whole family is organised by a single initial "gaseous" analytic ring, produced by the course's new lightweight method of imposing endomorphisms of the projective generator $P$.

Throughout, "ring" means commutative ring and everything lives in light condensed abelian groups. We use freely the solid theory $\text{Solid} \subseteq \text{CondAb}^{\text{light}}$ (Sections 5 and 6), the abstract notion of analytic ring (Section 8), and its localization over the adic/valuative spectrum (Sections 9 and 10).

## 12.1 Where we are, and two questions

*Remark* 12.1 (Completeness as a relative notion). The point of departure was condensed mathematics as a framework for combining homological algebra and functional analysis, in which one wanted a notion of *complete* module. The solid modules over $\mathbb{Z}$, over any finite type $\mathbb{Z}$-algebra $R$ (Section 11), and over any Huber pair gave one such notion, closely tied to Huber's adic spaces. The conceptual lesson was that completeness should not be an absolute notion fixed once and for all, but a *relative* one: it is part of the datum of what an analytic ring is, namely the choice of which modules count as complete (Definition 8.8). The programme is then to take analytic rings as building blocks and glue them, as one glues affine schemes into schemes, to obtain analytic spaces – and within that world to locate the various classical notions of analytic space.

Before building the general gluing formalism, one wants more examples of analytic rings, and in particular a theory that is not only nonarchimedean. This raises two questions, which will organise the lecture.

*Remark* 12.2 (Two questions).

1. Does $\mathbb{R}$ carry a natural analytic ring structure, suitable for real and complex geometry?
2. Is there a good way to *combine* the nonarchimedean and archimedean theories into one?

The reals plausibly support both an archimedean geometry and, via solid modules, a nonarchimedean one; the substance of question (2) is whether the two can be made to sit inside a single framework. Scholze develops one route to both, guided by a single key example.

## 12.2 The Tate elliptic curve

**Definition 12.3** (Tate elliptic curve)**.** Over the Laurent series ring $\mathbb{Z}(\!(q)\!)$, the *Tate elliptic curve* $E_q$ is defined, in the language of adic spaces, as the quotient of the analytic multiplicative group by the subgroup generated by multiplication by $q$:

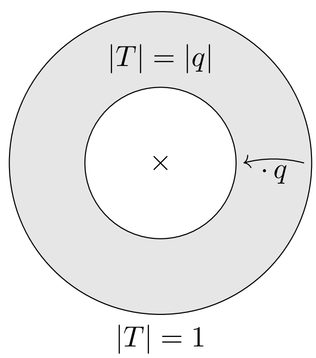
$$E_q \;=\; \mathbb{G}^{\mathrm{an}}_{m\,\mathbb{Z}(\!(q)\!)} \big/ \, q^{\mathbb{Z}},$$

together with its commutative group structure. Here $\mathbb{G}_m^{\mathrm{an}}$ is the analytic space over $\mathrm{Spa}(\mathbb{Z}(\!(q)\!), \mathbb{Z}[\![q]\!])$ whose points are the coordinates $T$ with

$$\big\{\, T \;\big|\; \exists n,\; |q|^n \le |T| \le |q|^{-n} \,\big\},$$

and $q^{\mathbb{Z}}$ acts by multiplication $T \mapsto qT$. Because $|q| < 1$, this action multiplies $|T|$ by $|q| \in (0,1)$; it is free and totally discontinuous, so the quotient is locally split and $E_q$ is a genuine analytic space.

The construction is Huber's; it appears already in [Hub94] as one of the first applications of adic spaces (there in the generality of Tate abelian varieties). Concretely, a fundamental domain for the $q^{\mathbb{Z}}$-action is an annulus, and $E_q$ is obtained by gluing its two boundary circles:




*Remark* 12.4 ($E_q$ is algebraic)*.* $E_q$ is proper (it is compact, obtained from the compact annulus by gluing), separated, and smooth (locally it is just $\mathbb{G}_m^{\mathrm{an}}$), hence a proper smooth connected curve of relative dimension 1 over $\mathbb{Z}(\!(q)\!)$. In any reasonable analytic geometry a proper smooth curve is automatically algebraic: the inverse of the ideal sheaf of a closed point (e.g. the identity section, which exists here) is an ample line bundle, and a Riemann–Roch type statement then produces enough functions to embed it. Thus $E_q$ is the analytification of an honest elliptic curve over the abstract ring $\mathbb{Z}(\!(q)\!)$.

$$E_q \text{ is a proper smooth curve over } \mathbb{Z}(\!(q)\!),\ \text{hence an elliptic curve.}$$

Scholze noted that the global algebraicity in this argument implicitly uses a GAGA statement over a nice noetherian base, and that over a more general base Spa of a Huber pair the passage from a local (formal-model) ample bundle to a global one is not obviously automatic; here the identity section makes it work.

**Proposition 12.5** (Weierstrass equation)**.** *Removing the origin, $E_q$ has the Weierstrass form*

$$E_q \setminus \{0\}: \qquad y^2 + xy \;=\; x^3 - a_4 x - a_6,$$

*where (having inverted* 2 *to eliminate one term)*

$$a_4 \;=\; \sum_{n\ge1} 5\,n^3\,\frac{q^n}{1-q^n}, \qquad a_6 \;=\; \sum_{n\ge1} \frac{7n^5+5n^3}{12}\,\frac{q^n}{1-q^n}$$

*are power series in q.*[3]

*Remark* 12.6 (Polynomial growth of the coefficients). Expanding $\frac{1}{1-q^n} = 1 + q^n + q^{2n} + \cdots$ and collecting, one may write $a_4 = \sum_N c_N q^N$ (and likewise $a_6$); what struck Scholze on first seeing these formulas is that the resulting coefficients $c_N$ have at worst *polynomial* growth in $N$. This has two consequences.

- One may specialise $q$ to any $q \in \mathbb{C}$ with $0 < |q| < 1$, obtaining an actual complex torus $\mathbb{C}^*/q^{\mathbb{Z}}$; the analytic description $\mathbb{G}_m/q^{\mathbb{Z}}$ makes literal sense there. So the formal object specialises to any intermediate radius, not just to a formal neighbourhood.
- Convergence for $0 < |q| < 1$ is equivalent to *subexponential* growth of the coefficients:

$$\text{convergence for } 0 < |q| < 1 \quad \Longleftrightarrow \quad \text{subexponential growth of coefficients.}$$

Polynomial growth is strictly stronger than subexponential. There is a clear geometric reason the coefficients must grow subexponentially (the curve is defined over the open unit disk), but it is not clear how to see the sharper polynomial bound geometrically. Pinning down the smallest ring over which $E_q$ lives – one enforcing polynomial growth – is exactly what the “gaseous” structure of Theorem 12.22 will achieve.

### 12.3 Berkovich spectra and the geometry of $\mathbb{Z}((q))$

We now study the geometry of $\mathrm{Spa}(\mathbb{Z}((q)), \mathbb{Z}[\![q]\!])$, the adic spectrum of continuous valuations. The topologically nilpotent unit $q$ is a convenient handle: it lets one gauge the size of any function by comparison with $|q|$. In particular there is a unique order-preserving $|q| \mapsto \frac{1}{2}$ map $\Gamma \cup \{0\} \to \mathbb{R}_{\geq 0}$ out of the value group, so each continuous valuation acquires a real-valued absolute value. For most valuations this is injective, the exception being certain rank-2 valuations carrying a little extra infinitesimal information not remembered by the real value. This gives a comparison map to the Berkovich spectrum.

**Definition 12.7** (Berkovich spectrum)**.** For a Banach ring $R$, the *Berkovich spectrum* is

$$\mathcal{M}^{\mathrm{Berk}}(R) \;=\; \Big\{ |\cdot|\colon R \to \mathbb{R}_{\geq 0} \;\Big|\; |0| = 0,\ |1| = 1,\ |xy| = |x||y|,\ |x+y| \leq |x| + |y|,\ |x| \leq \|x\|_R \Big\},$$

the set of multiplicative seminorms bounded by the Banach norm $\|\cdot\|_R$, topologised as a subspace of $\prod_{x\in R}[0, \|x\|_R]$. Since each factor is a compact interval and all the defining conditions are closed, $\mathcal{M}^{\mathrm{Berk}}(R)$ is a compact Hausdorff space.

Note that Berkovich allows the ordinary triangle inequality $|x + y| \leq |x| + |y|$, not only the strong (nonarchimedean) one, so the theory is not restricted to the nonarchimedean setting.

**Proposition 12.8** (Adic spectrum as maximal Hausdorff quotient)**.** *The comparison map*

$$\mathrm{Spa}\big(\mathbb{Z}((q)), \mathbb{Z}[\![q]\!]\big) \;\longrightarrow\; \mathcal{M}^{\mathrm{Berk}}\big(\mathbb{Z}((q))\big)$$

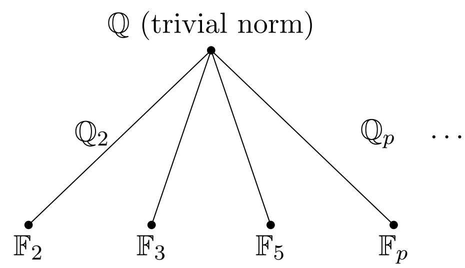
*exhibits the target as the* maximal Hausdorff quotient *of the source. The only difference between the two is the presence of higher-rank valuations in the adic spectrum, which produce very infinitesimal changes; up to those, the two spaces coincide.*

[3] Copied by Scholze from Wikipedia. The board index reads $n \geq 0$, but the $n = 0$ term is ill-defined, so the intended range is $n \geq 1$. With that range these are the standard Tate-curve coefficients: writing $s_k(q) = \sum_{n\geq 1} \frac{n^k q^n}{1-q^n}$, one has $a_4 = 5s_3(q)$ and $a_6 = \frac{7s_5(q)+5s_3(q)}{12}$, so that $y^2 + xy = x^3 - 5s_3\, x - \frac{5s_3+7s_5}{12}$ (cf. Silverman, *Advanced Topics in the Arithmetic of Elliptic Curves*, Ch. V).

The base of both spaces is the Berkovich spectrum of $\mathbb{Z}$, which – in contrast to the nearly point-like scheme-theoretic spectrum – is a genuinely interesting space. Here $\mathbb{Z}$ carries the norm with $\|0\|_{\mathbb{Z}} = 0$ and $\|n\|_{\mathbb{Z}} = 1$ for $n \neq 0$ (every integer lies in the unit ball), the nonarchimedean choice.

**Proposition 12.9** ($\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$)**.** *With the norm $\|n\|_{\mathbb{Z}} \in \{0, 1\}$, the Berkovich spectrum $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$ is a tree: a generic point (the trivial, "0-or-1", absolute value) from which, for each prime $p$, a ray emanates – an interval of $p$-adic absolute values $\|\cdot\|_p^{\alpha}$, $\alpha \in (0, \infty)$ – ending at a closed point $\mathbb{F}_p$ (the trivial absolute value on $\mathbb{F}_p$). The points correspond to maps to complete Banach fields:*

$$\mathbb{Q} \text{ (trivial abs. value)}, \qquad \mathbb{Q}_p \text{ (abs. value with } |p| \in (0,1)), \qquad \mathbb{F}_p \text{ (trivial abs. value)}.$$




*Remark* 12.10 (Fibres over $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$). The space $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z}(\!(q)\!))$ fibres over $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$, and one can describe the fibres. The statement is essentially Ostrowski's theorem, which determines the multiplicative seminorms once $|q|$ is fixed.

- Over the closed point $\mathbb{F}_p$, the fibre is a single point: the base change is $\mathbb{F}_p(\!(q)\!)$, already a nonarchimedean field with $|q|$ fixed to $\frac{1}{2}$.
- Over the ray of $\mathbb{Q}_p$-absolute values, the fibre is a *punctured open unit disk over* $\mathbb{Q}_p$. The map to the ray is a radius map. Its generic point is $\mathbb{Q}(\!(q)\!)$; the fibre over a fixed point on the ray is an annulus (a locus where $|q|$ is fixed).

Geometrically, $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z}(\!(q)\!))$ has one punctured open unit disk for each place; the pieces are glued so that moving *towards the centre* of a disk lands at the common generic point $\mathbb{Q}(\!(q)\!)$, while moving *towards the boundary* lands at the corresponding $\mathbb{F}_p(\!(q)\!)$. Thus $\mathbb{F}_p(\!(q)\!)$ sits near the boundary of the annulus, and as one moves inward along the ray towards $\mathbb{Q}$ the annuli move towards the origin.

## 12.4 An archimedean variant of $\mathbb{Z}(\!(q)\!)$

The tree picture suggests enlarging $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$ by allowing the *usual* archimedean norm on $\mathbb{Z}$ in addition to the nonarchimedean one. Then, besides the $p$-adic rays, there is an archimedean ray

$$\mathbb{Z} \to \mathbb{R} \xrightarrow{|\cdot|^{\alpha}} \mathbb{R}_{\geq 0}, \qquad \alpha \in (0, 1],$$

where $|\cdot|$ is the usual absolute value of $\mathbb{R}$. Unlike the $p$-adic rays (parametrised by $\alpha \in (0, \infty)$), the archimedean ray stops at $\alpha = 1$: raising the usual absolute value to a power $\alpha > 1$ violates the triangle inequality. So the real place "stops in the middle" compared to the others.

**Definition 12.11** (The Banach ring $\mathbb{Z}(\!(q)\!)_{1/2}$)**.** Endow the ring $\mathbb{Z}[q^{\pm 1}]$ with the norm

$$\Big| \sum_{n \in \mathbb{Z}} a_n q^n \Big| \;=\; \sum_n |a_n| \, \frac{1}{2^n}$$

(usual absolute value on $\mathbb{Z}$, and $|q| = \frac{1}{2}$) and complete it to a Banach ring

$$\mathbb{Z}(\!(q)\!)_{1/2} \;=\; \Big\{\, f = \sum_n a_n q^n \;\Big|\; \sum_n |a_n|\, \tfrac{1}{2^n} < \infty \,\Big\}.$$

Its elements converge for $q \in \mathbb{C}$ with $0 < |q| \le \frac{1}{2}$, and each defines a holomorphic function of $q$ on that region, continuous up to the boundary.

The Berkovich spectrum $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z}(\!(q)\!)_{1/2})$ has, at its archimedean part, a genuinely complex-analytic piece: the disk

$$\big\{\, q \in \mathbb{C} \;\big|\; 0 < |q| \le \tfrac{1}{2} \,\big\} \big/ \operatorname{Gal}(\mathbb{C}/\mathbb{R}) \;\longrightarrow\; \mathcal{M}^{\mathrm{Berk}}\big(\mathbb{Z}(\!(q)\!)_{1/2}\big),$$

where one quotients by complex conjugation because conjugate values give the same absolute value. This is the picture in which one would like to combine archimedean and nonarchimedean geometry: parts of the space are literally real- or complex-analytic, others nonarchimedean, all inside one space. The one awkward feature is that fixing $|q| = \frac{1}{2}$ makes the archimedean punctured disk stop at radius $\frac{1}{2}$, although from the point of view of the Tate curve there was no reason to stop before $|q| = 1$.

## 12.5 Free complete modules, and the liquid structure

*Remark* 12.12 (The question, and the answer). Can one endow $\mathbb{Z}(\!(q)\!)_{1/2}$ with a natural analytic ring structure? The answer is yes: the *liquid* analytic ring structure [CS20]. It is slightly better to work with

$$\mathbb{Z}(\!(q)\!)_{>1/2} \;=\; \bigcup_{r>1/2} \mathbb{Z}(\!(q)\!)_r,$$

functions converging on some disk of radius slightly larger than $\frac{1}{2}$. This is Scholze's answer to question (2) of Remark 12.2: the liquid structure combines the archimedean and nonarchimedean settings. This was very much on Clausen and Scholze's minds when they first found the solid theory and tried to go further; the eventual answer was the liquid construction.

The 2019 mindset for producing an analytic ring structure is to *describe the free complete modules.* For a light profinite set $S = \varprojlim_i S_i \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$ we compare three descriptions of the free module.

**Example 12.13** (Solid and abstract free modules)**.** For $S = \varprojlim_i S_i$:

1. The *solid* free module is the inverse limit

   $$\mathbb{Z}[S]^{\square} \;\xrightarrow{\sim}\; \varprojlim_i \mathbb{Z}[S_i] \;=\; \underline{\mathrm{Hom}}\big(\mathrm{Cont}(S,\mathbb{Z}),\mathbb{Z}\big),$$

   the $\mathbb{Z}$-valued measures on $S$ (Remark 5.24): a measure assigns an integer to each clopen subset, additively.

2. The *abstract* (uncompleted) free module sits inside it as the $\ell^0$-bounded part,

   $$\mathbb{Z}[S] \;\xrightarrow{\cong}\; \bigcup_{n>0} \varprojlim_i \; \mathbb{Z}[S_i]_{\ell^0 \le n}, \tag{12.1}$$

   where $\mathbb{Z}[S_i]_{\ell^0\le n}$ consists of the sums of at most $n$ basis elements (with multiplicity). Here Scholze's $\ell^0$-norm counts the total number of basis vectors used; on a discrete finite free group it agrees with $\ell^1$, but he prefers to call it $\ell^0$.

The liquid free module is a third possibility, interpolating between the uncompleted and solid ones by a convergence (norm) condition.

**Definition 12.14** (Liquid free module)**.** Endow each $\mathbb{Z}(\!(q)\!)_r[S_i]$ with the norm

$$\Big\| \sum_{\substack{n\in\mathbb{Z}\\ s\in S_i}} a_{n,s}\, q^n[s] \Big\| \;=\; \sum_{n,s} \|a_{n,s}\|\, r^n.$$

Then the free liquid module on $S$ over $\mathbb{Z}(\!(q)\!)_{>1/2}$ is

$$\mathbb{Z}(\!(q)\!)_{>1/2}[S]^{\mathrm{liq}_q} \;=\; \bigcup_{r>1/2}\,\bigcup_{c>0}\, \varprojlim_i \;\mathbb{Z}(\!(q)\!)_r[S_i]_{\|\cdot\|\le c}. \tag{12.2}$$

Once the radius $r$ and the size bound $c$ are fixed, the inverse limit of the bounded parts is compact Hausdorff, so the whole is a well-behaved condensed set. The hard theorem [CS20] is that this proposal really satisfies the axioms of an analytic ring; the description (12.2) is what suggests itself as soon as one asks for a notion of complete module over this Banach ring.

## 12.6 Analytic ring structures on $\mathbb{R}$

Answering question (1) of Remark 12.2, one specialises the liquid ring to the reals. Send $q \mapsto t$ with $0 < t < \frac{1}{2}$; the map $\mathbb{Z}(\!(q)\!)_{>1/2} \to \mathbb{R}$ defines a point of the archimedean Berkovich space lying over $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$, hence over some power $|\cdot|^\alpha$ of the usual absolute value of $\mathbb{R}$. The value of $t$ determines $\alpha \in (0,1]$ by the relation

$$t^\alpha = \tfrac{1}{2},$$

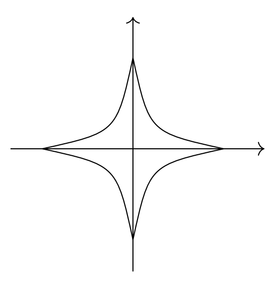
whose meaning is precisely that the point maps to the parameter $\alpha$ on the archimedean ray of $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$.

**Definition 12.15** ($\alpha$-liquid ring structure on $\mathbb{R}$)**.** For $0 < \alpha \le 1$ one obtains the *$\alpha$-liquid* analytic ring structure on $\mathbb{R}$, whose complete modules are forced by requiring completeness only after forgetting the linear structure. Its free modules are

$$\mathbb{R}[S]^{\alpha\text{-liq}} \;=\; \bigcup_{\beta<\alpha}\,\bigcup_{c>0}\, \varprojlim_i \;\mathbb{R}[S_i]_{\ell^\beta\le c}, \qquad \Big\| \sum_i x_i[s_i] \Big\|_{\ell^\beta} \;=\; \sum_i |x_i|^\beta, \quad 0<\beta<1. \tag{12.3}$$

The union over $\beta < \alpha$ is filtered, the bounded loci $\{\|\cdot\|_{\ell^\beta} \le c\}$ growing as $\beta$ increases.

*Remark* 12.16 (Non-local convexity). For $0 < \beta < 1$ the quantity $\|\cdot\|_{\ell^\beta}$ satisfies the triangle inequality but not the usual scaling axiom of a norm (it scales by the $\beta$-th power); it is not a norm in the standard sense. The $\ell^\beta$ "balls" are *not convex*: in two dimensions the unit ball is a four-pointed concave star. These $\ell^\beta$ with $\beta < 1$ sit to the left of the usual locally convex $\ell^p$ ($1 \le p \le \infty$), and the liquid theory is forced to use such non-locally-convex spaces.



*Remark* 12.17 (Warning: Radon measures do not work). The naive guess is that the complete modules over $\mathbb{R}$ should be the Radon measures, i.e. that the free module should be

$$\mathbb{R}[S]^{\wedge} \;:=\; \mathcal{M}^{\mathrm{Radon}}(S) \;=\; \underline{\mathrm{Hom}}\big(\mathrm{Cont}(S,\mathbb{R}),\mathbb{R}\big).$$

This *does not* define an analytic ring structure. The reason is that complete modules must be stable under extensions, and locally convex topological vector spaces are not: there is a natural extension of a Banach space by $\mathbb{R}$ (a one-dimensional space) – i.e. with $\mathbb{R}$ as the subspace – whose total space is not locally convex. Scholze attributed this three-space phenomenon to Ribe, arising from the entropy functional; it is a meaningful object, not a pathology, but it destroys the Radon-measure option.[4] Once one allows non-locally-convex spaces into the picture the theory works again – but one is forced into them.

The naive Radon-measure structure is really the opposite endpoint: taking the $\ell^1$ (locally convex) construction reproduces the Radon measures, which are related to the classical theory of complete locally convex complex vector spaces in the same way that solid modules are related to the classical theory of complete (pro-discrete) modules.

### 12.7 Analytic ring structures on $\mathbb{Q}_p$

One may instead specialise $\mathbb{Z}((q))_{>1/2} \to \mathbb{Q}_p$. Naively one might expect nothing new at the nonarchimedean places – that the extra archimedean convergence condition is invisible there and one simply recovers the solid structure. This is false: one gets genuinely new structures.

**Definition 12.18** ($\alpha$-liquid ring structure on $\mathbb{Q}_p$)**.** Specialising to $\mathbb{Q}_p$ (sending $q$ to a suitable power of $p$, so that any $\alpha$ is allowed), one obtains for every $0 < \alpha < \infty$ – i.e. for every point of the $p$-adic ray of $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$ – an *$\alpha$-liquid* analytic ring structure on $\mathbb{Q}_p$, with free modules

$$\mathbb{Q}_p[S]^{\alpha\text{-liq}} \;=\; \bigcup_{\beta<\alpha}\,\bigcup_{c>0}\,\varprojlim_i\; \mathbb{Q}_p[S_i]_{\ell^\beta\le c},$$

using the $\ell^\beta$-norm of (12.3). This should be compared with the solidification, which is the $\ell^\infty$ (i.e. $\alpha=\infty$) endpoint:

$$\mathbb{Q}_p[S]^{\square} \;=\; \bigcup_{c>0}\,\varprojlim_i\; \mathbb{Q}_p[S_i]_{\ell^\infty\le c}.$$

So even over $\mathbb{Q}_p$, where the solid theory already worked well, there is a whole new world of analytic ring structures, one for each $\alpha \in (0,\infty)$.

### 12.8 The spectrum of structures, and the gaseous ring

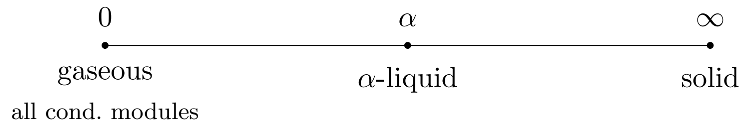
*Remark* 12.19 (Solid, liquid, gaseous). The various structures organise into a family parametrised by $\alpha \in [0,\infty]$:

$$\underset{\text{all condensed modules}}{\text{gaseous }(0)} \qquad \cdots\; \underset{\alpha}{\alpha\text{-liquid}} \;\cdots\; \underset{\infty}{\text{solid}}. \tag{12.4}$$

The class of complete modules *grows* as $\alpha$ decreases: being solid is a much stricter condition than being $\alpha$-liquid. At the endpoint $\alpha = 0$ one recovers *all* condensed modules – a naive $\ell^0$-condition (a

[4] The transcript's date ("around 1918") is certainly a mishearing. The non-locally-convex three-space example is due to M. Ribe, *Examples for the nonlocally convex three space problem*, Proc. Amer. Math. Soc. **73** (1979), 351–355: a nontrivial twisted sum $0 \to \mathbb{R} \to X \to \ell^1 \to 0$ built from the entropy function $t \mapsto t\log t$. The general twisted-sum method is due to Kalton–Peck (1979).

bound on the number of nonzero coefficients) reproduces all condensed modules, by a variant of the description in (12.1). The three regimes are Scholze's "solid / liquid / gaseous" states of matter.




*Remark* 12.20 (The course's method: impose endomorphisms of $P$). The 2019 mindset (describe the free complete modules, then prove the axioms by hand) is replaced in this course by a lighter method. To define an analytic ring structure one instead finds natural endomorphisms of the compact projective generator

$$P \;=\; A[\mathbb{N} \cup \{\infty\}]/[\infty]$$

– the free complete module on a null sequence – that one *wants* to become isomorphisms. Because in light condensed abelian groups $P$ is compact and internally projective (Theorem 3.15 and Definition 5.5), imposing such isomorphisms automatically defines an analytic ring structure. This makes new structures very easy to produce.

*Remark* 12.21 (What the Tate curve needs). To carry out the construction of the Tate elliptic curve one needs only two properties, both with a clear meaning in terms of the underlying condensed ring:

1. $q$ is a topologically nilpotent unit – i.e. the map from the free ring on a null sequence, sending the generator $1 \mapsto q$, is available; and
2. $1 - q \cdot \text{shift}\colon P \to P$ is an isomorphism.

There is an evident universal (initial) example: take the free ring on a topologically nilpotent unit and impose (2).

**Theorem 12.22** (The gaseous analytic ring)**.** *There is an initial analytic ring $A = (A^{\triangleright}, \mathrm{Mod}_A)$ with a topologically nilpotent unit $q$ and with* $1 - q \cdot \text{shift}$ *an automorphism of $P$ – the* gaseous *analytic ring structure. Its underlying ring is the ring of Laurent series with polynomially bounded coefficients,*

$$A^{\triangleright}(*) \;=\; \Big\{ \sum_n a_n q^n \in \mathbb{Z}(\!(q)\!) \;\Big|\; (a_n) \text{ has at most polynomial growth} \Big\}, \tag{12.5}$$

*and its free modules on $S = \varprojlim_i S_i$ are*

$$A[S] \;=\; \bigcup_{k>0} \bigcup_{n>0} \varprojlim_i \; \mathbb{Z}(\!(q)\!)[S_i], \tag{12.6}$$

*where the coefficient of $q^m$ is required to have $\ell^0$-norm at most $m + m^k$ (polynomial growth of degree $k$ in the exponent $m$, the summand $m$ allowing negative Fourier coefficients).*[5]

*Remark* 12.23 (The Tate curve over the smallest ring). Doing analytic geometry over the gaseous ring, one can repeat the construction of the Tate curve and see, for geometric reasons, that it lives over exactly this smallest ring (12.5) – the one enforcing the polynomial growth observed in Remark 12.6, rather than merely subexponential growth. Specialising back, the gaseous analytic

[5]Scholze wavered between $\ell^0$ and $\ell^1$ for the per-degree bound; the board reads $\ell^0$, and on a discrete free module the two agree (Example 12.13), so the distinction is immaterial here. The precise description, deferred at this point, is worked out in Section 14: there the coefficient of $q^m$ is required to lie in the $\ell^1$-ball $\mathbb{Z}[S_i]_{\leq (n+m)^k}$ (Theorem 14.3).

ring sits close to, but not quite at, the $\alpha = 0$ end of the spectrum (12.4). Base change of the free-module description (12.6) to $\mathbb{Q}_p$ is a somewhat nasty but instructive exercise, left for a future lecture.

# 13 Generalities on Analytic Rings: Completion, Colimits, and Frobenius

Having spent the previous lectures building the solid theory and its Huber-pair variants, and having seen Scholze introduce the liquid and gaseous structures that also work archimedeanly (Section 12), Clausen now steps back to develop some general operations on the *category* of analytic rings. The immediate motivation is that manipulating the new (liquid, gaseous, ...) examples is much easier with a good handle on how analytic rings are constructed, mapped, and glued; but the material should be recorded in any case. There are three parts.

- A *completion* functor turning a "pre-analytic ring" (the axioms of an analytic ring minus the requirement that the unit be complete) into an honest analytic ring, and its derived ($\mathbb{E}_\infty$/animated) enhancement.
- *Colimits* of analytic rings: computed by taking the colimit of pre-analytic rings and completing. Filtered colimits need no completion; pushouts (completed tensor products) are the substantive case.
- A theorem that *Frobenius is a map of analytic rings*, via a free-module criterion for maps of analytic rings and a Tate-cohomology computation that is special to the light setting. As a payoff one gets derived symmetric powers on the derived category, hence a good theory of animated analytic rings.

The lecture closes with a general "killing an algebra object" construction producing left adjoints to reflective subcategories, which uniformly recovers solidification, localization, and completion.

Throughout, "ring" means commutative ring, everything lives in light condensed abelian groups, and we use freely the abstract notion of analytic ring and its derived category (Section 8), the solid theory (Sections 5 and 6), and its localization over the (valuative) spectrum (Sections 9 and 10). We write $R^{\triangleright}$ for the underlying condensed ring of an analytic ring $R$ (the "triangle" notation of Sections 1 and 8).

## 13.1 Analytic and pre-analytic rings

We first recall the definition in the form Clausen will use, then relax it.

**Definition 13.1** (Analytic ring)**.** An *analytic ring* is a pair $(R^{\triangleright}, \mathrm{Mod}_R)$ where $R^{\triangleright}$ is a condensed (commutative) ring and

$$\mathrm{Mod}_R \;\subseteq\; \mathrm{Mod}_{R^{\triangleright}}$$

is a full subcategory of condensed $R^{\triangleright}$-modules, closed under limits, colimits and extensions, such that

$$\underline{\mathrm{Ext}}^i_{R^{\triangleright}}(M, N) \;\in\; \mathrm{Mod}_R \qquad \text{for all } N \in \mathrm{Mod}_R,\ M \in \mathrm{Mod}_{R^{\triangleright}},\ i \geq 0, \tag{13.1}$$

and such that, finally,

$$R^{\triangleright} \;\in\; \mathrm{Mod}_R\,. \tag{$\star$}$$

We call $(\star)$ the *completeness* axiom.

**Definition 13.2** (Pre-analytic ring)**.** A *pre-analytic ring* is a pair $(R^\triangleright, \mathrm{Mod}_R)$ as in Definition 13.1 except that we drop $(\star)$: the unit object $R^\triangleright$ need not be complete.

*Remark* 13.3 (Why relax the definition). The completeness axiom $(\star)$ is exactly the clause that is awkward to verify directly on an interesting condensed ring $R^\triangleright$: it forces one to control Ext-groups in $\mathrm{Mod}_{R^\triangleright}$, i.e. to understand the homological algebra of condensed $R^\triangleright$-modules before any computation can begin – hard, for instance, for a liquid base like a ring of convergent Laurent series. In practice one instead maps a *simpler* ring into $R^\triangleright$ (e.g. $\mathbb{Z}[T, T^{-1}]$ for the Laurent case) on which a pre-analytic structure is easy to write down, and then *completes*. We saw the same phenomenon for adic spaces: the module category attached to a rational open was first specified by imposing conditions – a pre-analytic ring – and one then had to change $R^\triangleright$ to (essentially) the value of Huber's structure sheaf to make $(\star)$ hold. The completion procedure below is what carries out this last step.

**Definition 13.4** (Maps)**.** A *map of pre-analytic rings* $(R^\triangleright, \mathrm{Mod}_R) \to (S^\triangleright, \mathrm{Mod}_S)$ is a map of underlying condensed rings $R^\triangleright \to S^\triangleright$ such that restriction of scalars sends $\mathrm{Mod}_S$ into $\mathrm{Mod}_R$: for $N \in \mathrm{Mod}_S$, the underlying $R^\triangleright$-module lies in $\mathrm{Mod}_R$. Equivalently, writing $\widehat{(-)}_R\colon \mathrm{Mod}_{R^\triangleright} \to \mathrm{Mod}_R$ and $\widehat{(-)}_S\colon \mathrm{Mod}_{S^\triangleright} \to \mathrm{Mod}_S$ for the completion (localization) functors, the condition says that base change followed by $S$-completion descends to a functor

$$- \otimes_R S : \ \mathrm{Mod}_R \ \longrightarrow \ \mathrm{Mod}_S,$$

which is then automatically symmetric monoidal. (Concretely, this uses the universal property of the localization $\widehat{(-)}_R$: a map that becomes an isomorphism after $R$-completion also becomes one after $S$-completion.)

Since analytic rings are pre-analytic rings, there is a fully faithful inclusion

$$\{\text{analytic rings}\} \ \hookrightarrow \ \{\text{pre-analytic rings}\}. \tag{13.2}$$

## 13.2 Completion

**Proposition 13.5** (Completion is a left adjoint)**.** *The inclusion* (13.2) *has a left adjoint. On objects it sends a pre-analytic ring* $(R^\triangleright, \mathrm{Mod}_R)$ *to*

$$(R^\triangleright, \mathrm{Mod}_R) \ \longmapsto \ (\widehat{R^\triangleright}, \mathrm{Mod}_R),$$

*where* $\widehat{R^\triangleright}$ *is the completion of the unit – the image of* $R^\triangleright$ *under the left adjoint* $\widehat{(-)}_R\colon \mathrm{Mod}_{R^\triangleright} \to \mathrm{Mod}_R$ *to the inclusion – and* $\mathrm{Mod}_R$ *is the same module category, now viewed inside* $\mathrm{Mod}_{\widehat{R^\triangleright}}$.

*Proof sketch.* There are three things to check: that $\widehat{R^\triangleright}$ is a condensed ring; that $\mathrm{Mod}_R$ is a full subcategory of $\mathrm{Mod}_{\widehat{R^\triangleright}}$; and that $(\widehat{R^\triangleright}, \mathrm{Mod}_R)$ satisfies the axioms of a (pre-)analytic ring, including $(\star)$.

$\widehat{R^\triangleright}$ *is a commutative condensed ring.* This is completely formal. The completion functor $\widehat{(-)}_R\colon \mathrm{Mod}_{R^\triangleright} \to \mathrm{Mod}_R$ is symmetric monoidal and colimit-preserving, and its right adjoint is the inclusion $\mathrm{Mod}_R \hookrightarrow \mathrm{Mod}_{R^\triangleright}$. The right adjoint of a symmetric monoidal functor is lax symmetric monoidal, hence carries the unit object of $\mathrm{Mod}_R$ to a commutative algebra object of $\mathrm{Mod}_{R^\triangleright}$; that algebra is precisely $\widehat{R^\triangleright}$. A fortiori $\widehat{R^\triangleright}$ is a condensed commutative ring. By the same formal fact, the right adjoint sends any object of $\mathrm{Mod}_R$ to a module over $\widehat{R^\triangleright}$, so the inclusion factors

$$\mathrm{Mod}_R \ \hookrightarrow \ \mathrm{Mod}_{\widehat{R^\triangleright}} \ \longrightarrow \ \mathrm{Mod}_{R^\triangleright},$$

the second map being restriction of scalars along $R^\triangleright \to \widehat{R^\triangleright}$.

$$\begin{tikzcd} \mathrm{Mod}_R \arrow[rr, hook] \arrow[dr] & & \mathrm{Mod}_{R^\triangleright} \\ & \mathrm{Mod}_{\widehat{R^\triangleright}} \arrow[ur] & \end{tikzcd}$$

$\mathrm{Mod}_R \hookrightarrow \mathrm{Mod}_{\widehat{R^\triangleright}}$ *is fully faithful.* For $M, N \in \mathrm{Mod}_R$ one compares $\mathrm{Hom}_{R^\triangleright}(M, N)$ with $\mathrm{Hom}_{\widehat{R^\triangleright}}(M, N)$. Since $N$ is an $\widehat{R^\triangleright}$-module,

$$\mathrm{Hom}_{\widehat{R^\triangleright}}(M, N) = \mathrm{Hom}_{\widehat{R^\triangleright}}\left(\widehat{R^\triangleright} \otimes_{R^\triangleright} M,\, N\right);$$

as $N$ is complete one may apply $R$-completion inside, and, completion being symmetric monoidal and idempotent, this collapses to $\mathrm{Hom}_{\widehat{R^\triangleright}}(M, N) = \mathrm{Hom}_{R^\triangleright}(M, N)$. Hence $\mathrm{Mod}_R \hookrightarrow \mathrm{Mod}_{\widehat{R^\triangleright}}$ is full and faithful.

*The axioms.* Closure under limits and colimits is automatic since the forgetful functor $\mathrm{Mod}_{\widehat{R^\triangleright}} \to \mathrm{Mod}_{R^\triangleright}$ preserves them. The completeness axiom $(\star)$ holds by construction: $\widehat{R^\triangleright} \in \mathrm{Mod}_R$. Closure under extensions and the internal-Ext axiom (13.1) are again Ext-computations; the fully-faithful argument upgrades to them, but only *provided the derived completion of* $R^\triangleright$ *agrees with the underived completion* $\widehat{R^\triangleright}$ – see Remark 13.6. ☐

*Remark* 13.6 (Why one is forced into the derived setting)*.* The Ext-part of Proposition 13.5 is subtle. The derived completion $\widehat{R^\triangleright}$ need not sit in degree 0: its $H_0$ is the underived completion, but it can carry higher homology (this is exactly the phenomenon behind the non-sheafy adic spaces of Section 10). The formal argument for the Ext-axioms goes through cleanly *if* one replaces $\widehat{R^\triangleright}$ by its derived version and uses a notion of *derived analytic ring.* So it is convenient to introduce that notion in the middle of this proof.

## 13.3 Derived analytic rings and the abelian comparison

**Definition 13.7** (Analytic $\mathbb{E}_\infty$-ring)**.** An *analytic* $\mathbb{E}_\infty$*-ring* (or *derived analytic ring*) is a pair $(R^\triangleright, D_{\geq 0}(R))$ where $R^\triangleright$ is a condensed connective $\mathbb{E}_\infty$-ring and

$$D_{\geq 0}(R) \;\subseteq\; D_{\geq 0}(R^\triangleright)$$

is a full subcategory (of the connective part of the derived category), closed under all limits and colimits, such that

$$\underline{\mathrm{Hom}}_{R^\triangleright}(M, N) \;\in\; D_{\geq 0}(R) \qquad \text{for } N \in D_{\geq 0}(R),\; M \in D_{\geq 0}(R^\triangleright).$$

It is a *pre-analytic* $\mathbb{E}_\infty$-ring if we do not require the unit; an (honest) *analytic* one if in addition $R^\triangleright \in D_{\geq 0}(R)$. Only the connective part is specified; note that closure under the loop functor in $D_{\geq 0}$ – shift down and truncate – forces the homology-wise description below.

**Proposition 13.8** (Derived vs. abelian structures)**.** *For any connective condensed* $\mathbb{E}_\infty$*-ring* $R^\triangleright$ *there is a bijection*

$$\{\textit{pre-analytic ring structures on } R^\triangleright\} \\ \xrightarrow{\sim} \{\textit{pre-analytic ring structures on } \pi_0 R^\triangleright \ (\textit{old, abelian sense})\}.$$

*In the forward direction one restricts to the heart,*

$$D_{\geq 0}(R) \longmapsto D_{\geq 0}(R) \cap D_{=0}(R^{\triangleright}) = D_{=0}(\pi_0 R^{\triangleright}), \tag{13.3}$$

*and in the backward direction one takes homology termwise,*

$$\mathrm{Mod}_R \longmapsto \{M \in D_{\geq 0}(R^{\triangleright}) : H_i M \in \mathrm{Mod}_R \ \forall\, i \geq 0\}. \tag{13.4}$$

*An analytic $\mathbb{E}_\infty$-ring structure maps to an analytic (abelian) one; the converse holds when $R^{\triangleright} = \pi_0 R^{\triangleright}$, and in general the analytic structures on $R^{\triangleright}$ correspond to those abelian structures on $\pi_0 R^{\triangleright}$ for which every homotopy group $\pi_i R^{\triangleright}$ (not only $\pi_0$) is complete.*

*Remark* 13.9 (Consistency in the classical case). When $R^{\triangleright} = \pi_0 R^{\triangleright}$ is concentrated in degree 0 (a classical ring), the two notions overlap and Proposition 13.8 says they agree: the derived definition of a (pre-)analytic ring structure matches the naive abelian-level definition of Section 8.

Proposition 13.8 finishes the proof of Proposition 13.5. There is no obstruction to proving the completion proposition in the derived setting (Remark 13.6); one then transports the resulting analytic-ring structure on the derived $\widehat{R^{\triangleright}}$ down to its $\pi_0$ by the recipe (13.3). The proof of Proposition 13.8 itself was given in an earlier iteration of the course [CS20].

*Remark* 13.10 (Warning: the derived category can change). An analytic ring structure on $\pi_0 R^{\triangleright}$ produces both an abelian category and, via (13.4), a derived category. If $(\widehat{R^{\triangleright}}, \mathrm{Mod}_R)$ is the (abelian) completion of $(R^{\triangleright}, \mathrm{Mod}_R)$ from Proposition 13.5, it is *not* in general true that

$$D(\widehat{R^{\triangleright}}, \mathrm{Mod}_R) = D(R^{\triangleright}, \mathrm{Mod}_R);$$

this holds only if one uses *derived* completion on the left-hand side and that derived completion happens to live in degree 0. When the derived completion has higher homology one should genuinely work with a derived analytic ring rather than its $\pi_0$. This is the mechanism behind the derived structure sheaves of the exotic rational localizations of Section 10: even though one has an underlying ordinary analytic ring by taking $\pi_0$, the correct object is the derived analytic ring. Finally, maps of derived analytic rings are the same as maps of the underlying derived rings, which on hearts recover maps in the old abelian sense.

### 13.4 Colimits of analytic rings

The completion functor of Proposition 13.5 lets us compute colimits.

**Proposition 13.11** (Colimits: colimit of pre-analytic rings, then complete)**.** *Colimits in analytic rings are computed by taking the colimit in pre-analytic rings and completing. The colimit of a diagram $(R_i^{\triangleright}, \mathrm{Mod}_{R_i})_{i\in I}$ of pre-analytic rings is*

$$\varinjlim_{i\in I}(R_i^{\triangleright}, \mathrm{Mod}_{R_i}) = \Big(\varinjlim_i R_i^{\triangleright},\ \mathrm{Mod}_{\varinjlim R_i}\Big), \tag{13.5}$$

*where $\varinjlim_i R_i^{\triangleright}$ is the colimit in condensed commutative rings and*

$$\mathrm{Mod}_{\varinjlim R_i} = \big\{M \in \mathrm{Mod}_{\varinjlim R_i^{\triangleright}} : \forall i,\ M|_{R_i^{\triangleright}} \in \mathrm{Mod}_{R_i}\big\}, \tag{13.6}$$

*i.e. the intersection over $i$ of the conditions of lying in $\mathrm{Mod}_{R_i}$ after restriction of scalars.*

That (13.5) is the colimit in pre-analytic rings is immediate from the definition of a map (Definition 13.4): a map out of the colimit is a map of the underlying rings out of $\varinjlim_i R_i^{\triangleright}$ with the correct completeness property, and (13.6) is tailored to encode exactly that. Checking the pre-analytic axioms is elementary (limits and colimits because the relevant forgetful functors preserve them; the Ext-axiom directly). That colimits of analytic rings are then computed by completing is formal, the completion being a left adjoint.

**Example 13.12** (Filtered colimits need no completion)**.** For a *filtered* diagram the completion is unnecessary: $(\varinjlim_i R_i^{\triangleright}, \mathrm{Mod}_{\varinjlim R_i})$ is already complete. Indeed, to test $(\star)$ one must check that the unit $\varinjlim_i R_i^{\triangleright}$ lies in $\mathrm{Mod}_{R_i}$ for every $i$; but from $i$ onwards it is a filtered colimit of objects of $\mathrm{Mod}_{R_i}$ (cofinally), and $\mathrm{Mod}_{R_i}$ is closed under filtered colimits, so it lies there. Hence the pre-analytic colimit is already an analytic ring. For instance, for a discrete commutative ring $R$ written as a filtered colimit $R = \varinjlim_i R_i$,

$$\mathrm{Solid}_R \;=\; \varinjlim_{i\in I} \mathrm{Solid}_{R_i} \qquad \text{(colimit in analytic rings)},$$

as we already saw in Section 11.

*Remark* 13.13 (Sifted suffices; the only difficulty is finite). One need not say "filtered" in Example 13.12: *sifted* colimits are just as easy. Since general colimits decompose into sifted colimits and finite coproducts (just as they decompose into filtered colimits and pushouts), the completion functor for a relative tensor product is the same as for the absolute one, and the only genuine difficulty is the finite case. In other words $A \otimes_R B$ is no harder than $A \otimes B$; the substance is entirely in the pushout.

**Example 13.14** (Pushouts and completed tensor products)**.** For a span $A \leftarrow R \rightarrow B$ of analytic rings, the pushout is the completion of the pre-analytic ring with underlying ring $A^{\triangleright} \otimes_{R^{\triangleright}} B^{\triangleright}$ and module category

$$\big\{\, M \in \mathrm{Mod}_{A^{\triangleright}\otimes_{R^{\triangleright}} B^{\triangleright}} \;:\; M|_{A^{\triangleright}} \in \mathrm{Mod}_A \;\text{ and }\; M|_{B^{\triangleright}} \in \mathrm{Mod}_B \,\big\}. \tag{13.7}$$

Here the completion is essential, since $A^{\triangleright} \otimes_{R^{\triangleright}} B^{\triangleright}$ has no reason to lie in $\mathrm{Mod}_A$ or $\mathrm{Mod}_B$.

*Remark* 13.15 (The pushout completion iterates). The completion functor in Example 13.14 a priori involves *iterating* $A$-completion (of the underlying $A^{\triangleright}$-module) and $B$-completion (of the underlying $B^{\triangleright}$-module), alternately, along a countable sequential colimit. In practice it is usually not so bad, but that is what one must do in general. The same discussion applies verbatim in the derived/animated context, interpreting $\otimes$ as $\otimes^{\mathbb{L}}$ and Mod as $D_{\geq 0}$.

**Example 13.16** (Rational opens as pushouts)**.** In the solid $(R, R^+)$-theory of Section 9, a rational open $U \subseteq \mathrm{Spv}(R, R^+)$ gives a new analytic ring $(\mathcal{O}(U), \mathcal{O}^+(U))$: the completion of the pre-analytic structure on $R$ in which one enforces the relations describing $U$ – for $U = U(f_1, \dots, f_n/g)$, that $g$ act invertibly and each $f_i/g$ be a solid variable on all modules. Pushouts then compute intersections:

$$(\mathcal{O}(U), \mathcal{O}^+(U)) \;\otimes_{(R,R^+)}\; (\mathcal{O}(V), \mathcal{O}^+(V)) \;=\; (\mathcal{O}(U\cap V), \mathcal{O}^+(U \cap V)),$$

directly from (13.7) (in the discrete case $\otimes_{(R,R^+)}$ is literally $\otimes_R$). Geometrically these pushouts are the important construction: they are what should correspond to pullbacks (here, intersections of opens), and the need to complete is the additional subtlety of analytic geometry over ordinary algebraic geometry.

## 13.5 Frobenius is a map of analytic rings

Fix a prime $p$.

**Theorem 13.17** (Frobenius)**.** *Let $R$ be an analytic ring. Then the Frobenius map*

$$\mathrm{Frob}\colon\ R^{\triangleright} \longrightarrow R^{\triangleright}/p, \qquad x \longmapsto x^p,$$

*is a map of analytic rings $R \to R/p$. Here $R^{\triangleright}/p$ means $\widehat{R^{\triangleright}}/p$, and $R/p$ carries the induced analytic ring structure*

$$R/p \;=\; \big(R^{\triangleright}/p,\ \mathrm{Mod}_{R^{\triangleright}/p}(\mathrm{Mod}_R)\big), \tag{13.8}$$

*i.e. a complete $R^{\triangleright}/p$-module is an $R^{\triangleright}/p$-module that is complete when viewed as an $R^{\triangleright}$-module.*

To lighten notation one may assume $R^{\triangleright}$ has characteristic $p$ (is an $\mathbb{F}_p$-algebra), so that $\mathrm{Frob}\colon R^{\triangleright} \to R^{\triangleright}$. By (13.8) the content is:

$$M \in \mathrm{Mod}_R \implies \mathrm{Frob}_* M \in \mathrm{Mod}_R,$$

where $\mathrm{Frob}_* M$ is $M$ with $R^{\triangleright}$-action restricted along Frobenius. This is not obvious: the axioms of an analytic ring are all categorical closure properties in linear algebra (limits, colimits, Ext), and say nothing about Frobenius. One needs another handle on maps of analytic rings.

### A free-module criterion for maps

**Proposition 13.18** (Necessary data)**.** *If $(R^{\triangleright}, \mathrm{Mod}_R) \to (S^{\triangleright}, \mathrm{Mod}_S)$ is a map of analytic rings, then for every light profinite set $T$ one gets an $R^{\triangleright}$-linear map of free complete modules*

$$R[T] \longrightarrow S[T], \tag{13.9}$$

*functorial in $T$. (By hypothesis the free complete $S$-module $S[T]$ lies in $\mathrm{Mod}_R$ after restriction; hence the map $R^{\triangleright}[T] \to S[T]$ from the uncompleted free module factors through the completion $R[T]$. It suffices to have this for a generating class of $T$, e.g. the Cantor set.) Moreover the maps (13.9) are compatible with maps into the ground ring: for $t\colon T \to R^{\triangleright}$ a map of condensed sets, the square*

$$\begin{tikzcd} R[T] \arrow[r] \arrow[d] & S[T] \arrow[d] \\ R^{\triangleright} \arrow[r] & S^{\triangleright} \end{tikzcd} \tag{13.10}$$

*commutes, where $R[T] \to R^{\triangleright}$ is the extension of $t$ (using completeness of $R^{\triangleright}$) and the right and lower maps are the evident ones.*

**Lemma 13.19** (Sufficiency of the free-module data)**.** *Conversely, if $R^{\triangleright} \to S^{\triangleright}$ is a map of condensed rings for which there exist $R^{\triangleright}$-linear maps $R[T] \to S[T]$, functorial in $T$ and satisfying the compatibility of (13.10), then $R^{\triangleright} \to S^{\triangleright}$ is a map of analytic rings $R \to S$.*

The point of Lemma 13.19 is that one need not build the base-change functor $\mathrm{Mod}_S \to$ ($S$-modules from $R$-modules) by hand: producing the maps on free objects, plus the one compatibility, is enough.

*Idea.* $\mathrm{Mod}_R$ is monadic over condensed sets: the forgetful functor $\mathrm{Mod}_R \to \mathrm{Mod}_{R^{\triangleright}} \to$ (condensed sets) is the right adjoint of a localization followed by a forgetful functor, so it satisfies the hypotheses of the Barr–Beck monadicity theorem, and $\mathrm{Mod}_R$ is the category of algebras over the free-complete-$R$-module monad. To show every $S$-module is an $R$-module it suffices to produce a map of monads from the free-$R$-module monad to the free-$S$-module monad; its first datum is exactly a map of free objects (13.9), and unwinding the compatibility with the monad structure reduces to the single commuting square (13.10). The full argument is in the course notes [CS20]. □

By Proposition 13.18 and Lemma 13.19, to prove Theorem 13.17 it suffices to construct, for every light profinite set $T$, a *Frobenius-semilinear* map $R[T] \to R[T]$, functorial in $T$ and compatible as in (13.10).

### Recollection on Frobenius

Let $C_p$ denote the cyclic group of order $p$, and write $\widehat{H}^0(C_p, -) = \pi_0\big((-)^{tC_p}\big)$ for degree-zero Tate cohomology.

*Recollection* 13.20 (The abelian Frobenius). For any abelian group $M$ there is a natural map

$$M \;\longrightarrow\; \widehat{H}^0\big(C_p,\, M^{\otimes p}\big) \;=\; \mathrm{coker}\big(N\colon (M^{\otimes p})_{C_p} \to (M^{\otimes p})^{C_p}\big) \;=\; \pi_0\big(M^{\otimes p}\big)^{tC_p}, \tag{13.11}$$

where $C_p$ permutes the tensor factors of $M^{\otimes p}$ and $N$ is the norm (transfer) map. It is induced by the diagonal

$$m \;\longmapsto\; [\,m \otimes \cdots \otimes m\,] \;\in\; (M^{\otimes p})^{C_p}/N\big((M^{\otimes p})_{C_p}\big),$$

which is $C_p$-fixed. Although $m \mapsto m^{\otimes p}$ is not additive, the cross-terms in the image of $m+n$ are each norms, hence vanish in the Tate quotient, so (13.11) is a group *homomorphism*. (A nice exercise, if one has not done it.)

If $M = R$ is a commutative ring, composing with the ($C_p$-equivariant) multiplication $R^{\otimes p} \to R$, on whose target $C_p$ acts trivially, gives

$$R \;\longrightarrow\; \pi_0\big(R^{\otimes p}\big)^{tC_p} \;\longrightarrow\; \pi_0\, R^{tC_p} \;=\; R/p, \qquad m \longmapsto m^p,$$

the usual Frobenius: as $C_p$ acts trivially on $R$, the norm is multiplication by $p$ and $R^{tC_p} = R/p$. If $M$ is an $R$-module, the map $M \to \pi_0(M^{\otimes p})^{tC_p}$ is a map of abelian groups, and in fact a map of $R$-modules once one Frobenius-twists the right-hand side – the abelian-group Frobenius is linear over the ring Frobenius.

### Reduction to a key lemma

This suggests applying the abelian Frobenius to the free module $R[T]$, viewed merely as a sheaf of abelian groups. For a light profinite set $T$ one gets a sequence

$$\begin{tikzcd}
R[T] \arrow[r] & \pi_0\big(R[T]^{\otimes p}\big)^{tC_p} \arrow[r] & \pi_0\big(R[T]^{\otimes_R p}\big)^{tC_p} \arrow[r, equal] & \pi_0\big(R[T^p]\big)^{tC_p} \\
& & & \pi_0\big(R[T]\big)^{tC_p} \arrow[u, "\Delta"', "\sim"] \arrow[r, equal] & R/p\,[T]
\end{tikzcd} \tag{13.12}$$

Here the first $\otimes$ is over $\mathbb{Z}$ (the uncompleted, section-wise tensor of sheaves of abelian groups), the middle arrow is the base change to $\otimes_R$, and the Künneth identity for free modules gives $R[T]^{\otimes_R p} = R[T^p]$ with $C_p$ acting by permuting the factors of $T^p$. The diagonal embedding $T \hookrightarrow T^p$ induces a $C_p$-equivariant map $\Delta\colon R[T] \to R[T^p]$ (trivial action on the source), hence $\pi_0(R[T])^{tC_p} = R/p\,[T] \to \pi_0(R[T^p])^{tC_p}$.

**Claim 13.21.** $\Delta$ *induces an isomorphism* $\pi_0(R[T])^{tC_p} \xrightarrow{\sim} \pi_0(R[T^p])^{tC_p}$.

Granting Claim 13.21, the composite in (13.12) lands isomorphically in $\pi_0(R[T])^{tC_p} = R/p\,[T]$ (trivial action, so the Tate construction is reduction mod $p$), producing the required Frobenius-semilinear map $R[T] \to R/p\,[T]$, functorial in $T$ and compatible with maps to the ground ring. This proves Theorem 13.17.

*Remark* 13.22 (What is new in the light setting). In the previous iteration of this material Claim 13.21 had to be imposed as an extra *axiom* on analytic rings. In the light-profinite setting it can instead be *proved* for arbitrary $R$; that is the point of the following lemma.

Claim 13.21 is the case $S = T^p$ (with $S^{C_p} = T$, the diagonal) of:

**Lemma 13.23** (Fixed points and Tate cohomology)**.** *Let $S$ be a light profinite set with a $C_p$-action, and $S^{C_p} \subseteq S$ its (closed, again light profinite) subset of fixed points. Then the inclusion $R[S^{C_p}] \hookrightarrow R[S]$ induces an isomorphism on Tate cohomology $\widehat{H}^*(C_p, -)$, in particular on $\pi_0$.*

*Proof. Step 1: the inclusion is injective, and it suffices to kill the quotient.* Any inclusion of light profinite sets admits a retraction (Proposition 2.14); applying it to $S^{C_p} \hookrightarrow S$ and hitting with $R[-]$ shows $R[S^{C_p}] \hookrightarrow R[S]$ is (split) injective. By the long exact sequence in Tate cohomology it therefore suffices to show that

$$R[S]/R[S^{C_p}] \quad \text{has vanishing Tate cohomology.} \tag{13.13}$$

*Step 2: the quotient depends only on the complement.* For $T \subseteq S$ a closed light profinite subset, $R[S]/R[T]$ depends only on the (locally light profinite, i.e. locally compact) complement $S \setminus T$. Precisely: given $S' \to S$ with pullback $T' = S' \times_S T$, if $S' \to S$ is an isomorphism over $S \setminus T$ then

$$R[S']/R[T'] \xrightarrow{\sim} R[S]/R[T].$$

This holds because the square

$$\begin{tikzcd} T' \arrow[r, hook] \arrow[d] & S' \arrow[d, "\text{iso over } S\setminus T"] \\ T \arrow[r, hook] & S \end{tikzcd}$$

is a pushout in (light) condensed sets – a small exercise with the Grothendieck topology. Intuitively one is free to replace $T$ by any larger closed set, i.e. to choose a different compactification of $S \setminus T$.

*Step 3: free actions on such spaces are induced.* If $X$ is a $\sigma$-compact totally disconnected locally compact Hausdorff space ($\sigma$-compact = countable union of compacts) on which $C_p$ acts with no fixed points, then

$$X \;\cong\; \coprod_{C_p} Y, \qquad C_p \text{ permuting the copies.} \tag{13.14}$$

Indeed $X \to X/C_p$ is a covering map (by local compactness, Hausdorffness and freeness of the action), and $X/C_p$ is again $\sigma$-compact, totally disconnected, locally compact Hausdorff, hence a countable disjoint union of light profinite sets. Over each such profinite piece the cover is trivial (total disconnectedness makes it locally, hence globally, trivial), and trivializing gives (13.14) with $Y = X/C_p$.

*Conclusion.* Combining Steps 2 and 3: the complement $S \setminus S^{C_p}$ is locally light profinite with a free $C_p$-action (a light profinite set minus a light closed subset is $\sigma$-compact totally disconnected LCH, by lightness), so it is $\coprod_{C_p} Y$; recompactifying by compactifying $Y$ and taking the $C_p$ copies, Step 2 identifies

$$R[S]/R[S^{C_p}] \;\cong\; \bigoplus_{C_p} X',$$

an *induced* $C_p$-module. Induced modules have vanishing Tate cohomology, giving (13.13). □

*Remark* 13.24 (Frobenius, symmetric powers, and animated analytic rings). Theorem 13.17 is not merely cute. Suppose $(R^{\triangleright}, D_{\geq 0}(R))$ is a derived pre-analytic ring and we equip $R^{\triangleright}$ with the structure of an *animated* (condensed) commutative ring. This produces the derived symmetric power functors $\mathrm{Sym}^i$ on $D_{\geq 0}(R^{\triangleright})$ – the functors that build free animated rings, i.e. the monad describing animated rings over $R^{\triangleright}$. The Frobenius theorem implies that the $\mathrm{Sym}^i$ descend to the localization $D_{\geq 0}(R)$: if $M \to N$ becomes an isomorphism in $D_{\geq 0}(R)$ (i.e. after completion), then so does $\mathrm{Sym}^i M \to \mathrm{Sym}^i N$, so $\mathrm{Sym}^i$ is defined on $D_{\geq 0}(R)$. The deduction from the Frobenius statement uses Dold–Puppe's computation of the stable derived functors of the symmetric power functors [DP61]; details are in [CS20]. Consequently one can carry out the completion of Proposition 13.5 in the animated context as well, yielding a good category of *animated analytic rings*: pairs $(R^{\triangleright}, D_{\geq 0}(R))$ with $R^{\triangleright}$ an animated condensed ring and $D_{\geq 0}(R)$ an analytic ring structure on the underlying $\mathbb{E}_\infty$-ring.

## 13.6 Killing an algebra object

The last topic is a general mechanism, of which solidification is the motivating case. Recall that solid $\mathbb{Z}$ was presented as the analytic ring structure in which $M \in \mathrm{Solid}_{\mathbb{Z}}$ iff $\underline{\mathrm{Hom}}(P, M) \xrightarrow{1-\sigma} \underline{\mathrm{Hom}}(P, M)$ is an isomorphism, where $P = \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/[\infty]$ and $\sigma$ is the shift. This is equivalent to the vanishing

$$\mathrm{RHom}(A, M) = 0, \qquad A \;:=\; P/(T-1),$$

where $T = \sigma$ acts via the ring structure on $P$ (measures on $\mathbb{N}$, with multiplication from addition; a priori $A$ is the derived quotient, but multiplication by $T-1$ is injective, and in any case the class of modules is unchanged). Thus $A$ is an algebra object of the ambient category $D(\mathrm{CondAb}^{\mathrm{light}})$, and one is "killing" it.

**Definition 13.25** (General context)**.** Let $(\mathcal{C}, \otimes)$ be a presentable stable symmetric monoidal $\infty$-category with $\otimes$ commuting with colimits in each variable. Let $A \in \mathcal{C}$ be an "algebra object" in a very weak sense: a multiplication $\mu\colon A \otimes A \to A$ and a unit $\eta\colon \mathbf{1} \to A$ such that the composite

$$A \;\xrightarrow{\eta\otimes A}\; A \otimes A \;\xrightarrow{\mu}\; A \tag{13.15}$$

is an isomorphism (one-sided unitality). No associativity is required. Define

$$\mathcal{D} \;\subseteq\; \mathcal{C}, \qquad \mathcal{D} = \{M \in \mathcal{C} \;:\; \underline{\mathrm{Hom}}(A, M) = 0\},$$

the objects for which $A$ is "internally 0". The goal is, in favorable situations, to give a formula for the left adjoint to the inclusion $\mathcal{D} \subseteq \mathcal{C}$.

**Example 13.26** (Three instances)**.**

1. $\mathcal{C} = D(\mathrm{CondAb}^{\mathrm{light}})$, $A = P/(T-1)$: then $\mathcal{D} = D(\mathrm{Solid}_{\mathbb{Z}})$; similarly $\mathrm{Solid}_{\mathbb{Z}[T]}$ arises by killing the corresponding algebra.
2. $\mathcal{C} = D(R)$ for an ordinary ring $R$, $A = R/f$ (the cofiber of $\cdot f$): then $\mathcal{D} = D(R[1/f])$, i.e. inverting $f$.
3. $\mathcal{C} = D(R)$, $A = R[f^{-1}] = R[1/f]$: then $\mathcal{D} = D(R)^{\wedge}_f$, the $f$-complete objects, and the left adjoint is derived $f$-completion.

**Construction 13.27** (The functor $F$ and its iteration)**.** Let $C = \mathrm{fib}(\mathbf{1} \to A)$ be the fiber of the unit, and define

$$F : \mathcal{C} \to \mathcal{C}, \qquad F(X) = \underline{\mathrm{Hom}}(C, X).$$

The map $C \to \mathbf{1}$ induces a natural transformation

$$X = \underline{\mathrm{Hom}}(\mathbf{1}, X) \longrightarrow \underline{\mathrm{Hom}}(C, X) = F(X).$$

Iterating – applying $F$ to the previous *map* (not the instance of the previous map at $F(X)$) – gives a sequence

$$X \longrightarrow F(X) \xrightarrow{F(-)} F^2(X) \longrightarrow F^3(X) \longrightarrow \cdots, \qquad F^\infty(X) := \varinjlim_n F^n(X). \tag{13.16}$$

**Claim 13.28.** *If $M \in \mathcal{D}$, then $\underline{\mathrm{Hom}}(-, M)$ applied to $X \to F(X)$ is an isomorphism.*

*Proof.* It suffices that the fiber of $X \to F(X)$ be an $A$-module. Since $A = \mathrm{cofib}(C \to \mathbf{1})$, the fiber of $\underline{\mathrm{Hom}}(\mathbf{1}, X) \to \underline{\mathrm{Hom}}(C, X)$ is, up to a shift, $\underline{\mathrm{Hom}}(A, X)$, which carries an $A$-module structure (acting on the $A$-variable). Now any $A$-module is a retract of $A \otimes$ (itself), by the unitality axiom (13.15); and $\underline{\mathrm{Hom}}(A \otimes -, M) = \underline{\mathrm{Hom}}(-, \underline{\mathrm{Hom}}(A, M)) = 0$ because $M \in \mathcal{D}$. Hence homing out of an $A$-module into $M$ vanishes, so $\underline{\mathrm{Hom}}(-, M)$ sees $X \to F(X)$ as an isomorphism. ☐

**Proposition 13.29** (The left adjoint)**.** *Suppose either*

1. *the colimit* (13.16) *stabilizes for all $X$ (every map from some point on is an isomorphism), or*
2. $\underline{\mathrm{Hom}}(A, -)$ *commutes with (sequential) colimits.*

*Then $X \mapsto F^\infty(X)$ is the left adjoint to the inclusion $\mathcal{D} \subseteq \mathcal{C}$.*

*Proof.* Two things must be checked.

*(a)* $\mathrm{Hom}(X, -) \simeq \mathrm{Hom}(F^\infty X, -)$ *on $\mathcal{D}$.* Just as for $X \to F(X)$, each map $F^n X \to F^{n+1} X$ is an isomorphism on $\underline{\mathrm{Hom}}(-, M)$ for $M \in \mathcal{D}$: applying $F$ preserves the property used in Claim 13.28 (the fiber is an $A$-module, and internally homing into an $A$-module gives an $A$-module). Homing out of the colimit is the inverse limit of homing out of the terms, so $X \to F^\infty X$ is an isomorphism on $\underline{\mathrm{Hom}}(-, M)$; here one uses hypothesis (1) or (2) to pull the internal Hom inside the colimit.

*(b)* $F^\infty X \in \mathcal{D}$. One checks $\underline{\mathrm{Hom}}(A, F^\infty X) = 0$. Under the hypothesis $\underline{\mathrm{Hom}}(A, -)$ commutes with the colimit, and the transition maps $\underline{\mathrm{Hom}}(A, F^n X) \to \underline{\mathrm{Hom}}(A, F^{n+1} X)$ are null-homotopic (internally homing $A$ shifts one step along the sequence), so the colimit vanishes. ☐

*Remark* 13.30 (Recovering the classical formulas). The formula $F^\infty$ specializes to familiar left adjoints (Example 13.26).

- For $A = R/f$ (inverting $f$) the left adjoint is

$$L(M) = \varinjlim \big(M \xrightarrow{f} M \xrightarrow{f} M \xrightarrow{f} \cdots\big) = M[1/f],$$

  which $F^\infty$ reproduces as a sequential colimit.

- For $A = R[1/f]$ ($f$-completion) the left adjoint is

$$L(M) \;=\; \varprojlim_n M/f^n \;=\; F(M),$$

  where the iteration stabilizes after the first term: the inverse limit is already seen at $F(M)$, and the subsequent maps are isomorphisms.

- For solidification, $F(X)$ is explicitly a quotient of null sequences in $X$ modulo the telescoping relations (those forcing a null sequence to be summed); iterating computes the solidification of a general module, and one can check it recovers the classical formulas.

*Remark* 13.31 (When the colimit stabilizes). The stabilization hypothesis (1) of Proposition 13.29 holds, for example, when $A$ is idempotent – as in Example 13.26(3), $A = R[1/f]$. By contrast $A = R/f$ in Example 13.26(2) is *not* idempotent, and there one uses hypothesis (2), that $\underline{\mathrm{Hom}}(A, -)$ commutes with colimits. (Solidifying $R[1/f]$ would instead be the other operation, not the algebraic inverting of $f$.)

# 14 The Gaseous Analytic Ring, Gaseous Real Vector Spaces, and the Tate Elliptic Curve

Scholze picks up the *gaseous* analytic ring structure introduced at the end of Section 12 and analyses it, using the general machinery Clausen developed on Wednesday (Section 13): the completion of a pre-analytic ring, and the "killing an algebra object" formula for the completion functor. The lecture has three parts, matching the plan Scholze wrote down:

1. the *gaseous base ring* – a version of arithmetic Laurent series $A^{\triangleright} \subset \mathbb{Z}((q))$, cut out by a polynomial-growth condition on the coefficients;
2. *gaseous* $\mathbb{R}$*-vector spaces* – the specialisation to the real numbers, and its relation to the $\alpha$-liquid theory of Section 12;
3. a first pass at the *construction of the Tate elliptic curve* over the gaseous base ring, where one begins to need genuine analytic geometry (and the language of analytic stacks, still to be introduced).

Throughout, "ring" means commutative ring and everything lives in light condensed abelian groups. We use freely the solid theory (Sections 5 and 6), the abstract notion of analytic ring and its completion (Sections 8 and 13), the compact projective generator $P = \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/[\infty]$ and its internal projectivity (Theorem 3.15 and Definition 5.5), and the "killing an algebra object" construction of Section 13. As in Section 12 the guiding example is the Tate elliptic curve, and the goal is to pin down the *smallest* analytic ring over which it can be defined.

## 14.1 The gaseous base ring

*Remark* 14.1 (Motivation and desiderata). The motivation is to find a "minimal" analytic ring $(A^{\triangleright}, \mathrm{Mod}_A)$ over which the Tate elliptic curve $E_q$ can be defined. Two properties are needed of such a ring, both with a transparent meaning in terms of the underlying condensed ring:

1. there is a topologically nilpotent unit $q \in A^{\triangleright}$; and

2. if $P \otimes_{\mathbb{Z}} A^{\triangleright}$ is the free $A^{\triangleright}$-module on a null sequence, then

$$1 - q \cdot \mathrm{shift} \;:\; P \otimes_{\mathbb{Z}} A^{\triangleright} \;\longrightarrow\; P \otimes_{\mathbb{Z}} A^{\triangleright} \tag{14.1}$$

is an isomorphism.

Condition (2) means that for a null sequence $(a_n)_n$ in $A^{\triangleright}$ one can form the sum $\sum_{n\geq 0} a_n q^n$: the inverse of $1 - q \cdot \mathrm{shift}$ is $1 + q \cdot \mathrm{shift} + q^2 \cdot \mathrm{shift}^2 + \cdots$, and evaluating on the zeroth coordinate produces exactly such a geometric series with null-sequence coefficients. This is a genuine but *weaker* completeness condition than solidity: for solid modules one imposed the same isomorphism with $q = 1$ (and did not ask that 1 be a topologically nilpotent unit), which forces summability of *all* null sequences, not only those weighted by powers of a topologically nilpotent $q$ (Definition 5.5). Some completeness condition is essential – with none, one would work with all condensed modules and complete no infinite series at all.

Following the lightweight method of Remark 12.20, one turns the two desiderata into an analytic ring by imposing that a natural endomorphism of $P$ become an isomorphism. There is an evident initial (universal) example.

**Proposition 14.2** (The initial ring)**.** *There is an initial analytic ring with the properties* (14.1) *of Remark 14.1. The same proof produces an initial* animated *such ring.*

*Construction.* Write

$$\mathbb{Z}[\widehat{q}] \;:=\; \mathbb{Z}\big[q^{\mathbb{N}\cup\{\infty\}}\big]/(q^{\infty} = 0),$$

the free condensed ring on a topologically nilpotent element $q$; as a condensed abelian group $\mathbb{Z}[\widehat{q}] = P$, the free module on the null sequence $(q^n)_n$ (measures on $\mathbb{N}$, with the ring structure from addition, cf. Remark 7.1). Inverting $q$ gives the free condensed ring on a topologically nilpotent *unit*

$$A_0^{\triangleright} \;:=\; \mathbb{Z}[\widehat{q}][\widehat{q}^{-1}].$$

This is the underlying ring on which desideratum (1) is tautological: a map $A_0^{\triangleright} \to A'^{\triangleright}$ is the same as a topologically nilpotent unit of $A'^{\triangleright}$. For desideratum (2), form $P \otimes_{\mathbb{Z}} A_0^{\triangleright}$ with its endomorphism $1 - q \cdot \mathrm{shift}$ and declare

$$\mathrm{Mod}_A \;:=\; \Big\{ M \in \mathrm{Mod}_{A_0^{\triangleright}} \;\Big|\; 1 - q \cdot \mathrm{shift} \text{ is an isomorphism on } \underline{\mathrm{Hom}}\big(P \otimes_{\mathbb{Z}} A_0^{\triangleright},\, M\big) \Big\}. \tag{14.2}$$

Exactly as for solid modules (Proposition 5.7), the internal projectivity of $P$ makes $\mathrm{Mod}_A \subseteq \mathrm{Mod}_{A_0^{\triangleright}}$ closed under all the operations required of an analytic ring, so $(A_0^{\triangleright}, \mathrm{Mod}_A)$ is a *pre-analytic* ring in the sense of Definition 13.2. It is not analytic, because $A_0^{\triangleright} \notin \mathrm{Mod}_A$: the base ring $A_0^{\triangleright}$ is uncompleted. Conditions (1)+(2) on an analytic ring are precisely the datum of a map out of $(A_0^{\triangleright}, \mathrm{Mod}_A)$, so the completion of $(A_0^{\triangleright}, \mathrm{Mod}_A)$ in the sense of Proposition 13.5 is the required initial analytic ring $(A^{\triangleright}, \mathrm{Mod}_A)$. The animated statement is proved the same way; it will turn out below that the derived completion already lives in degree 0, so the animated initial object agrees with the classical one. □

We now compute this analytic ring $(A^{\triangleright}, \mathrm{Mod}_A) =: A$.

**Theorem 14.3** (Structure of the gaseous base ring)**.** *The initial analytic (animated) ring A of Proposition 14.2 is described as follows, and in particular sits in degree* 0*.*

1. *$A^{\triangleright}$ and all free modules $A^{\triangleright}[T]$, for $T \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$, sit in degree* 0 *and are quasiseparated.*

*2. $A^{\triangleright}$ is the subring of $\mathbb{Z}(\!(q)\!)$ of Laurent series with polynomially bounded coefficients:*

$$A^{\triangleright}(*) \;=\; \Big\{ \textstyle\sum_{n\in\mathbb{Z}} a_n q^n \;\Big|\; a_n = 0 \textit{ for } n \ll 0,\; |a_n| \textit{ has at most polynomial growth} \Big\}, \tag{14.3}$$

*and, as a light condensed ring,*

$$A^{\triangleright} \;=\; \bigcup_{k>0,\; n>0}\; \prod_{m\geq -n} \Big(\mathbb{Z}\cap\big[-(m+n)^k,\, (m+n)^k\big]\Big)\, q^m.$$

*3. For $S = \varprojlim_i S_i \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$, the free complete module is the space of $\mathbb{Z}((q))$-valued measures on $S$ with polynomially bounded coefficients,*

$$A[S] \;\subseteq\; \mathbb{Z}(\!(q)\!)[S]^{\square} \;=\; \Big(\varprojlim_i \mathbb{Z}[\widehat{q}][S_i]\Big)[\widehat{q}^{-1}] \;=\; \mathcal{M}\big(S, \mathbb{Z}(\!(q)\!)\big),$$

*namely*

$$\begin{aligned} A[S] = \Big\{ \mu = \textstyle\sum_{m\in\mathbb{Z}} \mu_m q^m \;\Big|\; & \mu_m \in \mathbb{Z}[S] \subseteq \mathbb{Z}[S]^{\square}, \\ & |\mu_m| \textit{ at most polynomial} \Big\} \\ = \bigcup_{k>0,\; n>0}\; \varprojlim_i \prod_{m\geq -n} & \mathbb{Z}[S_i]_{\leq (n+m)^k}\, q^m. \end{aligned}$$

*Here each $\mu_m$ is an* honest *(finite) sum of Dirac measures – for a fixed power of $q$ no infinite sum is allowed – and $\mathbb{Z}[S_i]_{\leq N}$ denotes sums of at most $N$ basis elements (see Notation 14.4).*

*Notation* 14.4 (Bounded part of a free module). For a finite set $I$ write

$$\mathbb{Z}[I]_{\leq n} \;:=\; \Big\{ \textstyle\sum_{i\in I} a_i\,[i] \;\Big|\; \sum_i |a_i| \leq n \Big\} \;\subseteq\; \mathbb{Z}[I],$$

the (compact, light profinite) locus of measures of $\ell^1$-norm at most $n$.

*Remark* 14.5 (Where does polynomial growth come from?). The polynomial-growth condition in (14.3) is the surprising output: in defining the gaseous structure one imposed only that $1 - q\cdot\mathrm{shift}$ be invertible, with no growth condition visible anywhere. That this is what comes out is exactly what happened to Clausen and Scholze when they first wrote down the condition and computed. Note that polynomial growth is what makes $A^{\triangleright}$ the *minimal* ring supporting $E_q$: the coefficients of the Tate-curve equation have polynomial (not merely subexponential) growth (Remark 12.6).

**Sketch of the computation**

*Sketch of Theorem 14.3.* The computation runs through the completion formula from the end of Section 13. First rewrite the completeness condition (14.2) as the vanishing of RHom out of a single algebra object. Remembering the ring structure on $P = \mathbb{Z}[\widehat{q}]$, the shift is multiplication by the nilpotent generator, so

$$B \;:=\; \big(P \otimes_{\mathbb{Z}} A_0^{\triangleright}\big)/(1 - q\cdot\mathrm{shift}) \;=\; \mathbb{Z}[\widehat{x},\widehat{q}][\widehat{q}^{-1}]/(1-qx)$$

(writing $x$ for the generator of the source copy of $P$ and $q$ for the unit of $A_0^{\triangleright}$; multiplication by $x$ is the shift). It is a commutative ring, and

$$M \in \mathrm{Mod}_A \quad\Longleftrightarrow\quad \mathrm{RHom}_{A_0^{\triangleright}}(B, M) = 0,$$

because this RHom is literally the cone of the map $1 - q \cdot \text{shift}$ on $\underline{\mathrm{Hom}}(P \otimes_{\mathbb{Z}} A_0^{\triangleright}, M)$, and these internal Homs sit in degree 0 by internal projectivity of $P$. Moreover $\mathrm{RHom}_{A_0^{\triangleright}}(B, -)$ commutes with all colimits. Thus $B$ is an algebra object of the "killing" type of Definition 13.25, and the derived completion in the pre-analytic ring $(A_0^{\triangleright}, \mathrm{Mod}_A)$ is given by the sequential-colimit formula of Construction 13.27 and Proposition 13.29: with $I := \mathrm{fib}(\mathbf{1} \to B)$,

$$M \;\longmapsto\; \operatorname{colim}\Big(M \to \mathrm{RHom}(I, M) \to \mathrm{RHom}(I \otimes I, M) \to \cdots\Big), \qquad M \in D(A_0^{\triangleright}). \tag{14.4}$$

*The key computation.* Everything reduces to computing internal Homs out of $P$. One may proceed without inverting $q$ (and invert it at the end); doing so replaces $A_0^{\triangleright}$ by $P = \mathbb{Z}[\widehat{q}]$ and drops the terms with negative powers of $q$. Writing $P = \mathbb{Z}[\widehat{x}]$ (free ring on a topologically nilpotent $x$) and $y$ for the dual polynomial variable:

**Lemma 14.6** (Internal Homs out of $P$)**.** *For $S = \varprojlim_i S_i \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$,*

$$\underline{\mathrm{Hom}}_{\mathbb{Z}}\big(P, \mathbb{Z}[\widehat{q}]\big) = \bigcup_{n>0} \varprojlim_m \Big(\big(\mathbb{Z}[\widehat{q}]/\widehat{q}^m\big)_{\le n}[y]\Big),$$
$$\underline{\mathrm{Hom}}_{\mathbb{Z}}\big(P, \mathbb{Z}[\widehat{q}][S]\big) = \bigcup_{n>0} \varprojlim_m \Big(\big((\mathbb{Z}[\widehat{q}]/\widehat{q}^m)[S_i]\big)_{\le n}[y]\Big).$$

*Equivalently these are the subspaces of "bounded" power series*

$$\underline{\mathrm{Hom}}_{\mathbb{Z}}\big(P, \mathbb{Z}(\!(q)\!)\big) = \mathbb{Z}[y][\![q]\!]^{\mathrm{bdd}} \subseteq \mathbb{Z}[y][\![q]\!], \qquad \underline{\mathrm{Hom}}_{\mathbb{Z}}\big(P, \mathbb{Z}[\widehat{q}][S]\big) = \mathbb{Z}[y][S][\![q]\!]^{\mathrm{bdd}} \subseteq \mathbb{Z}[y][S][\![q]\!], \tag{14.5}$$

*where* bdd *means the coefficient of each $y$-monomial satisfies the $\le n$ boundedness property.*

Since the free condensed abelian groups on light profinite sets are unions over $n$ of loci of norm $\le n$, this boundedness pulls through the internal Hom; the dual variable $y$ appears because $\underline{\mathrm{Hom}}$ out of the free ring on the nilpotent $x$ is computed by polynomials in the dual variable.

*From $\underline{\mathrm{Hom}}$ to the Koszul complex.* With the boxed complex $I = \mathrm{fib}(\mathbf{1} \to B)$, formal manipulation of (14.5) gives, in homological degrees 1 and 0,

$$\mathrm{RHom}_{\mathbb{Z}[\widehat{q}]}\big(I,\, \mathbb{Z}[\widehat{q}]\big) \;=\; \Big[\; \mathbb{Z}[y][\![q]\!]^{\mathrm{bdd}} \;\xrightarrow{\;q-y\;}\; \mathbb{Z}[y][\![q]\!]^{\mathrm{bdd}} \;\Big], \tag{14.6}$$

and for the $k$-fold tensor power the answer is a Koszul complex,

$$\mathrm{RHom}_{\mathbb{Z}[\widehat{q}]}\big(I^{\otimes k},\, \mathbb{Z}[\widehat{q}]\big) \;=\; \mathbb{Z}[y_1, \ldots, y_k][\![q]\!]^{\mathrm{bdd}} /^{\mathbb{L}} \big(q - y_1, \ldots, q - y_k\big). \tag{14.7}$$

It is crucial here that each individual term is computed by the full (derived) Koszul complex, because there *is* higher homology at each finite stage; only the colimit is well behaved (nice, quasiseparated, in degree 0). Feeding (14.7) into (14.4) and doing the same with $\mathbb{Z}[\widehat{q}][S]$ in place of $\mathbb{Z}[\widehat{q}]$ yields

$$A^{\triangleright} \;=\; \operatorname*{colim}_k\; \mathbb{Z}[y_1, \ldots, y_k][\![q]\!]^{\mathrm{bdd}} /^{\mathbb{L}} \big(q - y_1, \ldots, q - y_k\big), \tag{14.8}$$

and analogously for $A[S]$.

*The growth condition.* Finally one analyses $H_0$ (or rather its quasiseparated quotient): passing to $H_0$ means setting each $y_i = q$, i.e. computing the image of

$$\mathbb{Z}[y_1, \ldots, y_k][\![q]\!]^{\mathrm{bdd}} \;\longrightarrow\; \mathbb{Z}[\![q]\!], \qquad y_i \longmapsto q.$$

For a fixed $n$, each occurrence of $q$ in a monomial can be routed either to a $q$ or to one of $y_1, \dots, y_k$, and each $y_i$ may be used at most $n$ times; the number of monomials of degree $< m$ in $k$ variables is $\binom{m+k-1}{k}$, so the coefficient of $q^m$ in the image can be any integer in

$$\left[ -n\tbinom{m+k-1}{k},\ n\tbinom{m+k-1}{k} \right],$$

which grows in $m$ like a polynomial of degree $k$. Taking the union over $k$ and $n$ produces exactly the polynomially-growing Laurent coefficients of (14.3). Whenever a null sequence of coefficients "grows and decays normally", the many available monomials let one split each summand into bounded pieces; showing the Koszul complex collapses to degree 0 amounts to showing two such splittings differ by a bounded correction in the next term of the complex, which needs to increase $k$ slightly but then works. The argument is of the same flavour as the well-behavedness proofs in the liquid theory, but several orders of magnitude easier. $\square$

*Remark* 14.7 (Individual terms versus the colimit). As in the solid case, the individual terms of the colimit (14.4)–(14.8) are badly behaved – spread out, with higher homology, not quasiseparated. It is only the colimit that is nice. Solidification itself is the degenerate case of this formula that one would obtain by trying to use (14.4) directly, and it is famously confusing to run in that form; the gaseous case is where the colimit visibly does something.

### 14.2 Gaseous real vector spaces

The same lightweight recipe, applied with the topologically nilpotent unit $\frac{1}{2} \in \mathbb{R}$ in place of the formal $q$, produces an analytic ring structure on $\mathbb{R}$. Throughout this subsection everything is light (the qualifier is dropped).

**Definition 14.8** (Gaseous real vector space)**.** A light condensed $\mathbb{R}$-vector space $V$ is *gaseous* if

$$1 - \tfrac{1}{2} \cdot \mathrm{shift} \;:\; \underline{\mathrm{Hom}}_{\mathbb{Z}}(P, V) \;\longrightarrow\; \underline{\mathrm{Hom}}_{\mathbb{Z}}(P, V) \tag{14.9}$$

is an isomorphism.

*Remark* 14.9 (Independence of the parameter). The condition (14.9) is independent of the choice of $\frac{1}{2} \in (0,1)$: any real number in $(0,1)$ would do. Intuitively, making the number smaller weakens the condition, because the series one is asked to sum decays faster; but one may always split off finitely many leading terms and telescope, so the class of complete modules does not change. (This is unlike the liquid case, where the analogous exponent $\alpha$ genuinely matters.)

*Remark* 14.10 (Gaseous is the weakest condition). Every $\alpha$-liquid (or $p$-liquid) real vector space is gaseous. So gaseous vector spaces form an extremely general class of condensed real vector spaces – the $\alpha = 0$ regime of the spectrum (12.4) – yet still workable for many purposes.

The following proposition ties Definition 14.8 back to the gaseous base ring of Theorem 14.3: gaseous real vector spaces are the modules of the analytic ring obtained from $A$ by specialising $q \mapsto \frac{1}{2}$.

**Proposition 14.11** (Specialisation $q \mapsto \frac{1}{2}$)**.** *Let $A_{\mathrm{gas},\mathbb{R}}$ denote the analytic ring of gaseous real vector spaces (it is an analytic ring structure by internal projectivity of $P$, as usual). It is obtained from the gaseous base ring $A$ of Theorem 14.3 by the map $1 - q \cdot \mathrm{shift} \mapsto 1 - \frac{1}{2} \cdot \mathrm{shift}$, i.e. by setting $q = \frac{1}{2}$; the completed ring $A^{\triangleright} \otimes_{q \mapsto \frac{1}{2}}$ is exactly $\mathbb{R}$ (the polynomial-growth Laurent series, evaluated at $q = \frac{1}{2}$, converge to real numbers). Consequently the free gaseous real vector space on a light* *profinite* $S = \varprojlim_i S_i$ *is*

$$\mathbb{R}[S]^{\mathrm{gas}} \;=\; \bigcup_{k>0} \bigcup_{c>0} \varprojlim_i \Big\{ \textstyle\sum_{s\in S_i} x_s\,[s] \in \mathbb{R}[S_i] \;\Big|\; \textstyle\sum_s f_k(x_s) \le c \Big\}, \tag{14.10}$$

*for the functions* $f_k$ *of Definition 14.12.*

Because $q$ has become the constant $\frac{1}{2}$, the polynomial-growth condition of (14.3) can no longer be read off directly on the coefficients; rephrasing it on $\mathbb{R}$ produces the "crazy" description (14.10), governed by a sequence of gauge functions.

**Definition 14.12** (The gauge functions $f_k$)**.** The $f_k\colon \mathbb{R} \to \mathbb{R}_{\ge 0}$ are any increasing functions such that, for $|x|$ small,

$$f_k(x) \;=\; \big|\log|x|\big|^{-k}. \tag{14.11}$$

For the transition maps of (14.10) to be defined one wants $f_k$ increasing and *concave*, so that

$$f_k(x+y) \;\le\; f_k(x) + f_k(y);$$

only the behaviour near 0 matters (in an admissible infinite sum almost all terms are tiny), so away from 0 one extends $f_k$ arbitrarily, e.g. linearly. The function $|\log|x||^{-k}$ is itself increasing and concave near 0 but eventually misbehaves – at $x = 1$ it blows up – which is why one only prescribes it near the origin.

*Remark* 14.13 (Which sums can be formed: liquid versus gaseous)*.* The gauge functions encode which infinite sums a complete module can evaluate.

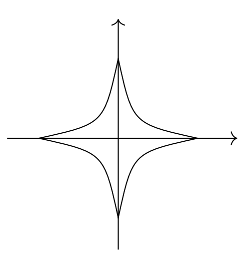
- In the $\alpha$-liquid theory ($\alpha \in (0,1]$) one can form $\sum_{n\ge 0} x_n$ precisely when the sequence is *$\beta$-summable* for some $\beta < \alpha$, i.e. $\exists\,\beta < \alpha$ with $\sum_n |x_n|^{\beta} < \infty$.
- In the gaseous theory one can form $\sum_n x_n$ only when

  $$\exists\, k \text{ with } \sum_n f_k(x_n) < \infty,$$

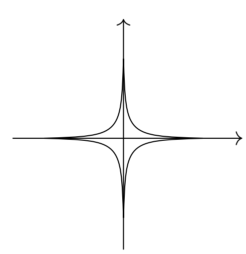
  which is equivalent (up to reordering) to *quasi-exponential decay* of the terms:

  $$\exists\, \varepsilon > 0,\ \exists\, C > 0 : \quad |x_n| \;\le\; C \cdot \left(\tfrac{1}{2}\right)^{n^{\varepsilon}}. \tag{14.12}$$

The equivalence is a direct computation: substituting $|x_n| = (\frac{1}{2})^{n^{\varepsilon}}$ into (14.11) turns the logarithm into $n^{\varepsilon}$, so $f_k(x_n) \sim n^{-\varepsilon k}$, which is summable once $\varepsilon k > 1$. Thus the gaseous condition is a much stronger constraint on the summable sequences than any liquid one – one must have (nearly) exponential decay, not just power-law decay.



liquid: $\{\sum |x_s|^{\beta} \le c\}$



gaseous: $\{\sum f_k(x_s) \le c\}$

*Remark* 14.14 (Gaseous suffices for complex geometry). The gaseous theory would serve just as well as the liquid theory for the analytic geometry over $\mathbb{C}$ developed in Clausen–Scholze's notes on complex geometry [CS20]: the only structural input there is that the algebras of over-convergent holomorphic functions are nuclear, and this remains true in the gaseous story. Indeed restriction of holomorphic functions from a disk to a smaller one rescales the power-series coefficients by a fixed factor $< 1$, so the transition maps have exponential decay – exactly the kind of decay (14.12) the gaseous theory is built to handle.

*Remark* 14.15 (Light versus all condensed sets). Asked whether non-light condensed sets are still used: originally extremally disconnected sets were crucial, as they supplied enough projectives; for the present theory one largely forgets the full category of condensed sets. It remains occasionally convenient to have the ambient framework of all condensed sets – light objects embed into it, and one can test isomorphisms on extremally disconnected sets. But for analytic geometry the light setting was chosen because of the internal projectivity of $P$, used throughout. One can ask whether liquid and gaseous real vector spaces can also be defined on *all* condensed modules; probably yes, but the arguments here use internal projectivity of $P$ and would need to be redone more carefully.

## 14.3 Some gaseous geometry: the Tate elliptic curve

Finally Scholze does a first piece of genuine analytic geometry over the gaseous base ring, constructing the Tate elliptic curve. The construction uses the language of analytic stacks – the eventual subject of the course, still to be introduced – but the idea can already be stated.

*Remark* 14.16 (Goal and idea). The goal is to define the Tate elliptic curve $E_q$ over the gaseous base ring $A = (A^\triangleright, \mathrm{Mod}_A)$ of Theorem 14.3. The idea is that we have a fixed topologically nilpotent unit $q$, and (as one always does with such a $q$) one uses it to *gauge* the size of any other function $f$: one asks how powers $|f|^\alpha$ compare to powers $|q|^\beta$. Normalising $|q| = \frac{1}{2}$ turns this into an ordinary real-valued absolute value.

*Remark* 14.17 (The gauge morphism). The claim – to be made precise once analytic stacks are available – is that there is a morphism of analytic stacks

$$\mathbb{A}^1_A \xrightarrow{|T|} [0,\infty], \tag{14.13}$$

where $T \in A^\triangleright$ is the coordinate function; more precisely, on the projective line,

$$\mathbb{P}^1_A \xrightarrow{|T_1/T_2|} [0,\infty], \qquad [T_1 : T_2] \longmapsto |T_1/T_2|, \tag{14.14}$$

sending the two boundary points $0, \infty$ of $\mathbb{P}^1$ to the endpoints of the closed interval. To define such a gauge for an arbitrary function it suffices to do it for the universal one, the coordinate $T$ on $\mathbb{A}^1$.

**Definition 14.18** (Analytic $\mathbb{G}_m$ and the Tate curve)**.** Define the analytic multiplicative group as the preimage of the open interval under the gauge (14.14),

$$\mathbb{G}_{mA}^{\mathrm{an}} := |T|^{-1}\big((0,\infty)\big) = \bigcup_n \{\, |q|^n \le |T| \le |q|^{-n} \,\},$$

the rising union of annuli of all radii (this is what one starts from to define the Tate curve). Multiplication by $q$ gives a free, totally discontinuous action on $\mathbb{G}_{mA}^{\mathrm{an}}$, and the Tate elliptic curve is the quotient

$$E_q := \mathbb{G}_{mA}^{\mathrm{an}}/q^{\mathbb{Z}}.$$

The gauge morphism is equivariant for multiplication by $q$ upstairs and by $|q| = \frac{1}{2}$ on $(0, \infty)$, so it descends to the quotients, giving a commuting square:

$$\begin{tikzcd} \mathbb{G}_{mA}^{\mathrm{an}} \arrow[r, "|T|"] \arrow[d, "/q^{\mathbb{Z}}"'] & (0,\infty) \arrow[d, "/(\frac{1}{2})^{\mathbb{Z}}"] \\ E_q \arrow[r] & (0,\infty)/(\frac{1}{2})^{\mathbb{Z}} = S^1 \end{tikzcd}$$

**Proposition 14.19** ($E_q$ is a proper smooth curve)**.** *The map $\mathbb{G}_{mA}^{\mathrm{an}} \to (0, \infty)$ is proper (it is the restriction of the proper $\mathbb{P}^1 \to [0, \infty]$ over the open locus). Since $(0, \infty) \to (0, \infty)/(\frac{1}{2})^{\mathbb{Z}} = S^1$ is proper and the $q^{\mathbb{Z}}$-action is free and totally discontinuous, the map $E_q \to S^1$ is proper. Locally on $S^1$ the quotient map is an isomorphism onto an open subset of $\mathbb{P}^1$, which is smooth; hence $E_q$ is a proper smooth curve over $A$ – the Tate elliptic curve.*

*Remark* 14.20 (Two roles of condensed mathematics; discrete rings suffice). The boundary points $0, \infty$ (and the interval $[0, \infty]$) are here regarded as *light condensed sets*, and there will be a functor from light condensed sets to analytic stacks giving them an incarnation in this world. Consequently the theory of analytic stacks will be tied to the condensed story in *two* ways: first, the analytic rings themselves are built on condensed sets (the role condensed mathematics has played so far); and second, the geometric objects $[0, \infty]$, $S^1$, ... will themselves be realised through condensed sets. Notably, the realisation of $E_q$ as an analytic stack will use only *discrete* rings – no interesting condensed structure at the ring level. This differs from the usual (adic-space) theory and will be developed once the formalism of analytic stacks is set up.

# 15 Toward Analytic Stacks: Motivation from Algebraic Geometry

Having built the theory of analytic rings and worked through a range of examples (solid, liquid, gaseous; Sections 8 and 12 to 14), Clausen turns to the next stage of the course: using analytic rings to build *geometric* objects, the *analytic stacks*. This lecture is entirely motivational – the precise definition of an analytic stack is deferred – and proceeds by analogy with classical algebraic geometry. Everything is kept deliberately loose; several statements are heuristic, and two live corrections (recorded below) reshaped the account of analytification and of GAGA over $\mathbb{Q}_p$ as the lecture went on. The plan:

- recall the modern (functor-of-points) paradigm for schemes and stacks, and the question it answers: how to single out a good class of geometric objects inside sheaves on affines;
- survey why one wants *stacks* – and objects even more general than Artin stacks – and arrive at the (perhaps surprising) answer that the right ambient category is *all* sheaves on affines, modulo set theory;
- transport this to analytic geometry: a Grothendieck topology on $(\mathrm{AnRing})^{\mathrm{op}}$, the desiderata one wants the resulting stacks to satisfy, the mechanism of *analytification*, and GAGA read as an isomorphism of stacks;
- address the set-theoretic technicalities via accessible presheaves.

Throughout, "ring" means commutative ring, and we work over a loosely-fixed base (usually $\mathbb{Z}$, occasionally a field). We write $\mathrm{Aff} = (\mathrm{CRing})^{\mathrm{op}}$ for affine schemes.

## 15.1 The paradigm of algebraic geometry

The paradigm is that commutative rings describe basic geometric objects, the *affine schemes*, via the (anti-)equivalence

$$R \longmapsto \operatorname{Spec} R, \qquad \mathrm{CRing} \simeq \mathrm{Aff}^{\mathrm{op}},$$

and that one wants to *glue* affine schemes together to make more general geometric objects.

*Remark* 15.1 (Two aspects of gluing). Gluing has two aspects: *(1) which gluings are allowed*, and *(2) how to identify the results of two different gluings* – more generally, what the maps between formal gluings are, i.e. what the category is. Aspect (2) is best confronted first: the projective line $\mathbb{P}^1 = \mathbb{A}^1_+ \cup_{\mathbb{G}_m} \mathbb{A}^1_-$ (glue two copies of $\mathbb{A}^1$ along $\mathbb{G}_m$ by inversion) must acquire symmetries such as $\mathrm{PGL}_2$ that do *not* respect the chosen decomposition, so the category must be told about maps invisible to any one cover.

The cleanest way to handle (2) is to *specify an ambient category* containing $\mathrm{Aff} = (\mathrm{CRing})^{\mathrm{op}}$, and single out a full subcategory of "allowable gluings of affine schemes" inside it.

*Remark* 15.2 (Classical vs. modern ambient categories). There are two approaches.

- *Classical (schemes).* Take the ambient category to be locally ringed spaces; $R \mapsto \operatorname{Spec} R$ is such, and a scheme is a locally ringed space locally isomorphic to some $\operatorname{Spec} R$. The allowable gluings of affine opens are gluings along open subsets in the locally-ringed-space sense.
- *Modern.* Do not try to be clever about the ambient category; build it formally from CRing. The universal category in which one can glue (i.e. freely adjoin colimits) is the category of presheaves PSh(Aff): giving a colimit-preserving functor out of it is the same as giving an ordinary functor from Aff. (There is a set-theoretic caution here – Aff is not small – taken up in Section 15.7.) One has *formally* allowed gluing but not yet said how to identify the results of two gluings; that is done by passing to sheaves Sh(Aff) for a Grothendieck topology. More covers in the topology means more maps get made (like the $\mathrm{PGL}_2$-symmetries of $\mathbb{P}^1$); too many covers destroys information, so the choice of topology is delicate.

**Example 15.3** (Schemes, modern-style)**.** For the Zariski topology one has

$$\{\text{schemes}\} \subseteq \mathrm{Sh}^{\mathrm{Zar}}(\mathrm{Aff})$$

as the full subcategory of Zariski sheaves that are *locally representable* in the sense of open covers; equivalently, one defines open inclusions in $\mathrm{Sh}^{\mathrm{Zar}}(\mathrm{Aff})$ by reduction (via pullback) to the affine case. This is the perspective we will take for analytic stacks. But schemes are too restrictive a model: more general geometric objects come up and are genuinely relevant.

## 15.2 Why stacks?

Often, moduli spaces in algebraic geometry have the form $X/G$ with $X$ a variety and $G$ a group variety (or scheme) acting on $X$.

**Example 15.4** (Moduli of elliptic curves)**.** Inverting 6 for simplicity, the moduli of elliptic curves is

$$\mathcal{M}_{\mathrm{ell},\,\mathbb{Z}[1/6]} = \operatorname{Spec}\bigl(\mathbb{Z}[1/6][A,B][\Delta^{-1}]\bigr)/\mathbb{G}_m,$$

parametrizing the elliptic curve in Weierstrass form $y^2 = x^3 + Ax + B$ (invert the discriminant $\Delta$); the only isomorphisms between two such curves are scalar substitutions with fixed weights on $x, y$, giving the $\mathbb{G}_m$-action. The same stack has other presentations, e.g. after adding level structure,

$\mathcal{M}_{\mathrm{ell},N}/\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$ (with $\mathcal{M}_{\mathrm{ell},N}$ a variety for $N$ large); the chosen Grothendieck topology must identify the two, so one at least allows étale covers – the classic choice.

*Remark* 15.5 (The naive quotient is bad). The quotients of Example 15.4 exist already in schemes, giving $\mathbb{A}^1_{\mathbb{Z}[1/6]}$, the "$j$-line" (via the $j$-invariant). But this is *not a good quotient*. On $\mathcal{M}_{\mathrm{ell}}$ there is a natural line bundle $\omega = \mathrm{Lie}(E_{\mathrm{univ}})^*$ (the cotangent space at the origin of the universal elliptic curve), which is $\mathbb{G}_m$-equivariant upstairs but does not descend to $\mathbb{A}^1$; and the universal elliptic curve itself cannot be defined over $\mathbb{A}^1$. The problem is that the action is *not free* (some elliptic curves have extra automorphisms), so the scheme-theoretic quotient collapses too much and forgets information.

*Remark* 15.6 (Groupoid quotients and Artin stacks). The solution is to take the quotient in a more refined (2-)category, that of (sheaves of) *groupoids*. The groupoid quotient $X/G$ pretends every action is free: its objects are points of $X$ and its morphisms are elements of $G$, so a fixed point acquires extra automorphisms, and the map $X \to X/G^{\mathrm{gpd}}$ always behaves like the total space of a $G$-bundle – fibres (taken as 2-categorical fibre products) are all isomorphic to $G$:

$$\begin{tikzcd} G \arrow[r] \arrow[d] & X \arrow[d] \\ * \arrow[r] & X/G^{\mathrm{gpd}} \end{tikzcd}$$

Line bundles on $X/G^{\mathrm{gpd}}$ are then exactly $G$-equivariant line bundles on $X$. This leads to *Deligne–Mumford* and, more generally, *Artin* stacks: full subcategories of étale sheaves on Aff that are, roughly, locally – in the sense of admitting a *smooth* cover by affine schemes – of the form $X/G$ for a smooth group scheme $G$ acting on an affine $X$. This is excellent for moduli theory; the theoretical underpinning is Artin's representability theorem [Art74].

*Remark* 15.7 (The constraint: smooth covers force finite type). Artin stacks are still constrained by the need for a smooth cover, which in particular forces *finite presentation*. An infinite-dimensional (non-finite-type) group scheme acting on something need not admit a quotient in this world.

There are many geometrically relevant $X \in \mathrm{Sh}^{\text{ét}}(\mathrm{Aff})$ that are *not* Artin stacks (nor schemes).

**Example 15.8** (Formal schemes)**.** For $R$ noetherian and $I \subseteq R$ an ideal, one wants the formal spectrum. Huber's theory (Section 8) and Grothendieck's theory both view it as a topological ring (localizing along different subsets); but another viewpoint is simply the union of the infinitesimal neighbourhoods,

$$\mathrm{Spf}(R^{\wedge}_I) \;=\; \bigcup_n \mathrm{Spec}(R/I^n), \tag{15.1}$$

so that giving finite-type data on the formal scheme – e.g. a vector bundle – is giving a compatible collection over all the thickenings:

$$\mathrm{Vect}(R^{\wedge}_I) \;=\; \varprojlim_n \mathrm{Vect}(R/I^n). \tag{15.2}$$

This is a very different gluing from those of schemes or Artin stacks: no smooth covers are in sight, only a union of infinitesimal thickenings.

**Example 15.9** (Moduli of formal groups)**.** For some moduli problems one genuinely quotients by an infinite-dimensional group. The example dear to homotopy theorists (and number theorists) is the moduli of 1-dimensional formal groups $\mathcal{M}_{\mathrm{FG}}$ (with the Lubin–Tate spaces as covers): the group to mod out by is coordinate changes on a 1-dimensional formal scheme, with infinitely many coefficients, hence infinite-dimensional. This does not fit the Artin framework and suggests moving to a finer topology, e.g. *fpqc* rather than étale, to accommodate infinite-type covers.

*Remark* 15.10 (Simpson's slogan). There is a remarkable class of examples originating with Carlos Simpson, which one might crudely summarize – Clausen stresses this is his own too-strong-to-be-literally-true heuristic, not a claim of Simpson's – as: *every linear-algebra category is* QCoh *of some (suitably generalized) stack*. Simpson's work produced many instances of natural linear-algebra categories realized as quasi-coherent sheaves on a stack.

**Example 15.11** (Some Simpson-type stacks)**.** Let $k$ be a field.

1. For an algebraic group $G/k$, the classifying stack $BG = */G^{\mathrm{gpd}}$ has

$$\mathrm{QCoh}(BG) \;=\; \mathrm{Rep}_G(k\text{-Vect}),$$

   $G$-equivariant $k$-vector spaces, i.e. representations of $G$.

2. For $\mathbb{A}^1/\mathbb{G}_m$ (scalar multiplication), $\mathbb{G}_m \subseteq \mathbb{A}^1$ is a $\mathbb{G}_m$-invariant open on which one takes $\mathbb{G}_m/\mathbb{G}_m = *$, while the closed complement is $0/\mathbb{G}_m = B\mathbb{G}_m$; so this stack has a point as an open substack and $B\mathbb{G}_m$ as its closed complement. Then

$$\mathrm{QCoh}^{\mathrm{flat}}(\mathbb{A}^1/\mathbb{G}_m) \;=\; \{\, k\text{-vector spaces with a } \mathbb{Z}\text{-indexed filtration} \,\}.$$

**Example 15.12** (The de Rham stack)**.** Let $k$ have characteristic 0 and $X/k$ a smooth variety. Simpson's de Rham stack $X^{\mathrm{dR}}$ is presented by quotienting $X$ not by a group action but by the equivalence relation identifying infinitesimally close points: one takes the formal completion of $X \times_k X$ along the diagonal,

$$(X \times_k X)^{\wedge}_X \rightrightarrows X \longrightarrow X^{\mathrm{dR}}, \tag{15.3}$$

the completion taken in the sense of (15.1). Then

$$\mathrm{Vect}(X^{\mathrm{dR}}) \;=\; \{\, \text{vector bundles on } X \text{ with flat connection} \,\},$$

Grothendieck's interpretation of a flat connection as descent data along infinitesimally close points, and the cohomology of the structure sheaf recovers de Rham cohomology,

$$R\Gamma(X^{\mathrm{dR}}, \mathcal{O}) \;=\; R\Gamma_{\mathrm{dR}}(X/k).$$

This is far from an Artin stack: one is modding out by a formal scheme giving an equivalence relation.

*Remark* 15.13 (Prismatic and related stacks). More recently, Bhatt–Lurie and Drinfeld define stacks whose QCoh captures coefficient systems for various $p$-adic cohomology theories in characteristic $p$, mixed characteristic, or in the limit over $\mathbb{Z}_p$ ([Dri24; BL22]). For de Rham cohomology in characteristic $p$ there is again a stack – using the divided-power envelope of the diagonal in place of the formal neighbourhood (15.3). Prismatic, de Rham, and crystalline cohomology, and the comparison theorems between them, then acquire a geometric explanation: an a priori linear-algebraic comparison of cohomology theories is promoted to an isomorphism of stacks, from which the comparison follows by passing to quasi-coherent sheaves and functoriality.

**Example 15.14** (Betti stacks and condensed sets)**.** Betti cohomology – singular/sheaf cohomology with constant coefficients on the analytic topological space underlying a complex variety – also has a stacky avatar. For $S$ a compact Hausdorff space, choose a surjection $T_0 \twoheadrightarrow S$ from a profinite set; then $T_1 := T_0 \times_S T_0$ is again profinite (a closed subset of $T_0 \times T_0$), so $S$ is the quotient of an equivalence relation in profinite sets. Applying Spec of the ring of continuous $\mathbb{Z}$-valued (or $k$-valued) functions gives a groupoid in affine schemes, whose quotient in fpqc sheaves is the Betti stack:

$$\operatorname{Spec} C(T_1, \mathbb{Z}) \rightrightarrows \operatorname{Spec} C(T_0, \mathbb{Z}) \longrightarrow S^{\mathrm{Betti}}. \tag{15.4}$$

Then quasi-coherent sheaves on $S^{\mathrm{Betti}}$ are ordinary sheaves of $\mathbb{Z}$-modules (abelian groups) on the topological space $S$, and coherent cohomology on $S^{\mathrm{Betti}}$ is topological cohomology of $S$. More generally a condensed set can be viewed as a stack, by mapping its profinite presentation into the world of stacks via this procedure:

$$\{\text{condensed sets}\} \longrightarrow \{\text{stacks}\}.$$

*Remark* 15.15 (Why the descent works)*.* The checks for (15.4): a surjection of profinite sets induces a faithfully flat map on continuous functions, and fibre products of profinite sets correspond to relative tensor products of the function rings. Both reduce, via filtered colimits, to finite sets, where $C(-, \mathbb{Z})$ is a finite product of copies of $\mathbb{Z}$ and everything is purely combinatorial. In fact any map of profinite sets is flat (surjective $\Rightarrow$ faithfully flat), and $\operatorname{Spec} C(T_0, \mathbb{Z})$ is $\underline{T_0}$, the inverse limit of the constant schemes on the finite quotients.

Faced with this baffling array of stacks – many bearing no resemblance to Artin stacks, yet wanted as convenient encodings of linear-algebraic and geometric phenomena – one asks:

*Remark* 15.16 (The question and its answer)*.* *Question.* How to define a reasonable full subcategory of $\mathrm{Sh}^{\mathrm{fpqc}}(\mathrm{Aff})$ containing all of the above examples?

*Answer.* $\mathrm{Sh}^{\mathrm{fpqc}}(\mathrm{Aff})$ itself – the whole category (modulo set theory, Section 15.7). It is not clear that there is any other answer. There is content in this answer, but it introduces nothing not already present in the question: one takes the full subcategory that is the whole thing.

## 15.3 Analytic stacks: the plan and the desiderata

*Remark* 15.17 (The plan)*.* We take the same approach for analytic stacks: define a Grothendieck topology on the affine analytic stacks $(\mathrm{AnRing})^{\mathrm{op}}$ – the opposite of analytic rings – and take sheaves with respect to it (again modulo set theory). We will not fix the precise topology today; instead we describe the phenomena the resulting theory should capture.

*Remark* 15.18 (Desiderata)*.* The examples we want (each producing an analytic stack):

1. *Adic spaces* (Huber). The building blocks – Huber pairs $(R, R^+)$ – already give analytic rings (Sections 8 and 9), and the localization of $D(R)$ over the Huber spectrum (Sections 9 and 10) is the input to gluing along rational open subsets. We want that gluing to yield an analytic space.

2. *Complex analytic spaces*, over a gaseous or $p$-liquid version of $\mathbb{C}$, with gluing along complex-analytic open subsets allowed.

3. *Algebraic stacks*: analytic geometry should generalize schemes. Over the uncompleted base $\mathbb{Z}$ (the trivial/maximal structure, with $D = D(R^{\triangleright})$ for all modules, cf. Remark 1.32) one can

do ordinary algebraic geometry; a slight modification of Remark 15.16 gives algebraic stacks as analytic stacks. Universally over any analytic ring, an algebraic object yields an analytic object by base change.

4. *Banach rings* $\rightsquigarrow$ an analytic stack matching Berkovich's theory. Interestingly, the stack attached to a general Banach ring is *not* affinoid but a genuine stack: e.g. $\mathbb{Z}$ with its usual archimedean norm goes to a stack corresponding to the Berkovich spectrum $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$ (a tree, cf. Proposition 12.9), not an affinoid.

5. *Coefficient systems*: analytic stacks whose QCoh captures various cohomology-coefficient systems (as in Remark 15.13).

6. *The Tate curve* and its uniformization over the gaseous base ring (Definition 14.18), together with machinery to prove it algebraic – a Riemann–Roch theorem (giving a projective embedding from the identity section) and a GAGA theorem.

## 15.4 Analytification: importing Huber's theory of pairs

We spend some time on how one item – GAGA – can be explained as an isomorphism of stacks, since it illustrates the mechanism.

*Remark* 15.19 (Recollection: the solid affine line). Adic geometry in the solid context (Section 7) got started once we saw that there is a nice "subset" of the affine line $\mathbb{A}^1$ over $\mathbb{Z}_\square$, namely the closed unit disk (which Remark 7.14 taught us to think of as *open*). Algebraically it came from an *idempotent* algebra – the power series ring $\mathbb{Z}[\![T]\!] = P^\square$, regarded in $\mathbb{Z}[T]$-modules in solid abelian groups (the analytic ring controlling $\mathbb{A}^1_{\mathbb{Z}_\square}$, Lemma 7.2 and Example 7.13). Moving it to infinity by $T \mapsto T^{-1}$ produced the closed unit disk, which lets one tie into the $(R, R^+)$-theory: one asks that the elements of $R^+$ actually land in the closed unit disk, not merely map to $\mathbb{A}^1$.

*Remark* 15.20 (The gaseous analogue: germs at the origin). Working instead over the gaseous ring $\mathbb{Z}[\widehat{q}^{\pm 1}]^{\mathrm{gas}}$ (so $q$ is inverted; cf. Theorem 14.3), one can again define a "subset" of $\mathbb{A}^1_{\mathbb{Z}[\widehat{q}^{\pm 1}]^{\mathrm{gas}}}$ corresponding to an idempotent algebra – morally, the functions convergent in *some unspecified* disk around the origin, i.e. germs of functions at 0. It is a filtered colimit

$$\varinjlim \big( P \xrightarrow{q\,\cdot} P \xrightarrow{q\,\cdot} \cdots \big) \tag{15.5}$$

of the basic object $P$ under multiplication by $q$. In contrast to the solid case, where $P$ ($= \mathbb{Z}[\![T]\!]$ once solidified) was already idempotent, here $P$ is *not* idempotent as an object of $\mathbb{Z}[T]$-modules in the gaseous category; but taking the colimit (15.5) – which shrinks the open unit disk down to the origin – makes it idempotent.

This idempotent algebra shares the formal properties of $\mathbb{Z}[\![T]\!]$ and lets us import Huber's theory of pairs into the gaseous context.

**Definition 15.21** (Analytification of an affine over a gaseous base)**.** Let $R$ be finite type over the underlying ring $\mathbb{Z}[\widehat{q}^{\pm 1}]^{\mathrm{gas}}(*)$, and equip it with the analytic ring structure $(R, \mathrm{Mod}_R(\text{gaseous modules}))$. An element $f \in R$ gives a section $\operatorname{Spec} R \xrightarrow{f} \mathbb{A}^1$. Moving the germ-at-0 idempotent algebra of Remark 15.20 to infinity gives a germ-at-$\infty$ idempotent algebra; one then passes to the subspace of $\operatorname{Spec} R$ on which $f$ lands away from infinity – concretely, where killing that idempotent algebra after base change along $f$ holds,

$$(\text{germ-at-}\infty) \;\otimes_R\; R/f \;=\; 0. \tag{15.6}$$

The result is a new analytic space $\mathrm{Spec}(R)^{\mathrm{an}}$, the *analytification*.

**Example 15.22** (Analytic $\mathbb{G}_m$)**.** For $R = \mathbb{Z}[T^{\pm 1}]$ the analytification $\mathrm{Spec}(R)^{\mathrm{an}} = \mathbb{G}_m^{\mathrm{an}}$ is exactly the object of Section 14 that one quotients by $q^{\mathbb{Z}}$ to build the Tate curve (Definition 14.18) – up to base change to a ring where $\mathbb{Z}$ is bounded, see Remark 15.23.

*Remark* 15.23 (Two corrections: which functions, and boundedness of the base)*.* As stated at the board, Definition 15.21 was corrected twice.

1. The analytification does *not* depend on imposing (15.6) for every $f \in R$ separately. It suffices to impose it for the universal coordinate (for $\mathbb{G}_m$: for $T$ and $T^{-1}$); one automatically gets the condition for all elements, by the general formalism. Imposing it naïvely for all $f$ is not even well-posed as a subset condition – the locus "away from $\infty$" is not closed under addition (the sum of two bounded functions need not be bounded).
2. For the analytification of $\mathrm{Spec}\, R$ itself to be well-behaved one must add the axiom that every element of the ground ring, in particular every integer – and 2 being bounded is enough – is bounded. Otherwise the analytification changes the base: forcing all scalars to be bounded is itself a nontrivial operation. With the axiom, $\mathrm{Spec}\, R$ is its own analytification. This axiom was already present in the earlier write-up of complex-analytic geometry [CS20]; the archimedean/Berkovich subtlety is exactly that the archimedean branch of $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$ runs all the way out, so an unbounded integer like 2 escapes the "away from $\infty$" locus.

*Remark* 15.24 (Solid and gaseous theories glue)*.* The gaseous construction makes sense not only over $\mathbb{Z}[\widehat{q}^{\pm 1}]^{\mathrm{gas}}$ but over $\mathbb{Z}[q]$ and $\mathbb{Z}[q]^{\mathrm{gas}}$: defining the structure only required writing down the endomorphism $1 - q \cdot \mathrm{shift}$ of $P$ and asking it to become an isomorphism, which needs neither $q$ topologically nilpotent nor $q$ a unit. Specializing $q$ interpolates the states of matter (cf. Remark 12.19):

$$\begin{aligned} q = 0 &\rightsquigarrow \text{uncompleted } \mathbb{Z}\text{-theory}, \\ q = 1 &\rightsquigarrow \text{solid theory}, \\ q \text{ top. nilpotent unit} &\rightsquigarrow \text{gaseous theory}. \end{aligned}$$

Working *away* from the locus of topological nilpotence one forces both $q$ and $q^{-1}$ to be gaseous, hence 1 gaseous, hence solid; so over that locus $\mathbb{Z}[q^{\pm 1}]^{\mathrm{gas}}$ restricts to $\mathbb{Z}[q^{\pm 1}]_{\square}$. Thus the solid and gaseous theories are glued together.

### 15.5 GAGA as an isomorphism of stacks

*Remark* 15.25 (A stack of analytifications)*.* Any variable $q$ declared to be gaseous produces (via the germ-at-0 colimit (15.5), moved to $\infty$) an idempotent algebra and hence a notion of analytification. Different choices of $q$ differing by a bounded unit – an element of $\mathbb{G}_m^{\mathrm{an}}$ – give the *same* algebra. So the parameter space of notions of analytification is the stack

$$\mathrm{Spec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]^{\,q\text{-gas}}\big)/\mathbb{G}_m^{\mathrm{an}}, \tag{15.7}$$

a quotient of (the spectrum of) an analytic ring by an equivalence relation. This is a new role of stacks in the theory: a stack parametrizing choices of analytic geometry over a given base ring. For an analytic ring $R$, a map

$$\mathrm{Spec}\, R \longrightarrow \mathrm{Spec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]^{\,q\text{-gas}}\big)/\mathbb{G}_m^{\mathrm{an}} \tag{15.8}$$

is a *gaseous structure on* $\mathrm{Spec}\, R$. (The quotient by $\mathbb{G}_m^{\mathrm{an}}$ encodes that a topologically nilpotent unit should only matter up to a bounded difference: given a Tate–Huber ring there exists such a unit, but it plays no essential role beyond a bounded ambiguity. Clausen and Scholze noted at the board that the group one divides by is a subtler object than the naïve $\mathbb{G}_m^{\mathrm{an}}$ – one modifies $q$ by units of norm 1 or by raising to a power – and left the precise identification to a later lecture. )

*Remark* 15.26 (Two functors and a natural transformation). Fix a gaseous structure (15.8) on $\operatorname{Spec} R$ such that $2 \in \mathbb{Z}$ is bounded (Remark 15.23). Then one gets a theory of analytification: two functors from $R$-schemes to analytic stacks over $R$, together with a natural transformation. For $X = \operatorname{Spec} A$ with $A$ an $R(*)$-algebra:

- on the one hand, view $X$ as an analytic stack over the uncompleted $\mathbb{Z}$ and base change to $R$, giving $X_R := X \times_{\mathbb{Z}} R$ (ordinary algebraic geometry over $R$);
- on the other hand, form the analytification $X_R^{\mathrm{an}}$ over $R$ (requiring the coordinate functions to land away from $\infty$).

Both glue over non-affine $X$, and there is a natural map

$$\begin{tikzcd} X_R^{\mathrm{an}} \arrow[d, "\mathrm{nt}"] \\ X_R = X \times_{\mathbb{Z}} R. \end{tikzcd} \tag{15.9}$$

**Theorem 15.27** (GAGA). *With notation as in Remark 15.26, and assuming every $f \in R(*)$ is bounded, if $X \to \operatorname{Spec}(R(*))$ is proper (and finitely presented), then the natural transformation* (15.9) *is an isomorphism:*

$$X_R^{\mathrm{an}} \xrightarrow{\sim} X_R.$$

*Remark* 15.28 (On the finite-presentation hypothesis). Clausen flagged the finite-presentation hypothesis as one he was unsure is needed for the isomorphism itself: it entered when he last thought about setting up the formalism with lower-shrieks and proper pushforward (to know, e.g., that resolving $R/I$ relatively over a noetherian $R$ gives compact projective generators that are products of copies of $R/I$). It may be inessential for the GAGA statement proper but relevant for proper pushforward ($f_* = f_!$); kept in to be safe.

*Remark* 15.29 (Formal and non-formal consequences). Theorem 15.27 implies *formally* that the derived categories agree,

$$D(X_R^{\mathrm{an}}) = D(X_R),$$

a version of GAGA; and *non-formally* (under further, quite general technical conditions on $R$) that, e.g.,

$$\mathrm{Vect}(X_R^{\mathrm{an}}) = \mathrm{Vect}(X_R) \implies \text{classical GAGA}.$$

*Remark* 15.30 (Special cases). All classical GAGA theorems are special cases of Theorem 15.27.

1. *Complex-analytic GAGA* (Serre [Ser56]): over $\mathbb{C}^{\mathrm{gas}}$, or $\mathbb{C}$ with a $p$-liquid structure, etc.
2. *Formal GAGA* (Grothendieck): for a complete noetherian ring with a proper scheme over it, coherent sheaves agree with coherent sheaves on the formal completion (compatible collections over the nilpotent thickenings, (15.2)). In Huber-pair terms one works over $\operatorname{Spa}(R_I^{\wedge}, R_I^{\wedge})$, which lives over $\mathbb{Z}_{\square}$ and inherits the closed-unit-disk notion of analytification.

3. *Non-archimedean GAGA* over $\mathbb{Q}_p$ (or any complete non-archimedean field): take the analytic ring $\mathrm{Spec}(\mathbb{Q}_{p,\square})$ with the gaseous structure corresponding to

$$\mathrm{Spec}(\mathbb{Q}_{p,\square}) \xrightarrow{q \mapsto p} \mathrm{Spec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]^{q\text{-gas}}\big), \tag{15.10}$$

$p$ being the standard choice of topologically nilpotent unit. This picks out the usual analytification of algebraic varieties over $\mathbb{Q}_p$.

*Remark* 15.31 (The $\mathbb{Q}_p$ excursion, corrected). Clausen initially suggested that choosing instead a gaseous structure on $\mathbb{Q}_{p,\square}$ factoring through $\mathbb{Z}_\square$ would give a different GAGA theorem, with the analytification of $\mathbb{A}^1$ becoming the closed unit disk. This was corrected in discussion. With that "$\mathbb{Z}_\square$" structure one does *not* get a GAGA statement at all: the required boundedness axiom (Remark 15.23) would force $1/p$ to be bounded, and forcing all scalars bounded in the solid $\mathbb{Q}_p$ setting kills everything (the 0 ring). (The false expectation "$\mathbb{A}^{1,\mathrm{an}}$ = closed disk" also fails because forcing $T$ analytic gives no reason for $T/p$ to be analytic, contradicting the $\mathbb{G}_m(\mathbb{Q}_p)$-rescaling symmetry.) The salvageable point: there *is* a second, genuine gaseous structure on $\mathbb{Q}_{p,\square}$ obtained by factoring through $\mathbb{Z}_{p,\square}$, and the difference between it and the standard one (15.10) is morally the difference between usual rigid geometry and an over-convergent version of rigid geometry.

## 15.6 Relations between examples

Beyond GAGA, one wants the category of analytic stacks to be large enough to house *maps* between the various classes of examples, whenever a comparison is expected. Some maps to be made sense of as morphisms of analytic stacks:

*Remark* 15.32 (Comparison maps).

1. The Tate curve maps to the topological circle,

$$\mathrm{Tate}_q \longrightarrow (0,\infty)/q^{\mathbb{Z}} = S^1,$$

from Scholze's uniformization (Definition 14.18 and the square there), now to be a map of analytic stacks: $S^1$ is an algebraic (indeed topological) stack via Example 15.14, and algebraic stacks embed into analytic stacks even over the initial analytic ring.

2. A complex-analytic space $X$ maps to its underlying topological space,

$$X \longrightarrow X(\mathbb{C}),$$

which explains (essentially by definition) the well-known fact that the coherent cohomology of $X$ localizes on the underlying topological space.

3. A Banach ring $R$ maps its (liquid) spectrum to the Berkovich spectrum,

$$\mathrm{Spec}^{\mathrm{liq}}(R) \longrightarrow \mathcal{M}^{\mathrm{Berk}}(R),$$

in the same way that the $(R, R^+)$ affine analytic stacks match Huber's picture.

The comparison isomorphisms one expects between these (e.g. Betti vs. de Rham in the complex-analytic setting) should likewise promote to isomorphisms of analytic stacks.

## 15.7 Set-theoretic technicalities

We at least pin down what the category of analytic *stacks* is (the Grothendieck topology is deferred).

*Remark* 15.33 (The problem). In the algebraic analogue, $\mathrm{Sh}^{\mathrm{fpqc}}(\mathrm{Aff})$ is a full subcategory of $\mathrm{Fun}(\mathrm{CRing}, \mathrm{Set})$ (or groupoids), but $\mathrm{Aff} = (\mathrm{CRing})^{\mathrm{op}}$ is not a small category, so morphisms in the functor category can involve more than a set's worth of data – a genuine pain.

**Definition 15.34** (Accessible presheaves)**.** Instead of $\mathrm{PSh}(\mathrm{Aff})$, consider the *accessible* presheaves

$$\mathrm{PSh}^{\mathrm{acc}}(\mathrm{Aff}) \;=\; \mathrm{Fun}^{\mathrm{acc}}(\mathrm{CRing}, \mathrm{Set}),$$

where a functor $F$ is *accessible* iff there exists a regular cardinal $\kappa$ such that $F$ commutes with $\kappa$-filtered colimits.

*Remark* 15.35 (Reading the cardinal). One should think of a regular cardinal $\kappa$ not as a set but as indexing the sets of smaller cardinality. For $\kappa = \aleph_0$ these are the finite sets and $\kappa$-filtered colimits are ordinary filtered colimits (so a functor commuting with filtered colimits is accessible). Larger $\kappa$ means *fewer* $\kappa$-filtered colimits, hence *more* functors commute with them: e.g. $\kappa = \aleph_1$ indexes countable sets, and a $\aleph_1$-filtered diagram is one in which every countable subset admits a cone.

*Remark* 15.36 (Accessible = small colimit of representables). For every $X \in \mathrm{Aff}$ the representable $h_X$ is accessible; in fact a presheaf is accessible iff it is a *small* colimit of representables $h_X$. Thus, thinking of affine schemes as building blocks and colimits as gluing, accessibility says precisely that one is only allowed to glue a set's worth of things at a time. Moreover fpqc sheafification preserves accessible presheaves (Waterhouse [Wat75]), so one may impose the sheaf condition without leaving the accessible world and without set-theoretic obstruction.

*Remark* 15.37 (What makes this work). Two non-trivial inputs:

1. CRing is presentable – in fact compactly generated; this is what makes the theory of accessible presheaves work (it is what lets "commutes with $\kappa$-filtered colimits" match "small colimit of representables");
2. there is an a priori cardinality bound on fpqc covers: every fpqc cover in Aff is an $\aleph_1$-filtered colimit of $\aleph_1$-compact (countably presented) fpqc covers. Equivalently, every flat cover of a ring is an $\aleph_1$-filtered colimit of countably-presented flat covers.

Such a bound is what lets one verify that fpqc sheafification preserves accessibility.

**Theorem 15.38** (Presentability of analytic rings)**.** *The category of analytic rings is presentable; in fact it is $\kappa$-compactly generated for $\kappa = (2^{\aleph_0})^+$ (so $\kappa$-small means of cardinality at most the continuum).*

This is what lets one formally avoid the set-theoretic difficulties of taking presheaves and sheaves on the large category $(\mathrm{AnRing})^{\mathrm{op}}$.

*Remark* 15.39 (An enrichment aside). The category of analytic rings is naturally enriched over condensed anima (its mapping spaces are condensed anima). Analytic *stacks* are, however, not enriched over analytic rings (nor obviously over themselves). They *do* carry a distinct enrichment over light condensed sets: an $S$-valued point of a stack is a map

$$S \times X \;\longrightarrow\; Y, \qquad S \text{ a light profinite (condensed) set,}$$

imported through the very trivial analytic rings of continuous functions (Example 15.14). This enrichment has essentially no relation to the condensed-anima enrichment on analytic rings, which Clausen notes is genuinely confusing – for instance Wiesława Nizioł (in recent work on $p$-adic coefficient systems) wants sheaves of solid abelian groups on the pro-étale site, so that profinite sets appear in two orthogonal directions at once.

# 16 Analytic Stacks and Six Functors: Proper Maps, Open Immersions, and the Grothendieck Topology

Having motivated analytic stacks last time (Section 15) as sheaves on the opposite category of analytic rings, Scholze now turns to the objective and the mechanism. The plan of Section 15 was: take the affine analytic spaces attached to analytic rings as basic building blocks, and specify a Grothendieck topology on $(\mathrm{AnRing})^{\mathrm{op}}$ that tells us which gluings are allowed. The whole content of the definition sits in *which* Grothendieck topology one chooses, and the thesis of this lecture is that this choice is dictated – once one insists on a good theory – by the requirement that the assignment $A \mapsto D(A)$ underlie a *six-functor formalism*. Setting up such a formalism, extracting proper maps and open immersions from it, and reading off the topology occupy the lecture.

Throughout "ring" means commutative ring and everything is (light) condensed. Two technical caveats are deferred to the end: the set-theoretic issue that $(\mathrm{AnRing})^{\mathrm{op}}$ is a large category (so "small sheaves" must be used, as in Section 15.7), and the issue of sheaves versus hypersheaves – one lands on something strictly in between. We suppress both for now.

## 16.1 Recollection: the same game, played before

*Remark* 16.1 (Analytic stacks by analogy with condensed sets). The construction to come is very close to what was done at the very beginning of the course. There, condensed (or light condensed) sets were defined as sheaves on profinite sets,

$$\mathrm{CondSet}^{\mathrm{light}} \;=\; \mathrm{Shv}\big(\mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})\big) \;=\; \mathrm{Shv}\big((\text{countable Boolean algebras})^{\mathrm{op}}\big),$$

starting from basic objects (profinite sets) and allowing oneself to glue them along a specified, rather general, Grothendieck topology – finite disjoint unions together with *all* surjections of profinite sets (Definition 2.16). We now play the same game with a much broader class of basic geometric objects, the affine analytic spaces attached to analytic rings. These model many different geometries at once – algebraic, $p$-adic, complex analytic – and the point of analytic stacks is to put all of them into a single world where spaces can be built out of pieces of different provenance. (One could equally have started from some nicer class of basic objects, e.g. smooth manifolds; that too would sit inside the world of analytic stacks.)

## 16.2 Desiderata: the primary invariant and the six functors

The primary invariant attached to an analytic ring is its derived category of complete modules.

*Remark* 16.2 (The primary invariant is a sheaf of $\infty$-categories). For an analytic (animated) ring $A = (A^{\triangleright}, \mathrm{Mod}_A)$ – most examples of interest are static, but the general formalism is much better behaved if animated rings are allowed – the invariant we care about is

$$D(A) \;\subseteq\; D(A^{\triangleright}), \qquad D(A) \;=\; \{\, M \in D(\mathrm{Cond}(A^{\triangleright})) \;:\; H_i(M) \in \mathrm{Mod}_A \;\; \forall i \,\},$$

the full subcategory of the derived category of condensed $A^{\triangleright}$-modules whose homology groups are all complete (Definitions 8.14 and 13.1). We want the assignment

$$A \;\longmapsto\; D(A)$$

to be a *sheaf*, and the only sensible way to phrase this is as a sheaf of $\infty$-categories: from now on $D(A)$ is always treated as an $\infty$-category. In particular, for any analytic stack $X$ one then *defines* its quasi-coherent derived category by descent,

$$D(X) \;=\; \varprojlim_{\mathrm{AnSpec}(A)\to X} D(A),$$

the limit over all affine analytic spaces mapping to $X$.

We want something more than a bare sheaf of $\infty$-categories, though: we want the whole assignment $X \mapsto D(X)$ to carry a six-functor formalism.

*Remark* 16.3 (The six operations). A six-functor formalism organizes six operations into three adjoint pairs:

$$\otimes \;\dashv\; \mathrm{RHom}, \qquad f^* \;\dashv\; f_*, \qquad f_! \;\dashv\; f^!,$$

of which the first two pairs are structure we already possess, and the third pair is the genuinely new ingredient. Concretely:

1. Each $D(X)$ carries a symmetric monoidal tensor product $\otimes$, and it is *closed* – there is an internal Hom object RHom with the usual adjunction $\mathrm{Hom}(A\otimes B, C) = \mathrm{Hom}(A, \mathrm{RHom}(B, C))$.
2. For every map $f\colon Y \to X$ there is a pullback $f^*\colon D(X) \to D(Y)$, symmetric monoidal, with right adjoint the pushforward $f_*\colon D(Y) \to D(X)$. These come for free from the functoriality of module categories: a map of rings lets one base-change modules, and the formalism is automatically compatible with base change.
3. For certain maps $f\colon Y \to X$ – the *shriekable* ones – there is a third functor $f_!\colon D(Y) \to D(X)$, *cohomology with compact support*, with right adjoint $f^!$.

*Remark* 16.4 (The two properties of $f_!$). The functor $f_!$ is required to satisfy two properties, which are exactly what pin down its meaning. First, *base change*: for any Cartesian square

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ {\scriptstyle f'}\big\downarrow & & \big\downarrow{\scriptstyle f} \\ X' & \xrightarrow[g]{} & X \end{array}$$

the natural comparison $g^* f_! \xrightarrow{\sim} f'_!\, g'^*$ is an isomorphism; in particular $f'$ is again in the class where $f_!$ is defined. Second, the *projection formula*: for $A \in D(X)$ and $B \in D(Y)$,

$$f_!\big(f^*A \otimes B\big) \;\cong\; A \otimes f_!\, B. \tag{16.1}$$

*Remark* 16.5 (Where this structure comes from, and coherent compact support). This is a structure arising across mathematics. The most classical instance takes a nice category of topological spaces (locally compact Hausdorff, say) and a suitable derived category of sheaves; then $f_!$ exists for *all* morphisms and is literally the derived functor of sections with compact support. Historically the formalism was first developed not there but for torsion étale sheaves on schemes (Grothendieck), and later recognized for $D$-modules and other settings. One place where it was *not* much developed is the coherent (quasi-coherent) setting, where one usually has no notion of coherent cohomology with compact support: there is Deligne's appendix to Hartshorne's *Residues and Duality* [Har66] sketching such a formalism, and a few scattered papers, but it is not a theory in wide use. Our setting will naturally include a well-behaved theory of coherent cohomology with compact support, and we very much want it.

*Remark* 16.6 (The isomorphism must be supplied: abstract six-functor formalisms). As Gabber would rightly object, the description above is imprecise: when one writes the projection-formula isomorphism (16.1), there is no canonical comparison between the two functors – these are *a priori* two different functors that happen to be identified, and the identification is data one must supply. Once one starts supplying such isomorphisms, coherence problems proliferate (base-change twice, compare the two ways, ask which compatibilities are forced, and so on), and it was for a while unclear what the minimal way to encode all the data in a six-functor formalism is. There is now a good $\infty$-categorical notion of an *abstract six-functor formalism*: work of Liu–Zheng [LZ24] in the setting of higher Artin stacks; work of Gaitsgory–Rozenblyum [GR17] in coherent cohomology (with different names for the functors, and no talk of compact-support cohomology, but setting up such a formalism), which is however phrased in $(\infty, 2)$-categorical language and correspondingly hard; and Mann's isolation of the key structures [Man22], further streamlined in Scholze's own course and notes on six-functor formalisms [Sch23]. A moral Scholze takes from that course:

*the definition of a six-functor formalism determines everything, including the "correct" choice of Grothendieck topology.*

## 16.3 Constructing $f_!$: proper maps, open immersions, shriekable maps

The abstract theory reduces the construction of $f_!$ to specifying two classes of morphisms and checking a short list of conditions – *conditions*, not data.

*Remark* 16.7 (The idea of the construction). One specifies:

- a class of *proper* morphisms, on which one *declares* $f_! = f_*$; here $f_!$ carries no new data, but one must check that this choice satisfies proper base change and the projection formula. Since $f_! = f_*$, there is always a natural base-change transformation and a natural projection-formula map already present, so declaring $f$ proper is just asking that these natural maps be isomorphisms – a condition, not a datum.
- a class of *open immersions*, on which one declares $f_!$ to be the *left* adjoint $j_!$ of the pullback $j^*$; again there is a natural comparison map in each of base change and the projection formula, and one asks that they be isomorphisms.

The general maps for which $f_!$ is then defined – the *shriekable* maps – are the composites of an open immersion followed by a proper map:

$$\begin{tikzcd} Y \arrow[r, "j"] \arrow[rr, bend right, "f"'] & \overline{Y} \arrow[r, "\overline{f}"] & X, \end{tikzcd} \qquad f_! := \overline{f}_* j_!, \tag{16.2}$$

with $j$ an open immersion (a "compactification" of $Y$ inside $\overline{Y}$) and $\overline{f}$ proper.

*Remark* 16.8 (Independence of the compactification). Written as (16.2) this is not yet a definition: a shriekable map admits many compactifications, and one must show $f_!$ is independent of the choice, canonically – and, since everything is $\infty$-categorical, not merely up to isomorphism but with the isomorphism itself unique up to coherent higher homotopy, and so on. This sounds like a real pain, but it is exactly what the abstract machinery handles.

**Theorem 16.9** (Liu–Zheng, Mann). *Under very minimalistic assumptions on the classes of proper maps, open immersions, and the resulting shriekable maps (essentially only that composites of shriekable maps are shriekable, i.e. can be compactified), the above data produces an abstract six-functor formalism. The assumptions include* no *condition whatsoever on uniqueness or domination* *of compactifications: for each shriekable map one need only produce* one *compactification, not compare two.*

*Remark* 16.10 (On the no-uniqueness point)*.* Gabber asked whether one must assume that two compactifications can be dominated by a third. One does not. (Something in this direction follows for free from the mere existence of compactifications: given two, the map to their fibre product is still a compactification.) The one genuinely needed input is that every shriekable map is such a composite – open immersion followed by proper – but only existence of a single factorization is required. A precise statement is in Scholze's six-functor notes [Sch23] and in Mann's thesis [Man22]. As a sample of what the formalism is for: it should let one prove Riemann–Roch-type statements and construct characteristic (Chern) classes in the analytic-stacks setting.

### 16.4 The category, and proper maps

We now apply these ideas to our setting.

*Notation* 16.11 (Affine analytic spaces)*.* Take the category

$$\mathcal{C} \;=\; (\mathrm{AnRing})^{\mathrm{op}} \;=\; \mathrm{Aff}\ \mathrm{AnStk},$$

the affine analytic (stacks) spaces. An object is written $\mathrm{AnSpec}(A)$; the symbol AnSpec carries no topological space – there is, unfortunately, no underlying space now – and is just a formal name for the geometric object attached to $A$, turning AnRing into a category of geometric objects by passing to the opposite.

We must decide which morphisms are proper and which are open immersions, and here one should discard any preconceived notion of properness from algebraic geometry and read off what the formulas demand.

**Definition 16.12** (Proper map)**.** A map $f\colon \mathrm{AnSpec}(B) \to \mathrm{AnSpec}(A)$ – i.e. a map of analytic rings $A \to B$ – is *proper* if the pushforward

$$f_*\colon D(B) \;\longrightarrow\; D(A),$$

which is simply the forgetful functor (restriction of scalars along $A \to B$: any $B$-module is an $A$-module), satisfies the projection formula.

*Remark* 16.13 (This is not the algebraic notion)*.* This properness is very different from the algebraic-geometry one: usually properness is finiteness-flavoured, whereas here it will be satisfied very generally. One could *a priori* demand more of $f_*$ (proper base change, etc.), but it turns out that once the projection formula holds, all the other good properties follow. In particular if one embeds ordinary commutative rings into the condensed setting via the trivial (maximal) analytic ring structure (Remark 1.32), then *every* map of ordinary rings is declared proper – which looks strange, but is exactly Huber's usage for adic spaces (see Remark 16.17).

To analyze properness, recall the two canonical operations on analytic ring structures over a fixed base.

*Remark* 16.14 (Induced structure and further completion)*.* Given a map of analytic rings $A = (A^{\triangleright}, D(A)) \to B = (B^{\triangleright}, D(B))$, any map factors canonically through the *induced analytic ring structure* on $B^{\triangleright}$: this is the pair

$$\big(B^{\triangleright},\ \{\, M \in D(B^{\triangleright}) \;:\; M|_{A^{\triangleright}} \in D(A)\,\}\big), \tag{16.3}$$

whose complete modules are those $B^{\triangleright}$-modules that are complete when restricted to $A$-modules. So an arbitrary map $A \to B$ is the composite of the passage $A \to$ (induced structure on $B^{\triangleright}$) followed by a *further completion* (a localization keeping the same underlying ring $B^{\triangleright}$ but shrinking the complete modules). It is precisely this further completion that corresponds geometrically to compactification – and the induced structures are exactly the proper maps.

**Proposition 16.15** (Properness = inducedness)**.** *The map $f\colon \mathrm{AnSpec}(B) \to \mathrm{AnSpec}(A)$ is proper if and only if $B$ carries the induced analytic ring structure from $A$, i.e.* (16.3) *holds.*

*Proof.* The projection formula asks that, for all $M \in D(A)$ and $N \in D(B)$,

$$f_!\big(f^*M \otimes_B N\big) \;\cong\; M \otimes_A f_! N, \qquad \text{i.e.} \quad (M \otimes_A B) \otimes_B N \;\cong\; M \otimes_A N,$$

where on the right the tensor is taken in $D(A)$ (recall $f_! = f_*$ is the forgetful functor). If $B$ has the induced structure, all tensor products can be computed in $D(A)$ and the identity is clear. Conversely, take $N = B^{\triangleright}$. Then the left-hand side is the base change $M \otimes_A B$ *in the sense of analytic rings*, while the right-hand side is $M \otimes_A B^{\triangleright}$ computed in $D(A)$; so the projection formula says

$$M \otimes_A B \;\cong\; M \otimes_A B^{\triangleright} \qquad (\text{tensor in } D(A)). \tag{16.4}$$

The special case $A^{\triangleright} = B^{\triangleright}$ is instructive: base change to $B$ is then just the further completion, and taking $M$ complete and $N = B^{\triangleright} = A^{\triangleright}$ neither side does anything ($M$ is tensored with the unit), so (16.4) reads $M \otimes_A B = M$, i.e. the further completion is trivial and $A = B$. In general (16.4) says that the base change in the sense of analytic rings agrees with the (induced) tensor in $D(A)$ – with no further completion needed – which is exactly the assertion that $B$ has the induced analytic ring structure. □

**Corollary 16.16** (Proper base change)**.** *The class of proper maps is stable under base change, and proper base change holds.*

*Proof.* Stability under base change is immediate from the description (16.3) (an induced structure stays induced after base change, a straightforward unravelling of the symbols); given stability, the projection formula after any base change gives proper base change. □

*Remark* 16.17 (Comparison with Huber). This matches Huber's notion of properness when applied to objects coming from Huber pairs $(A^{\triangleright}, A^{+})$ sitting inside the analytic rings (Sections 8 and 9). For a map of affinoids in which the upstairs ring of integral elements $A^{+}$ is the smallest possible one – the one determined (integrally) by the $A^{+}$ downstairs – Huber already calls the map proper, and it satisfies all the good properties a proper map should. So the seemingly strange Definition 16.12 is in fact familiar from the theory of adic spaces.

## 16.5 Open immersions and idempotent algebras

For the other class one again forgets the algebro-geometric picture.

**Definition 16.18** (Open immersion)**.** A map $j\colon \mathrm{AnSpec}(B) \to \mathrm{AnSpec}(A)$ is an *open immersion* if the pullback $j^*$ admits a *left* adjoint $j_!$ which is fully faithful (equivalently, $j^*$ is a localization) and satisfies the projection formula. (Scholze added the fully-faithfulness clause as an afterthought: without it one has merely a left adjoint, not an open immersion.)

*Remark* 16.19 (The non-example). Any open immersion in ordinary algebraic geometry that is not also a closed immersion is *not* an open immersion in this sense. For instance $\mathbb{G}_m = \operatorname{AnSpec}\mathbb{Z}[T^{\pm 1}] \hookrightarrow \operatorname{AnSpec}\mathbb{Z}[T] = \mathbb{A}^1$ with the trivial (uncompleted) ring structure: a left adjoint of $j^*$ would force pullback $= \otimes$-of-modules to commute with all (infinite) products, and in ordinary algebraic geometry this essentially never happens.

**Example 16.20** (The solid $\mathbb{G}_m$)**.** There is however a different route from algebraic to analytic geometry, through adic spaces and *relative solid* ring structures (Section 9), and along it open immersions do go to open immersions. The key example is

$$j:\ \operatorname{AnSpec}\bigl(\mathbb{Z}[T^{\pm 1}]_\square\bigr) \longrightarrow \operatorname{AnSpec}\bigl(\mathbb{Z}[T]_\square\bigr), \tag{16.5}$$

the solid $\mathbb{G}_m$ inside the solid affine line, which *is* an open immersion. A more primitive variant maps the solid $\mathbb{G}_m$ into the *induced* structure on the affine line,

$$j:\ \operatorname{AnSpec}\bigl(\mathbb{Z}[T^{\pm 1}]_\square\bigr) \longrightarrow \operatorname{AnSpec}\bigl((\mathbb{Z}[T],\mathbb{Z})_\square\bigr),$$

of which (16.5) is essentially a base change.

*Remark* 16.21 (Compact support: functions vanishing near infinity). Why should such a $j_!$ deserve the name "cohomology with compact support"? Consider the global sections (cohomology) of the structure sheaf of the affine line, $j_!\mathcal{O}_{\operatorname{AnSpec}(\mathbb{Z}[T])_\square}$: this should be compactly supported coherent cohomology of $\mathbb{A}^1$. A compactly supported function is one vanishing near $\infty$. A function is an element of $\mathbb{Z}[T]$; functions near $\infty$ are power series in $T^{-1}$, so "vanishing at $\infty$" means lying in the fibre of the restriction map

$$\mathbb{Z}[T] \longrightarrow (\text{functions near } \infty).$$

A nonzero polynomial is determined by its germ at $\infty$, so this map is *injective* – the naive kernel vanishes – but as a *complex* the fibre is still interesting: its cokernel is nontrivial, in fact a product of copies of $\mathbb{Z}$ (the coefficients of the strictly-negative powers of $T$). Thus $j_!\mathcal{O}$ is a genuine two-term complex, exhibiting compact-support cohomology even though no honest compactly-supported functions exist.

Concretely, the completion $j_*j^*$ is computed by an internal Hom out of the object $C' = j_!\mathbb{Z}[T]$, recovering the solidification formula of Sections 7 and 9.

**Proposition 16.22** (The completion $j_*j^*$)**.** *For the open immersion of Example 16.20, and $M \in D(\mathbb{Z}[T]_\square)$,*

$$j_*\,j^*M \;=\; \operatorname{RHom}_{\mathbb{Z}[T]}\bigl(C',\,M\bigr), \qquad C' \;:=\; j_!\,\mathbb{Z}[T] \;=\; \operatorname{fib}\bigl(\mathbb{Z}[T] \hookrightarrow \mathbb{Z}(\!(T^{-1})\!)\bigr), \tag{16.6}$$

*the two-term complex* $\bigl[\,\mathbb{Z}[T] \to \mathbb{Z}(\!(T^{-1})\!)\,\bigr]$ *(with $\mathbb{Z}[T]$ in homological degree* 0 *and the differential the inclusion into the "functions near $\infty$" algebra), whose only nonzero homology is $H_{-1}(C') = \mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T] \cong \prod_{\mathbb{N}}\mathbb{Z}$; in particular*

$$j_!\,\mathcal{O}_{\operatorname{AnSpec}(\mathbb{Z}[T])_\square} \;=\; j_!\,\mathbb{Z}[T] \;=\; C'.$$

*This is precisely the $\mathbb{Z}[T]$-solidification formula Clausen gave for the complement of the closed disk (Proposition 7.9, Equation* (9.9)*).*

Reading (16.6) the other way, tensoring against the relevant object gives the left adjoint $j_!$ as a $\otimes$-with-a-module, which is exactly what one needs to verify the projection formula.

**Proposition 16.23** (Open immersions = idempotent algebras)**.** *Let* $X = \mathrm{AnSpec}(A)$*. The open immersions* $j\colon Y = \mathrm{AnSpec}(B) \to X$ *are equivalent to idempotent commutative algebra objects* $C \in D(A)$ *such that* $\mathrm{RHom}_A(C, -)[1]$ *commutes with all limits and preserves connectivity (*$D_{\geq 0}(A)$*), via*

$$j \;\longmapsto\; C \;=\; \mathrm{cone}\big(j_!\mathbf{1} \to \mathbf{1}\big).$$

*Sketch.* Since a right adjoint to $j_!$ automatically exists (namely $j^*$, once $j_!$ is fully faithful – there is a localization here), one recovers $B$ as the reflective subcategory

$$D(B) \;=\; \{\, M \in D(A) \;:\; \mathrm{RHom}_A(C, M) = 0 \,\},$$

the objects killed by $C$ in the sense of Definition 13.25. To produce an idempotent algebra one needs only supply the unit $\mathbf{1} \to C$ and check that $C \otimes_A (\mathbf{1} \to C)$ becomes an isomorphism; the projection formula for $j_!$ does the rest. Indeed $j_!\mathbf{1}$ is itself an idempotent (co)algebra:

$$j_!\mathbf{1} \otimes_A j_!\mathbf{1} \;=\; j_!\big(j^* j_!\mathbf{1}\big) \;=\; j_!\mathbf{1},$$

so its cone $C$ is an idempotent algebra, describing a "space near infinity" – the complementary closed subset cut out by the near-infinity functions. The two conditions on $C$ (that the completion $\mathrm{RHom}_A(C, -)$ commute with all limits and preserve connectivity) are exactly what is needed for this reflective localization to define an analytic ring structure; all other analytic-ring axioms follow formally (Definition 13.25 and Proposition 13.29). □

*Remark* 16.24 ($j_!j^*$ via the projection formula). The projection formula gives $j_!j^*M = j_!\mathbf{1} \otimes_A M$, and dually $j_*j^*M = \mathrm{RHom}_A(j_!\mathbf{1}, M)$. So the whole new (localized) analytic ring structure is determined by the single object $j_!\mathbf{1}$, or equivalently by $C$ (recover $j_!\mathbf{1}$ as $\mathrm{fib}(\mathbf{1} \to C)$).

**Corollary 16.25** (Base change)**.** *The class of open immersions is stable under base change, and the functors* $j_!$*,* $j^*$ *commute with base change.*

## 16.6 Shriekable maps and the six-functor formalism

With proper maps and open immersions in hand, the shriekable maps are their composites – and here the factorization is canonical, because Proposition 16.15 already tells us which proper map to take.

**Definition 16.26** (Shriekable map)**.** A map $f\colon \mathrm{AnSpec}(B) \to \mathrm{AnSpec}(A)$ is *shriekable* if, in its canonical factorization through the induced structure,

$$\begin{tikzcd} \mathrm{AnSpec}(B) \arrow[r, "j"] \arrow[rr, bend right, "f"'] & \mathrm{AnSpec}\big(A \otimes_{A^{\triangleright}} B^{\triangleright}\big) \arrow[r, "\overline{f}"] & \mathrm{AnSpec}(A), \end{tikzcd}$$

the first map $j$ – from $B$ to the induced analytic ring structure on $B^{\triangleright}$ – is an open immersion. (The second map $\overline{f}$ is proper by Proposition 16.15, and $j$ automatically has $j_!$ shriekable-friendly; the only real condition is that $j$ be an open immersion.) One then sets $f_! = \overline{f}_* \, j_!$.

**Example 16.27** (Structure map of the solid $\mathbb{G}_m$)**.** Taking the solid structure on $\mathbb{Z}[T^{\pm 1}]$, the structure map

$$f\colon\;\; \mathrm{AnSpec}\big(\mathbb{Z}[T^{\pm1}]_{\square}\big) \;\longrightarrow\; \mathrm{AnSpec}(\mathbb{Z}_{\square})$$

is shriekable: the compactification $\overline{f}$ is exactly the induced structure, and $j$ is the open immersion of Example 16.20. More generally, any map of finite-type algebras equipped with the relative solid ring structure gives shriekable maps.

*Remark* 16.28 (Huber's "+-weakly finite type"). For a map of Huber pairs, being shriekable is exactly the condition Huber calls being of *+-weakly finite type*: that the upstairs ring of integral elements $A^+$ is generated, as a ring of integral elements, by finitely many new elements over the base. Huber introduced this class of morphisms (and the associated proper/open notions) precisely as the maps for which one can define compact-support functors in *torsion étale* cohomology of adic spaces; remarkably, they are now also the maps for which one gets compact-support functors in *coherent* cohomology.

**Proposition 16.29** (The six-functor formalism on affine analytic spaces)**.** *The data of proper maps (Definition 16.12), open immersions (Definition 16.18) and the resulting shriekable maps satisfies all the conditions required by Theorem 16.9, and hence produces a six-functor formalism*

$$X \longmapsto D(X) \qquad on \qquad \mathcal{C} = (\mathrm{AnRing})^{\mathrm{op}} = \mathrm{Aff\ AnStk}, \ \textit{with shriekable maps.}$$

*Remark* 16.30 (The small checks). Several small verifications are suppressed (each an easy check): that a composite of two shriekable maps is again shriekable – essentially by base change – and one compatibility between the lower-shriek and the lower-star (= lower-shriek) functors of the proper maps. Granting these, Proposition 16.29 is a corollary of the general existence result. This is what was promised, but not fully constructible, in the first Clausen–Scholze course on condensed mathematics: there one specialized to adic spaces with underlying ring a field and solid modules, said such a formalism should exist, and now it does – and from it proofs of Poincaré duality, Serre duality, and the like follow easily.

## 16.7 From affine analytic spaces to analytic stacks

So far we have a six-functor formalism on the affine objects. To reach genuine geometric objects – built by gluing affines, as schemes are built from affines and stacks from schemes – one wants to extend it to sheaves, and the extension itself selects the Grothendieck topology.

*Remark* 16.31 (The general extension question). Given a six-functor formalism $X \mapsto D(X)$, a functor $\mathcal{C} \to \mathrm{Cat}$ on some $\infty$-category $\mathcal{C}$, one wants to extend it (for some notion of sheaves) to the sheaves on $\mathcal{C}$,

$$\mathrm{Shv}\big(\mathcal{C},\ \mathrm{Anima}\big),$$

i.e. to stacks glued out of objects of $\mathcal{C}$, on which the extended $D$ becomes a sheaf of $\infty$-categories. This was in fact the original motivation of Liu–Zheng [LZ24], who wanted to extend torsion étale cohomology from schemes to (higher) Artin stacks. Scholze's six-functor notes [Sch23] analyze this question – following Mann's rephrasing [Man22], then rephrased again – and try to pin down the best Grothendieck topology, including a general definition of what Scholze there calls the "D-topology". He is not entirely happy with the precise combinatorics in full generality, but the takeaway is clean.

*Remark* 16.32 (The takeaway: universal descent). The covers should be exactly those that satisfy *universal $*$-descent* and *universal !-descent*. Precisely:

1. A map $f\colon Y \to X$ satisfies *$*$-descent* if the pullback exhibits $D(X)$ as the totalization of the Čech diagram of $f$,

$$D(X) \xrightarrow{p^*} \varprojlim_{\Delta} \Big( D(Y) \rightrightarrows D(Y \times_X Y) \Rrightarrow \cdots \Big),$$

all transition functors being upper-stars $T_1^*, T_2^*, \ldots$; it satisfies *universal* $*$*-descent* if this holds after any base change.

2. A shriekable map $f\colon Y \to X$ satisfies $!$*-descent* if the same holds with the upper-shriek functors $f^!$ in place of the upper-stars, and *universal* $!$*-descent* if this persists after any base change.

Here "any base change" should be read as base change staying within $\mathcal{C}$; the fully general (base change to any presheaf) formulation is more awkward and is what makes the abstract topology delicate.

**Theorem 16.33** (!-descent suffices)**.** *If a shriekable map $f\colon Y \to X$ in* Aff AnStk *satisfies* $!$*-descent, then it satisfies both universal* $!$*-descent and universal* $*$*-descent. In fact it satisfies something stronger: descent even at the level of the $(\infty, 2)$-category of module categories – the assignment $A \mapsto \mathrm{Pr}^L_{D(A)}$ of presentable stable $\infty$-categories linear over $D(A)$ satisfies descent. Asking for* $!$*-descent is equivalent to asking descent at this two-categorical level (which is in fact how the theorem is proved).*

**Definition 16.34** (The analytic-stacks topology)**.** Endow Aff AnStk with the Grothendieck topology generated by:

1. finite disjoint unions – whenever $I$ is finite and the $X_i$ are affine analytic spaces, $\coprod_{i\in I} X_i$ is covered by the individual $X_i$ (as for profinite sets, where one allowed finite disjoint unions); and

2. shriekable maps satisfying !-descent.

Analytic stacks are then (a suitable class of hyper-)sheaves on Aff AnStk for this topology.

*Remark* 16.35 (Descendability, $h$-covers, and how general this is). This is a very general class. Restricting to proper maps, !-descent (there $! = *$) is precisely the condition that the map of algebras be *descendable* in the sense of Mathew [Mat16] – a robust, higher-categorical notion of descent satisfied by virtually all faithfully flat maps that are countably presented (one reason to work in the light condensed setting: light rings are countably presented). Descendable maps form a broad class: they include all countably presented faithfully flat maps, all quotients by nilpotent ideals, and – more surprisingly – all $h$-covers in Voevodsky's $h$-topology. Over Noetherian affine schemes, for a finitely-presented map of Noetherian affines, descendability is equivalent to being an $h$-cover. (For usual commutative rings with the trivial structure one declares all maps of affines proper, which looks awkward but works: then all maps are shriekable, and !-descent picks out exactly the descendable, i.e. $h$-covering, maps.) For the $(\infty, 2)$-categorical descent statements one should consult Mathew's paper.

*Remark* 16.36 (The two deferred caveats). As flagged at the start, two technical points remain. First, set theory: as in Section 15.7 one must work with accessible presheaves, and check that the above topology has good approximation properties, i.e. that sheafification preserves accessibility. Second, hyper-completeness: as already in the light condensed setting one does not want plain sheaves but a certain class of hypersheaves, so Definition 16.34 really allows a controlled class of hypercovers – the same maneuver as before. Both are suppressed here.

*Remark* 16.37 (What "module categories" descent means). The strong form of Theorem 16.33 uses Lurie's category $\mathrm{Pr}^L$ of presentable $\infty$-categories and colimit-preserving functors [Lur17]; inside it, the stable ones linear over a base $D(A)$ form the objects $\mathrm{Pr}^L_{D(A)}$, morally $D(A)$-linear DG-categories (e.g. $D(B)$ for a $D(A)$-algebra $B$). One asks whether $A \mapsto \mathrm{Pr}^L_{D(A)}$ satisfies descent (a $*$-type descent:

base-change module categories, ask the presheaf to be a sheaf). This is naturally an $(\infty, 2)$-category, but for the descent question one may forget the non-invertible 2-morphisms and work with the underlying $(\infty, 1)$-category – those extra 2-morphisms automatically satisfy descent as well. In this sense the $(\infty, 2)$-categorical structure does not matter much, and 2-categorical $*$-descent is equivalent to 1-categorical !-descent one level down.

# 17 !-Descent, Span Categories, and $\mathrm{Pr}^L$

Clausen continues the discussion of !-descent begun by Scholze in Section 16. The aim of the whole enterprise is to build geometric objects – the *analytic stacks* – out of analytic rings, in the model of scheme theory: one declares the affine objects to be $(\mathrm{AnRing})^{\mathrm{op}}$ and then glues them for a Grothendieck topology. The topology chosen is the one of !-descent (Section 16), and today's business is to make it usable. Two goals:

1. prove !-descent results – identify natural presheaves on Aff that satisfy !-descent; and

2. give examples of !-covers, so that the descent results can be applied.

The bulk of the lecture is spent setting up the machinery that makes both possible: the encoding of six-functor formalisms via span categories (Liu–Zheng, Mann, after Gaitsgory–Rozenblyum), and Lurie's category $\mathrm{Pr}^L$ of presentable $\infty$-categories, in which !-descent becomes an ordinary (co)descent statement. Only Goal (1) is reached today; Goal (2) is postponed to the next lecture (after the Christmas break). Throughout, "ring" means commutative ring, and everything is (light) condensed.

## 17.1 Recollection: the setup

We start from analytic rings and formally declare the affine objects to be the opposite category

$$\mathrm{Aff} \;:=\; (\mathrm{AnRing})^{\mathrm{op}}.$$

An object is a formal symbol $X$; it corresponds to an analytic ring which we write $(\mathcal{O}(X), D(X))$, so that $\mathcal{O}(X)$ is the underlying animated condensed ring and $D(X) \subseteq D(\mathcal{O}(X))$ is the full subcategory of the (full) derived category of $\mathcal{O}(X)$-modules cut out by the completeness conditions. We regard the full derived category $D(X)$ – not its connective part $D_{\geq 0}(X)$ – as the primary object attached to $X$.

*Remark* 17.1 (Why the full derived category). For a single analytic ring both formalisms are equivalent, related by the natural $t$-structure on $D(X)$ (its connective part is $D_{\geq 0}(X)$, with heart the abelian category of complete modules). But the choice matters for descent: the full derived category $D(X)$ satisfies !-descent, whereas its connective part does not. Concretely, an analytic ring carries a $t$-structure on $D(X)$ – detected all the way down in the derived category of condensed abelian groups – but a general analytic *stack*, obtained by gluing local derived categories, will *not* carry a $t$-structure. So today $D(X)$ means the full, unbounded derived category.

*Remark* 17.2 (The working definition of $D(X)$). Clausen's operative description: there is a functor from (animated/simplicial) rings to $\mathbb{E}_1$-rings – algebra objects in the relevant $\infty$-category – and one takes modules over that algebra object, using the $\infty$-categorical formalism of algebras and modules. Equivalently one can form chain complexes of simplicial modules and totalize, but the module-over-an-algebra description is the one to compute with: the free objects are of the form "$(-) \otimes \mathcal{O}(X)$", and one resolves by these.

Recall the three classes of maps from Section 16. A map $f\colon X \to Y$ in Aff (i.e. a map of analytic rings $\mathcal{O}(Y) \to \mathcal{O}(X)$) is *!-able* if it factors as an open immersion followed by a proper map,

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[rr, bend right, "f"'] & X' \arrow[r, "\pi"] & Y, \end{tikzcd} \tag{17.1}$$

with the notions defined as follows.

**Definition 17.3** (Proper maps and open immersions)**.** Let $\pi\colon X' \to Y$ and $j\colon X \to X'$ be maps in Aff.

- $\pi$ is *proper* if $X'$ carries the *induced* analytic ring structure from $Y$: an object $M \in D(\mathcal{O}(X'))$ lies in $D(X')$ if and only if its image under the forgetful functor (restriction of scalars along $\mathcal{O}(Y) \to \mathcal{O}(X')$) lies in $D(Y)$. Given the ring $\mathcal{O}(X')$ this always defines an analytic ring structure, and these are exactly the proper maps (cf. Proposition 16.15).
- $j$ is an *open immersion* if the pullback $j^*\colon D(X') \to D(X)$ (which one always has) is a localization whose kernel is $\mathrm{Mod}_A(D(X'))$ for some algebra $A \in D(X')$.

*Remark* 17.4 (The idempotent algebra)*.* The algebra $A$ cutting out $\ker(j^*)$ is necessarily idempotent, and necessarily a compact object: by definition of a map of analytic rings the right adjoint of $j^*$ commutes with colimits, so $j^*$ is a localization in the honest category-theoretic sense (its right adjoint is fully faithful). Since that right adjoint $j_*$ exists as part of the analytic-ring formalism, being a localization forces the existence of a *left* adjoint as well.

The good behaviour of the two extreme classes is what feeds the formalism.

*Remark* 17.5 (Proper and open maps are "good")*.* For a proper map $\pi$, the right adjoint $\pi_*$ is *good*: it commutes with colimits, satisfies the projection formula (i.e. it is $D(Y)$-linear – pulling back an object of $D(Y)$, tensoring, and pushing forward agrees with pushing forward and then tensoring, the natural comparison map being an isomorphism), and commutes with base change (a pullback of a proper map is proper, and the formation of $\pi_*$ commutes with that base change). For an open immersion $j$ there is a left adjoint $j_\natural$ to $j^*$ which is good in the same sense.

**Theorem 17.6** (Six-functor formalism, Section 16)**.** *There is a six-functor formalism on* Aff *such that the collection of maps $f\colon X \to Y$ for which the extra functor $f_!\colon D(X) \to D(Y)$ is specified is the class of* !*-able maps, with $f_! = f_*$ for $f$ proper and $j_! = j_\natural$ for $j$ an open immersion.*

*Remark* 17.7 (Closure properties of !-able maps)*.* The class of !-able maps has good closure properties. For $X \xrightarrow{f} Y \xrightarrow{g} Z$:

- $f, g$ !-able $\Rightarrow$ $gf$ !-able;
- $g$ and $gf$ !-able $\Rightarrow$ $f$ !-able

(so "a map between !-able maps is !-able"), and !-able maps are stable under base change. Moreover, for maps $f_i\colon X_i \to Y$ $(i = 1, \dots, n)$ with common target, each $f_i$ is !-able if and only if the induced map $f\colon \coprod_i X_i \to Y$ is !-able. Here the finite disjoint union in Aff corresponds to the finite product in AnRing, defined naively coordinate-wise.

**Example 17.8** (Compactly supported cohomology of $\mathbb{A}^1$)**.** Take $Y = \mathrm{AnSpec}(\mathbb{Z}_\square)$ and $X = \mathrm{AnSpec}(\mathbb{Z}[T]_\square)$, with the natural map $X \to Y$. Factor it through $X' = \mathrm{AnSpec}\big((\mathbb{Z}[T], \mathbb{Z})_\square\big)$:

$$X \xrightarrow{j} X' \xrightarrow{\pi} Y,$$

where $\pi$ is proper and $j$ is an open immersion, with complementary idempotent algebra

$$A \;=\; \mathbb{Z}(\!(T^{-1})\!).$$

Then $j_!$ of the structure sheaf of $X$ is the two-term complex

$$j_!\,\mathcal{O}_X \;=\; \mathrm{fib}\big(\mathbb{Z}[T] \to \mathbb{Z}(\!(T^{-1})\!)\big), \tag{17.2}$$

which is also the formula for $\pi_*$ (itself just the forgetful functor, forgetting the $\mathbb{Z}[T]$-module structure). Intuitively $\mathbb{Z}[T]$ is functions on the affine line and $\mathbb{Z}(\!(T^{-1})\!)$ is functions localized near the missing point at infinity, so (17.2) is functions vanishing near $\infty$ – compactly supported cohomology of $\mathcal{O}$ on $\mathbb{A}^1$. This is what a six-functor formalism does: it specifies a well-behaved notion of (relative) compactly supported cohomology. The key property of $f_!$ is that it commutes with base change in complete generality; here the affine line, not proper in ordinary algebraic geometry, is compensated by the solid theory, in which one has a new, "proper" version of it.

Finally, recall the descent condition and the topology it generates.

**Definition 17.9** (!-descent and its topology)**.** A !-able map $f\colon X \to Y$ *satisfies* !*-descent* if the augmented cosimplicial diagram built from the upper-shriek functors $(-)^!$ (each $(-)^!$ the right adjoint of the corresponding $(-)_!$)

$$D(Y) \xrightarrow{\;f^!\;} D(X) \rightrightarrows D(X \times_Y X) \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} \cdots$$

induces an equivalence

$$D(Y) \;\xrightarrow{\;\sim\;}\; \varprojlim_{n\in\Delta}{}^{!}\; D\big(X^{\times_Y n}\big). \tag{17.3}$$

The *topology of* !*-descent* on Aff has covers the finite collections $\{f_i\colon X_i \to Y\}$ of !-able maps such that $f\colon \coprod_i X_i \to Y$ satisfies !-descent; equivalently, a sieve is covering if it contains finitely many !-able maps with this property.

*Remark* 17.10 (Not the naive $*$-descent). The naive condition would be $*$-descent – that an object of $D(Y)$ be the same, via $f^*$-pullback, as a compatible family along the Čech diagram. Instead one asks the stronger-looking condition that $f^!$-pullbacks glue. That this !-descent condition is in fact stronger, and implies $*$-descent, is one of the outputs proved today.

*Remark* 17.11 (Why the topology is well-defined). Checking that covers are stable under composition amounts to interchanging proper maps and open immersions, and is straightforward: for a !-able map there is a *canonical* candidate for the intermediate $X'$ in (17.1). Namely, $X$ carries its structure sheaf $\mathcal{O}(X)$; one takes $X'$ with the same $\mathcal{O}(X') = \mathcal{O}(X)$ and $D(X')$ the full subcategory of $D(\mathcal{O}(X))$ of objects whose image in $D(\mathcal{O}(Y))$ lands in $D(Y)$. This is the induced (proper) structure, and $X \to X'$ is then the open immersion; one checks it is the canonical factorization.

## 17.2 Six-functor formalisms via span categories

The existence of the six-functor formalism on affinoids (Theorem 17.6) follows rather immediately from work of Liu–Zheng [LZ24] as reinterpreted by Mann [Man22]. Following Gaitsgory–Rozenblyum [GR17], Mann encodes six-functor formalisms in terms of *span categories*. We set this up, since the precise encoding is what makes the arguments below run.

**Definition 17.12** (Span category)**.** Let $\mathcal{C}$ be an ($\infty$-)category with all pullbacks, and $S$ a class of maps in $\mathcal{C}$ stable under composition (hence containing the identities) and under pullback. The *span category* $\mathrm{Span}(\mathcal{C})^{S,\mathrm{all}}$ has the same objects as $\mathcal{C}$, but a morphism $X \to Y$ is a diagram (span)

$$\begin{tikzcd} & M \arrow[dl] \arrow[dr, "\in S"] & \\ X & & Y, \end{tikzcd}$$

with the right leg $M \to Y$ in $S$ (and the left leg $M \to X$ arbitrary). Composition of a span $X \leftarrow M \to Y$ with a span $Y \leftarrow N \to Z$ is given by pullback:

$$\begin{tikzcd} & & M \times_Y N \arrow[dl] \arrow[dr] & & \\ & M \arrow[dl] \arrow[dr, "\in S"'] & & N \arrow[dl] \arrow[dr, "\in S"] & \\ X & & Y & & Z. \end{tikzcd}$$

A precise construction of $\mathrm{Span}(\mathcal{C})^{S,\mathrm{all}}$ as an $\infty$-category (as a complete Segal space) is due to Barwick; the point of the notation is that it lets one encode a formalism where shriek-functors are defined for maps in $S$ and star-functors for all maps.

**Definition 17.13** (2-functor formalism)**.** A 2-*functor formalism* on $(\mathcal{C}, S)$ is a functor

$$D\colon\ \mathrm{Span}(\mathcal{C})^{S,\mathrm{all}} \longrightarrow \mathrm{Cat}_\infty, \tag{17.4}$$

$X \mapsto D(X)$, sending each span to a functor. Every morphism decomposes canonically as a composite of two special spans. The first – with only the $S$-leg nontrivial –

$$\begin{tikzcd} & X \arrow[dl, equal] \arrow[dr, "f\in S"] & \\ X & & Y \end{tikzcd} \qquad \rightsquigarrow \qquad f_!\colon D(X) \to D(Y), \tag{17.5}$$

gives, for $f\colon X \to Y$ in $S$, a functor $f_!$; the second – with only the "all"-leg nontrivial –

$$\begin{tikzcd} & Y \arrow[dl, "f"'] \arrow[dr, equal] & \\ X & & Y \end{tikzcd} \qquad \rightsquigarrow \qquad f^*\colon D(X) \to D(Y), \tag{17.6}$$

gives, for $f\colon Y \to X$, a functor $f^*$.

*Remark* 17.14 (2 is the essence of 6). The functoriality of $D$ (the relation attached to a composable pair, i.e. to a Cartesian square in $\mathcal{C}$) amounts to the *base-change formula* for the lower-shriek functors – $(-)_!$ commutes with $(-)^*$ across a Cartesian square – together with the compatibility of $(-)_!$ and $(-)^*$ under composition, and the higher coherences between these:

functoriality $\rightsquigarrow$ base-change formula for $f_!$ & compatibility of $(-)_!$, $(-)^*$ under composition.

This is only 2 of the 6 functors, but 2 is the essence of 6 in the following sense. Granting that $f^*$ admits a right adjoint one gets $f_*$ for free; granting that $f_!$ admits a right adjoint one gets $f^!$ for free – four functors, none of which needs to be encoded explicitly. The remaining two are a tensor product $\otimes$ on each $D(X)$ and its internal RHom.

To encode the tensor product and its interaction with the functors – the projection formulas, and compatibility with $(-)_!$ and $(-)^*$ – one uses the symmetric monoidal structure on the span category induced by the product in $\mathcal{C}$ (assume now $\mathcal{C}$ has a terminal object, so that finite products exist). This is Cartesian product in $\mathcal{C}$, but it is *not* the Cartesian product of $\mathrm{Span}(\mathcal{C})^{S,\mathrm{all}}$; it is some other symmetric monoidal structure. One then requires:

**Definition 17.15** (Lax symmetric monoidal encoding)**.** A tensor structure on the 2-functor formalism is the datum making $D\colon \mathrm{Span}(\mathcal{C})^{S,\mathrm{all}} \to \mathrm{Cat}_\infty$ *lax symmetric monoidal* with respect to the product-induced structure on the source and the Cartesian product of $\infty$-categories on the target. Concretely, the basic data is a natural functor

$$D(X) \times D(Y) \;\longrightarrow\; D(X \times Y) \tag{17.7}$$

(the direction of the laxness), together with compatibilities conveniently packaged by lax symmetric monoidality.

*Remark* 17.16 (Unwinding the lax structure)*.* Applied with $X = Y$ and pulled back along the diagonal, (17.7) yields a symmetric monoidal structure on each $D(X)$, and makes the $f^*$ symmetric monoidal:

$$\big\{\text{symmetric monoidal structure on } D(X)\big\} \;+\; \big\{f^* \text{ symmetric monoidal}\big\}.$$

Encoding the product in $\mathcal{C}$ (rather than merely specifying a monoidal structure on each $D(X)$ separately) is what carries the compatibility of the tensor with the lower-shriek functors – the projection formula – as extra data. Were $S$ the trivial class, (17.7) would be exactly the datum of a symmetric monoidal structure on each $D(X)$ with $f^*$ symmetric monoidal.

### 17.3 Lurie's category $\mathrm{Pr}^L$

There is a very convenient way to organize the passage from 3 functors to 6: a piece of higher-categorical magic due to Lurie, the category $\mathrm{Pr}^L$ ([Lur17]).

**Definition 17.17** ($\mathrm{Pr}^L$)**.** The objects of $\mathrm{Pr}^L$ are the *presentable* $\infty$-categories: those admitting all small colimits and controlled by a small subcategory (equivalently, $\kappa$-compactly generated for some regular cardinal $\kappa$, so that the whole category is obtained by formally adjoining sufficiently filtered colimits to a small subcategory). The morphisms are the colimit-preserving functors, equivalently the functors admitting a right adjoint.

The remarkable features of $\mathrm{Pr}^L$ are that both limits and colimits in it are computed naively on underlying $\infty$-categories.

**Proposition 17.18** (Limits and colimits in $\mathrm{Pr}^L$)**.** $\mathrm{Pr}^L$ *has all small limits, and the forgetful functor* $\mathrm{Pr}^L \to \mathrm{Cat}_\infty$ *preserves them.* $\mathrm{Pr}^L$ *also has all small colimits, computed by the contravariant functor*

$$\mathrm{Pr}^L \;\longrightarrow\; \mathrm{Cat}_\infty^{\mathrm{op}}, \qquad \mathcal{C} \;\mapsto\; \mathcal{C}, \qquad \big[\mathcal{C} \xrightarrow{F} \mathcal{D}\big] \;\mapsto\; \big[\mathcal{D} \xrightarrow{F^r} \mathcal{C}\big],$$

*sending a colimit-preserving functor to its right adjoint. This functor preserves colimits – i.e. colimits in* $\mathrm{Pr}^L$ *become limits in* $\mathrm{Cat}_\infty$ *– so colimits in* $\mathrm{Pr}^L$ *are also computed naively, as limits of the underlying categories with respect to the right adjoints of the functors in the diagram.*

*Remark* 17.19 (Why this is magical)*.* In $\mathrm{Cat}_\infty$ colimits are hard: an amalgamated product of groups, say, is a nontrivial construction. In $\mathrm{Pr}^L$ they are easy, being computed as underlying limits along right adjoints. This is exactly what tames !-descent.

**Corollary 17.20** (!-descent is codescent in $\mathrm{Pr}^L$)**.** *For a !-able map $f\colon X \to Y$, the shriek-limit condition* (17.3)*,*

$$\varprojlim_{n\in\Delta}{}^{!}\, D\big(X^{\times_Y n}\big) \xrightarrow{\ \sim\ } D(Y),$$

*is precisely* codescent *for the lower-shriek functors $(-)_!$ in $\mathrm{Pr}^L$: the colimit, in the sense of $\mathrm{Pr}^L$, of the Čech diagram of $D(X^{\times_Y \bullet})$ along the maps $(-)_!$ is $D(Y)$. (Recall the colimit in $\mathrm{Pr}^L$ is computed as the naive limit along the right adjoints $(-)^!$.) This is a much more convenient way to think about !-descent.*

There is also a symmetric monoidal structure on $\mathrm{Pr}^L$ – in Clausen's estimation, whichever of Lurie's books it appears in, it is the main theorem of that book.

**Proposition 17.21** (Tensor product and internal hom on $\mathrm{Pr}^L$)**.** $\mathrm{Pr}^L$ *carries a symmetric monoidal structure $\otimes$, characterized by the universal property*

$$\mathrm{Maps}_{\mathrm{Pr}^L}(\mathcal{C}\otimes\mathcal{D},\ \mathcal{E}) \ = \ \big\{\textit{functors } \mathcal{C}\times\mathcal{D}\to\mathcal{E} \textit{ commuting with colimits in each variable separately}\big\}.$$

*This $\otimes$ itself commutes with colimits in each variable separately, and the corresponding internal hom is*

$$\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{D}) \ = \ \mathrm{Fun}^L(\mathcal{C},\mathcal{D}), \tag{17.8}$$

*the $\infty$-category of colimit-preserving functors $\mathcal{C}\to\mathcal{D}$.*

*Remark* 17.22 (Mapping spaces and the $(\infty,2)$-structure). The mapping space in $\mathrm{Pr}^L$ is the maximal subgroupoid of the internal hom,

$$\mathrm{Maps}_{\mathrm{Pr}^L}(\mathcal{C},\mathcal{D}) \ = \ \mathrm{Fun}^L(\mathcal{C},\mathcal{D})^{\simeq}$$

(only remembering isomorphisms), but the full mapping *category* is recovered from the tensor structure via (17.8). In principle $\mathrm{Pr}^L$ is an $(\infty,2)$-category – there is a whole category of maps between two objects – but one need not remember the $(\infty,2)$-categorical structure, since it is also just the internal hom.

The tensor structure lets one speak of algebras and modules in $\mathrm{Pr}^L$.

**Proposition 17.23** (Presentably symmetric monoidal categories and their modules)**.** *A commutative algebra object $\mathcal{C}\in\mathrm{CAlg}(\mathrm{Pr}^L)$ is the same as a symmetric monoidal presentable $\infty$-category whose tensor product commutes with colimits in each variable – a* presentably symmetric monoidal $\infty$-category*. A basic example is $D(X)$ for an affinoid $X$. For such $\mathcal{C}$ one forms*

$$\mathrm{Mod}_{\mathcal{C}}(\mathrm{Pr}^L),$$

*the $\mathcal{C}$-linear presentable $\infty$-categories: presentable $\infty$-categories tensored over $\mathcal{C}$ (hence also enriched over $\mathcal{C}$ – for $M,N\in\mathrm{Mod}_{\mathcal{C}}(\mathrm{Pr}^L)$ there is a $\mathcal{C}$-internal hom, right adjoint to $\mathcal{C}\otimes M\to M$). The category $\mathrm{Mod}_{\mathcal{C}}(\mathrm{Pr}^L)$ has all limits and colimits, the forgetful functor $\mathrm{Mod}_{\mathcal{C}}(\mathrm{Pr}^L)\to\mathrm{Pr}^L$ preserves them, and it carries a relative tensor product $\otimes_{\mathcal{C}}$.*

**Example 17.24** (Modules over algebras in $\mathcal{C}$)**.** For $\mathcal{C}\in\mathrm{CAlg}(\mathrm{Pr}^L)$ and commutative algebra objects $R,S\in\mathrm{CAlg}(\mathcal{C})$, the categories $\mathrm{Mod}_R(\mathcal{C})$, $\mathrm{Mod}_S(\mathcal{C})$ of $R$- resp. $S$-modules in $\mathcal{C}$ are objects of $\mathrm{Mod}_{\mathcal{C}}(\mathrm{Pr}^L)$, and

$$\mathrm{Mod}_R(\mathcal{C}) \ \otimes_{\mathcal{C}} \ \mathrm{Mod}_S(\mathcal{C}) \ = \ \mathrm{Mod}_{R\otimes S}(\mathcal{C}). \tag{17.9}$$

$\mathrm{CAlg}(\mathrm{Pr}^L)$ has colimits, computed as usual by relative tensor products; and its limits are naive, computed in $\mathrm{Pr}^L$.

### 17.4 Analytic rings inside $\mathrm{Pr}^L$

With $\mathrm{Pr}^L$ in hand, the good way to encode a six-functor formalism is as a *lax symmetric monoidal functor*

$$\mathrm{Span}(\mathcal{C})^{S,\mathrm{all}} \longrightarrow (\mathrm{Pr}^L, \otimes),$$

now using the $\mathrm{Pr}^L$-tensor rather than the Cartesian product of $\mathrm{Cat}_\infty$. This amounts to the same formalism as before, together with the condition that $D(X) \times D(Y) \to D(X \times Y)$ commute with colimits in $D(X)$ and $D(Y)$ separately – best thought of as saying the formalism takes values in $\mathrm{Pr}^L$.

The link with our setting is the following observation.

**Proposition 17.25** (The functor $R \mapsto D(R)$)**.** *There is a functor*

$$\mathrm{AnRing} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}^L)_{D(\mathrm{CondAb}^{\mathrm{light}})/}, \qquad (R^{\triangleright}, D(R)) \mapsto D(R), \tag{17.10}$$

*sending an analytic ring to its (presentably symmetric monoidal) derived category, viewed as an object* under $D(\mathrm{CondAb}^{\mathrm{light}})$ *– the derived category of condensed abelian groups, which is the image of the initial analytic ring $\mathbb{Z}$ and hence maps by base change to every $D(R)$. This functor commutes with colimits and detects isomorphisms (is conservative).*

**Corollary 17.26** (Pushouts are relative tensor products)**.** *For a pushout $A \otimes_R B$ in analytic rings,*

$$D(A \otimes_R B) = D(A) \otimes_{D(R)} D(B),$$

*the relative tensor product in* $\mathrm{Pr}^L$*. Thus, although the pushout of analytic rings is a subtle operation from the ring point of view (one completes a pre-analytic ring structure, Example 13.14), on the level of derived categories it is completely naive: it is the Lurie relative tensor product.*

*Indication.* When $A, B$ are both proper over $R$, this is an instance of (17.9) (the module categories are literally $\mathrm{Mod}_{\mathcal{O}(A)}$, $\mathrm{Mod}_{\mathcal{O}(B)}$ in $D(R)$). In general any map of analytic rings factors as a proper map followed by a localization; granting the proper case, for localizations one identifies the two sides by comparing universal properties. (The localization need not have a left adjoint here, but this does not matter for the comparison.) ☐

*Remark* 17.27 (Why the functor is not fully faithful). The functor (17.10) is *not* fully faithful, for a technical reason: one cannot recover $R^{\triangleright}$ from $D(R)$ together with its $D(\mathrm{CondAb}^{\mathrm{light}})$-tensoring. One *can* recover the underlying condensed ring, and even the underlying $\mathbb{E}_\infty$-ring: taking the unit object $\mathbf{1} \in D(R)$ and its internal endomorphisms over $D(\mathrm{CondAb}^{\mathrm{light}})$,

$$\underline{\mathrm{End}}_{D(\mathrm{CondAb}^{\mathrm{light}})}(\mathbf{1}) \in \mathrm{CAlg}\big(D(\mathrm{CondAb}^{\mathrm{light}})\big),$$

a commutative algebra in $D(\mathrm{CondAb}^{\mathrm{light}})$, which is the underlying $\mathbb{E}_\infty$-ring of $R^{\triangleright}$. But in the *animated* setting this is not enough to recover the animated ring: one would have to remember additional structure – the non-abelian derived functors of symmetric powers – which modules over an animated ring carry but modules over an $\mathbb{E}_\infty$-ring (with only the symmetric monoidal structure) do not. There is, however, more than enough structure to detect isomorphisms, since one recovers the underlying object.

## 17.5 Dualizability of $D(X)$ over $D(Y)$

Fix $Y \in \mathrm{Aff}$ and consider the category $\mathrm{Aff}^!_{/Y}$ of !-able maps $X \to Y$. By the 2-of-3 property (Remark 17.7) every map in $\mathrm{Aff}^!_{/Y}$ is itself !-able, so the six-functor formalism restricts to a functor on the span category with $S = \mathrm{all}$,

$$\mathrm{Span}\big(\mathrm{Aff}^!_{/Y}\big)^{\mathrm{all,all}} \longrightarrow \mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L), \tag{17.11}$$

which is a priori lax symmetric monoidal, and is in fact symmetric monoidal. The following is the key structural output.

**Corollary 17.28** ($D(X)$ is self-dual over $D(Y)$)**.** *For $X \to Y$ !-able, the object $D(X) \in \mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L)$ is dualizable, and in fact canonically self-dual. With respect to this self-duality, if $X \xrightarrow{p} X'$ over $Y$, then the dual of the pullback*

$$p^*\colon D(X') \to D(X) \qquad \textit{is} \qquad p_!\colon D(X) \to D(X').$$

*Sketch.* Everything is checked universally in the span category $\mathrm{Span}(\mathcal{C})^{\mathrm{all,all}}$ and transported by the symmetric monoidal functor (17.11) (which sends dualizable objects to dualizable objects, and a duality datum to a duality datum). In the span category every object $X$ is self-dual (here in the slice over $Y$, where the monoidal unit is $Y$ and products are fibre products over $Y$): the unit and counit of the duality are the spans

$$Y \leftarrow X \xrightarrow{\Delta} X \times_Y X, \qquad\qquad X \times_Y X \xleftarrow{\Delta} X \to Y,$$

and the triangle identities are immediate. Under this self-duality, passing to the dual of a span is transposing the span, which converts the lower-shriek functor of $p$ into its transpose – and reading off the two special classes (17.5)–(17.6), the dual of $p^*$ is exactly $p_!$. $\square$

## 17.6 The descent proposition

We can now prove the equivalence of descent conditions that Scholze stated last time (Section 16) without proof.

**Proposition 17.29** (Forms of !-descent)**.** *For a !-able map $f\colon X \to Y$, the following are equivalent.*

1. *$f$ satisfies !-descent (Definition 17.9).*
2. *For all $M \in \mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L)$, the natural map is an equivalence*

$$M \xrightarrow{\sim} \varprojlim_{n\in\Delta}{}^{*}\, M \otimes_{D(Y)} D\big(X^{\times_Y n}\big)$$

*(the transition functors being upper-stars). In particular, for $M = D(Y)$ this is $*$-descent.*

3. (2-descent, Gaitsgory–Rozenblyum.) *The assignment $Y' \mapsto \mathrm{Mod}_{D(Y')}(\mathrm{Pr}^L)$ satisfies descent along $f$:*

$$\mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L) \xrightarrow{\sim} \varprojlim_{n\in\Delta} \mathrm{Mod}_{D(X^{\times_Y n})}(\mathrm{Pr}^L).$$

The upshot is strong: !-descent – a condition about the exotic $f^!$-pullback functors – automatically upgrades to $*$-descent (condition (2) with $M = D(Y)$), and even to descent one categorical level up (condition (3)), which is how the theorem is really proved.

*Proof.* *(1)⇔(2).* By Corollary 17.20, !-descent says exactly that

$$\varinjlim_n D\big(X^{\times_Y n}\big) \xrightarrow{\sim} D(Y) \qquad \text{in } \mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L),$$

the colimit in $\mathrm{Pr}^L$ (computed as the shriek-limit (17.3)). Apply $\underline{\mathrm{Hom}}(-, M)$ for $M \in \mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L)$: since internal hom turns this colimit into a limit,

$$M \xrightarrow{\sim} \varprojlim_n \underline{\mathrm{Hom}}\big(D(X^{\times_Y n}), M\big) = \varprojlim_n D\big(X^{\times_Y n}\big) \otimes_{D(Y)} M, \tag{17.12}$$

where the last identification uses the self-duality of $D(X^{\times_Y n})$ (Corollary 17.28). The manner in which it is self-dual converts the lower-shriek transition maps into upper-star maps, so (17.12) is precisely (2). (This is a proof that (1)⇔(2); the outstanding delicate point is producing the coherent identification of duals universally in the span category, which can certainly be done; there is also an independent argument avoiding it.)

*(2)⇒(3).* (Mathew [Mat16].) The base-change functor $\mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L) \to \varprojlim_n \mathrm{Mod}_{D(X^{\times_Y n})}(\mathrm{Pr}^L)$ has a right adjoint: send a compatible system $(M_n)$ to the limit $\varprojlim_n M_n$ of the underlying $D(Y)$-modules. One checks the unit and counit are isomorphisms. That the *unit* is an isomorphism is exactly (2). For the *counit*: it is an isomorphism after base change to $D(X)$, because

(i) $D(X)$ is dualizable over $D(Y)$ (Corollary 17.28), so one can pull the limit out of the tensor with $D(X)$;

(ii) the cover is split on pullback to $D(X)$ (the base-changed Čech object acquires an extra degeneracy, as in ordinary descent theory);

(iii) base change is compatible: the right adjoint (the forgetful functor $\mathrm{Mod}_{D(X)} \to \mathrm{Mod}_{D(Y)}$) commutes with base change, base change meaning pushout in $\mathrm{CAlg}(\mathrm{Pr}^L)$ – purely algebraic.

Finally, (2) implies that $- \otimes_{D(Y)} D(X)$ detects isomorphisms, so the counit is an isomorphism, and we are done.

*(3)⇒(2).* Condition (3) is the statement that the base-change functor is an equivalence; unwinding what it means for its unit to be an isomorphism gives exactly (2), so (3) is manifestly stronger. (One can also see the passage more explicitly: from (3) in the $(\infty,1)$-sense one gets, for $M, N \in \mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L)$,

$$\mathrm{Maps}_{\mathrm{Mod}_{D(Y)}(\mathrm{Pr}^L)}(M, N) = \varprojlim_n \mathrm{Maps}\big(M \otimes D(X^{\times_Y n}),\, N \otimes D(X^{\times_Y n})\big),$$

the set of objects of the relevant internal hom. Tensoring the source with presheaves on $\Delta^n$ (with values in $D(Y)$) and letting $n$ vary recovers not just the set of objects but the whole mapping category – the complete Segal space of $\mathrm{Fun}^L_{D(Y)}(M, N)$ – which is the $(\infty,2)$-categorical strengthening. In particular one gets a strong form of (3): descent even at the level of $\mathrm{Fun}^L_{D(Y)}$, i.e. full faithfulness in the $(\infty,2)$-sense, whose essential surjectivity part already follows from the $(\infty,1)$-statement.) □

**Corollary 17.30** (!-descent is stable under base change)**.** *If $f$ satisfies !-descent, then so does any pullback of $f$. This follows from (2) together with the symmetric monoidality of* (17.11)*: taking $M = D(Y')$ for a base change $Y' \to Y$ gives $*$-descent for the pullback, and $M$ may be taken to be any $D(Y')$-module, so condition (2) is manifestly stable under base change. (Concretely, the pullback of analytic rings is already computed by a relative tensor product, Corollary 17.26.) This stability is what allows the !-descent condition to define a Grothendieck topology in the first place.*

**Corollary 17.31** (Subcanonicity)**.** *The topology of* !*-descent is subcanonical; equivalently, the affinoids embed fully faithfully into the category of sheaves. Thus one obtains very strong* $*$*-descent results as consequences of* !*-descent.*

*Remark* 17.32 (What remains). This completes Goal (1): natural presheaves satisfying !-descent, and the strong descent consequences. Goal (2) – producing *examples* of !-covers, so that these results can be applied – is taken up in the next lecture (scheduled after the Christmas break).

# 18 Checking !-Covers: Descendability, Closed and Open Covers, Zariski Descent

After the Christmas break, Clausen returns to the !-topology on the category of affine analytic spaces $\mathrm{Aff} = (\mathrm{AnRing})^{\mathrm{op}}$ set up in Sections 16 and 17. The plan of Section 17 left two goals; Goal (1), identifying presheaves that satisfy !-descent, was reached there. Today is Goal (2): how does one *check* that a given map is a !-cover, and what are the examples? The engine is a single proposition reducing !-descent to a finitary condition on the unit object, refined in two directions (proper maps and universally locally acyclic maps), from which closed covers, open covers, and finally Zariski (and étale, fpqc, $h$-…) covers all fall out. The lecture closes by re-running the same definitions inside ordinary algebraic geometry, where they reproduce Mathew's notion of descendability. Throughout "ring" means commutative ring and everything is (light) condensed.

## 18.1 Recollection: the setup

We recall the framework of Sections 16 and 17. An analytic ring is a pair

$$R = (R^{\triangleright},\, D(R)),$$

where $R^{\triangleright}$ is a condensed animated ring and $D(R) \subseteq D(R^{\triangleright}) = \mathrm{Mod}_{R^{\triangleright}}(D(\mathrm{CondAb}^{\mathrm{light}}))$ is a full subcategory of the (full, unbounded) derived category of $R^{\triangleright}$-modules, cut out by the completeness axioms (Definitions 8.8 and 13.1); we take the whole $D(R)$, not its connective part, as the primary invariant. The category of affine analytic spaces is the opposite,

$$\mathrm{Aff} \;:=\; (\mathrm{AnRing})^{\mathrm{op}}. \tag{18.1}$$

Three classes of maps were singled out. For $f\colon X \to Y$ in Aff – equivalently a map of analytic rings $R^{\triangleright} \to S^{\triangleright}$ where $X = \mathrm{AnSpec}(S)$, $Y = \mathrm{AnSpec}(R)$ – one has the pullback $f^*\colon D(Y) \to D(X)$, and:

**Definition 18.1** (Proper maps and open immersions)**.**

- $f$ is *proper* if $f^*$ admits a *good* right adjoint $f_*$, where "good" means that $f_*$ commutes with base change (pullback), commutes with colimits, and satisfies the projection formula. Equivalently, $X$ carries the *induced* analytic ring structure from $Y$: $D(X) \subseteq D(S^{\triangleright})$ is exactly the objects whose restriction of scalars to $D(R^{\triangleright})$ lies in $D(Y)$ (Proposition 16.15). So the analytic ring structure is unchanged in passing from $Y$ to $X$; only the underlying ring changes.
- $f$ is an *open immersion* if $f^*\colon D(Y) \to D(X)$ is a localization (its right adjoint is fully faithful) and has a good left adjoint. Equivalently, $f^*$ is localization away from an idempotent algebra $A \in D(Y)$ ("functions on the closed complement"), with

$$\ker\bigl(f^*\bigr) \;=\; \mathrm{Mod}_A\bigl(D(Y)\bigr).$$

**Definition 18.2** (!-able maps and $f_!$)**.** A map $f\colon X \to Y$ is *!-able* if it factors as an open immersion followed by a proper map,

$$\begin{tikzcd} X \arrow[r, hook, "j"] \arrow[rr, bend right, "f"'] & \overline{X} \arrow[r, "\pi"] & Y, \end{tikzcd}$$

$j$ an open immersion, $\pi$ proper. All three classes have good closure properties: closed under composition and base change, containing all isomorphisms, and closed under passage to diagonals (equivalently: given two maps to a common target both in the class, any map between them lies in the class). The six-functor formalism of Section 16 then produces $f_!$ exactly on the !-able maps, with $f_! = f_*$ for $f$ proper and $j_! =$ the good left adjoint of $j^*$ for $j$ an open immersion (equivalently $j^! = j^*$).

*Remark* 18.3 (The natural transformation $f_! \to f_*$)*.* There is a natural transformation $f_! \to f_*$ in general. For an open immersion it comes from the natural map from the left adjoint of $j^*$ to its right adjoint, present for any localization; one then composes and checks independence of the factorization. A cleaner construction passes to the diagonal, using that the diagonal of an open immersion is an isomorphism (as in ordinary algebraic geometry).

The descent condition and the topology it generates were the subject of Section 17.

**Definition 18.4** (!-cover)**.** A !-able map $f\colon X \to Y$ is a *!-cover* if the natural functor to the totalization of the Čech nerve, formed with the *upper-shriek* transition functors, is an equivalence:

$$D(Y) \xrightarrow{\sim} \varprojlim_{n\in\Delta}{}^{!}\, D\big(X^{\times_Y n}\big).$$

**Theorem 18.5** (Equivalent forms, Section 17)**.** *For !-able $f\colon X \to Y$, the following are equivalent.*

1. *$f$ is a !-cover.*
2. (!-codescent.) *The colimit, formed in $\mathrm{Pr}^L$ (equivalently in $\mathrm{Mod}_{D(Y)}\,\mathrm{Pr}^L$) along the lower-shriek functors, recovers $D(Y)$:*
$$\varinjlim_{n\in\Delta^{\mathrm{op}}} D\big(X^{\times_Y n}\big) \xrightarrow{\sim} D(Y). \tag{18.2}$$
3. (∗-descent with coefficients.) *For every $M \in \mathrm{Mod}_{D(Y)}\,\mathrm{Pr}^L$, the ∗-pullbacks glue:*
$$M \xrightarrow{\sim} \varprojlim_{n\in\Delta}{}^{*}\, M \otimes_{D(Y)} D\big(X^{\times_Y n}\big).$$
4. (2-descent.) *The assignment of module ∞-categories descends:*
$$\mathrm{Mod}_{D(Y)}\,\mathrm{Pr}^L \xrightarrow{\sim} \varprojlim_{n\in\Delta}{}^{*}\ \mathrm{Mod}_{D(X^{\times_Y n})}\,\mathrm{Pr}^L.$$

*Taking $M = D(Y)$ in (3) gives ordinary ∗-descent; so !-descent, a condition about the exotic $(-)^!$-functors, upgrades to ∗-descent and even to descent one categorical level up. Here $\mathrm{Mod}_{D(Y)}\,\mathrm{Pr}^L$ is the collection of presentable ∞-categories tensored over the presentably symmetric monoidal ∞-category $D(Y)$, with the naive relative-tensor base-change functors.*

*Remark* 18.6 (Finitary versus infinitary). The topology is finitary by fiat: covers are finite families. One might instead allow arbitrary (infinite) families and ask for !-descent. Scholze has checked that this a priori infinitary version of the Grothendieck topology described below is in fact finitary – every cover admits a finite subcover. The point is that the unit object $\mathbf{1}_Y \in D(Y)$ is compact (Proposition 18.7), so an infinitary witness collapses to a finite stage. This is genuinely a feature of !-*descent*, not of $*$-descent: for $*$-descent alone one can have infinitary covers of profinite sets with no finite subcover (e.g. a large profinite set written as an infinite union of small ones), a phenomenon the set theorists have examined. The universally submersive family of *all* maps from a fixed profinite set $P$ into the Cantor set is another candidate infinitary cover whose cohomological (descent) behaviour is unclear.

## 18.2 Checking !-covers: a finitary criterion

The following proposition reduces the (infinitary, $\infty$-categorical) !-descent condition to a finitary membership statement, at the cost of an extra hypothesis for its converse.

**Proposition 18.7** (Criterion for a !-cover)**.** *Let $f\colon X \to Y$ be a !-able map.*

1. *If $f$ is a !-cover, then the unit object lies in the thick subcategory generated by the image of $f_!$:*

$$\mathbf{1}_Y \ (= \mathcal{O}_Y) \ \in \ \big\langle \operatorname{im}\big(f_!\colon D(X) \to D(Y)\big) \big\rangle, \tag{18.3}$$

*where $\langle - \rangle$ denotes closure under the finitary operations: cones, shifts, and retracts.*

2. *The converse holds provided* either
   (a) *$f$ is proper;* or
   (b) *$f$ is universally locally acyclic* (U.L.A.)*, meaning $f^!$ is good: it commutes with base change, commutes with colimits, and satisfies the projection formula in the form*

$$f^!(-) \ \cong \ f^!(\mathbf{1}_Y) \otimes f^*(-).$$

*Remark* 18.8 (The image of $f_!$ and compactness). Because $f_!$ is not fully faithful in general, its image has no closure properties on its own; one must pass to the generated thick subcategory. One could ask instead that $\mathbf{1}_Y$ lie in the *infinitary* (colimit-)closure of the image; since $\mathbf{1}_Y$ is compact in $D(Y)$, this is equivalent to the finitary membership (18.3). Condition (1) has the flavour of a purely set-theoretic surjectivity of $f$ – a "cover" condition, like a cover in some constructible topology – and by itself is much weaker than !-descent.

*Remark* 18.9 (The U.L.A. condition). The object $f^!(\mathbf{1}_Y)$ is the relative dualizing object of $f$. The good-ness of $f^!$ is exactly the assertion that this dualizing object is compatible with base change – an equisingularity property of the family $X \to Y$. Properness and U.L.A. are the analytic-geometry analogues of "closed" and "open" respectively (Remark 18.12), and one may combine (18.3) with either to conclude honest descent; one cannot mix open and closed, just as in topology finite closed covers and open covers each give descent but a mixture need not.

Part (1) alone does not imply !-cover: the converse genuinely needs (a) or (b).

**Example 18.10** (The converse fails: an open glued to its closed complement)**.** Take the induced solid structure on the affine line,

$$Y = \operatorname{AnSpec}\big((\mathbb{Z}[T], \mathbb{Z})_\square\big),$$

and let $X = U \amalg Z$ be the disjoint union of the open piece and its closed complement at infinity,

$$U = \mathrm{AnSpec}\big(\mathbb{Z}[T]_{\square}\big) \xrightarrow{\ j\ } Y, \qquad Z = \mathrm{AnSpec}\big(\mathbb{Z}(\!(T^{-1})\!)_{\square}\big) \xrightarrow{\ i\ } Y,$$

with $j$ an open immersion and $i$ a (proper) closed immersion. Here

$$j_!\big(\mathbb{Z}[T]\big) = \mathrm{fib}\big(\mathbb{Z}[T] \to \mathbb{Z}(\!(T^{-1})\!)\big), \qquad i_!\big(\mathbb{Z}(\!(T^{-1})\!)\big) = \mathbb{Z}(\!(T^{-1})\!),$$

the first being "functions vanishing near infinity" and the second $i_! = i_*$ the forgetful functor. The unit $\mathbf{1}_Y = \mathbb{Z}[T]$ is built from these two by a single cofiber sequence

$$j_!\,\mathbf{1}_U \;\longrightarrow\; \mathbf{1}_Y \;\longrightarrow\; i_*\,\mathbf{1}_Z, \qquad \text{i.e.} \quad \mathrm{fib}\big(\mathbb{Z}[T] \to \mathbb{Z}(\!(T^{-1})\!)\big) \to \mathbb{Z}[T] \to \mathbb{Z}(\!(T^{-1})\!),$$

so (allowing retracts) $\mathbf{1}_Y$ lies in the thick subcategory generated by $\mathrm{im}(f_!)$: (18.3) holds. Yet $f = (j, i)$ is *not* a !-cover, because the descent object splits as $D(U) \times D(Z)$ rather than $D(Y)$ (the global sections on $U \amalg Z$ are the product ring). Here $f$ mixes an open with a closed piece and is neither proper nor U.L.A., so neither converse hypothesis applies.

*Proof of Proposition 18.7.* Spelling out Definition 18.4: the comparison functor $D(Y) \to \varprojlim^!_n D(X^{\times_Y n})$ is a right adjoint (built from the $(-)^!$ functors), and by general category theory it itself has a left adjoint, informally

$$(M_n)_{n\in\Delta} \;\longmapsto\; \varinjlim_{n\in\Delta^{\mathrm{op}}} g_!\,M_n,$$

where for each $n$ we write $g = g_n\colon X^{\times_Y n} \to Y$ for the structure map. (Making this precise is due to Lurie.)

*(1).* Full faithfulness of the comparison amounts to the counit

$$\varinjlim_{n\in\Delta^{\mathrm{op}}} g_!\,g^!\,M \;\xrightarrow{\ \sim\ }\; M \tag{18.4}$$

being an isomorphism for all $M \in D(Y)$. Take $M = \mathbf{1}_Y$. The geometric realization (18.4) is a colimit over $\Delta^{\mathrm{op}}$, which one filters by the partial totalizations

$$\operatorname*{colim}_{n\in\Delta^{\mathrm{op}}_{\le d}} g_!\,g^!\,\mathbf{1}_Y, \tag{18.5}$$

each a *finite* colimit (a nice fact about the simplex category: a partial geometric realization is a finite colimit, expressible through iterated pushouts and the initial object). Since $\mathbf{1}_Y$ is *compact*, an isomorphism onto a filtered colimit factors through a finite stage: $\mathbf{1}_Y$ is a retract of some partial totalization (18.5). Every term there lies in the image of $f_!$, since each structure map $g$ factors through $X \to Y$; and the partial totalization is a finite colimit of such. Hence $\mathbf{1}_Y \in \langle \mathrm{im}(f_!)\rangle$.

*(2).* Conversely, the hypothesis (18.3) plus the projection formula for $f_!$ gives that *every* $M \in D(Y)$ is generated by objects in the image of $f_!$: tensoring (18.3) with $M$ and using $f_!(f^*M \otimes N) \cong M \otimes f_!N$ shows $M \in \langle \mathrm{im}\, f_!\rangle$ too. To prove the comparison is an equivalence we prove full faithfulness and (separately) the other unit/counit. For full faithfulness we may, by the previous sentence, assume $M = f_!N$; and we want the counit (18.4) to be an isomorphism. The maps $g$ are composites of pullbacks of $f$, so if $f$ has property (a) or (b), so does every $g$, and one has a base-change identity commuting $g_!$ with $g^!$ (or $g'_!$ with $g'^!$ for the various projections), by which one reduces to the split situation:

- *proper case:* $g^! = g_*$ and base change follows from $((-)^*, (-)_!)$-base change by passing to adjoints;
- *U.L.A. case:* $g^! = g^!(\mathbf{1}) \otimes g^*$ and base change follows directly from $((-)^*, (-)_!)$-base change.

(One must check the two a priori different base-change comparison maps agree, in order to transport one being an isomorphism to the other.) This gives full faithfulness. The remaining unit/counit is handled similarly, base-changing along $X \to Y$, where the cover becomes split and the statement is formal. $\square$

### 18.3 Special cases: closed covers

We now make Proposition 18.7 concrete, beginning with the "closed" side. Recall that Aff has proper maps and open immersions but no notion of closed immersion was singled out; here is one.

**Definition 18.11** (Closed immersion)**.** A map $f\colon X \to Y$ is a *closed immersion* if it is proper and $f^*\colon D(Y) \to D(X)$ is a localization. Since $f$ is proper, $D(X) = \mathrm{Mod}_A\, D(Y)$ for the algebra $A := f_*(\mathbf{1}_X) \in D(Y)$; being a closed immersion is then exactly the condition that $A$ be *idempotent*, i.e. the multiplication $A \otimes_{D(Y)} A \to A$ is an isomorphism.

*Remark* 18.12 (Open and closed are not complementary in Aff)*.* In Aff it is not generally true that open and closed immersions with a fixed target are in bijection: an open immersion need not have a closed complement in Aff, and a closed immersion need not have an open complement in Aff.

Given an open immersion $j\colon U \to Y$ one gets an idempotent algebra $A \in D(Y)$ with $D(U) = D(Y) /\!\!/ \mathrm{Mod}_A\, D(Y)$; but $A$ need not lie in $D_{\geq 0}(Y)$, and connectivity is exactly what is needed for $A$ to be the pushforward of an *animated* ring, hence to define a closed immersion in Aff. For example, with $Y = \mathrm{AnSpec}((\mathbb{Z}[X,Y], \mathbb{Z})_\square)$ and $U = \mathrm{AnSpec}(\mathbb{Z}[X,Y]_\square)$, the corresponding idempotent $A$ has a nonzero $H_{-1}$. (This is analogous to the scheme-theoretic phenomenon that the open complement of a closed subscheme of an affine scheme need not be affine, though it is still a scheme; here too one usually gets a closed complement that is an analytic space, just not an affine one.)

In the other direction, given a closed immersion $Z \to Y$ – an idempotent algebra $A \in D_{\geq 0}(Y)$ – to obtain a complementary open *in* Aff one needs the candidate localization functor

$$\underline{\mathrm{RHom}}_{D(Y)}\big(\mathrm{fib}(\mathbf{1}_Y \to A), -\big)\colon\ D(Y) \to D(Y)$$

to commute with filtered colimits and to preserve $D_{\geq 0}(Y)$ – precisely the axioms making it define an analytic ring structure. These subtleties will be largely fixed once one allows general (non-affine) analytic spaces and stacks.

**Proposition 18.13** (Finite closed covers)**.** *Let $Z_1, \dots, Z_n \to X$ be closed immersions in* Aff *(with idempotent algebras $A_i \in D_{\geq 0}(X)$, so $Z_i \cap Z_j$ has algebra $A_i \otimes A_j$, etc.). Then $\coprod_i Z_i \to X$ is a !-cover if and only if the structure sheaf satisfies the most naive form of descent: the augmented Čech complex*

$$\mathbf{1}_X \xrightarrow{\ \sim\ } \Big[\ \bigoplus_i \mathbf{1}_{Z_i} \longrightarrow \bigoplus_{i<j} \mathbf{1}_{Z_i \cap Z_j} \longrightarrow \cdots\ \Big] \tag{18.6}$$

*is an equivalence. The complex terminates at a finite stage.*

*Proof.* If (18.6) holds then $\mathbf{1}_X$ lies in the image of the lower-star (= lower-shriek, as the $Z_i \hookrightarrow X$ are proper) functors from the $\coprod Z_i$, so Proposition 18.7(1) is met; and since the map is proper, Proposition 18.7(2a) upgrades this to a !-cover. Conversely, a !-cover satisfies $*$-descent for $D(-)$, so

$D(X) = \varprojlim$; plugging in the unit object of $D(X)$ and reading off the base-change-then-right-adjoint yields exactly the structure-sheaf descent (18.6). $\square$

## 18.4 Special cases: open covers

Dually, for open immersions the criterion becomes vanishing of a product of the complementary idempotents.

**Proposition 18.14** (Finite open covers)**.** *Let $U_1, \ldots, U_n \to X$ be open immersions, with complementary idempotent algebras $A_1, \ldots, A_n \in D(X)$ (so $A_i$ describes the closed complement of $U_i$). Then*

$$\coprod_i U_i \to X \quad \text{is a !-cover} \quad \iff \quad A_1 \otimes \cdots \otimes A_n = 0 \quad \text{in } D(X).$$

*Every open cover has a finite refinement, so no generality is lost in taking $n$ finite: the vanishing $A_1 \otimes \cdots \otimes A_n = 0$ of an algebra means $\mathbf{1} = 0$ in it, and by compactness of the unit this is witnessed at a finite stage.*

Intuitively the $U_i$ cover $X$ precisely when the intersection of the complementary closed sets is empty, i.e. $A_1 \otimes \cdots \otimes A_n = 0$.

*Proof for $n = 2$.* The general case reduces to $n = 2$ by an induction using the appropriate categorical form of the statement (the union $U_1 \cup U_2$ need not itself lie in Aff). For $n = 2$, !-descent for the cover is a Mayer–Vietoris pullback square (the infinite Čech totalization collapses, as the $j$'s are monomorphisms),

$$\begin{tikzcd} D(X) \arrow[r, "j_1^!"] \arrow[d, "j_2^!"'] & D(U_1) \arrow[d, "j^!"] \\ D(U_2) \arrow[r, "j^!"'] & D(U_1 \cap U_2), \end{tikzcd}$$

all maps upper-shriek. Reading off the induced statement for the unit object, this is the assertion that $\mathbf{1}_X$ is the total fiber of the Mayer–Vietoris diagram of $j_{1,!}j_1^!\mathbf{1},\ j_{2,!}j_2^!\mathbf{1},\ j_{12,!}j_{12}^!\mathbf{1}$; writing each $j_!j^!\mathbf{1}$ in terms of the corresponding idempotent (recall $j^! = j^*$ for an open immersion, and $j_!\mathbf{1} = \mathrm{fib}(\mathbf{1} \to A)$), the cofiber sequence

$$j_{12,!}\, j_{12}^!\, \mathbf{1} \ \longrightarrow\ j_{1,!}\, j_1^!\, \mathbf{1} \oplus j_{2,!}\, j_2^!\, \mathbf{1} \ \longrightarrow\ \mathbf{1}_X$$

is directly equivalent, on unravelling, to $A_1 \otimes A_2 = 0$. (Open immersions are U.L.A., so Proposition 18.7(2b) applies and the membership statement upgrades to a genuine !-cover.) $\square$

## 18.5 Zariski (and étale, fpqc, ...) covers

We now produce genuine examples of !-covers by importing them from algebraic geometry. There is a functor from ordinary commutative rings to analytic rings sending $R$ (viewed as a discrete condensed ring) to the pair

$$R \ \longmapsto\ \big(R,\, D^{\mathrm{cond}}(R)\big),$$

where $D^{\mathrm{cond}}(R)$ is the full (large) derived category of *all* condensed $R$-modules – no solidification is imposed. This works verbatim for animated rings, and on opposite categories gives

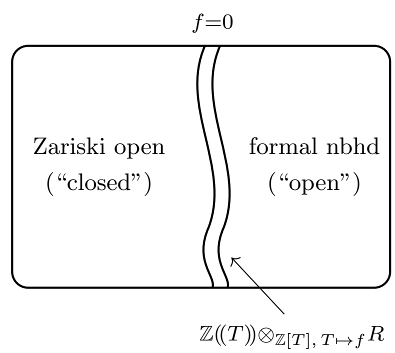
$$\mathrm{AffDerSch} \ \longrightarrow\ \mathrm{Aff}, \tag{18.7}$$

a functor commuting with finite limits (in particular sending the terminal object to the terminal object): relative tensor products of animated rings go to relative tensor products of analytic rings.

**Proposition 18.15** (Zariski descent)**.** *Under* (18.7)*, Zariski open covers of an affine (derived) scheme go to* !*-covers.*

*Proof.* A principal open $D(g) \subseteq \operatorname{Spec} R$, given by inverting $g$, goes to a *closed immersion* in Aff: inverting $g$ produces an idempotent algebra, and that algebra defines a closed cover in the sense of Proposition 18.13. The condition (18.6) to be checked is then just ordinary Zariski descent of the structure sheaf, which holds. $\square$

*Remark* 18.16 (Why Zariski opens look closed: the fuzz picture)*.* That a Zariski open goes to a *closed* immersion is Clausen's recurring heuristic. In algebraic geometry one pictures the vanishing locus $\{f = 0\}$ as closed and the principal open $\{f \neq 0\}$ as its open complement. But the localization $R[1/f]$ does not literally delete $\{f = 0\}$; derived-ly it deletes the whole formal neighbourhood (the locus $\{f^n = 0\}$ for all $n$), so the deleted region acquires "fuzz". This fuzz makes the Zariski open behave like a *closed* subset with a boundary – a tubular neighbourhood – while the formal neighbourhood that was removed plays the role of the open complement:




Once one is in the solid world one can *name* this boundary: base-change $\mathbb{Z}(\!(T)\!)$ along $\mathbb{Z}[T] \to R$, $T \mapsto f$, to get an idempotent algebra in $D\big((R[1/f])_{\mathbb{Z}_\square}\big)$; removing this closed boundary leaves an honest open, the complement $D\big((R[1/f], \mathbb{Z}[1/f])_\square\big)$. So there are two ways to embed schemes into adic/analytic spaces:

$$R \longmapsto (R, \mathbb{Z})_\square \qquad \text{and} \qquad R \longmapsto (R, R)_\square.$$

The first, base change to $\mathbb{Z}_\square$, is the one above: Zariski opens look closed. The second always removes the boundaries, and there Zariski opens go genuinely to open immersions.

*Remark* 18.17 (The Grothendieck topology and the functor of topoi)*.* One obtains a Grothendieck topology on Aff by declaring a sieve over $X \in \mathrm{Aff}$ to be a !-cover if it contains finitely many !-able maps $X_i \xrightarrow{f_i} X$ such that $\coprod_i X_i \xrightarrow{f} X$ is a !-cover in the sense of Definition 18.4. That this defines a topology rests, beyond formal properties of finite disjoint unions, on stability of !-covers under base change – itself a consequence of the $\mathrm{Pr}^L$-colimit description (18.2), since base change $\mathrm{Mod}_{D(Y)} \mathrm{Pr}^L \to \mathrm{Mod}_{D(Y')} \mathrm{Pr}^L$ commutes with colimits. In particular one gets a functor

$$\mathrm{Shv}_{\mathrm{Zar}}\big(\mathrm{AffDerSch}; \mathrm{An}\big) \longrightarrow \mathrm{Shv}_!\big(\mathrm{Aff}; \mathrm{An}\big),$$

which is in fact a pullback of topoi (it commutes with colimits and finite limits). The target is essentially the category of analytic stacks, modulo two technicalities deferred here: (i) one works with sheaves valued in anima (homotopy types / Kan complexes / $\infty$-groupoids), and wants an intermediate class of hyperdescent conditions strictly between plain-set descent and full hyperdescent; (ii) the set-theoretic point that CRing is not small, so one restricts to accessible presheaves and checks sheafification preserves accessibility (cf. Section 15.7).

*Remark* 18.18 (The topology is inexplicit)*.* More general covers descend as well; for instance étale covers give !-covers. The essential difficulty going forward is that the !-topology is quite inexplicit: unlike the Zariski or étale topologies, whose covers are finitely generated with standard models, one has little a priori control over what a !-cover can look like, so it is harder to rule out pathologies locally. Proving full faithfulness of the passage from classical analytic geometries into analytic stacks – easy in the affine case at hand – will in more complicated situations require exactly the compactness/generation considerations of Proposition 18.7.

## 18.6 Appendix: relation to Mathew's descendability

Much of this was motivated by Akhil Mathew's theory of *descendability* [Mat16]. The discussion above is very categorical, and one may re-run the same definitions inside ordinary algebraic geometry: replace analytic rings by the pairs

$$\big(R,\, D(R)\big), \qquad R \text{ a (derived) commutative ring, } D(R) \text{ its usual derived category,} \tag{18.8}$$

so that Aff becomes affine (derived) schemes.

*Remark* 18.19 (Everything is proper)*.* In this world a great simplification occurs: *every* map is proper. Indeed for $R \to S$ one has $D(S) = \mathrm{Mod}_S(D(R))$ almost by definition, which is the induced structure. Hence every map is !-able. Beware that the resulting functors are not the classical ones: $f^!$ is simply the right adjoint of $f_*$ (which exists for general categorical reasons), *not* the upper-shriek of Grothendieck duality. For instance for $\pi\colon \operatorname{Spec} k[T] \to \operatorname{Spec} k$, the functor $\pi^!$ here is not the Grothendieck-duality one, and $\pi_! = \pi_*$.

**Proposition 18.20** (!-covers = descendable maps)**.** *In the setting* (18.8)*, a map $f\colon X \to Y$ is a !-cover if and only if the unit lies in the thick subcategory generated by the image of $f_*$,*

$$\mathbf{1}_Y \;\in\; \big\langle\, f_*\colon D(X) \to D(Y) \,\big\rangle, \tag{18.9}$$

*which is exactly Mathew's definition of* descendability*. Consequently any descendable map $R \to S$ (in Mathew's sense) gives a* !*-cover in* Aff *under the functor* (18.7)*.*

This holds because, every map being proper, Proposition 18.7(2a) makes the membership condition (18.9) equivalent to being a !-cover.

**Example 18.21** (Descendable maps)**.** Mathew gives many examples, all of which therefore yield !-covers:

- étale covers, and more generally *countably presented* faithfully flat maps $A \to B$ of ordinary rings (fpqc covers). The countable-presentation hypothesis is apparently (probably genuinely) needed. For simplicial rings the analogous condition should ask $\pi_0$ to be faithfully flat, with a further condition on the higher $\pi_i$; Clausen did not work out the precise derived statement.
- $R \to R/I$ for $I$ a nilpotent ideal. Here descendability holds on the level of *derived* categories; it does *not* say $D(R) = D(R/I)$, since the descent is computed derived-ly and produces terms like $R/I \otimes^{\mathbb{L}}_R D(-)$ in the colimit, not $R/I$ itself.

- a proper surjective map of Noetherian schemes is descendable (extending the affine discussion to schemes); and more generally any finitely presented $h$-cover, combining the flat and proper cases.

By contrast $R \to \pi_0 R$ is not descendable in general: with $R = \mathbb{Q}[x]$, $|x| = 2$, inverting the degree-2 generator gives a ring module that vanishes modulo $x$ but is nonzero. (The simplicial analogue of a nilpotent-ideal quotient requires $R$ to be truncated – finitely many homotopy groups, an Artinian-type condition – before $R \to \pi_0(R)/I$ becomes descendable.)

*Remark* 18.22 (Independence of the ambient framework). For a map of commutative rings one may ask whether descendability in ordinary commutative algebra is equivalent to being a !-cover after applying (18.7) into the condensed world. A priori the condensed side could be a !-cover for some exotic reason involving condensed modules; but in fact the two coincide. This uses a further characterization of descendability (Mathew): one asks not merely for a limit diagram from the cosimplicial ring $B,\ B \otimes_A B, \ldots$, but for a *uniformly* stable one – the $n$-indexed tower obtained from the cosimplicial object should be *pro-isomorphic* to the constant pro-object $A$ (so the vanishing is witnessed by a fixed finite shift at every stage). This pro-object is the old discrete one, and the usual $D(R)$ embeds fully faithfully into $D^{\mathrm{cond}}(R)$, hence $\mathrm{Pro}(D(R)) \hookrightarrow \mathrm{Pro}(D^{\mathrm{cond}}(R))$ fully faithfully, so a pro-isomorphism upstairs is detected downstairs. (For the pro-zero part one passes to fibers and may work in the homotopy category.)

*Remark* 18.23 (Mathew's general setting). Mathew works in the generality of a presentably symmetric monoidal $\infty$-category $\mathcal{C} \in \mathrm{CAlg}(\mathrm{Pr}^L)$ and an algebra object $A \in \mathrm{CAlg}(\mathcal{C})$, comparing $\mathcal{C}$ with $\mathrm{Mod}_A(\mathcal{C})$:

$$\mathcal{C} \in \mathrm{CAlg}(\mathrm{Pr}^L), \quad A \in \mathrm{CAlg}(\mathcal{C}), \qquad \mathcal{C} \longrightarrow \mathrm{Mod}_A(\mathcal{C}).$$

In our language every such map is proper, so his descendability is precisely the proper-case of !-descent. This specializes to the condensed world, but only ever addresses the proper maps, not the general !-able ones – which is the extra mileage of the present formalism.

*Remark* 18.24 (A decomposition formula, for perfect objects). In answer to a closing question: the "decomposition formula" from the previous lecture is not expected to hold for all objects, but should hold for a suitable finiteness class such as the perfect complexes, with a similar intuition. This kind of condition will resurface when the six-functor formalism on Aff is extended to general analytic stacks, where – unlike the affine case settled today – full faithfulness of the comparison from classical analytic geometries is far from immediate.

# 19 Analytic Stacks: Definition, Examples, and the Tate Elliptic Curve

Having spent Sections 16 to 18 building the !-topology on $\mathrm{Aff} = (\mathrm{AnRing})^{\mathrm{op}}$, Scholze now finally defines an *analytic stack* – and, as promised, it is not difficult: a sheaf on Aff for the !-topology, packaged as an accessible functor AnRing $\to$ Anima. The bulk of the lecture is a tour of examples, showing that essentially every existing flavour of geometry – derived schemes, adic spaces, complex- and real-analytic spaces, topological/$C^k$/$C^\infty$ manifolds, condensed anima – embeds into analytic stacks, often in several inequivalent ways related by natural comparison maps (one incarnation of GAGA being an isomorphism of stacks). The lecture closes by returning to the Tate elliptic curve $E_q$, now constructible over the gaseous base ring via a "norm" morphism $\mathbb{P}^1 \to [0, \infty]$.

Throughout, "ring" means commutative ring and everything is (light) condensed.

## 19.1 The definition

Recall (Section 15) that AnRing, the $\infty$-category of analytic rings, is presentable: Clausen showed it has all colimits and is generated under colimits by a set's worth of objects (the $\kappa$-compact analytic rings for $\kappa = (2^{\aleph_0})^+$, Theorem 15.38). This is what makes the following definition sensible.

**Definition 19.1** (Analytic stack)**.** An *analytic stack* is an accessible functor

$$X\colon \ \mathrm{AnRing} \longrightarrow \mathrm{Anima}$$

that commutes with finite products and satisfies the following descent condition: for every !-hypercover $\mathrm{AnSpec}(A_\bullet) \to \mathrm{AnSpec}(A)$ *for which* the associated diagram of derived categories is a limit,

$$D(A) \xrightarrow{\sim} \varprojlim_{\Delta}{}^{!} D(A_\bullet), \tag{19.1}$$

the space of $A$-points is the corresponding limit of the $A_\bullet$-points:

$$X(A) \xrightarrow{\sim} \varprojlim_{\Delta} X(A_\bullet). \tag{19.2}$$

Here *accessible* means $X$ commutes with $\kappa$-filtered colimits for some (equivalently, all sufficiently large) regular cardinal $\kappa$; the word is present only to circumvent the set-theoretic issue that AnRing is not small (Section 15.7). Commuting with finite products (including the empty product) says geometrically that $X$ turns finite disjoint unions of affines into products. We write AnStk for the $\infty$-category of analytic stacks.[6]

The subtlety is that descent is imposed only along the hypercovers that already satisfy the derived condition (19.1). Recall (Section 17 and Corollary 17.20) that (19.1) is equivalently the statement that $D(A)$ is the colimit of the $D(A_\bullet)$ along the lower-shriek functors, computed in $\mathrm{Pr}^L$:

$$D(A) \xrightarrow{\sim} \varinjlim_{\Delta^{\mathrm{op}}}{}^{!} D(A_\bullet) \quad \text{in } \mathrm{Pr}^L,$$

a condition stable under base change.

*Remark* 19.2 (An intermediate topos)*.* The definition selects, up to set theory, an $\infty$-topos lying strictly *between* !-sheaves and !-hypersheaves. For Čech nerves of !-covers the condition (19.1) holds automatically (by the very definition of a !-cover, Definition 18.4), so every analytic stack is a !-sheaf. For hypercovers one would normally demand descent against *all* hypercovers; here one asks it only against those hypercovers for which the derived categories already glue, i.e. satisfy (19.1). Restricting the class of hypercovers this way is exactly what is needed so that the primary invariant $D(X)$ (below) will itself satisfy descent.

*Remark* 19.3 (Colimits and sheafification)*.* One can form colimits in AnStk: take the colimit in accessible presheaves Fun(AnRing, Anima) – still an accessible functor – and then enforce the sheaf condition by sheafifying. Sheafification preserves accessibility (and set-theoretic size); this is the analogue for the !-topology of Waterhouse's theorem [Wat75] for the fpqc topology (Remark 15.36). Constructing this sheafification requires knowing that the pullback of a hypercover satisfying (19.1) again satisfies it, which follows from the same base-change argument as in Sections 17 and 18.

[6] This is the definition given in the lectures, but it is in fact incorrect (if one wishes to obtain a topos). See [ABLBRCS26, Definition 4.2.1] and [Sch26, Fact 1.21] for the corrected version.

## 19.2 Examples

**Example 19.4** (Affine analytic stacks)**.** For any $A \in \mathrm{AnRing}$ the functor of points

$$\mathrm{AnSpec}(A)\colon\; B \longmapsto \mathrm{Hom}_{\mathrm{AnRing}}(A, B)$$

is an analytic stack. It manifestly commutes with finite products (it lands in mapping *spaces*, already in Anima, so nothing needs sheafifying), and the descent condition (19.2) is precisely that $A$ is recovered as the limit of the $A_\bullet$ – both on the level of rings and, using (19.1), on the level of derived categories. Thus one obtains a fully faithful embedding

$$(\mathrm{AnRing})^{\mathrm{op}} = \mathrm{Aff} \;\hookrightarrow\; \mathrm{AnStk}\,. \tag{19.3}$$

The affine analytic stacks are the basic building blocks; all others arise by gluing them, i.e. as colimits.

*Remark* 19.5 (Reading the accessibility condition)*.* The accessibility clause in Definition 19.1 is exactly the condition that $X$ be a small colimit of objects in the essential image of (19.3):

$$X \text{ accessible} \iff X \text{ is a small colimit of affines } \mathrm{AnSpec}(A).$$

The colimit here is taken in presheaves (before enforcing the sheaf condition); sufficiently filtered colimits already preserve the descent condition, so one may sheafify afterwards. This is the same dichotomy as for condensed sets (Section 2), which one may think of either as built from profinite sets under small colimits or as the accessible sheaves.

The point of imposing so strong a topology is that analytic stacks with completely different presentations – built as colimits of affines in a priori unrelated ways – can nonetheless be identified. A strong topology is what makes such non-obvious isomorphisms hold; we will see a GAGA instance below.

**Example 19.6** (Derived schemes)**.** There is a functor

$$\mathrm{DerSch} \longrightarrow \mathrm{AnStk}, \qquad \mathrm{Spec}(A) \longmapsto \mathrm{AnSpec}(A), \tag{19.4}$$

where a derived scheme is regarded as a sheaf for the descendable topology on affine derived schemes and $A$ is given the *trivial* analytic ring structure (all condensed $A$-modules complete). Since descendable covers go to !-covers (Propositions 18.15 and 18.20), this is well-defined. It is fully faithful on derived schemes; the proof uses Bhatt's Tannaka duality.[7]

**Example 19.7** (Adic spaces)**.** Assuming sheafiness throughout (Definition 10.7), there is a functor

$$\{\text{adic spaces}\} \longrightarrow \mathrm{AnStk}, \qquad \mathrm{Spa}(A, A^+) \longmapsto \mathrm{AnSpec}\big((A, A^+)_\square\big). \tag{19.5}$$

It is fully faithful on quasi-projective objects; one does not know a full faithfulness statement in general, because there is no good version of Tannaka duality for adic spaces.

[7]Scholze named "Bhatt's Tannaka duality" without a precise reference. Editorially, the relevant statement is Bhatt's *Algebraization and Tannaka duality* [Bha14], where a qcqs (derived) scheme or algebraic space $X$ is recovered from the symmetric monoidal $\infty$-category $\mathrm{QCoh}(X)$, so that maps into $X$ correspond to suitable tensor functors on quasi-coherent complexes.

*Remark* 19.8 (Two embeddings of schemes into adic/analytic spaces)*.* Composing Sch $\hookrightarrow$ {adic spaces} with (19.5) gives *different* functors, according to the choice of ring of integral elements. For a discrete ring $A$,

$$\begin{tikzcd} & \operatorname{Spa}(A,\mathbb{Z}) \arrow[r, mapsto] & \operatorname{AnSpec}\big((A,\mathbb{Z})_\square\big) \\ \operatorname{Spec}(A) \arrow[ur, mapsto] \arrow[dr, mapsto] & & \\ & \operatorname{Spa}(A,A) \arrow[r, mapsto] & \operatorname{AnSpec}\big((A,A)_\square\big). \end{tikzcd}$$

Both are fully faithful embeddings. They differ sharply in their geometry, as already flagged by Clausen (Remark 18.16): under the first, $(A,\mathbb{Z})_\square$, *every* map of affines becomes proper and open immersions become closed immersions (Zariski opens "look closed"); under the second, $(A,A)_\square$ (the "relatively solid" structure), open immersions go to open immersions, matching the usual scheme-theoretic notions. The topmost row $(A,\mathbb{Z})_\square$ is itself obtained from the basic embedding into $\operatorname{AnSpec}(\mathbb{Z}_\square)$ by base change from $\operatorname{AnSpec}(\mathbb{Z})$ with the trivial structure. More generally, for any $A \in \operatorname{AnRing}$ one gets a functor

$$\operatorname{DerSch}_{A^\triangleright(*)} \longrightarrow \operatorname{AnStk}_{/A},$$

from derived schemes over the underlying discrete ring $A^\triangleright(*)$ to analytic stacks over $A$, by using the previous functors and base-changing along $\operatorname{AnSpec}(A) \to \operatorname{AnSpec}(A^\triangleright(*))$. For instance, taking $A = \mathbb{C}$ with its gaseous structure, ordinary schemes over $\mathbb{C}$ map to analytic stacks over $\mathbb{C}_{\text{gas}}$.

**Example 19.9** (Complex-analytic spaces)**.** There is a functor

$$\{\text{complex-analytic spaces}\} \longrightarrow \operatorname{AnStk}_{/\mathbb{C}_{\text{gas}}}, \qquad X = \bigcup_K K \longmapsto \bigcup_{K\subseteq X} \operatorname{AnSpec} \mathcal{O}(K)^t,$$

where $K$ ranges over the *compact Stein* subsets of $X$. Complex-analytic geometers do not usually isolate a notion of "affinoid", but they could: any complex-analytic space is a union of compact Stein subsets, and these behave very much like affinoid objects. A typical compact Stein is the vanishing locus of an ideal in a closed polydisc,

$$V(I) \subseteq \mathbb{D}^n = \{(z_1,\dots,z_n) \in \mathbb{C}^n : \ |z_i| \le 1\}. \tag{19.6}$$

On such a $K$ one puts the algebra of *overconvergent* holomorphic functions

$$\mathcal{O}(K)^t \ = \ \varinjlim_{U\supseteq K} \mathcal{O}(U), \tag{19.7}$$

the colimit over open neighbourhoods $U$ of $K$ of the holomorphic functions on $U$; it carries a natural (dual nuclear Fréchet, DNF) topology, hence a condensed ring structure that is a gaseous $\mathbb{C}$-algebra, and one gives AnSpec the induced analytic ring structure. The functor is fully faithful on quasi-projective objects.

*Remark* 19.10 (Noetherianity of $\mathcal{O}(K)$)*.* It is a theorem of the 1970s–80s that for $K$ suitably close to a polydisc – in particular for $K = V(I)$ a Zariski-closed subset of a polydisc as in (19.6) – the ring $\mathcal{O}(K)^t$ is Noetherian and excellent, and regular when $X$ is a manifold, so that coherent sheaves on $X$ are described locally by finitely generated modules over these Noetherian rings. There is a subtlety: for a more general compact Stein (e.g. one whose slice is a profinite subset of the polydisc, giving infinitely many connected components of a zero locus) Noetherianity can fail; it is the Zariski-closed-in-a-polydisc case that is Noetherian.[8]

*Remark* 19.11 (Other manifold flavours)*.* The same kind of definition works for real-analytic manifolds, smooth ($C^\infty$) manifolds, $C^k$-manifolds, and $C^0$ = topological manifolds; one uses in each case the appropriate over-convergent functions on neighbourhoods (over-convergent continuous functions being a slightly awkward but workable notion). In each case one may take real- or complex-valued functions, giving two functors with one the base change of the other along $\mathbb{R} \to \mathbb{C}$.

**Example 19.12** (Condensed anima)**.** Light condensed anima also map to analytic stacks:

$$\mathrm{CondAn}^{\mathrm{light}} \longrightarrow \mathrm{AnStk}\,.$$

This comes not from the condensed structure on the rings but from the test objects: a light profinite set $S \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$ is sent to the spectrum of its ring of continuous (= locally constant) $\mathbb{Z}$-valued functions,

$$S \longmapsto \mathrm{Spec}\ \mathrm{Cont}(S, \mathbb{Z}) \longmapsto \mathrm{AnSpec},$$

composed with the derived-scheme functor (19.4) (with its trivial structure). For $S$ finite this is a finite disjoint union of copies of $\mathrm{Spec}\,\mathbb{Z}$; in general $S = \varprojlim_i S_i$ and the spectrum is the corresponding limit. A surjection $S' \twoheadrightarrow S$ induces a faithfully flat map $\mathrm{Cont}(S, \mathbb{Z}) \to \mathrm{Cont}(S', \mathbb{Z})$, and *lightness* guarantees it is countably presented, hence descendable (Example 18.21), so a !-cover; this is why condensed anima, defined as hypersheaves of anima on $\mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$, map to analytic stacks. The intermediate topology between sheaves and hypersheaves reappears here: one checks that a hypercover of light profinite sets goes to something satisfying exactly the descent condition (19.1).

## 19.3 Incarnations and a GAGA isomorphism

The examples above give *several* incarnations of the same underlying object as an analytic stack, related by natural maps. Take the two-sphere $S^2$, worked over the gaseous complex numbers throughout.

*Remark* 19.13 (Incarnations of $S^2 \cong \mathbb{P}^1_{\mathbb{C}}$)*.* There are natural maps

$$\begin{array}{c} (S^2)^{\mathrm{top.mfd}}_{\mathbb{C}_{\mathrm{gas}}} = \mathrm{AnSpec}\ \mathrm{Cont}(S^2, \mathbb{C})_{\mathrm{gas}} \\ \downarrow \\ (S^2)^{\mathbb{R}\text{-an}}_{\mathbb{C}_{\mathrm{gas}}} = \mathrm{AnSpec}\ C^\omega(S^2, \mathbb{C})_{\mathrm{gas}} \\ \downarrow \\ (\mathbb{P}^1_{\mathbb{C}})^{\mathbb{C}\text{-an}}_{\mathbb{C}_{\mathrm{gas}}} \\ \sim \big\downarrow \mathrm{GAGA} \\ (\mathbb{P}^1_{\mathbb{C}})^{\mathrm{alg}}_{\mathbb{C}_{\mathrm{gas}}}\,. \end{array} \tag{19.8}$$

[8] Scholze gave no precise attribution. Editorially: Noetherianity of $\mathcal{O}(K)$ for $K$ a compact Stein set is due to Frisch [Fri67], and excellence to Bingener [Bin78] and Scheja–Storch [SS72]; cf. the appendix of Clausen–Scholze's lectures on condensed mathematics and complex geometry.

The top two are affine: $S^2$ as a topological manifold gives AnSpec of the (gaseous) algebra of continuous functions, and as a real-analytic manifold gives AnSpec of the algebra $C^\omega(S^2,\mathbb{C})$ of real-analytic functions. The complex-analytic $\mathbb{P}^1$ is *not* affine: it is glued from two copies of AnSpec $\mathcal{O}(\mathbb{D})^t$ (over-convergent holomorphic functions on a disc) along the over-convergent holomorphic functions on the circle $S^1$. The algebraic $\mathbb{P}^1$ is likewise glued from two copies of AnSpec of the polynomial algebra $\mathbb{C}[T]$. In addition, one may treat $S^2$ purely as a condensed set and base-change to $\mathbb{C}_{\mathrm{gas}}$ (Example 19.12); this incarnation is secretly defined over $\mathbb{Z}$ (it uses locally constant $\mathbb{Z}$-valued functions on a profinite presentation $S^2 = \widetilde{S}/\sim$), and there is a natural map from the continuous-function incarnation at the top all the way down to it.

*Remark* 19.14 (GAGA as an isomorphism of stacks). The marked arrow in (19.8) is an *isomorphism*: the complex-analytic $\mathbb{P}^1$ and the algebraic $\mathbb{P}^1$ agree as analytic stacks. This is one incarnation of GAGA – and a strong one, an isomorphism of the geometric objects themselves (in particular of their derived categories), not merely a comparison of coherent sheaves or their cohomology. The two sides are built in genuinely different ways – one from polynomial algebras, one from algebras of over-convergent holomorphic functions – and it is precisely the strong (!-)topology imposed in Definition 19.1 that forces such an isomorphism to hold. One proves it by covering both by two discs, checking the two discs form a descendable (!-)cover, and unravelling what it means for the glued objects to coincide. (The GAGA *isomorphism* of stacks is logically prior to, and does not by itself contain, statements like Riemann–Roch, which require knowing the object is algebraic.)

*Remark* 19.15 (The sheaf $X \mapsto D(X)$ and $\mathrm{Pr}^L$-valued sheaves). The assignments

$$\mathrm{AnSpec}(A) \;\longmapsto\; D(A) \;\longmapsto\; \mathrm{Pr}^L_{D(A)} = \mathrm{Mod}_{D(A)}(\mathrm{Pr}^L) \;\in\; \mathrm{Cat}_\infty$$

satisfy descent (this was the design goal of the !-topology), so they induce, for every analytic stack $X \in \mathrm{AnStk}$, an $\infty$-category $D(X)$ of quasi-coherent sheaves and a 2-category $\mathrm{Pr}^L_X$ of sheaves of categories over $X$, via the intermediate (sheaf-but-not-quite-hypersheaf) descent condition. Both are part of a six-functor formalism to be developed later. The different incarnations of Remark 19.13 have very different module theories: for the top (continuous-function) incarnation $D(X)$ is modules over the algebra, whereas base-changing $D\big((S^2)^{\mathrm{top}}_{\mathbb{C}_{\mathrm{gas}}}\big)$ to AnSpec($\mathbb{C}$) recovers the derived category of sheaves of $\mathbb{C}$-vector spaces on the underlying topological space $S^2$:

$$D(X) \otimes_{D(\mathbb{C}_{\mathrm{gas}})} D(\mathbb{C}) \;\simeq\; D\big(\mathrm{Shv}(X,\mathbb{C})\big).$$

## 19.4 Construction of the Tate elliptic curve

We return to the example that opened the discussion of analytic stacks (Sections 12 and 14): the Tate elliptic curve $E_q$. With the language of analytic stacks in place, it can now be constructed properly.

*Remark* 19.16 (The gaseous base ring). Recall (Theorem 14.3) the analytic ring

$$A := \mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}},$$

initial (as an analytic, or animated analytic, ring) among those endowed with a topologically nilpotent *unit* $q$ such that, with $P = \mathbb{Z}[\mathbb{N} \cup \{\infty\}]/[\infty]$,

$$1 - q\cdot\mathrm{shift}\colon\; P \otimes_{\mathbb{Z}} \mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}} \;\xrightarrow{\sim}\; P \otimes_{\mathbb{Z}} \mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}$$

is an isomorphism. Its underlying condensed ring is the ring of Laurent series in $q$ with integer coefficients of polynomial growth,

$$A^{\triangleright}(*) = \Big\{\, f = \textstyle\sum_n a_n q^n \;\Big|\; a_n \in \mathbb{Z},\; a_n = 0\ (n \ll 0),\; \exists k:\ |a_n| = O(n^k)\ (n \to \infty) \,\Big\}.$$

*Remark* 19.17 (Goal and outline). The goal is to define an elliptic curve $E_q$ over the discrete ring $A^{\triangleright}(*)$ such that, as an analytic stack over $A$, its base change is the Tate uniformisation

$$(E_q)_A^{\mathrm{sch}} \;=\; \mathbb{G}_{m,A}^{\mathrm{an}} \big/ q^{\mathbb{Z}}.$$

The outline (Section 14) is:

1. Produce a "norm" morphism of analytic stacks $(\mathbb{P}^1)_A^{\mathrm{sch}} \to [0,\infty]$ (in the precise sense of Definition 19.19 below), and define the analytic $\mathbb{G}_m$ as the preimage of the open interval,

$$\mathbb{G}_{m,A}^{\mathrm{an}} \;=\; |\cdot|^{-1}\big((0,\infty)\big) \;\subseteq\; (\mathbb{P}^1)_A^{\mathrm{sch}}.$$

2. Form the quotient $\mathbb{G}_{m,A}^{\mathrm{an}}/q^{\mathbb{Z}}$ *in analytic stacks*.
3. ("Algebraisation".) Show the quotient is a proper smooth curve; it therefore lies in the image of $\mathrm{Sch}_{A^{\triangleright}(*)} \to \mathrm{AnStk}_A$, i.e. is the analytification of an honest elliptic curve $E_q$ over $A^{\triangleright}(*)$.

*Remark* 19.18 (On the algebraisation step). Step 3 is an algebraisation theorem: a proper smooth curve over $A$, at least one carrying a section, is the analytification of a scheme. Producing the ample line bundle needed is not automatic even in genus one, and here it uses GAGA (Remark 19.14); in the present framework GAGA is far more general than Serre's and does not by itself deliver, say, Riemann–Roch, for which one wants to know a priori that the object is algebraic.

The heart of Step 1 is the notion of a norm, which is not a norm on the underlying set of $A$ but a much more geometric datum.

**Definition 19.19** (Norm on an analytic ring)**.** Let $A \in \mathrm{AnRing}$. A *norm* on $A$ is a morphism of analytic stacks

$$|\cdot|\colon\; (\mathbb{P}^1)_A^{\mathrm{sch}} \;\longrightarrow\; [0,\infty],$$

where $[0,\infty]$ and $\mathbb{P}^1$ over $A$ arise from the condensed-set and scheme functors base-changed to $A$ (one must fix a value for $|q|$; the normalisation is $|q| = \frac{1}{2}$), subject to:

1. *Values at* $0,1,\infty$*:* the constant sections $0,1,\infty \in \mathbb{P}^1$ go to the constant sections $0,1,\infty \in [0,\infty]$.
2. *Topologically nilpotent* $\Rightarrow |\cdot| \le 1$*:* the topologically nilpotent locus of $\mathbb{A}^1$, i.e. the monomorphism

$$\mathrm{AnSpec}\big(A[T^{\mathbb{N}\cup\{\infty\}}]/(T^\infty = 0)\big) \;\longrightarrow\; \mathrm{AnSpec}\; A[T],$$

   has the property that its composite to $[0,\infty]$ factors over $[0,1] \subseteq [0,\infty]$. (This last is automatically a condition, not extra data, since the maps involved are monomorphisms.)

3. *Multiplicativity:* let $X \subseteq (\mathbb{P}^1)^3$ be the closure of $\{xy = z\} \subseteq \mathbb{G}_m^3$, and let $\overline{X} \subseteq [0,\infty]^3$ be the closure of $\{xy = z\} \subseteq (0,\infty)^3$. Then the norm-cubed carries $X$ into $\overline{X}$:

$$\begin{tikzcd} X_A^{\mathrm{sch}} \arrow[r] \arrow[d, "|\cdot|_A^3"'] & \big[(\mathbb{P}^1)^3\big]_A^{\mathrm{sch}} \arrow[d, "|\cdot|^3"] \\ \overline{X} \arrow[r, hook] & {}^3. \end{tikzcd} \tag{19.9}$$

Working with $\mathbb{P}^1$ rather than $\mathbb{A}^1$ is deliberate: even on $\mathbb{A}^1$ one wants to allow functions of infinite norm, so one might as well allow $\infty$ on the source; and taking closures in (19.9) is what handles the degenerate products $0 \cdot \infty$ cleanly. Equivalently, one may require the norm to be equivariant for inversion on $\mathbb{P}^1$ and on $[0, \infty]$ and ask for honest multiplicativity on the preimage of $[0, \infty) \subseteq \mathbb{P}^1$ (which lies inside $\mathbb{A}^1$, but is in general a proper subset, since a point of $\mathbb{A}^1$ can still have infinite norm).

*Remark* 19.20 (No triangle inequality). There is deliberately *no* additivity or triangle-inequality condition. In particular $|2|$ need not satisfy $|2| \leq 2$; there is a theorem that on the gaseous base ring $|2|$ can even be $\infty$. Concretely $|2|$ is a function on $\mathrm{AnSpec}(A)$ valued in $[0, \infty]$, and it is surjective, so there is a locus where one is forced to have $|2| = \infty$.

**Theorem 19.21** (Existence and uniqueness of the norm)**.** *There is a unique norm on* $\mathrm{AnSpec}\ \mathbb{Z}[q^{\pm 1}]_{\mathrm{gas}}$ *taking* $q$ *to* $\frac{1}{2}$*. (Any value in* $(0,1)$ *would do equally: since no property of the target was used beyond its order structure, one rescales* $[0, \infty]$ *by an exponent.)*

The construction rests on a Tannaka-duality description of maps into a locally compact Hausdorff space, treated as an analytic stack.

**Lemma 19.22** (Tannaka duality for locally compact Hausdorff spaces)**.** *Let* $X$ *be a finite-dimensional (metrizable) locally compact Hausdorff space, regarded as an analytic stack* $X^{\mathrm{cond}}$ *via Example 19.12, and let* $Y \in \mathrm{AnStk}$*. Then*

$$\mathrm{Hom}(Y, X^{\mathrm{cond}}) \;=\; \left\{ \begin{array}{l} F\colon D(X, \mathbb{Z}) \to D(Y) \textit{ tensor functors, commuting with} \\ \textit{colimits, } D(\mathbb{Z})\textit{-linear, such that !-locally on } Y, \\ \textit{for every closed } Z \subseteq X,\ F(i_{Z!}\mathbb{Z}) \in D(Y) \textit{ is connective} \end{array} \right\}.$$

*Remark* 19.23 (Reading the lemma). On $X$ there are many idempotent algebras: for each closed subset $Z \subseteq X$ the constant sheaf $i_{Z!}\mathbb{Z} = \mathbb{Z}_Z$ (which for a closed immersion equals the pushforward, $i_! = i_*$) is an idempotent commutative algebra in $D(X, \mathbb{Z})$, and these generate $D(X, \mathbb{Z})$ under colimits. Hence a colimit-preserving tensor functor $F$ is determined by the idempotent algebras $F(\mathbb{Z}_Z)$, and the only conditions to impose are that their tensor behaviour matches the intersection pattern $Z \cap Z'$ of closed subsets (this is what "tensor functor" encodes) together with a connectivity constraint. The connectivity of $F(\mathbb{Z}_Z)$ is what forces the corresponding idempotent algebra downstairs to come from an honest (AnRing-level) closed immersion; if $X$ were profinite, closed subsets would be intersections of clopens, so the $F(\mathbb{Z}_Z)$ would be retracts of the unit and $Y$ affine forces them connective automatically, but in general the connectivity must be imposed !-locally on $Y$. The mapping object is in fact a set, not merely an anima.

*Remark* 19.24 (Sketch: constructing the norm). By Lemma 19.22, to give the norm $|\cdot|\colon (\mathbb{P}^1)^{\mathrm{sch}}_A \to [0, \infty]$ it suffices to specify, compatibly, the idempotent algebras corresponding to the closed subsets of $[0, \infty]$. By the multiplicativity/inversion symmetry it is enough to give the preimages of the intervals $[0, r]$, $r \in \mathbb{R}$ – i.e. the loci $\{|T| \leq r\}$ – as over-convergent closed discs

$$\mathrm{AnSpec}\ A\{T\}_r \;=\; D(0, r)^t \;\subseteq\; (\mathbb{A}^1)^{\mathrm{sch}}_A \;\subseteq\; (\mathbb{P}^1)^{\mathrm{sch}}_A,$$

where

$$A\{T\}_r \;=\; \left\{ \textstyle\sum_{n \geq 0} a_n T^n \;\middle|\; a_n \in A,\ \exists\, r' > r :\ |a_n|\, r'^{\,n} \to 0 \right\} \;=\; \varinjlim_t\ A\big[(q^t T)^{\mathbb{N} \cup \{\infty\}}\big] \big/ \big((q^t T)^{\infty} = 0\big),$$

the colimit running over (rational) exponents $t$, with the normalisation $|q| = \frac{1}{2}$. The idea is to start from the free algebra on a null sequence and multiply the basis vectors by suitable powers of $q$ to enforce the over-convergence condition. One then checks – and this is the one computation genuinely using the gaseous structure – that the tensor products of these idempotent algebras (glued at 0 and $\infty$) reproduce exactly the intersection behaviour of the intervals in $[0, \infty]$, which by Lemma 19.22 yields the desired norm. Clausen promised a fuller account of this computation next time.

*Remark* 19.25 (Analytic spaces versus analytic stacks). In closing discussion, the question arose whether one can axiomatise a notion of *analytic space* inside analytic stacks – a single umbrella definition specialising to all the classical theories. This is exactly the intent of the whole framework: the analytic stack *is* the general notion, and one should not be put off by the change of name from "space" to "stack". Scholze and Clausen were never satisfied with any proposed axioms cutting out the "non-stacky" objects; a candidate condition floated was that the functor of points be a colimit of affines along *monomorphisms*. Merely asking that the functor take values in sets already fails – even affine objects $\mathrm{AnSpec}(A)$ are not set-valued, since they can detect derived structure.

# 20 Normed Analytic Rings

Having defined analytic stacks in Section 19, Clausen spends the first part of the lecture revisiting two points from Scholze's exposition – the functor from light condensed anima to analytic stacks, and the description of maps into (the analytic stack attached to) a compact Hausdorff space – filling in the argument that a finite-dimensional compact Hausdorff space is recovered Tannakially from its category of sheaves. The rest of the lecture develops the notion of a *norm* on an analytic ring (Definition 19.19). Clausen gives an axiomatisation that differs slightly from Scholze's: one of Scholze's multiplicativity conditions is rephrased, and a new condition (Axiom 4 below) is added, placing the "basic disk" $\mathrm{Spec}(P)$ between the open and the closed unit disk. Whether Axiom 4 follows from the others is not known in general (it does over a base that solidifies to 0). The lecture closes with a classification: giving a norm together with a topologically-small element $q$ is the same as a map to $\mathrm{Spec}(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}})$ times a value $N(q) \in (0, 1)$; the gaseous base ring of Section 14 is exactly what controls norms.

Throughout, "ring" means commutative ring and everything is (light) condensed.

## 20.1 The functor from condensed anima

Recall (Example 19.12) the functor

$$\mathrm{CondAn}^{\mathrm{light}} \longrightarrow \mathrm{AnStk}\,. \tag{20.1}$$

It is induced on the generators – light profinite sets $T \in \mathrm{ProFin}_{\mathbb{N}}(\mathrm{Fin})$ – by

$$T \longmapsto \mathrm{AnSpec}\big(\mathrm{Cont}(T, \mathbb{Z})\big), \tag{20.2}$$

where $\mathrm{Cont}(T, \mathbb{Z})$ is the *discrete* ring of continuous (= locally constant) $\mathbb{Z}$-valued functions on $T$, endowed with the maximal (trivial, "uncompleted") analytic ring structure – i.e. all condensed modules, or all derived condensed modules, over that discrete ring are declared complete.

Two facts make this functor well-behaved.

- The assignment (20.2) sends hypercovers of light profinite sets to !-hypercovers satisfying !-descent (equivalently !-codescent, a purely formal $\mathrm{Pr}^L$-fact of Lurie, independent of the shape of the diagram; Theorem 18.5 and Corollary 17.20).

- It preserves finite limits, in particular pullbacks. A pullback of light profinite sets is a filtered inverse limit of pullbacks of finite sets, which goes to a filtered colimit of the corresponding situation for finite sets; the finite case is clear.

Consequently the induced functor (20.1) is colimit-preserving and finite-limit-preserving.

*Remark* 20.1 (Algebro-geometric analogue). Gregorić [Gre24] studies the analogue of (20.1) with analytic stacks replaced by fpqc sheaves in ordinary algebraic geometry over a field, developing a Stone-type dictionary between condensed mathematics and algebraic geometry in some detail; Clausen recommends it.

*Remark* 20.2 (No underlying topological space). Asked how to recover a topological space from an analytic stack, Clausen was non-committal: there is an underlying light condensed set, and one could map that to topological spaces, but there is no evident well-defined "underlying topological space" functor on arbitrary analytic stacks. One can try to extract one using monomorphisms (open substacks) of analytic stacks; there are several inequivalent attempts. For a *normed* analytic ring (§20.3 below) there is, however, a good way of extracting underlying topological spaces for objects over it.

## 20.2 Maps into a compact Hausdorff space

Let $K$ be a compact Hausdorff space that is metrizable and finite-dimensional – equivalently, $K$ embeds as a closed subset of a finite power of the closed unit interval,

$$K \;\hookrightarrow\; [0,1]^N.$$

Viewing $K$ as a light condensed set (hence a light condensed anima), we get an associated analytic stack, abusively also denoted $K$. Scholze made two claims about it (Lemma 19.22); we reprove them.

**Theorem 20.3** (Sheaf description and Tannakian points)**.** *Let $K$ be metrizable, finite-dimensional compact Hausdorff.*

1. *The derived category of the associated analytic stack is sheaves on the topological space $K$ valued in the derived category of the trivial base ring:*

$$D(K) \;=\; \mathrm{Shv}\big(K;\ D^{\mathrm{cond}}(\mathbb{Z})\big), \qquad D^{\mathrm{cond}}(\mathbb{Z}) = D(\mathrm{CondAb}^{\mathrm{light}}),$$

*and after base change to an analytic ring $R$ one replaces $D^{\mathrm{cond}}(\mathbb{Z})$ by $D(R)$.*

2. *For an analytic ring $R$, maps of analytic stacks $\mathrm{Spec}(R) \to K$ are the same as symmetric monoidal, colimit-preserving, $D(\mathbb{Z})$-linear functors*

$$F\colon\ \mathrm{Shv}\big(K;\ D(\mathbb{Z})\big) \;\longrightarrow\; D(R) \tag{20.3}$$

*such that, !-locally on $\mathrm{Spec}(R)$, one has $F\big(\mathrm{Shv}(K; D(\mathbb{Z})_{\geq 0})\big) \subseteq D(R)_{\geq 0}$.*

*Both sides of* (20.3) *are sets, not merely anima.*

Here $\mathrm{Shv}(K; D(\mathbb{Z}))$ uses *discrete* coefficients (the ordinary derived category of abelian groups). The finite-dimensionality of $K$ enters only through part (2), and only in comparing $D(\mathbb{Z})$-valued with $D^{\mathrm{cond}}(\mathbb{Z})$-valued sheaves; part (1) and the reduction to idempotent algebras below work for an arbitrary compact Hausdorff $K$.

*Remark* 20.4 (Both sides are sets). The left-hand side of (20.3) is a set because the diagonal $K \xrightarrow{\Delta} K \times K$ is a monomorphism in $\mathrm{CondAn}^{\mathrm{light}}$, and (20.1) preserves this pullback; so two maps $\mathrm{Spec}(R) \to K$ agree in at most one way. The right-hand side is a set because of $D(\mathbb{Z})$-linearity: such an $F$ is the same as a colimit-preserving functor out of $\mathrm{Shv}(K; \mathrm{Anima})$, which is generated under colimits by the representables, hence determined on those.

*Remark* 20.5 (Reduction to idempotent algebras). The functor $F$ in (20.3) is determined by where it sends the constant sheaves $\mathbb{Z}_Z = i_{Z!}\mathbb{Z}$ on closed subsets $Z \subseteq K$; each such must go to an idempotent commutative algebra in $D(R)$. Thus the right-hand side is

$$\left\{\begin{matrix} \text{maps } \{\text{closed subsets of } K\} \;\longrightarrow\; \{\text{idempotent algebras in } D(R)\} \\ \text{sending filtered } \bigcap \text{ to filtered } \varinjlim, \text{ and } \bigcup \text{ to the Mayer–Vietoris construction} \end{matrix}\right\}$$

together with the connectivity condition. The idempotent algebras in $D(R)$ form a poset (a priori an $\infty$-category, but one checks it is a poset), so a map of posets satisfying these conditions is unique if it exists. The identification of $\mathbb{Z}_Z$ with the "derived take-away of $K$" near $Z$ is where finite-dimensionality is used.

We indicate the two directions.

*The forward map.* Given $f\colon \mathrm{Spec}(R) \to K$ one takes $f \mapsto f^*$. When $K$ is profinite, a map $\mathrm{Spec}(R) \to K$ is by construction the same as a ring map $\mathrm{Cont}(K, \mathbb{Z}) \to R^{\triangleright}$ (a filtered colimit of the finite case $\mathrm{Cont}(K_n, \mathbb{Z})$), and there the connectivity condition is automatic: the unit $\mathbb{Z}_K$ goes to the unit of $D(R)$, hence to something connective; every clopen subset then goes to a retract of the unit, and clopen subsets generate the connective part under colimits, so everything in $\mathrm{Shv}(K; D(\mathbb{Z})_{\geq 0})$ lands in $D(R)_{\geq 0}$.

*The converse.* Given $F$, choose a surjection $T_0 \twoheadrightarrow K$ from a profinite set and form its Čech nerve $T_\bullet$ (each $T_i \subseteq T_0^{\times(i+1)}$ is again profinite, being closed in a product of profinite sets). Applying $F$ to the pushforwards $\pi_{i*}\mathbb{Z}$ gives a cosimplicial diagram of commutative algebras in $D(R)_{\geq 0}$; by a Künneth computation – and because the maps $\pi_i$ are proper (any map of compact Hausdorff spaces is proper), so proper base change applies – the whole diagram is the Čech nerve of a single map, and $R\pi_{0*}\mathbb{Z}$ is concentrated in degree 0. Set

$$R' \;:=\; \Big(F\big((\pi_0)_*\mathbb{Z}\big),\; \mathrm{Mod}_{F((\pi_0)_*\mathbb{Z})}\big(D(R)\big)\Big),$$

the analytic ring on the algebra $F((\pi_0)_*\mathbb{Z}) \in \mathrm{CAlg}(D(R)_{\geq 0})$ with the induced structure. One produces a map $\mathrm{Spec}(R') \to \mathrm{Spec}(R)$ (proper: only the underlying ring changes) together with a map from its Čech nerve to that of $T_0 \to K$, and hence, by descent, a map $\mathrm{Spec}(R) \to K$ – *provided* $\mathrm{Spec}(R') \to \mathrm{Spec}(R)$ is a !-cover.

*Remark* 20.6 (The descendability reduction). As $\mathrm{Spec}(R') \to \mathrm{Spec}(R)$ is proper, Proposition 18.7(2a) reduces !-cover to the finitary membership $R^{\triangleright} \in \langle \mathrm{im}(D(R') \to D(R))\rangle$. Applying $F$, it suffices that

$$\mathbb{Z}_K \;\in\; \Big\langle \mathrm{im}\big(\mathrm{Shv}(T_0; D(\mathbb{Z})) \xrightarrow{\pi_*} \mathrm{Shv}(K; D(\mathbb{Z}))\big)\Big\rangle,$$

i.e. that $\pi_*\mathbb{Z}_{T_0}$ is a *descendable* commutative algebra object (Proposition 18.20) in $\mathrm{Shv}(K; D(\mathbb{Z}))$. One is free to choose the profinite cover, so it suffices to find one profinite cover for which descendability holds. By pullback (and by Künneth) one reduces to $K = [0, 1]^N$, and it is enough to treat $K = [0, 1]$.

*Remark* 20.7 (The Cantor cover of the interval and finite-dimensionality). For $K = [0,1]$ write the standard Cantor cover as an inverse limit of compact Hausdorff spaces rather than of finite sets: at each stage take the disjoint union of the two dyadic halves of $[0,1]$ (then of the four quarters, etc.), each mapping onto $[0,1]$; the inverse limit is the Cantor set, surjecting onto $[0,1]$. The pushforward from the Cantor set is the sequential colimit of the pushforwards from these finite stages. There is a general fact that descendability of a sequential colimit follows from descendability with a *uniform* exponent of nilpotence at each finite stage; at each stage one has a Mayer–Vietoris cover of the closed pieces with only double (no triple) intersections, giving exponent $\approx 2$, and the colimit $\approx 3$ or 4. For $[0,1]^N$ the exponent grows with $N$ but stays finite – this is exactly where finite-dimensionality is needed; for the Hilbert cube $[0,1]^{\mathbb{N}}$ the estimate degenerates and one can expect nothing.

*Remark* 20.8 (Two further remarks).

- A posteriori (from the proof, not a priori), the condition in Theorem 20.3(2) is equivalent to asking that the connectivity estimate hold after a *proper* cover of $\mathrm{Spec}(R)$, rather than only !-locally.
- If $Z \subseteq K$ is closed with open complement $U$, then the associated analytic stacks are each other's complements in the naive way: the functor of points of $U$ sends $R$ to those maps $\mathrm{Spec}(R) \to K$ whose pullback to $Z$ is the empty analytic stack, and dually. (Check on profinite sets.)

The passage from $F((\pi_0)_*\mathbb{Z})$ as an $\mathbb{E}_\infty$-algebra to an *animated* algebra requires an extra argument (derived symmetric powers; work in progress with Maxime Ramzi, on a categorical characterisation of animated commutative rings – what structure on the module category of an $\mathbb{E}_\infty$-ring promotes it to an animated ring, something like derived symmetric power functors); Clausen deferred it. In characteristic 0 it is automatic.

### 20.3 Norms on analytic rings

We now define norms. Scholze introduced the notion last time (Definition 19.19); Clausen's axioms differ cosmetically (multiplicativity) and substantively (Axiom 4). Fix an analytic ring $R$. Over any analytic ring one can build the *algebraic* projective line $\mathbb{P}^1_R$ (base change of $\mathbb{P}^1_{\mathbb{Z}}$ with its trivial structure), and the closed interval $[0,\infty]$ is a condensed set, hence an analytic stack.

**Definition 20.9** (Norm)**.** A *norm* on $R$ is a map of analytic stacks

$$N\colon\ \mathbb{P}^1_R \longrightarrow [0,\infty]$$

subject to Axioms 1–4 below.

Before listing the axioms, note what a norm does to a ring element. For $f \in R^{\triangleright}(*)$ one gets a section $\mathrm{Spec}(R) \to \mathbb{A}^1_R \subseteq \mathbb{P}^1_R$, and composing with $N$ a map

$$N(f)\colon\ \mathrm{Spec}(R) \longrightarrow [0,\infty].$$

So the norm of an element is *not* a real number but a map $\mathrm{Spec}(R) \to [0,\infty]$: think of it as a family of residue fields, with a real number for each, possibly varying. The datum is geometric: given a map $R \to R'$ of analytic rings, a norm on $R$ pulls back to a norm on $R'$ (by composition), and persists.

**Axiom 1.** $N(0)\colon \mathrm{Spec}(R) \to [0,\infty]$ factors through $\{0\}$ (the analytic stack of the singleton point $\{0\} \subseteq [0,\infty]$); i.e. the norm of 0 is the constant function 0.

*Remark* 20.10 (The zero ring). This geometric phrasing sidesteps the usual annoyance of whether $N(1)$ should be 1 or 0 over the zero ring: for the zero ring $\mathrm{Spec}(R) = \emptyset$, so $N(1)$ is simultaneously 1 and 0 (the empty map factors through both), and no inconsistency arises.

**Axiom 2** (compatibility with inversion)**.** Writing $T$ for the coordinate on $\mathbb{P}^1$, the square

$$\begin{tikzcd} \mathbb{P}^1_R \arrow[r, "N"] \arrow[d, "T\mapsto T^{-1}"'] & {[0,\infty]} \arrow[d, "\lambda\mapsto\lambda^{-1}"] \\ \mathbb{P}^1_R \arrow[r, "N"'] & {[0,\infty]} \end{tikzcd}$$

commutes, both vertical maps exchanging 0 and $\infty$.

Axioms 1 and 2 together imply that $N(\infty)$ factors through $\{\infty\}$, that $N(1)$ factors through $\{1\}$, and (by the same reasoning) $N(-1)$ through $\{1\}$.

**Definition 20.11** (Analytic affine line)**.** Given a norm $N$ on $R$, the associated *analytic affine line* is the open locus where the norm is a finite real number:

$$\mathbb{A}^{1,\mathrm{an}}_R \;:=\; N^{-1}\big([0,\infty)\big) \;\subseteq\; \mathbb{P}^1_R.$$

**Axiom 3** (multiplicativity)**.** The square

$$\begin{tikzcd} \mathbb{A}^{1,\mathrm{an}}_R \times \mathbb{A}^{1,\mathrm{an}}_R \arrow[r, "N\times N"] \arrow[d, "(T,S)\mapsto TS"'] & {[0,\infty) \times [0,\infty)} \arrow[d, "(\lambda,\lambda')\mapsto\lambda\lambda'"] \\ \mathbb{A}^{1,\mathrm{an}}_R \arrow[r, "N"'] & {[0,\infty)} \end{tikzcd}$$

commutes; i.e. $N(TS) = N(T)N(S)$ on the analytic affine line.

The multiplication $(T,S) \mapsto TS$ needs justification: it is a priori defined only on the algebraic affine line, and we must know it restricts to $\mathbb{A}^{1,\mathrm{an}}_R$.

**Lemma 20.12** (Multiplication is defined on $\mathbb{A}^{1,\mathrm{an}}_R$)**.** *$\mathbb{A}^{1,\mathrm{an}}_R \subseteq \mathbb{P}^1_R$ is contained in the algebraic affine line $\mathbb{A}^1_R$ (recall $D(\mathbb{A}^1_R) = \mathrm{Mod}_{R[T]}(D(R))$), and multiplication carries $\mathbb{A}^{1,\mathrm{an}}_R \times \mathbb{A}^{1,\mathrm{an}}_R$ into it.*

*Sketch.* Since $N(\infty) = \infty$ (Axioms 1–2), the $\infty$-section of $\mathbb{P}^1$ is an algebra over $N^*(\mathbb{Z}_{\{\infty\}})$, so $\mathbb{A}^{1,\mathrm{an}}_R$ misses the $\infty$-section. It remains to see that a stack over $\mathbb{A}^1$ missing the 0-section factors through $\mathbb{G}_m$; this is a special case of the following general claim.

*Claim.* If $X$ is an analytic stack over $\mathbb{A}^1_R$ whose pullback to the 0-section is the empty stack, then $X \to \mathbb{A}^1_R$ factors through $(\mathbb{G}_m)_R$. Indeed, one may assume $X = \mathrm{Spec}(A)$ affine and think of the map via the pullback functor $f^*\colon D(\mathbb{A}^1_R) \to D(A)$. Missing the 0-section means $f^*$ kills the structure sheaf of the origin $\mathcal{O}_{\mathbb{A}^1}/T$; hence it kills everything built from it under colimits, so it does not distinguish an object from the result of inverting $T$, i.e. $f^*$ factors through $D\big((\mathbb{G}_m)_R\big)$. □

For the fourth axiom we need the "basic disk". Recall from Sections 5 and 14 the free complete module on a null sequence,

$$P \;=\; R[\mathbb{N} \cup \{\infty\}]/R[\infty],$$

which carries a ring structure (measures on $\mathbb{N}$, multiplication from addition on $\mathbb{N}$; Remark 7.1) and a ring map $R[T] \to P_R$ sending $T$ to the Dirac measure at 1. Set

$$D \;:=\; \mathrm{Spec}(P), \qquad D_R \;:=\; \mathrm{Spec}(P_R).$$

The map $R[T] \to P_R$ gives $D_R \to \mathbb{A}^1_R$; it is *not* a monomorphism (so $D_R$ is not literally a subset of $\mathbb{A}^1_R$), but it is proper (hence !-able) – a genuine disk that is not a subspace.

**Axiom 4.** With $D_R \to \mathbb{P}^1_R$ as above:

(4a) the composite $D_R \to \mathbb{P}^1_R \xrightarrow{N} [0, \infty]$ factors through $[0, 1] \subseteq [0, \infty]$; and

(4b) the map $D_R \to \mathbb{P}^1_R$ is an isomorphism over the open unit disk $N^{-1}\big([0, 1)\big)$; i.e. in the square

$$\begin{tikzcd} D_R \arrow[r] \arrow[dr] & \mathbb{P}^1_R \arrow[d, "N"] \\ & {[0,\infty]} \end{tikzcd} \qquad \begin{tikzcd} N^{-1}\big([0,1)\big) \arrow[r, hook] \arrow[d] & \mathbb{P}^1_R \arrow[d, "N"] \\ {[0,1)} \arrow[r, hook] & {[0,\infty]} \end{tikzcd}$$

the pullback of $D_R$ over $N^{-1}([0, 1))$ is $N^{-1}([0, 1))$ itself.

So $D_R$ is "sandwiched between the closed and the open unit disk": its norm is $\leq 1$, and it agrees with $\mathbb{P}^1_R$ (is a genuine subspace) over the open unit disk, while blowing up (failing to be a monomorphism) along the boundary $N = 1$.

*Remark* 20.13 (Motivation for Axiom 4: $P_R$ as a disk between open and closed). The ring map $R[T] \to P_R$ induces an isomorphism modulo $T^n$ for every $n$, so $D_R \to \mathbb{A}^1_R$ is an isomorphism on the formal neighbourhood of the origin and blows up as one moves away from it. In most examples $P_R$ lives in the heart of $D(R)$ and one has injections

$$R[T] \;\hookrightarrow\; P_R \;\hookrightarrow\; R[[T]],$$

so that $P_R$ is a space of sequences

$$P_R \;=\; \big\{(r_0, r_1, \dots) \;:\; \text{termwise multiplication by a null sequence yields a "summable" sequence}\big\},$$

where the notion of summability depends only on the analytic ring structure. (For intuition one may take $R = \mathbb{R}$ and "summable" = absolutely summable – not literally a special case, but close enough.) This is forced by the universal property: maps $P_R \to M$ in $D(R)$ correspond to null sequences in $M$, paired by summation. For $R = \mathbb{C}$ with its gaseous or liquid structure (Definition 12.15), one gets

$$\mathcal{O}^{\mathrm{hol}}\big(\{|z| \leq 1\}\big) \;\longrightarrow\; P_{\mathbb{C}} \;\longrightarrow\; \mathcal{O}^{\mathrm{hol}}\big(\{|z| < 1\}\big),$$

the summability condition being weaker than overconvergence on the closed unit disk (a power series with a value at 1 has summable coefficients) but stronger than convergence on the open disk. So $P$ sits strictly between the open and closed unit disk of radius 1; Axiom 4 axiomatises exactly this placement.

*Remark* 20.14 (Status of Axiom 4). Axiom 4 was not among Scholze's conditions (Definition 19.19); Clausen adds it, and it is not known whether it follows from Axioms 1–3. It *does* follow when the base solidifies to 0 – e.g. if $\mathrm{Spec}(R)$ base-changed to $\mathrm{Spec}(\mathbb{Z}_\square)$ is the empty analytic stack, as happens for any $R$ whose underlying ring is an algebra over $\mathbb{R}^\triangleright$. Non-archimedean Huber-ring norms are not solid, so there could a priori be pathological norms violating Axiom 4 there; but the usual norms all satisfy it, and Clausen takes it as an axiom. There is no additivity or triangle-inequality condition anywhere – $N$ may be archimedean or non-archimedean at will.

*Remark* 20.15 (What a norm supplies: overconvergent disks). For $r \in \mathbb{R}_{\geq 0}$, the pullback $N^*(\mathbb{Z}_{[0,r]})$ is an idempotent algebra in $D(\mathbb{P}^1_R)$, to be interpreted as *overconvergent* functions on the closed disk of radius $r$. Overconvergence is forced: the constant sheaf on $[0, r]$ is the filtered colimit

$$\mathbb{Z}_{[0,r]} \;=\; \varinjlim_{r'>r} \mathbb{Z}_{[0,r']},$$

and $N^*$ (a pullback) commutes with colimits, so any version of "functions on the disk of radius $r$" receiving a map from these is forced to be the overconvergent one. The data of these idempotent algebras, together with the inversion Axiom 2, determines $N$: the closed subsets of $[0, \infty]$ are easy to classify, one need only give the loci $\{N \leq r\}$ and (by inversion) $\{N \geq r\}$, subject to simple compatibilities.

*Remark* 20.16 (Why norms, not just a base ring). Over a bare analytic ring the only geometry one knows how to do is to import algebraic geometry (Proposition 18.15). A norm supplies the extra structure – disks of prescribed radius inside the affine line, which turn out to be !-able subsets – needed to do something resembling traditional analytic geometry (open and closed disks of a given radius). Without a norm there is no notion of the radius of a disk.

## 20.4 Classifying norms

We now classify norms on a given analytic ring $R$. The first step lets us assume a topologically-small unit exists.

**Lemma 20.17** (A small element exists locally)**.** *Given a norm $N\colon \mathbb{P}^1_R \to [0, \infty]$, the locus $N^{-1}\big((0, 1)\big) \to \operatorname{Spec}(R)$ is a cover in the !-topology. Equivalently, working locally on* $\operatorname{Spec}(R)$ *we may assume there is an element $q \in R^{\triangleright}(*)$ with $N(q) \in (0, 1)$.*

*Sketch.* It suffices, and is stronger, to show $N^{-1}(\{\frac{1}{2}\}) \to \operatorname{Spec}(R)$ is a cover. For simplicity assume $N^*(\mathbb{Z}_{\{1/2\}})$ is connective, so that $N^{-1}(\{\frac{1}{2}\})$ is affine, and proper over $\mathbb{A}^1_R$, hence over $\operatorname{Spec}(R)$ (in general one refines by a proper descendable cover by an affine and combines the generation statements; we omit this). By the !-cover criterion for proper maps (Proposition 18.7(2a)) we must see $R^{\triangleright} \in \langle N^*(\mathbb{Z}_{\{1/2\}})\rangle$.

*Claim: $R^{\triangleright}$ is a direct retract of $N^*(\mathbb{Z}_{\{1/2\}})$.* In $D(\mathbb{P}^1_R)$ the constant sheaf on $[0, \infty]$ is the pullback (equivalently pushout) of the constant sheaves on $[0, \frac{1}{2}]$ and $[\frac{1}{2}, \infty]$ glued along the point $\{\frac{1}{2}\}$:

$$\begin{tikzcd}
\mathcal{O}_{\mathbb{P}^1_R} \arrow[r] \arrow[d] & N^*(\mathbb{Z}_{[0,1/2]}) \arrow[d] \\
N^*(\mathbb{Z}_{[1/2,\infty]}) \arrow[r] & N^*(\mathbb{Z}_{\{1/2\}}).
\end{tikzcd}$$

Pushing forward to $D(R)$, the cohomology of $\mathcal{O}_{\mathbb{P}^1_R}$ is $R^{\triangleright}$ in degree 0, and the pushout presents $R^{\triangleright}$ as a homotopy pushout of the three pushed-forward corners. To retract $R^{\triangleright}$ off $N^*(\mathbb{Z}_{\{1/2\}})$ one builds a map $R^{\triangleright} \to N^*(\mathbb{Z}_{\{1/2\}})$ using the two evaluations at 0 and $\infty$ (each landing in $R^{\triangleright}$), which agree on restriction, so give a map splitting the unit – concretely, picking out the constant term of a Laurent series. Hence $R^{\triangleright}$ is a retract of $N^*(\mathbb{Z}_{\{1/2\}})$, which is thus descendable of index 0: the descent is direct, by a retract. $\square$

The next result says a topologically nilpotent unit visible to the norm must come from the gaseous base ring of Section 14,

$$\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}},$$

the initial analytic ring with a topologically nilpotent unit $q$ for which $1 - q \cdot \mathrm{shift}$ is invertible on $P \otimes_{\mathbb{Z}} (-)$ (Theorem 14.3 and Remark 19.16).

**Proposition 20.18** (Small elements are gaseous)**.** *Given a norm on $R$ and $q \in R^{\triangleright}(*)$ with $N(q) \in (0,1)$, the ring map $\mathbb{Z}[T] \to R^{\triangleright}$, $T \mapsto q$, factors uniquely through $\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}$.*

*Sketch. Uniqueness.* This has nothing to do with norms: it is the claim that

$$\mathrm{Spec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}\big) \longrightarrow \mathbb{A}^1_{\mathbb{Z}}$$

is a monomorphism (so the space of factorisations is contractible if nonempty). This is not obvious from the construction of the gaseous ring, which inverted $q$ and imposed gaseousness on $\mathbb{Z}[\widehat{q}] = P$, an object that is *not* idempotent. Instead present $\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}} = \mathrm{Mod}_P\big(\mathbb{Z}[q^{\pm 1}]_{\mathrm{gas}}\big)$, i.e. view it as $P$-modules in the polynomial ring with a *gaseous* variable $q$. Then $P$ *is* idempotent in $D(\mathbb{Z}[q^{\pm 1}]_{\mathrm{gas}})$:

$$P \otimes P \xrightarrow{\sim} P \quad \text{in } D\big(\mathbb{Z}[q^{\pm 1}]_{\mathrm{gas}}\big),$$

which after base change reduces to the computation

$$P_{\mathbb{Z}[q^{\pm 1}]_{\mathrm{gas}}}/(T - q) \;=\; \mathbb{Z}[q^{\pm 1}]_{\mathrm{gas}}$$

(here $T$ is the variable coming from $P$ and $q$ the unit); modding out by $T - q$ returns the underlying ring, checked via a short exact sequence in the free-module description of $P$.

*Existence.* From $N(q) \in (0,1)$: being in $(0,1)$ means being away from $N^{-1}([1,\infty])$, hence away from the "disk $D$ sitting at $\infty$" (the image of $D$ under inversion), which by Lemma 20.12's argument forces $q$ to be gaseous; and being in $[0,1)$ forces $q$ topologically nilpotent (it comes from $D$, by Axiom 4). Together these say $q$ factors through $\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}$. □

**Theorem 20.19** (Classification of norms)**.** *Let $R$ be an analytic ring. The map recording $q$ and its norm,*

$$\left\{\begin{matrix} N\colon \mathbb{P}^1_R \to [0,\infty] \text{ a norm on } R, \\ q \in R^{\triangleright}(*) \text{ with } N(q) \in (0,1) \end{matrix}\right\} \xrightarrow{\;\sim\;} \left\{\begin{matrix} \mathrm{Spec}(R) \xrightarrow{(q, N(q))} \\ \mathrm{Spec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}\big) \times (0,1) \end{matrix}\right\} \tag{20.4}$$

*is an equivalence.*

So, up to a cover of $\mathrm{Spec}(R)$, giving a norm on $R$ is the same as giving, for a chosen element $q$: the value $N(q)\colon \mathrm{Spec}(R) \to (0,1)$, and a witness that $q$ comes from the gaseous theory (the map to $\mathrm{Spec}(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}})$).

*Remark* 20.20 (Both sides are "sets" relative to $\mathbb{A}^1$). Both sides of (20.4) map to $\mathbb{A}^1$ (recording $q$), with fibres that are sets; they are "as much sets as each other" but neither is individually a set, since $q$ is a derived datum (chosen in $\pi_0\big(R^{\triangleright}(*)\big)$, with the higher structure coming along automatically or requiring construction).

*Sketch.* After a cover of $\mathrm{Spec}(R)$ one may assume $q$ admits compatible $n$-th roots for all $n$: the map $\mathbb{Z}[T] \to \mathbb{Z}[T^{1/n} : \text{all } n]$ is fpqc and countably presented, hence descendable (Example 18.21), and both sides of (20.4) are sheaves, so we may work locally. Rescaling the norm we may also assume

$N(q) \equiv \frac{1}{2}$; then one checks $N(q^{1/n}) = (\frac{1}{2})^{1/n}$ (one inclusion by taking $n$-th powers, the other by passing to complementary opens), so in particular roots of unity have norm 1.

For $\alpha \in \mathbb{G}_m$ write $D_\alpha$ for the pullback of $D \to \mathbb{A}^1_R$ along multiplication by $\alpha$, $\mathbb{A}^1_R \xrightarrow{\cdot\alpha} \mathbb{A}^1_R$; this is a disk "of radius $\alpha$". If $\alpha = q\beta$ with $q$ topologically nilpotent, one gets a map $D_\alpha \xrightarrow{\cdot q} D_\beta$ (using that $P$ is a Hopf algebra, encoding multiplication) – an inclusion of disks. Applying this to $\alpha = q^{m/n}$, $m/n \in \mathbb{Q}$, produces disks of radius $N(q)^{m/n}$ with inclusions between them; taking overconvergent versions, the blow-up-along-the-boundary defects wash out in the colimit and one recovers exactly the datum a norm must supply. This shows the map is a monomorphism; conversely, running the construction (translate the fake disk by powers of $q$, form the inclusions, make them overconvergent, check the axioms) produces a norm from the right-hand data, giving surjectivity. $\square$

*Remark* 20.21 (Relation to Berkovich norms; and what norms are for). The assignment $R \mapsto \{\text{norms on } R\}$ is itself an analytic stack (it fits the definition, and Lemma 20.17 exhibits a cover). Only Tate (analytic) rings admit nontrivial norms: a discrete ring has no nontrivial theory of norms. For a Tate–Huber ring, a norm in this sense amounts to a choice of $|q| \in (0,1)$ for a pseudo-uniformizer $q$, yielding a non-archimedean norm on each residue field, chosen continuously; the space of norms is a torsor under rescaling ($N \mapsto N^\alpha$) of a fixed one, matching the rescaling on the Berkovich spectrum. For a global object this is far from a single Berkovich seminorm: it is a whole compatible family. Modding out by these exponential rescalings (so that one can no longer speak of a disk of a *fixed* radius, but much else survives) gives a larger stack that also accommodates $\mathbb{Z}_\square$ and objects over it; this is to be discussed later.

*Remark* 20.22 (Reconstruction from normed points). By analogy with the theory of diamonds – where a discrete (or general) ring is substantially recovered from how it maps to perfectoid Tate–Huber rings, and Scholze remembers only such maps – one can restrict attention to normed analytic rings and recover an object essentially from its functor of points on normed analytic rings alone.

# 21 Six Functors for Analytic Stacks

The bulk of this lecture sets up the six-functor formalism for analytic stacks and, following Liu–Zheng and Mann, shows that the formalism itself *dictates* the Grothendieck topology: there is a smallest class of "shriekable" morphisms of analytic stacks to which the six operations extend. Along the way Scholze gives inductive definitions of *proper* and *cohomologically smooth* maps and computes $f_!$ in examples (an interval, the real line). The lecture opens by discharging a promise from Sections 19 and 20–the explicit construction of the norm on the gaseous base ring–and closes by returning to the Tate elliptic curve $E_q$, now shown to be proper and smooth over the gaseous base and then *algebraised* into an honest scheme.

Throughout, "ring" means commutative ring and everything is (light) condensed.

## 21.1 Construction of the norm

We first construct the norm on the gaseous base ring

$$A \;=\; \mathbb{Z}[\widehat{q}^{\pm 1}]_{\text{gas}}, \qquad |q| = \tfrac{1}{2},$$

promised by Clausen in Section 20 (existence half of Theorem 20.19). By Proposition 20.18 one may assume all rational powers of $q$ are available. The task is to produce the norm $N\colon \mathbb{P}^1_A \to [0,\infty]$ of

Definition 20.9; by Remark 20.15 this amounts to specifying, compatibly, the idempotent algebras cutting out the loci $\{|T| \le r\}$, i.e. the *overconvergent* functions on the closed disc of radius $r$.

As Clausen explained, the norm axioms force the answer up to a single natural guess. The naive candidate is the free algebra on a topologically nilpotent generator obtained by rescaling the coordinate $T$ by a power of $q$; this is not quite right, but becomes correct after passing to the overconvergent (colimit) version.

**Definition 21.1** (Overconvergent closed disc)**.** For $r \in \mathbb{R}_{>0}$ set

$$\mathcal{O}\big(\{|T| \le r\}\big)^t \;:=\; \varinjlim_{t\,:\,2^t > r} A\big[\widehat{q^tT}\big],$$

where $A[\widehat{s}]$ denotes the free $A$-algebra on a topologically nilpotent element $s$ (the ring $P_A$ of Remark 7.1, with $s$ acting as the shift), and the colimit runs over rational $t$. The superscript $t$ records that this is the *overconvergent* structure sheaf, as in Equation (19.7).

The scaling is dictated by wanting $|T| = r$ and $|q^tT| < 1$ (so that $q^tT$ is genuinely topologically nilpotent): since $|q| = \frac{1}{2}$, the condition $(\frac{1}{2})^t\, r = |q^tT| < 1$ reads $2^t > r$. This is the only possible choice. One must then check that these algebras are idempotent and that (together with the analogous algebras "at $\infty$", obtained by replacing $T$ by $T^{-1}$) they assemble into a map $\mathbb{P}^1_A \to [0,\infty]$; all of these are straightforward computations. We record the one that matters.

**Proposition 21.2** (Discs of different radii are disjoint)**.** *For $r < s$ the closed disc $\{|T| \le r\}$ and the complement of the open disc $\{|T| \ge s\}$ do not intersect: with the algebra $\mathcal{O}(\{|T| \ge s\})^t = \varinjlim A[T, \widehat{q^{s'}T^{-1}}]$ obtained by inversion,*

$$A\big[\widehat{q^tT},\, T^{-1}\big] \;\otimes_{A[T^{\pm 1}]}\; A\big[T,\, \widehat{q^sT^{-1}}\big] \;=\; 0 \qquad (t < 0,\; s < 0).$$

*Sketch.* Writing $x = q^tT$ and $y = q^sT^{-1}$, one has $xy = q^{t+s}$, so the tensor product is the free algebra on the two topologically nilpotent generators $x, y$ subject to the single relation coming from their product,

$$A[\widehat{x}, \widehat{y}]/\big(1 - q^a xy\big) \;=\; P_A/\big(1 - q^a\,\mathrm{shift}\big), \qquad a = -(t+s) > 0.$$

But the gaseous ring structure was *defined* by declaring $1 - q\cdot\mathrm{shift}$ (and hence, since the exponent is immaterial, $1 - q^a\cdot\mathrm{shift}$) to be an isomorphism on the compact projective generator $P$ (Theorem 14.3). Thus the quotient vanishes. $\square$

*Remark* 21.3 (Two honest-subspace variants of the disc)*.* The basic disc used above,

$$D_A \;=\; \mathrm{AnSpec}\big(A[\widehat{T}]\big) \;\longrightarrow\; \mathbb{P}^1_A,$$

is proper but not a monomorphism (Definition 20.11). Scholze noted a cleaner perspective: instead of forcing $T$ topologically nilpotent one can force it merely *gaseous*, i.e. pass to the analytic ring $A[T]^{T\text{-gas}}$ obtained by enforcing $1 - T\cdot\mathrm{shift}$ to be an isomorphism on $P$ base-changed to it. Then $\mathrm{AnSpec}(A[T]^{T\text{-gas}})$ *is* an honest subspace of $\mathbb{P}^1_A$, sitting between the open and closed unit discs,

$$N^{-1}\big([0,1)\big) \;\subseteq\; \mathrm{AnSpec}\big(A[T]^{T\text{-gas}}\big) \;\subseteq\; N^{-1}\big([0,1]\big).$$

One may also take the intersection of $D_A$ with this gaseous disc, or the variant $A[\widehat{T}, T]^{T\text{-gas}}$; all three are again sandwiched between the overconvergent closed disc and the open disc, so for the *overconvergent* construction the choice is immaterial. This gives a calculation-free route to idempotency: two of the candidates are sandwiched in each other within the colimit, and since one term is always a monomorphism the idempotency for the other follows formally. The one genuinely necessary computation is Proposition 21.2, that the closed disc $\{|T| \leq r\}$ around 0 and the disc $\{|T| \geq s\}$ around $\infty$ do not meet when $r < s$.

## 21.2 Six functors: recollection and correspondences

Recall the two functors on analytic rings assembled so far. The primary invariant is

$$A \longmapsto D(A), \qquad A \longmapsto \mathrm{Pr}^L_{D(A)} = \mathrm{Mod}_{D(A)}(\mathrm{Pr}^L),$$

a functor AnRing $\to$ CAlg($\mathrm{Cat}_\infty$) landing in symmetric monoidal $\infty$-categories: the algebra structure supplies the tensor product $\otimes$, which is closed, so there is an internal RHom; and functoriality in the ring gives, for each map of rings, a base-change (pullback) functor $f^*$ with right adjoint $f_*$. These are four of the six operations. On the class of !-able (shriekable) maps (Definitions 16.26 and 18.2) we constructed in addition the lower-shriek $f_!$ and its right adjoint $f^!$.

A convenient way to package all this is the notion of a six-functor formalism via correspondence categories.

**Definition 21.4** (Correspondence category)**.** Let $\mathcal{C}$ be an $\infty$-category with all finite limits and let $E$ be a class of morphisms stable under composition (including identities) and base change. The *correspondence category* $\mathrm{Corr}(\mathcal{C}, E)$ is the symmetric monoidal $\infty$-category with the same objects as $\mathcal{C}$, whose morphisms $X \to Y$ are spans

$$\begin{tikzcd} & W \arrow[dl] \arrow[dr, "\in E"] & \\ X & & Y \end{tikzcd}$$

with the right leg in $E$, composed by forming the fibre product

$$\begin{tikzcd} & & W \times_Y W' \arrow[dl] \arrow[dr, "\in E"] & & \\ & W \arrow[dl] \arrow[dr, "\in E"] & & W' \arrow[dl] \arrow[dr, "\in E"] & \\ X & & Y & & Z \end{tikzcd}$$

(the outer legs form the composite span), and with symmetric monoidal structure $X \otimes Y = X \times Y$ given by the product in $\mathcal{C}$.

**Definition 21.5** (Three- and six-functor formalisms)**.** A 3-*functor formalism* on $(\mathcal{C}, E)$–encoding $\otimes$, $f^*$ and $f_!$ (for $f \in E$)–is a lax symmetric monoidal functor

$$D\colon \big(\mathrm{Corr}(\mathcal{C}, E), \otimes\big) \longrightarrow \big(\mathrm{Cat}_\infty, \times\big),$$

the lax structure being the natural maps $D(X) \times D(Y) \to D(X \times Y)$ (in one direction only, not necessarily isomorphisms). Applied to the empty product, the lax structure endows $D(*) = D(\text{final object})$ with a distinguished (unit) object, and every $D(X)$ receives a unit by pullback from $*$. It is a 6-*functor formalism* if RHom, $f_*$ and $f^!$ (for $f \in E$) exist as the corresponding right adjoints.

Producing such a functor directly is a formidable amount of coherence data. Conveniently, Liu–Zheng gave an extremely simple combinatorial recipe (the difficulty being entirely in the proof that it works), which makes these objects easy to construct in practice. For $\mathcal{C} = (\mathrm{AnRing})^{\mathrm{op}}$ and $E$ the shriekable maps, $A \mapsto D(A)$ (equivalently $A \mapsto \mathrm{Pr}^L_{D(A)}$) is such an example.

## 21.3 The topology dictated by the formalism

We now extend the formalism from affine analytic spaces to all analytic stacks, enlarging the class of shriekable maps as little as possible.

**Theorem 21.6** (Liu–Zheng, Mann)**.** *There is a (unique) smallest class $\widetilde{E}$ of morphisms in* AnStk *with the following properties.*

1. *$E \subseteq \widetilde{E}$, where $E$ is the class of shriekable maps of analytic rings.*
2. (Stable under disjoint unions.) *If $f_i \colon X_i \to Y$ lie in $\widetilde{E}$, then so does $f = \coprod_i f_i \colon \coprod_i X_i \to Y$.*
3. ($\widetilde{E}$ local on the target.) *If $f \colon X \to Y$ in* AnStk *admits a cover $\coprod_i Y_i' \twoheadrightarrow Y$ such that each base change $f_i' = f \times_Y Y_i' \colon X \times_Y Y_i' \to Y_i'$ lies in $\widetilde{E}$, then $f \in \widetilde{E}$.*
4. ($\widetilde{E}$ local on the source.) *If $f \colon X \to Y$ admits a cover $g \colon \widetilde{X} \twoheadrightarrow X$ with $g \in \widetilde{E}$ of universal !-descent and with the composite $\widetilde{X} \to Y$ in $\widetilde{E}$, then $f \in \widetilde{E}$.*
5. *If $f \colon X \to Y \in \widetilde{E}$ with $Y$ affine, then there is a cover $g \colon \coprod_i X_i \twoheadrightarrow X$ with $g \in \widetilde{E}$, all $X_i$ affine, and each composite $X_i \to Y$ in $E$.*
6. *The 6-functor formalism extends uniquely from $\big((\mathrm{AnRing})^{\mathrm{op}}, E\big)$ to $(\mathrm{AnStk}, \widetilde{E})$.*

The theorem follows from results of Liu–Zheng [LZ24] and Mann [Man22]; this is a general fact about six-functor formalisms, independent of the specifics of the present one, and an account is given in Scholze's notes on six functors [Sch23] in a slightly different setup.

*Remark* 21.7 (Localising on source and target)*.* Property (3) lets one check shriekability after localising on the target, and property (4) after localising (by a shriekable cover) on the source; iterating the two–each new shriekable cover produces further maps to be covered–one reduces to affine situations. Consequently, checking that a given map is shriekable is in practice easy: essentially every morphism is, and the local criteria make it routine to verify. The lower-shriek on $Y$ is then reconstructed from the one on a cover $Y'$: since $f_!$ commutes with base change, defining it after base change to $Y'$ automatically produces the descent datum along all further base changes.

**Example 21.8** (Reconstructing $f_!$ from a cover)**.** Suppose $X$ is stacky, presented by a cover $\widetilde{X} \twoheadrightarrow X$ over $Y$ for which shriek functors are already available. Then

$$D(X) \;=\; \varprojlim_{\Delta}{}^{!}\Big[\, D(\widetilde{X}) \rightrightarrows D(\widetilde{X} \times_X \widetilde{X}) \Rrightarrow \cdots \Big] \;=\; \varinjlim_{\Delta^{\mathrm{op}}}{}^{!}\Big[\, D(\widetilde{X}) \rightrightarrows D(\widetilde{X} \times_X \widetilde{X}) \Rrightarrow \cdots \Big] \quad \text{in } \mathrm{Pr}^L,$$

the first a limit along the upper-shriek transition functors and the second the same diagram read as a colimit along the lower-shriek functors in $\mathrm{Pr}^L$ (Corollary 17.20). As all the $D(\widetilde{X}^{\times_X n})$ carry lower-shriek functors to $D(Y)$ compatible with the transition maps, and $D(Y)$ is presentable, the colimit yields a unique $f_! \colon D(X) \to D(Y)$.

**Example 21.9** (The interval, and compactly supported cohomology)**.** Consider the interval mapping to a point, $[0,1] \to *$, which lives in condensed sets and hence (Example 19.12) in analytic stacks. By Clausen's descendable cover (Section 20) there is a Cantor set surjecting onto $[0,1]$; base-changing $[0,1] \to *$ along it gives a light profinite set over the Cantor set, which is a basic flat cover (a map of light profinite sets, in $E$). Hence

$$\{0,1\}^{\mathbb{N}} \xrightarrow{\;\in \widetilde{E}\;} [0,1] \xrightarrow{\;f\;} *,$$

with the composite $\{0,1\}^{\mathbb{N}} \to *$ (a map of light profinite sets) in $E$, exhibits $f$ as shriekable. Although we only defined shriek functors for maps of light profinite sets, the formalism now produces $f_!$ for the interval, and one observes

$$f_! \;=\; f_*,$$

as one expects since $[0,1] \to *$ is "proper"; this $f_!$ is compactly supported cohomology of $[0,1]$.

**Example 21.10** (The real line)**.** Locally compact Hausdorff spaces are handled the same way. The real line is covered by the disjoint union of the closed intervals,

$$\coprod_n [-n,n] \xrightarrow{\in \widetilde{E}} \mathbb{R},$$

which is surjective (as light condensed sets, $\mathbb{R}$ is this union) and lies in $\widetilde{E}$: covering by disjoint unions of profinite sets exhibits it as a disjoint union of closed subsets, each already in $\widetilde{E}$. The formalism thus yields an $f_!$ for $\mathbb{R} \to *$, and one checks it is the usual lower-shriek. Indeed $D(\mathbb{R})$ is the colimit $\varinjlim_n D([-n,n])$ along the lower-shriek maps, so it suffices to describe $f_!$ on each $[-n,n]$, where it is compactly supported cohomology. This abstract procedure recovers a great deal of the geometric intuition about lower-shriek.

## 21.4 Proper and smooth maps

We isolate two important classes: proper maps (for which $f_! = f_*$) and cohomologically smooth maps (for which $f^!$ is a twist of $f^*$).

**Definition 21.11** (Proper map)**.** A morphism $f\colon X \to Y$ in $\widetilde{E}$, assumed quasicompact and quasiseparated, is *proper* if one of the following (inductive) conditions holds.

1. (Relatively representable in affines and proper.) There is a cover $\coprod_i Y_i' \twoheadrightarrow Y$ by affines such that each $X_i' = X \times_Y Y_i'$ is affine and $X_i' \to Y_i'$ is proper in the earlier sense of Definition 16.12, i.e. $X_i'$ carries the induced analytic ring structure. In this case one obtains a canonical equivalence $f_! \cong f_*$.
2. (Proper diagonal.) The diagonal $\Delta_f\colon X \to X \times_Y X$ is proper (in this same sense, inductively).

*Remark* 21.12 (The comparison map $f_! \to f_*$). Under condition (1) the identification $f_! \cong f_*$ is definitional. In general it is built from a proper diagonal. Consider

$$\begin{tikzcd}
X \arrow[r, "\Delta_f"] \arrow[dr, "f"'] & X \times_Y X \arrow[r, "p_2"] \arrow[d, "p_1"] & X \arrow[d, "f"] \\
& X \arrow[r, "f"'] & Y.
\end{tikzcd}$$

Base change on the right-hand square gives $f^* f_! = p_{2!} p_1^*$, and, inserting the unit $\mathbf{1} \to \Delta_{f*}\Delta_f^*$ of the adjunction $\Delta_f^* \dashv \Delta_{f*}$ (and using $\Delta_{f!} = \Delta_{f*}$, available since $\Delta_f$ is proper),

$$f^* f_! \;=\; p_{2!}p_1^* \;\longrightarrow\; p_{2!}\Delta_{f*}\Delta_f^* p_1^* \;\cong\; p_{2!}\Delta_{f!}\Delta_f^* p_1^* \;=\; \mathrm{id},$$

whose adjoint is a natural transformation $f_! \to f_*$. One *requires* this to be an isomorphism; by a diagram chase (in the six-functor notes [Sch23]) it suffices to check it on the unit, $f_!(\mathbf{1}) \xrightarrow{\sim} f_*(\mathbf{1})$, and then it holds after every base change. The class of proper maps is stable under base change and composition. Scholze noted the definition should be restricted to quasicompact, quasiseparated maps: all maps in the two conditions are automatically QCQS (the basic ones are, and diagonals are), whereas passing to arbitrary disjoint unions $\coprod_i (X_i \to Y_i)$ with the senses tending to infinity would fail.

*Remark* 21.13 ($f^!$ of the unit; a caveat for schemes). Asked for an explicit description of $f^!(\mathbf{1})$ for $f\colon X \to *$: in the examples above (interval, real line) $f_!$ is the usual lower-shriek, so $f^!$ is the usual dualising complex, and in essentially any situation where a dualising complex already exists, $f^!$ agrees with it. One caveat: if a scheme is embedded into analytic stacks with its *trivial* analytic ring structure (Example 19.6), then every map is proper, $f_! = f_*$, and $f^!$ is the poorly behaved right adjoint of $f_*$–not the Grothendieck-duality shriek. To recover the usual dualising complex of a finite-type scheme one should instead use the relatively-solid embedding (Remark 19.8).

**Definition 21.14** (Cohomologically smooth map)**.** A morphism $f\colon X \to Y$ in $\widetilde{E}$ is *cohomologically smooth* if, after any base change, the natural map

$$f^*(-) \otimes f^!(\mathbf{1}) \xrightarrow{\sim} f^!(-) \tag{21.1}$$

is an isomorphism, $f^!(\mathbf{1}) \in D(X)$ is $\otimes$-invertible, and its formation commutes with base change.

The map (21.1) is adjoint to the projection-formula counit $f_!\big(f^*(-) \otimes f^!(\mathbf{1})\big) = (-) \otimes f_! f^!(\mathbf{1}) \to (-)$. Unlike properness, smoothness is not stable under passing to diagonals, and one does not expect $f^! = f^*$ on the nose–only up to the invertible twist $f^!(\mathbf{1})$, the relative dualising object. Cohomologically smooth maps are stable under base change and composition, and cohomological smoothness is local on the source (a cohomologically smooth cover of a cohomologically smooth map descends).

## 21.5 The Tate elliptic curve, revisited

Return to the gaseous base $A = \mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}$, now equipped with the norm of Section 21.1. The analytic multiplicative group is the preimage of the open interval,

$$\mathbb{G}_{m}{}^{\mathrm{an}}_{A} \;=\; N^{-1}\big((0,\infty)\big) \;\subseteq\; \mathbb{P}^1_A,$$

and the open inclusion $j\colon \mathbb{G}_{m}{}^{\mathrm{an}}_{A} \hookrightarrow \mathbb{P}^1_A$ is shriekable; it is in fact an open immersion, with $j^! = j^*$ and $j^* j_! = \mathrm{id}$, and it is cohomologically smooth. (Everything here reduces, by localising at 0 and $\infty$, to the usual six operations on locally compact Hausdorff spaces.) The projection $\pi\colon \mathbb{P}^1_A \to \mathrm{AnSpec}(A)$ is proper: it is not relatively representable in affines, but its diagonal is (it is the diagonal of $T_1$ inside $\mathbb{P}^1$, an affine situation), and one checks $\pi_!$ and $\pi_*$ of the unit agree, exactly as in algebraic geometry. Moreover $\pi$ is cohomologically smooth–since a morphism of six-functor formalisms (here schemes into analytic stacks) preserves cohomologically smooth maps, the algebraic smoothness of $\mathbb{P}^1$ implies the analytic one.

The action of $q$ by $T \mapsto qT$ is free and totally discontinuous, and one forms the quotient in analytic stacks. Rescaling so that $|q| = \frac{1}{2}$, the norm descends to $(0,\infty)/(\frac{1}{2})^{\mathbb{Z}} \cong S^1$, giving

$$E_{q,A} = \mathbb{G}_{m}{}^{\mathrm{an}}_{A}/q^{\mathbb{Z}} \xrightarrow{\text{proper}} (0,\infty)/(\tfrac{1}{2})^{\mathbb{Z}} = S^1 \xrightarrow{\text{proper}} \mathrm{AnSpec}(A).$$

The first map is proper (checked by base change over $(0,\infty)$) and the second is proper, so $E_{q,A} \to \mathrm{AnSpec}(A)$ is proper. The quotient map $\mathbb{G}_{m}{}^{\mathrm{an}}_{A} \to E_{q,A}$ is locally split, so $E_{q,A}$ is locally isomorphic to $\mathbb{G}_{m}{}^{\mathrm{an}}_{A}$ and hence cohomologically smooth. Thus $E_{q,A}$ is a proper smooth curve, the *Tate elliptic curve*, carrying the unit section $1 \in \mathbb{G}_m^{\mathrm{an}}$.

*Remark* 21.15 (Open immersions are not affine-representable). Asked whether open immersions of analytic stacks are relatively representable in affines: they are not. Were $j$ representable in affines it would in particular be quasicompact, whereas an open immersion such as ${\mathbb{G}_m}_A^{\mathrm{an}} \hookrightarrow \mathbb{P}^1_A$ need not be. The open immersions of the present formalism are far more general than the classical ones.

## 21.6 Algebraisation

It remains to see that the Tate curve is an honest (algebraic) elliptic curve.

**Theorem 21.16** ($E_q$ is algebraic). *The Tate elliptic curve $E_{q,A} \in \mathrm{AnStk}_A$ lies in the essential image of the fully faithful functor $\mathrm{Sch}_{A^\triangleright(*)} \to \mathrm{AnStk}_A$ from schemes over the underlying discrete ring; i.e. it is the analytification of an elliptic curve over $A^\triangleright(*)$.*

This is an instance of a general algebraisation principle, whose proof is the classical construction of Proj of a section ring, transported to the analytic-stack setting.

**Proposition 21.17** (General algebraisation setup). *Let $A \in \mathrm{AnRing}$ and let $X \in \mathrm{AnStk}_{/A}$ be proper. Suppose $X$ carries a line bundle $L$ such that, after replacing $L$ by a tensor power, one has:*

1. *$L$ is globally generated–there is a surjection $\mathcal{O}_X^{\oplus N} \twoheadrightarrow L$ (equivalently, a surjection on $H^0$, no higher cohomology intervening);*
2. *$H^i(X, L^{\otimes n}) = 0$ for all $i > 0$ and $n > 0$;*
3. *the global sections are discrete over $A^\triangleright(*)$: $R\Gamma(X, L^{\otimes n}) \in D_{\mathrm{disc}}(A^\triangleright(*)) \subseteq D(A)$.*

*Then*

$$X^{\mathrm{alg}} \;:=\; \mathrm{Proj} \bigoplus_{n \geq 0} R\Gamma(X, L^{\otimes n}) \;\in\; \mathrm{DerSch}_{A^\triangleright(*)}$$

*is a quasicompact quasiseparated proper scheme over $A^\triangleright(*)$, with a proper map $f\colon X \to X^{\mathrm{alg}}_A$. If $L$ is ample enough that $f$ is an isomorphism, then $X \xrightarrow{\sim} X^{\mathrm{alg}}_A$ is the analytification of $X^{\mathrm{alg}}$.*

*Remark* 21.18 ($f_*\mathcal{O} = \mathcal{O}$). The crucial observation is that $f_*(\mathbf{1}) = \mathbf{1}$, i.e. $f_*\mathcal{O}_X = \mathcal{O}_{X^{\mathrm{alg}}_A}$. This may be checked after pullback to an affine chart $U^{\mathrm{alg}}_A = D^+_1(\xi) \subseteq X^{\mathrm{alg}}_A$ (the non-vanishing locus of a section $\xi \in H^0(X, L^{\otimes n})$), and there on global sections. The relevant global sections are a filtered colimit along multiplication by $\xi$,

$$\varinjlim_{k,\, \times\xi} R\Gamma\big(X, L^{\otimes kn}\big) \;=\; \varinjlim_{k,\, \times\xi} R\Gamma\big(X^{\mathrm{alg}}, L^{\otimes kn}\big),$$

i.e. the localisation inverting $\xi$; since $X^{\mathrm{alg}}$ was built from exactly these graded pieces, the two sides agree, giving $f_*\mathcal{O} = \mathcal{O}$.

*Remark* 21.19 (From $f_*\mathcal{O} = \mathcal{O}$ to an isomorphism). Because $f$ is proper and $f_*(\mathbf{1}) = \mathbf{1}$, the map $f$ is surjective: $f_*(\mathbf{1}) = \mathbf{1}$ is preserved by every base change (proper base change), so after pullback to any affine the target ring embeds into the pushed-forward sections, forcing $f$ surjective. To upgrade surjectivity to an isomorphism it suffices that the diagonal $\Delta_f$ be an isomorphism (a surjection of sheaves with iso diagonal is iso). The identity $f_*(\mathbf{1}) = \mathbf{1}$ also shows, via the projection formula, that the pullback $f^*\colon D(X^{\mathrm{alg}}_A) \hookrightarrow D(X)$ is fully faithful (the unit of $f^* \dashv f_*$ is an equivalence, $f_*f^*(-) = (-) \otimes f_*\mathbf{1} = (-)$), and likewise after any base change. Hence it is enough to show that the structure sheaf of the diagonal $\Delta_X$–a coherent sheaf on $X^{\mathrm{alg}}_A \times X^{\mathrm{alg}}_A$, supported on a Zariski-closed immersion–lies in the essential image of $f^* \times f^*$; then it is pulled back from $X^{\mathrm{alg}} \times X^{\mathrm{alg}}$ and $\Delta_f$ is an isomorphism.

*Remark* 21.20 (The case of $E_q$). For the Tate curve one takes for $L$ the ample line bundle attached to the zero (unit) section. All global sections $R\Gamma(E_q, L^{\otimes n})$ are then computed inductively–$R\Gamma(E_q, \mathcal{O})$ directly, and each further power adds copies of the base ring from the zero section–and the required diagonal condition is verified using the group structure: the diagonal is translated onto the zero section, and the zero section is the vanishing locus one already controls. This produces the ample bundle and the isomorphism $E_{q,A} \xrightarrow{\sim} E_q^{\mathrm{alg}}$, proving Theorem 21.16.

*Remark* 21.21 (Properness here is not Grothendieck's). In the closing discussion Scholze stressed that the notion of proper used above (Definition 21.11) is not literally Grothendieck's finite-type + separated + universally closed. In particular, for schemes embedded into analytic stacks *every* map is proper in this sense; so the section ring $\bigoplus_n R\Gamma(X, L^{\otimes n})$ may be far from finitely generated (its degree-zero part alone can be an arbitrary ring). When $X$ genuinely comes from a more classical space–for instance a complex-analytic space, which embeds into analytic stacks–the analytic notion of properness specialises to the usual one (it implies quasicompactness), and algebraisation can succeed only when $X$ is proper in the universally-closed sense. The GAGA-type isomorphisms in this world hold for these "funny" analytic stacks, of which schemes and analytic spaces are full subcategories.

# 22 Toward Berkovich Spaces: Localization over Topological Spaces and the Stack of Norms

This is the penultimate lecture of the course. Clausen returns to the programme announced in Section 1: to see that the traditional frameworks of analytic geometry sit inside the theory of analytic stacks. Having already treated adic spaces (Section 10) and objects like the Tate curve (Sections 19 and 21), he now moves toward *Berkovich spaces.* The lecture has two halves. First a piece of setup: several inequivalent notions of how to *localize* an analytic stack over a topological space, phrased through locales and the six-functor formalism, and how the notion used earlier (for metrizable finite-dimensional compact Hausdorff spaces) sits among them. Second, the Berkovich spectrum $\mathcal{M}^{\mathrm{Berk}}(A)$ of a Banach ring, and its reincarnation as an analytic stack $\mathcal{N}$ of *norms*; the bulk of the time is spent computing the "image" of $\mathcal{N}$ over $\mathbb{Z}$, which turns out to be an *extended* Berkovich space of $\mathbb{Z}$, with the archimedean branch prolonged past the usual absolute value to a compactifying point.

Throughout, "ring" means commutative ring and everything is (light) condensed.

*Remark* 22.1 (Derived Huber pairs). Asked about the sheafiness defect of adic spaces (Section 10) and whether one must pass to some notion of *derived* Huber pair / derived adic space, Clausen said this is still in progress and possibly admits several formulations; a definition of derived Huber pairs appears in Rodríguez Camargo's recent paper on the analytic de Rham stack in rigid geometry [Cam24].

## 22.1 Localizing an analytic stack over a topological space

We have already seen (Theorem 20.3) one way to localize an analytic stack over a space. Let $S$ be a metrizable, finite-dimensional compact Hausdorff space – or a locally compact Hausdorff space locally of this form – viewed (Example 19.12) as an analytic stack. Given $X \in \mathrm{AnStk}$ one may simply ask for a map

$$f\colon X \longrightarrow S \qquad \text{in AnStk}\,. \tag{22.1}$$

As discussed, this equips $D(X)$ with a family of idempotent algebras indexed by the closed subsets $Z \subseteq S$ (the $i_{Z*}$ of constant sheaves) – or, dually, idempotent coalgebras $j_! \mathbf{1}_U$ indexed by the open complements – and these algebra objects localize $D(X)$ over $S$. For the comparison with Berkovich spaces the metrizability/finite-dimensionality hypotheses, though satisfied in practice, are an annoyance, so we introduce weaker notions of localization along a topological space and relate them to (22.1).

To any analytic stack $X$ one attaches its underlying *locale*

$$\mathrm{Loc}(X) \;:=\; \{\, Y \subseteq X \,\} \;=\; \{\text{monomorphisms into } X \text{ in } \mathrm{AnStk}\},$$

the poset of subobjects of $X$. This is the general construction of the underlying locale of an ($\infty$-)topos, retaining only the monomorphisms; the topos axioms guarantee the requisite unions and finite intersections (the category of analytic stacks has all limits and colimits, so one forms unions as images of coproducts). A monomorphism means, concretely, that the diagonal $Y \to Y \times_X Y$ is an isomorphism.

*Remark* 22.2 (A possible size issue that does not matter)*.* It is not clear (Clausen did not investigate) whether $\mathrm{Loc}(X)$ is a set; AnStk is not presentable, only closed under the same exactness properties as an $\infty$-topos. This is harmless for what follows: in the definitions below one maps *into* $\mathrm{Loc}(X)$ from an honest locale $S$, and, $S$ and $X$ being fixed, the resulting data is a set – there are no nontrivial automorphisms of any one of these maps, so no anima appear.

Now fix a locale $S$ (for a topological space, its poset of opens; we abuse $U \subseteq S$ to mean $U$ an element of this poset). There are three increasingly strong notions of localizing $X$ over $S$.

1. A map of locales (geometric morphism)

$$\pi\colon \mathrm{Loc}(X) \longrightarrow S, \tag{22.2}$$

   i.e. for every open $U \subseteq S$ a monomorphism $\pi^{-1}U \subseteq X$, sending unions to unions and finite intersections to finite intersections.

2. The stronger requirement that each inclusion of opens $U \subseteq V \subseteq S$ be sent to an *open immersion*

$$\pi^{-1}U \;\subseteq\; \pi^{-1}V \;\subseteq\; X$$

   from the perspective of the (extended) six-functor formalism of Section 21 – that is, a shriekable, cohomologically smooth monomorphism. (For a monomorphism the relative dualizing object is canonically the unit, so cohomological smoothness is exactly the condition of being an open immersion.) Equivalently, one factors (22.2) through the quotient locale $\mathrm{Loc}^{\varphi}(X)$ of those monomorphisms which are open immersions,

$$\mathrm{Loc}(X) \longrightarrow \mathrm{Loc}^{\varphi}(X) \longrightarrow S. \tag{22.3}$$

3. If $S$ is metrizable, finite-dimensional and compact Hausdorff, one may ask for an actual map $f\colon X \to S$ in AnStk as in (22.1).

**Proposition 22.3.** (3) $\Rightarrow$ (2) $\Rightarrow$ (1).

*Indication.* (2) $\Rightarrow$ (1) is tautological. For (3) $\Rightarrow$ (2) one checks on the analytic stack attached to $S$ that every inclusion of opens is an open immersion in the six-functor sense: this is because our six-functor formalism, restricted to (locally) compact Hausdorff spaces, recovers the usual one (Section 21), and the property of being an open immersion is stable under base change. (Upgrading (1) to (2) is just the check that certain inclusions are open immersions; upgrading (2) to the strongest notion (3), on the other hand, imposes a genuine *connectivity* condition on the idempotent algebras, as before.) $\square$

*Remark* 22.4 (Unions of open immersions). For $\mathrm{Loc}^{\varphi}(X)$ in (22.3) to be a locale one needs a union of open immersions to be an open immersion. This holds: a union along open immersions is a cover by shriekable maps, so it acquires a lower-shriek functor, and cohomological smoothness may be checked locally. It follows from the extension procedure for six-functor formalisms of Section 21.

**Example 22.5** (Huber pairs and the adic spectrum). Let $(R, R^+)$ be a Huber pair, with associated solid analytic ring $(R, R^+)^{\square}$ (Definition 9.5), and set $X = \mathrm{AnSpec}((R, R^+)^{\square})$. Then $\mathrm{Loc}(X)$ localizes over Huber's topological space $\mathrm{Spa}(R, R^+)$ of continuous valuations: for a rational open $U = U(f_1, \ldots, f_n/g)$ the preimage is again an affine analytic stack, the one enforcing $g$ invertible and $f_1/g, \ldots, f_n/g$ power-bounded (solid),

$$\pi^{-1}U \;=\; \mathrm{AnSpec}\big(R\langle f_1, \ldots, f_n/g\rangle^{\square}\big),$$

which is a monomorphism (on derived categories, a localization). This refines Theorem 9.18: the proof given there in fact showed the stronger statement that any cover of a rational open in $\mathrm{Spa}(R, R^+)$ pulls back under $\pi$ to an open cover in the six-functor sense.

*Remark* 22.6 (Rational opens need not be open immersions). In general it is *not* true that a rational open pulls back to an open immersion. Consider

$$(\mathbb{Z}_p, \mathbb{Z}_p)^{\square} \longrightarrow (\mathbb{Q}_p, \mathbb{Z}_p)^{\square},$$

i.e. inverting $p$ while keeping the ring of integral elements. On derived categories,

$$D\big((\mathbb{Z}_p, \mathbb{Z}_p)^{\square}\big) \longrightarrow D\big((\mathbb{Q}_p, \mathbb{Z}_p)^{\square}\big)$$

is the inclusion of the full subcategory on which $p$ acts invertibly (the map $p\colon X \xrightarrow{\sim} X$). From the six-functor perspective this is a *closed* immersion – a proper monomorphism – not an open one: only the underlying ring changes, the analytic ring structure is unchanged.

*Remark* 22.7 (The Tate case, and a modified topology). If $R$ is Tate, this pathology does not arise: inverting a function is a rising union of rational subsets cut out by inequalities,

$$\{f \neq 0\} \;=\; \bigcup_{\varepsilon} \{\, |f| \geq \varepsilon \,\},$$

each of which *is* open. (This subset is not quasicompact, though it is a quasicompact rational domain in the adic space – the discrepancy is that the map to $\mathrm{Spa}(A, A^+)$ is not literally the naive one, being built via Spa of the rational localizations and Huber's retraction.) There is, in addition, another topology on $\mathrm{Spa}(R, R^+)$ with the same constructible subsets – incomparable to Huber's, a spectral space, the opposite of one of Huber's topologies – in which Zariski-open subsets are re-declared closed, so that open pulls back to open even outside the Tate case. This modification is discussed in the "GAGA redux" chapter of Clausen–Scholze's *Lectures on Analytic Geometry* [CS20].

*Remark* 22.8 (Passing to inverse limits). If $(S_i)_{i\in I}$ is an inverse system of compact Hausdorff spaces, then to give compatible maps of locales $\mathrm{Loc}(X) \to S_i$ for all $i$ is the same as to give a single map

$$\mathrm{Loc}(X) \longrightarrow \varprojlim_i S_i,$$

since inverse limits of compact Hausdorff spaces agree with the corresponding limits of locales here; the same holds for $\mathrm{Loc}^{\varphi}$. Combined with the fact that *every* compact Hausdorff space is an inverse limit of metrizable finite-dimensional ones – potentially in many different ways, but often with a natural choice, as will occur for Berkovich spaces – this lets one bootstrap from the metrizable finite-dimensional case (where (3) is available) to general compact Hausdorff spaces, at the cost of finite-dimensionality. Asking for the strongest notion, an actual map $X \to \varprojlim_i S_i$ of analytic stacks, still requires the connectivity condition ($\mathrm{Loc}^{\varphi}$-locally all idempotent algebras connective) and is not automatic; Clausen expects it to be genuinely stronger, though produced no counterexample.

## 22.2 Berkovich spectra: recollection

We recall the classical definition to fix the target.

**Definition 22.9** (Banach ring)**.** A *Banach ring* $(A, |\cdot|)$ is a commutative ring $A$ together with a map $|\cdot|\colon A \to \mathbb{R}_{\geq 0}$ satisfying

$$(1)\ |0| = 0, \qquad (2)\ |x+y| \leq |x| + |y|, \qquad (3)\ |xy| \leq |x|\,|y|, \qquad (4)\ A = 0 \text{ or } |1| = 1,$$

together with $|-1| = |1|$ (equivalently the triangle inequality also for $x - y$), and such that $A$ is complete with respect to the metric $d(x,y) = |x-y|$ (completeness in particular forces $|f| = 0 \iff f = 0$).

**Example 22.10.**

1. For $S$ a compact Hausdorff space, $A = C(S;\mathbb{C})$ with the sup-norm
$$|f| \;=\; \sup_{s\in S} |f(s)|_{\mathrm{usual}}.$$
2. For $R$ a complete Tate–Huber ring with pseudo-uniformizer (topologically nilpotent unit) $\pi \in R$, a norm $|\cdot|\colon R \to \mathbb{R}_{\geq 0}$ with $|\pi| = \frac{1}{2}$, with $|R_0| \leq 1$ for a ring of definition $R_0$, and multiplicative in $\pi$:
$$|\pi f| \;=\; |\pi|\,|f| \qquad (f \in R).$$
(It is multiplicative when one factor is $\pi$, not in general.) E.g. $\mathbb{Q}_p$ with $\pi = p$, or $\overline{\mathbb{Q}}_p$ with $\pi = p$.

**Definition 22.11** (Berkovich spectrum)**.** To a Banach ring $(A, |\cdot|)$ Berkovich attaches

$$\mathcal{M}^{\mathrm{Berk}}(A, |\cdot|) \;=\; \big\{\, |\cdot|_x\colon A \to \mathbb{R}_{\geq 0} \text{ a multiplicative seminorm with } |f|_x \leq |f|\ \forall f \in A \,\big\}, \quad (22.4)$$

the set of multiplicative seminorms bounded by the given norm; "multiplicative" includes $|1|_x = 1$ (so the point is nonzero) and $|0|_x = 0$. It is topologized as a closed subset of a product of intervals,

$$\mathcal{M}^{\mathrm{Berk}}(A, |\cdot|) \;\hookrightarrow\; \prod_{f\in A} [0, |f|], \qquad |\cdot|_x \;\longmapsto\; \big(|f|_x\big)_{f\in A},$$

and is compact Hausdorff. For $A = 0$ it is empty.

**Example 22.12.** For $A = C(S,\mathbb{C})$ the seminorms $|f|_x = |f(x)|$, $x \in S$, are points of $\mathcal{M}^{\mathrm{Berk}}(A)$; if one moreover demands that $|\cdot|_x$ restrict to the usual norm on $\mathbb{C}$, these are *all* the points (Gelfand's theorem), recovering $S$.

## 22.3 The analytic stack of norms

The goal is to make the "same definition" as (22.4) – the set of multiplicative seminorms bounded by $|\cdot|$ – but inside analytic stacks rather than topological spaces, using the notion of a *norm* on an analytic ring from Section 20 (Definition 20.9).

**Definition 22.13** (Norm; recollection)**.** For $X \in \mathrm{AnStk}$, a *norm* on $X$ is a map of analytic stacks

$$N\colon\ \mathbb{P}^1_X \ \longrightarrow\ [0,\infty]$$

which is multiplicative on $N^{-1}\big((0,\infty)\big)$, satisfies $N(T^{-1}) = N(T)^{-1}$ (compatibility of inversion on $\mathbb{P}^1$ with $\lambda \mapsto \lambda^{-1}$ on $[0,\infty]$), and satisfies the condition on how the basic disk $P$ sits: $P$ lies between the open unit disk $N^{-1}\big([0,1)\big)$ and the closed unit disk $N^{-1}\big([0,1]\big)$ (Definition 20.9, Axiom 4). Here $T$ is the coordinate on $\mathbb{P}^1$.

*Notation* 22.14 (The stack of norms). Write $\mathcal{N}(X)$ for the set of norms on $X$. This is genuinely a set, as it is a set of maps in the locally small category AnStk. Norms satisfy descent, so $X \mapsto \mathcal{N}(X)$ is represented by an analytic stack, again denoted $\mathcal{N}$: thus $\mathcal{N}(X) = \mathrm{Hom}_{\mathrm{AnStk}}(X, \mathcal{N})$.

*Remark* 22.15 (The gaseous base is a cover). There is a cover, in the Grothendieck topology on analytic stacks,

$$\mathrm{AnSpec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}\big) \ \twoheadrightarrow\ \mathcal{N}, \tag{22.5}$$

classifying the universal norm on an analytic ring $(R, D(R))$ equipped with an element $q \in R^{\triangleright}$ whose norm is exactly $\frac{1}{2}$: a priori $N(q)$ is a map $\mathrm{Spec}(R) \to [0,\infty]$ with image possibly an interval, and one demands that it factor through the singleton $\{\frac{1}{2}\}$, a stringent condition whose universal solution is the gaseous base ring $\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}$ of Sections 14 and 20. The map (22.5) is a cover because, given any norm, one may locally find such an element $q$ (Lemma 20.17).

## 22.4 The image of $\mathcal{N}$: an extended Berkovich space of $\mathbb{Z}$

We now explore what $\mathcal{N}$ looks like. Over an arbitrary analytic ring the only ring elements available are the integers, so for $n \in \mathbb{Z}$ we get $N(n)\colon \mathrm{Spec}(R) \to [0,\infty]$, and hence a map of locales

$$\mathrm{Loc}(\mathcal{N}) \ \xrightarrow{\ \pi\ }\ \prod_{n\in\mathbb{Z}} [0,\infty].$$

Thus $\mathcal{N}$ localizes over this (large, non-finite-dimensional) product.

**Definition 22.16** (Image)**.** The *image* of $\mathcal{N}$ is the complement of the largest open subset $U \subseteq \prod_{n\in\mathbb{Z}}[0,\infty]$ with $\pi^{-1}U = \emptyset$.

We will see that the fibre over any point of the image is non-empty, so at least in this case the image is closed. Recall first the classical picture.

*Remark* 22.17 (The Berkovich space of $\mathbb{Z}$). The Berkovich spectrum $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$ (all points avoiding $\infty$) embeds in $\prod_{n\in\mathbb{Z}}[0,\infty)$ as the multiplicative norms $|\cdot|\colon \mathbb{Z} \to [0,\infty)$ with $|1| = 1$, $|0| = 0$ and $|x+y| \le |x| + |y|$. As a topological space it is a tree (Figure 22.1): a central point, the trivial norm $|\cdot|_0$ (value 1 on everything nonzero); an archimedean branch ending at the usual absolute value $|\cdot|_\infty$, with $|\cdot|_\infty^{1/2}$ at its midpoint; and, for each prime $p$, a branch of the powers $|\cdot|_p^\alpha$, $\alpha \in (0,\infty)$, whose far end ($\alpha \to \infty$) is the pullback of the trivial norm on $\mathbb{F}_p$ (value 0 on multiples of $p$, 1 elsewhere), the usual $p$-adic norm ($|p| = 1/p$) sitting partway along. It is a small, one-dimensional subspace of the huge product.

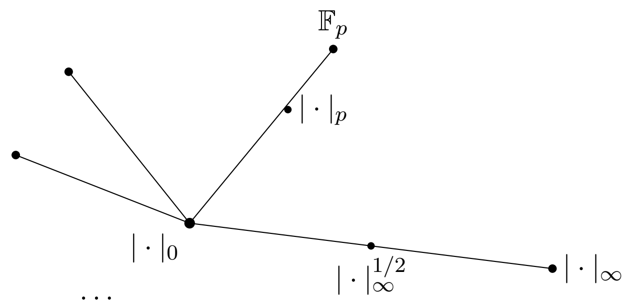



Figure 22.1: The Berkovich tree $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$: trivial norm at the centre, one branch per prime ending at $\mathbb{F}_p$, and an archimedean branch ending at the usual absolute value.

**Theorem 22.18** (The image is an extended Berkovich space)**.** *The image of $\mathcal{N}$ over $\mathbb{Z}$ is a larger subset than $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$: it agrees with the tree on every non-archimedean branch, but the archimedean branch is prolonged past $|\cdot|_\infty$ all the way to $\infty$ and then one-point compactified. In symbols*

$$\mathrm{Im}(\mathcal{N}) \;=\; \Big(\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})^{\mathbb{R}_{\geq 1}} \times \mathbb{R}_{>0}\Big)^{+}, \tag{22.6}$$

*the one-point compactification of $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$ with the exponent-rescaling action of $\mathbb{R}_{\geq 1}$ extended to $\mathbb{R}_{>0}$; the compactifying "strange point" is the norm with*

$$N(0) = 0, \quad N(\pm 1) = 1, \quad N(n) = \infty \;\; \textit{for } n \neq 0, \pm 1.$$

*Remark* 22.19. In particular the triangle inequality can fail on $\mathrm{Im}(\mathcal{N})$: at the strange point $N(1) = 1$ but $N(2) = \infty$; and already $|\cdot|_\infty^\alpha$ for $\alpha > 1$, a genuine new point of the prolonged archimedean branch, violates it. In (22.6) the superscript $\mathbb{R}_{\geq 1}$ is Clausen's shorthand for the exponent-rescaling action $N \mapsto N^\alpha$ by $\alpha \geq 1$ applied to $\mathcal{M}^{\mathrm{Berk}}(\mathbb{Z})$: this action does nothing on the non-archimedean branches (each is already a full ray) but extends the archimedean branch beyond $|\cdot|_\infty$, after which the $(-)^+$ one-point compactification adds the single point at infinity.

To justify Theorem 22.18 we analyse the branches one at a time via the function $N(p)$ for a fixed prime $p$.

**Proposition 22.20** (Interior of the $p$-adic branch)**.** *For the map $N(p)\colon \mathcal{N} \to [0, \infty]$,*

$$\mathcal{N}_{0<|p|<1} \;:=\; N(p)^{-1}\big((0,1)\big) \;=\; \mathrm{Spec}\big(\mathbb{Q}_p^{\mathrm{gas}}\big) \times (0,1).$$

*For a fixed $\lambda \in (0,1)$ the fibre $\mathrm{Spec}(\mathbb{Q}_p^{\mathrm{gas}}) \times \{\lambda\}$ carries a normed analytic ring structure whose notion of overconvergent functions on the disk of radius $r$ is the usual one from non-archimedean geometry, for the normalization of the $p$-adic absolute value with $|p| = \lambda$.*

*Indication.* The universal analytic stack equipped with a variable of norm in $(0,1)$ is $\mathrm{Spec}(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}) \times (0,1)$ (Remark 22.15). Imposing that the variable become $p$ – modding out by $q - p$ – yields, as in Section 14, the gaseous analytic ring structure on $\mathbb{Q}_p$; the second factor $(0,1)$ is unchanged. □

**Proposition 22.21** (Interior of the archimedean branch)**.** *The locus where $|p|$ is strictly between 1 and $\infty$ is*

$$\mathcal{N}_{1<|p|<\infty} \;=\; \mathrm{Spec}\big(\mathbb{R}^{\mathrm{gas}}\big) \times (0,1),$$

*and it is independent of $p$. For fixed $\lambda \in (0,1)$ the universal norm on* $\mathrm{Spec}(\mathbb{R}^{\mathrm{gas}}) \times \{\lambda\}$ *is given by the usual overconvergent holomorphic functions of complex geometry, with respect to the power $|\cdot|_\infty^\alpha$ of the usual absolute value where $\alpha$ is chosen so that*

$$|\tfrac{1}{p}|_\infty^\alpha \;=\; \lambda.$$

*Indication.* That $1 < |p| < \infty$ means $p$ is invertible and $0 < |1/p| < 1$; applying the argument of Proposition 22.20 to the variable $1/p$ gives $\mathrm{Spec}(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}) \times (0,1)$, and setting $q = 1/p$ produces the gaseous analytic ring structure on $\mathbb{R}$ (Section 14). Independence of $p$ is visible from the description, which never referred to $p$; this is the analytic-stacks incarnation of Ostrowski's classification – once $|p|$ is pinned in this range the norm of every other integer is determined. □

*Remark* 22.22. The subtlety of exactly which summability condition defines $\mathbb{R}^{\mathrm{gas}}$ (some quasi-exponential decay, Remark 14.13) – a space sitting strictly between the overconvergent functions on the closed unit disk and the holomorphic functions on the open disk – washes out once one passes to *over*convergent functions on a disk of a given radius: the answer is just the usual ring of overconvergent holomorphic functions, up to a scaling of the absolute value coming from the chosen value of $N(q)$.

### Triangle inequalities

The image computation rests on the following universal claims, in which 2 may be replaced by any integer $> 1$.

**Proposition 22.23** (Universal triangle inequalities)**.** *Let $(R, N)$ be a normed analytic ring.*

1. *If $N(2) \leq 1$, then $N$ satisfies the non-archimedean triangle inequality:*

$$\mathcal{N}_{|2|\leq 1} \;\subseteq\; \mathcal{M}^{\mathrm{Berkna}}(\mathbb{Z}). \tag{22.7}$$

2. *If $N(2) \leq 2$, then $N$ satisfies the usual triangle inequality.*
3. *If $N(2) < \infty$, then there is a constant $C > 0$ with*

$$|x + y| \;\leq\; C\,\big(|x| + |y|\big). \tag{22.8}$$

These are statements about maps of analytic stacks, not about literal real-valued norm functions. For instance (22.8) is the assertion that for $A, B < \infty$ the addition map, which carries the product of disks $\mathbb{P}^1_{|T|\leq A} \times \mathbb{P}^1_{|S|\leq B}$ into the affine line $\mathbb{A}^1_R \subseteq \mathbb{P}^1_R$ (Lemma 20.12), fits into a commuting square

$$\begin{tikzcd}
\mathbb{P}^1_{|T|\leq A} \times \mathbb{P}^1_{|S|\leq B} \arrow[r] \arrow[d, "(T,S)\mapsto T+S"'] & {[0,\, C(A+B)]} \arrow[d, hook] \\
\mathbb{P}^1_R \arrow[r, "N"'] & {[0, \infty].}
\end{tikzcd}$$

*Remark* 22.24 (Proof of the triangle claims)*.* By the cover (22.5) one may assume there is $q$ with $0 < N(q) < 1$, so that one works over the gaseous base $\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}$. In the universal case, the condition

$N(2) \leq 1$ makes one live over an idempotent algebra in $D(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\text{gas}})$, namely the overconvergent closed unit disk modded by $T = 2$,

$$\mathcal{O}_{\mathbb{Z}[\widehat{q}^{\pm 1}]_{\text{gas}}}\big(\{|T| \leq 1\}\big)/(T-2).$$

A calculation identifies this, on underlying rings, with

$$\mathbb{Z}(\!(q)\!) \ \cap \ \mathcal{O}_{q,0}^{\text{hol}}$$

– Laurent series with integer coefficients that are meromorphic at the origin, i.e. converge on some unspecified open disk around 0. This description is manifestly independent of 2, and one checks the non-archimedean triangle inequality by hand (it amounts to the existence of a map of idempotent algebras lying over the addition map on the polynomial ring). The claim $N(2) \leq 2$ is proved similarly, setting $T^{-1} = \frac{1}{2}$ instead: one gets a version of functions converging on the open disk in which the Laurent tails force convergence out to the boundary; it in turn implies the $N(2) < \infty$ claim.

*Remark* 22.25 (Proof of Theorem 22.18)*.* The image is computed by stratifying along $N(2)\colon \mathcal{N} \to [0,\infty]$ into $\{|2| \leq 1\}$, $\{1 < |2| < \infty\}$ and $\{|2| = \infty\}$: a closed stratum and its open complement need not form a cover, but they still detect isomorphisms – in particular emptiness – so it suffices to work over each stratum and take the union of the images. Over $\{|2| \leq 1\}$, (22.7) places us in the non-archimedean Berkovich space; over $\{1 < |2| < \infty\}$ we are in the (already-classified) archimedean locus $\operatorname{Spec}(\mathbb{R}^{\text{gas}}) \times (0,1)$; over $\{|2| = \infty\}$ one needs $N(n) = \infty$ for all $n \neq 0, \pm 1$, which follows because 2 was arbitrary (if some other $n$ were finite, one would land in a lower stratum). The reverse containment – that each point is realized by an actual normed analytic ring – holds because the interiors were realized above, the central trivial norm by non-archimedean geometry over a Laurent series ring $\mathbb{Z}(\!(q)\!)$ with $q$ of trivial norm, and the end-points by non-archimedean geometry over $\mathbb{F}_p$.

## 22.5 The stack of norms as a stack; rescaling

*Remark* 22.26 (Only Tate rings have norms; rescaling)*.* The assignment $R \mapsto \{\text{norms on } R\}$ is itself an analytic stack $\mathcal{N}$, with the cover (22.5). Only Tate (analytic) rings admit nontrivial norms; a discrete ring has no nontrivial theory of norms. For a Tate–Huber ring a norm amounts to a choice of $|q| \in (0,1)$ for a pseudo-uniformizer, and the space of norms is a torsor under the rescaling $N \mapsto N^\alpha$ of a fixed one – matching the rescaling on the Berkovich spectrum, though for a global object it is a whole compatible family rather than a single seminorm. Over $\mathbb{Z}_\square$ the reals vanish, so no norm with $N(q) \in (0,1)$ exists there and $\mathcal{N}$ is a genuinely different kind of base. Modding out by these exponential rescalings gives a quotient stack $\mathcal{N}/\mathbb{R}_{>0}$, to which any Tate adic space admits a unique map. This is to be taken up next time.

*Remark* 22.27 (The fibres are genuinely stacky at the ends)*.* On the interior of a branch the fibre feels like a space – the unit interval base-changed to a gaseous $\mathbb{Q}_p$ or $\mathbb{R}$ (Propositions 22.20 and 22.21). But toward the outer end-points (and the central point) it is genuinely a stack: over the boundary point of a $p$-branch the fibre carries e.g. $\mathbb{F}_p(\!(q)\!)$ with its notion of overconvergent functions on a disk, and rescaling the norm changes *which* disk of a given radius one is labelling – so the fibre is not affine, and the $\mathbb{R}_{>0}$-action, though split as a set, carries real geometric content.

*Remark* 22.28 (Administrative)*.* There is no lecture on the Friday; Scholze and others give related talks in person. The Berkovich discussion continues the following Wednesday, and the last class is the Friday after – the course concludes then (the IHÉS schedule listing two further weeks is to be corrected).

# 23 Berkovich Spaces as Analytic Stacks

Clausen returns to the Berkovich world of Section 22 and promotes the Berkovich spectrum of a Banach ring to an honest analytic stack $\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)$, sitting over the classical topological space $\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|)$ and carrying a structure sheaf of $\infty$-categories – a theory of quasi-coherent sheaves on the Berkovich space. The stack is cut out inside $\mathcal{N} \times \mathrm{Spec}(R, \mathrm{triv})$ (norms $\times$ trivially structured $R$) by the condition that the tautological multiplicative norm be bounded by the given Banach norm. The heart of the lecture is a theorem: under a mild hypothesis ($\star$) (existence of a topologically nilpotent unit whose norm strictly multiplies), $\mathrm{Spec}^{\mathrm{Berk}}$ is *affine*, corresponds to a *gaseous* analytic ring structure on the condensed ring $R$ depending only on $R$ (not on the norm), and its tautological norm is the expected one, given by overconvergent functions on discs. The proof runs through a trace-class calculational trick and Serre duality on $\mathbb{P}^1$. A short closing section globalises to a notion of Berkovich analytic space.

Throughout, "ring" means commutative ring and everything is (light) condensed.

## 23.1 Recollection: Berkovich spaces

We recall the classical setup (Definitions 22.9 and 22.11). A *Banach ring* $(R, |\cdot|)$ is a ring $R$ with a norm $|\cdot|\colon R \to \mathbb{R}_{\geq 0}$ that is submultiplicative and satisfies the triangle inequality,

$$|xy| \leq |x|\,|y|, \qquad |x+y| \leq |x| + |y|, \qquad |-x| = |x|, \qquad |1| = 1 \ \ (\text{unless } R = 0),$$

with respect to which $R$ is complete.

To $(R, |\cdot|)$ Berkovich attaches a compact Hausdorff space, a subset of a product of intervals,

$$\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|) \ \subseteq \ \prod_{f \in R} [0, |f|],$$

consisting of the *multiplicative* seminorms bounded by $|\cdot|$. A point is denoted $x$, but literally it is a valuation $|\cdot|_x\colon R \to \mathbb{R}_{\geq 0}$ which is strictly multiplicative and satisfies the triangle inequality,

$$|fg|_x = |f|_x\,|g|_x, \qquad |f+g|_x \leq |f|_x + |g|_x, \qquad |1|_x = 1.$$

In general $R$ is not complete for a given $|\cdot|_x$; indeed there can be many elements of norm 0.

*Remark* 23.1 (Residue fields and the three cases). Completing $R$ with respect to a point $|\cdot|_x$ produces a complete valued field

$$\widehat{\kappa(x)},$$

the *residue field* of $x$ in Berkovich's sense. There are essentially three cases.

1. $\widehat{\kappa(x)}$ is archimedean; by Ostrowski this means $(\mathbb{R}, |\cdot|^{\alpha})$ or $(\mathbb{C}, |\cdot|^{\alpha})$ with $\alpha \in (0,1]$. In the archimedean case there is no variety among complete normed fields, only the choice of exponent.

2. $|\cdot|_x$ is non-archimedean but $\kappa(x)$ is discrete; the extreme case is the trivial norm $|\cdot|_0$.

3. $\widehat{\kappa(x)}$ is a genuine non-archimedean field with non-discrete induced topology, carrying a normalised absolute value $|\cdot|^{\alpha}$, $\alpha \in (0, \infty)$ (no preferred normalisation absent a choice of pseudo-uniformizer and a real number).

These are the possible residue-field behaviours one sees.

## 23.2 The Berkovich spectrum as an analytic stack

The goal is to promote $\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|)$ to an analytic stack, using the *stack of norms* $\mathcal{N}$ of Sections 20 and 22. Recall (Definition 20.9 and Notation 22.14) that a map from an analytic stack $X$ to $\mathcal{N}$ is the same as a map

$$N\colon\ \mathbb{P}^1_X \ \longrightarrow\ [0,\infty]$$

satisfying the norm axioms – the most important being multiplicativity, which must be phrased with some care but in the end is just multiplicativity.

We also investigated the geometry of $\mathcal{N}$ last time: it lies over the *extended* Berkovich spectrum of the integers (Theorem 22.18, Figure 22.1). On the classical Berkovich spectrum one imposes the usual triangle inequality, which is exactly what prevents taking an arbitrary power $\alpha$ of the archimedean absolute value; dropping it, an arbitrary power behaves like a norm but really only a *quasi*-norm (a constant appears in front of the triangle inequality). In $\mathcal{N}$ no triangle inequality is imposed, and one obtains the non-strict version in which all powers of the usual archimedean norm are allowed, together with a strange limit point where the norm takes the value $\infty$ on natural numbers. Along the interior of the archimedean segment one is seeing the gaseous $\mathbb{R}$-theory; along the interior of a $p$-adic segment the gaseous $p$-adic numbers, with the norm changing but the underlying analytic ring not; the outer ends carry $\mathbb{F}_p$ and other characteristic-$p$ phenomena, while the archimedean end carries characteristic 0. Morally the points of $\mathcal{N}$ correspond to (minimal choices of) complete valued fields – reals, $p$-adics, discrete $\mathbb{F}_p$, discrete $\mathbb{Q}$ – a substitute for the notion of multiplicative valuation. But one still has to feed in the Banach ring $R$ to get something non-trivial.

**Definition 23.2** (Berkovich spectrum)**.** For a Banach ring $(R, |\cdot|)$, let

$$\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)\ \subseteq\ \mathcal{N} \times \mathrm{Spec}(R, \mathrm{triv})$$

be the analytic stack whose $X$-points are pairs

$$\big(\, N\colon \mathbb{P}^1_X \to [0,\infty], \quad R \to \mathcal{O}(X)\,\big)$$

– a norm $N$ on $X$ together with a map of condensed rings from $R$ to the functions on $X$ – such that for all $f \in R$ the composite $N(f)\colon X \to [0,\infty]$ (evaluate the coordinate of $\mathbb{P}^1$ at the section of $\mathbb{A}^1$ determined by $f$, then apply $N$) lands inside the subspace $[0, |f|]$:

$$N(f)\ \in\ [0, |f|] \qquad \forall f \in R. \tag{23.1}$$

Here $\mathrm{Spec}(R, \mathrm{triv})$ is the affine analytic stack of the Banach ring $R$, viewed as a light condensed ring via its topology and equipped with the *trivial* (maximal) analytic ring structure $D(R, \mathrm{triv}) = D^{\mathrm{cond}}(R)$: a map $X \to \mathrm{Spec}(R, \mathrm{triv})$ is exactly a map of condensed rings $R \to \mathcal{O}(X)$.

Condition (23.1) is what ties the norm factor to the ring factor.

**Theorem 23.3.** $\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)$ *is an analytic stack, and it localizes along the Berkovich spectrum* $\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|)$ *in the sense of Section 22: there is a map of locales*

$$\mathrm{Loc}^{\mathrm{op}}\big(\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)\big)\ \longrightarrow\ \mathcal{M}^{\mathrm{Berk}}(R, |\cdot|) \tag{23.2}$$

*from the locale of open substacks – monomorphisms into* $\mathrm{Spec}^{\mathrm{Berk}}$ *that are shriekable and cohomologically smooth – to the opens of the compact Hausdorff space* $\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|)$ *in its usual topology.*

*Remark* 23.4 (Structure sheaf and comparison to Poineau). In particular one obtains a structure sheaf – indeed a structure sheaf of $\infty$-categories, i.e. a theory of quasi-coherent sheaves – on the usual Berkovich space. When $\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|)$ is metrizable and finite-dimensional one can even ask for a genuine map of analytic stacks to it (Theorem 20.3); this is basically all cases, and requires only checking a connectivity property of the idempotent algebras attached to closed subsets, which is a condition verifiable in practice. Clausen noted it would be interesting to compare with Poineau, who described structure sheaves in certain cases over the integers (e.g. the Berkovich line over $\mathbb{Z}$) more or less by hand [Poi08; Poi12]. The present approach instead *defines* an object which is a priori a structure sheaf and satisfies $\infty$-categorical descent automatically, at the cost of then having to compute its values. For rings like the integers this computation is quite feasible; for a more arbitrary Banach ring it is less obvious. As in the Huber theory, extracting the "correct" rings of functions on closed or open discs likely requires working in a derived sense, with derived rings complete in the appropriate structure.

*Remark* 23.5 (Relation to the adic spectrum). When $R$ is Tate one also has the localization over Huber's Spa (Section 22), and a commutative diagram in which the Huber space maps to $\mathrm{Spec}^{\mathrm{Berk}}$ and the solid analytic ring maps to the trivially-structured $R$. The Tate case is examined in more detail below.

### Charts

To see that $\mathrm{Spec}^{\mathrm{Berk}}$ is an analytic stack (only an accessibility statement is needed, since it is defined as a functor of points) we exhibit charts. Recall (Remark 22.15) the chart for the stack of norms: the gaseous base ring gives a cover

$$\mathrm{Spec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}\big) \longrightarrow \mathcal{N}, \tag{23.3}$$

the universal locus inside $\mathbb{P}^1$ where $|T| = \frac{1}{2}$ (an element of norm $\frac{1}{2}$ has been adjoined). Pulling back (23.3) along the structure map $\mathrm{Spec}^{\mathrm{Berk}} \to \mathcal{N}$ produces a chart $Y$ (a universal annulus):

$$\begin{tikzcd} Y \arrow[r] \arrow[d] & \mathrm{Spec}\big(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}\big) \arrow[d] \\ \mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|) \arrow[r] & \mathcal{N}. \end{tikzcd}$$

**Proposition 23.6** (The chart is affine)**.** *The chart $Y$ is affine: concretely, one takes* $\mathrm{Spec}(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}) \times \mathrm{Spec}(R, \mathrm{triv})$ *and passes to the closed subsets given by the idempotent algebras obtained as $N(f)^{-1}$ of the closed subsets* $[0, |f|] \subseteq [0, \infty]$. *On underlying analytic rings*

$$Y \;=\; \big(R \otimes_{\mathbb{Z}} \mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}\big)^{\mathrm{gas}}. \tag{23.4}$$

For a completely general $R$ the difficult part of computing $Y$ is already the first step: forming the tensor product (23.4) and identifying the resulting analytic ring, in particular its underlying condensed ring. This involves computing the gaseous localization of $R$ and then applying the gaseous localization once more (recall from Section 14 that the gaseous localization is a sequential colimit of Koszul complexes). Once that ring is known, passing to the idempotent algebras cutting out the discs is not the hard part.

*Remark* 23.7 (Descent happens at the zeroth stage). The descent for the cover $Y \to \mathrm{Spec}^{\mathrm{Berk}}$ is straightforward. As we saw when checking that this map is a cover – it is proper and descendable – the unit object (structure sheaf) downstairs is already a direct *retract* of the pushforward of the structure sheaf on $Y$. So the structure sheaf on $\mathrm{Spec}^{\mathrm{Berk}}$ is read off as the retract: passing from $\mathrm{Spec}^{\mathrm{Berk}}$ up to $Y$ adjoins an extra variable $q$ with certain analytic properties, and one simply takes the possible zeroth $q$-coefficients of the ring computed on $Y$ (a linear retraction, exactly as constant coefficients sit inside Laurent series). This is a retraction of modules, not of rings.

**The map to the Berkovich spectrum**

On $\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)$ one has the universal norm $N$, and by fiat, for every $f \in R$, a map to the structure sheaf. Together these give

$$\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|) \longrightarrow \prod_{f \in R} [0, |f|]. \tag{23.5}$$

Because we enforced $N(f) \leq |f|$ for every element ((23.1)), in particular $N(2) \leq 2$, which by the universal triangle inequalities of Proposition 22.23 forces $N$ to satisfy the triangle inequality. Hence (23.5) lands inside the Berkovich spectrum $\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|) \subseteq \prod_{f \in R}[0, |f|]$, giving the underlying map of (23.2).

## 23.3 Affineness under a topologically nilpotent unit

In general $\mathrm{Spec}^{\mathrm{Berk}}$ is *not* affine.

**Example 23.8** (The integers with the archimedean norm)**.** For general $(R, |\cdot|)$ the stack $\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)$ is not affine. For instance

$$\mathrm{Spec}^{\mathrm{Berk}}(\mathbb{Z}, |\cdot|_\infty) \;=\; \mathcal{N}_{|2| \leq 2},$$

the locus $\{|2| \leq 2\}$ inside the stack of norms ($|\cdot|_\infty$ is the maximal norm on $\mathbb{Z}$, so this gives the biggest possible Berkovich spectrum). It really is a stack, as one sees at the points living at the boundaries.

We isolate the hypothesis that repairs this.

**Definition 23.9** (Assumption $(\star)$)**.** $(R, |\cdot|)$ satisfies $(\star)$ if (assuming $R \neq 0$) there exists $\pi \in R$ with

$$|\pi| < 1, \qquad \pi \text{ a unit}, \qquad |\pi f| = |\pi|\,|f| \;\; \forall f \in R.$$

The last condition (the norm strictly multiplies against $\pi$) is equivalent, for $R \neq 0$, to $|\pi^{-1}| = |\pi|^{-1}$, as one sees from submultiplicativity.

**Example 23.10** (When $(\star)$ holds)**.**

- Any non-discrete valued field admits such a norm.
- If $(R, |\cdot|)$ satisfies $(\star)$ and $\varphi \colon R \to R'$ is a contractive homomorphism ($|\varphi(f)|_{R'} \leq |f|_R$ for all $f \in R$), then $R'$ satisfies $(\star)$ with respect to $\varphi(\pi)$: the image is again a unit of norm $< 1$, and $|\varphi(\pi)^{-1}| = |\varphi(\pi)|^{-1}$ is more evidently preserved.
- Any Tate–Huber ring has a norm defining its topology and satisfying $(\star)$, with $\pi$ a pseudo-uniformizer; some mixed-characteristic examples also arise.

Note $(\star)$ fails for $\mathbb{Z}$ with $|\cdot|_\infty$ (Example 23.8).

**Theorem 23.11.** *If* $(\star)$ *holds, then* $\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)$ *is affine, and corresponds to an analytic ring structure on the condensed ring* $R$*. Moreover this analytic ring structure depends only on the condensed ring* $R$*, not on the choice of norm satisfying* $(\star)$*.*

### Producing the gaseous structure

*Proof of affineness.* Take $\pi$ as in $(\star)$. It gives a ring map $\mathbb{Z}[q] \to R$, $q \mapsto \pi$. Since $|\pi| < 1$, $\pi$ is topologically nilpotent (its powers tend to 0), so the map factors through the free condensed ring on a topologically nilpotent element $\mathbb{Z}[\widehat{q}]$; since $\pi$ is a unit it factors further:

$$\begin{tikzcd} \mathbb{Z}[q] \arrow[r, "q\mapsto\pi"] \arrow[d] & R \\ \mathbb{Z}[\widehat{q}^{\pm 1}] \arrow[ur, "q\mapsto\pi"'] & \end{tikzcd}$$

We must check that $R$ is $q$-gaseous. Recall that the gaseous structure (Section 14) is defined by requiring the map

$$1 - q \cdot \mathrm{shift}\colon\ \mathrm{Null}(R) \xrightarrow{\ \sim\ } \mathrm{Null}(R)$$

to be an isomorphism, where mapping the universal null sequence $P$ into the Banach space $R$ produces precisely the space of null sequences $\mathrm{Null}(R)$ in $R$, and $1 - q \cdot \mathrm{shift}$ is the endomorphism induced by $1 - T \cdot q\colon P \to P$. Its inverse is visibly the summation: if $(f_n)$ is a null sequence, one can form

$$\sum_n f_n \pi^n \ \in\ R$$

(the limit of a Cauchy sequence, using $|\pi| < 1$ and the triangle inequality). Thus $R$ is $q$-gaseous, and we take the induced analytic ring structure from $\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}}$. Write $R^{\mathrm{gas}}$ for the resulting analytic ring (underlying condensed ring $R$).

Now recall that on $\mathrm{Spec}(\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}})$ there is a universal norm $N$ with

$$N(q)\ \in\ \{|\pi|\}$$

(a singleton subspace of $[0, \infty]$, not merely a value; here $R \neq 0$, so $0 < |\pi| < 1$). By functoriality of norms under pullback we obtain a norm $N$ on $R^{\mathrm{gas}}$, i.e. on $\mathrm{Spec}(R^{\mathrm{gas}})$, with induced analytic ring structure. The content of the theorem is the following.

**Claim 23.12.** *The normed analytic ring* $\big(R^{\mathrm{gas}}, N\big)$ *represents* $\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)$*.*

Over $\mathrm{Spec}(R^{\mathrm{gas}})$ we have the map to $\mathrm{Spec}(R, \mathrm{triv})$ and the norm $N$ with $N(\pi) \in \{|\pi|\}$, and it is straightforward from the construction that $\mathrm{Spec}(R^{\mathrm{gas}})$ is universal with respect to that structure. The difference from $\mathrm{Spec}^{\mathrm{Berk}}$ is that the latter imposes a condition on the norm of *every* element, whereas $R^{\mathrm{gas}}$ was built imposing a condition only on the norm of $\pi$. So the claim reduces to the equivalence, for a norm $N$ on $R^{\mathrm{gas}}$,

$$N(\pi) \in \{|\pi|\} \quad \Longleftrightarrow \quad N(f) \in [0, |f|] \ \ \forall f \in R. \tag{23.6}$$

One direction is easy: given the right-hand side, apply it to $\pi$ and to $\pi^{-1}$ to recover $N(\pi) = |\pi|$. The substance is that pinning down $N(\pi)$ already constrains the norm of every element; this we prove by computing the universal norm on $\mathrm{Spec}(R^{\mathrm{gas}})$. $\square$

### Computing the universal norm

A norm in our sense on an analytic ring implicitly specifies the *overconvergent* functions on the disc of arbitrary radius $c$ centred at the origin. The claim is that these are the usual ones from Berkovich theory.

**Proposition 23.13** (Overconvergent functions on a disc)**.** *For $0 < c$, the value of the norm on* $\mathrm{Spec}(R^{\mathrm{gas}})$ *is given by*

$$\mathcal{O}_R\big(\{|T| \le c\}\big) \;=\; \varinjlim_{r>c} \mathcal{O}_R^{|T|\le r}, \qquad \mathcal{O}_R^{|T|\le r} \;=\; \Big\{\textstyle\sum_n f_n T^n \;:\; \sum_n |f_n|\, r^n < \infty\Big\}, \tag{23.7}$$

*the filtered colimit (in condensed rings) over radii $r > c$ of the universal Banach ring in which the norm is $\le r$, i.e. power series over $R$ whose coefficients decay fast enough that the substitution $T \mapsto r$ converges.*

To make the calculation, rather than expanding $\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}} \to \mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}} \otimes^{\mathbb{L}}_{\mathbb{Z}} R$ naively as a filtered colimit of copies of $P$ – where it is not even easy to see concentration in degree 0 – we use a purely formal trick.

**Lemma 23.14** (Trace-class colimit trick)**.** *Let $(\mathcal{C}, \otimes)$ be a closed symmetric monoidal $\infty$-category, and let*

$$\cdots \to X_3 \to X_2 \to X_1$$

*be a tower in $\mathcal{C}$ in which each transition map is* trace class *– a map $X \to Y$ is trace class if it comes from a map $\mathbf{1} \to X^\vee \otimes Y$, where $X^\vee = \underline{\mathrm{Hom}}(X, \mathbf{1})$ is the internal dual (no dualizability assumed). Then for all $Y \in \mathcal{C}$,*

$$\varinjlim_n X_n^\vee \otimes Y \;\xrightarrow{\;\sim\;}\; \varinjlim_n \underline{\mathrm{Hom}}(X_n, Y),$$

*an equality of internal (ind-)objects; if $\mathcal{C}$ has colimits and $\otimes$ commutes with colimits – as in our examples – the quotation marks disappear and this is a genuine colimit.*

*Idea.* Given a trace-class map $X \to Y$, tensoring the datum $\mathbf{1} \to X^\vee \otimes Y$ with $X$ and using evaluation $X \otimes X^\vee \to \mathbf{1}$ produces the map $X \to Y$; dually it produces backward maps between the two ind/pro-systems raising the index by one step, and these witness the isomorphism. □

Applying Lemma 23.14 with $Y = \mathbf{1}$, one recognises the structure-sheaf object as a colimit over the duals $P^\vee$ with trace-class transition maps; this reduces the computation to null sequences in $R$ and a filtered colimit of such (any two versions of the unit disc agree once made overconvergent, by sandwiching). To identify the dual of the overconvergent disc with functions on the open complement one uses Serre duality.

*Remark* 23.15 (Serre duality on $\mathbb{P}^1$). Working over the gaseous base (the choice is immaterial), algebraic Serre duality in the six-functor formalism says that $\mathbb{P}^1 \to \mathrm{Spec}$ is smooth and proper with dualizing object $\omega^1[1]$. Taking the (internal) linear dual of the overconvergent disc then identifies it with the ring of functions on the open complement. The finiteness needed for the double duality is arranged by an overconvergence trick: the overconvergent disc is an ind-object, whose dual is the pro-object obtained by *increasing* the radius; viewing that pro-object as an inverse limit of overconvergent discs with the non-strict inequalities and dualizing termwise gives back the overconvergent disc as a colimit of duals of $P$'s – exactly the shape needed for Lemma 23.14.

It remains to supply the trace-class hypothesis; it holds for soft reasons.

**Lemma 23.16** (Trace-class claim)**.** *Let $X$ be a topological space and $Z \subseteq Z'$ closed subsets such that there is an open $U$ with*

$$Z \subseteq U \subseteq Z'$$

*(two closed subsets separated by an open subset). Then the restriction map between the pushforwards of the constant sheaves is trace class in the derived category of sheaves on $X$ with values in $D$(base).*

Here the coefficient category is $D$ of the base ring, not the derived category of $X$. Applying this on the interval $[0, \infty]$ – where a closed disc of one radius sits inside an open disc inside a closed disc of larger radius – every transition map in (23.7) is trace class; pulling back along the symmetric monoidal functor $D(\mathrm{Shv}([0, \infty])) \to D(\mathbb{P}^1)$ gives a trace-class map on $\mathbb{A}^1$, and a further trick shows its image under the forgetful functor to the base is trace class as well (using that trace-class maps form a two-sided ideal). Hence the restriction maps between the overconvergent discs are trace class, and Proposition 23.13 follows, giving the presentation (23.7) that lets one calculate.

**Proposition 23.17** (The universal norm is the expected one)**.** *The norm $N$ on $\mathrm{Spec}(R^{\mathrm{gas}})$ is given by the usual overconvergent functions on discs over $R$. In particular, to prove the non-trivial direction of (23.6) – that $N(f) \in [0, c]$ for all $f$ (with $c = |f|$) – it suffices to show*

$$\mathcal{O}_R\big(\{|T| \leq c\}\big)/(T - f) \;=\; R. \tag{23.8}$$

*Proof.* The map is evaluation $T \mapsto f$. As Scholze observed, one does not even need to analyse the short exact sequence of Banach spaces: mere existence of the map suffices. Indeed $\mathcal{O}_R(\{|T| \leq c\})$ is an idempotent algebra over $\mathbb{A}^1$ (over the polynomial ring on one generator), so base-changing along $R[T]/(T - f) = R$ makes it an idempotent algebra over $R$; and $R$ itself is an idempotent algebra over $R$. Two idempotent algebras over $R$ are equal as soon as there is an algebra map between them, and the unit map (evaluation $T \mapsto f$) provides one. Hence (23.8). □

*Remark* 23.18 (Why overconvergence is essential). The overconvergence is doing real work. In the non-archimedean (solid) setting the affinoid algebra of the closed disc – without overconvergence – is already idempotent, as we saw in the solid theory. In the archimedean case it is not: the closed-disc algebra of convergent (rather than overconvergent) functions is not idempotent (Weierstrass division only gives convergence on a smaller disc), so the argument above would fail without passing to the overconvergent colimit.

## 23.4 Independence of the norm

Two things remain in Theorem 23.11: that the analytic ring $\mathrm{Spec}(R^{\mathrm{gas}})$ is independent of the choice of $|\cdot|$ and $\pi$, and (for the record) that the tautological norm on it is independent of $|\cdot|$ up to rescaling. The first is achieved by an intrinsic description of the gaseous structure.

**Proposition 23.19** (Intrinsic description)**.** *$R^{\mathrm{gas}}$ is the initial analytic ring $A$ with a map* $\mathrm{Spec}(A) \to \mathrm{Spec}(R, \mathrm{triv})$ *(i.e. a map $R^{\mathrm{triv}} \to A$) such that for all topologically nilpotent units $\pi \in R$, $A$ is $\pi$-gaseous – i.e. the ring map $\mathbb{Z}[q^{\pm 1}] \to A^{\triangleright}$, $q \mapsto \pi$, yields a map of analytic rings*

$$\mathbb{Z}[\widehat{q}^{\pm 1}]_{\mathrm{gas}} \longrightarrow A \qquad (\textit{equivalently } D(A) \subseteq \{\pi\textit{-gaseous modules}\}).$$

*This condition depends only on the topological ring $R$, not on the norm.*

*Proof.* If $\pi$ is topologically nilpotent then there is $n > 0$ with $|\pi^n| < 1$, and being $\pi$-gaseous is invariant under replacing $\pi$ by a power (a remark of Scholze's, made earlier); so we may assume $0 < |\pi| < 1$. Then for the universal norm built over $R^{\text{gas}}$,

$$N(\pi) \in (0,1), \tag{23.9}$$

which by the axioms of a normed analytic ring already implies $\pi$ is gaseous. Now $R^{\text{gas}}$ was built to satisfy ($\star$) only for some fixed $\pi_0$; the computation (23.9) shows it satisfies the gaseous condition for *all* choices of $\pi$, giving the initial/universal property. Since the condition never referred to the norm, $R^{\text{gas}}$ depends only on the condensed ring $R$. □

**Proposition 23.20** (The norm, up to rescaling)**.** *The norm $N$ on* $\mathrm{Spec}(R^{\text{gas}})$ *is independent of the choice of $|\cdot|$ up to exponentiation by a map*

$$\alpha\colon\ \mathrm{Spec}(R^{\text{gas}}) \longrightarrow \mathbb{R}_{>0}.$$

*Proof.* Let $N$ come from $|\cdot|$ (satisfying ($\star$) for $\pi$) and $N'$ from $|\cdot|'$ (satisfying ($\star$) for $\pi'$, with $0 < |\pi'|' < 1$). Then

$$N(\pi')\colon\ \mathrm{Spec}(R^{\text{gas}}) \longrightarrow (0,1), \tag{23.10}$$

and since $\mathbb{R}_{>0}$ acts simply transitively on $(0,1)$ by exponentiation there is $\alpha$ with

$$N(\pi')^{\alpha} = \{|\pi'|'\}. \tag{23.11}$$

Then $N^{\alpha}$ and $N'$ are two norms on $\mathrm{Spec}(R^{\text{gas}})$ with the same (fixed) value on $\pi'$, so by the classification of norms (Theorem 20.19) they are equal: $N' = N^{\alpha}$. □

*Remark* 23.21 (Ofer's question). Is the rescaling $\alpha\colon \mathrm{Spec}(R^{\text{gas}}) \to \mathbb{R}_{>0}$ pulled back from the Berkovich spectrum $\mathcal{M}^{\text{Berk}}(R, |\cdot|)$ – i.e. does it factor

$$\begin{tikzcd} & \mathbb{R}_{>0} \\ \mathrm{Spec}(R^{\text{gas}}) \arrow[ur, "\alpha"] \arrow[r] & \mathcal{M}^{\text{Berk}}(R, |\cdot|) \arrow[u, dashed, "?"'] \end{tikzcd}$$

The answer is yes, and by construction: $\alpha$ is determined by $N(\pi')$ ((23.11)), and the Berkovich spectrum records the norms of *all* elements of $R$, in particular that of $\pi'$. So $N(\pi')$, and hence $\alpha$, is pulled back from $\mathcal{M}^{\text{Berk}}(R, |\cdot|)$.

### 23.5 Globalization

Globalizing is now trivial. We may simply make a definition.

**Definition 23.22** (Berkovich analytic space)**.** A *Berkovich analytic space* is a triple $(X, S, \pi)$ consisting of an analytic stack $X$, a locally compact Hausdorff space $S$, and a map of locales

$$\pi\colon\ \mathrm{Loc}^{\mathrm{op}}(X) \longrightarrow S,$$

such that, locally on $S$ for the local section topology, it is isomorphic to

$$\big(\mathrm{Spec}^{\text{Berk}}(R, |\cdot|),\ \mathcal{M}^{\text{Berk}}(R, |\cdot|),\ \pi\big)$$

for some Banach ring $(R, |\cdot|)$.

The point is that working locally on $S$ automatically means working locally on $X$: by definition of a map of locales, an open cover of $S$ pulls back to a cover of $X$ – indeed a cover in the Grothendieck topology defining analytic stacks, even an open cover in the sense of the six-functor formalism.

*Notation* 23.23 (Local section topology). On the category of locally compact Hausdorff spaces, a family $\{X_i \to S\}$ is a covering for the *local section topology* if for every point of $S$ there is an open neighbourhood, an index $i$, and a section of the pullback of $X_i$ to that neighbourhood; the maps $X_i \to S$ themselves are arbitrary (not required to be open inclusions). This is the same Grothendieck topology as the one generated by the usual open covers – every covering sieve equals the sieve generated by some open cover of $S$ – but it is convenient for expressing the sense in which a locally compact Hausdorff space "is locally" compact Hausdorff: the affine models $\mathcal{M}^{\mathrm{Berk}}(R, |\cdot|)$ are compact Hausdorff, and open subsets of, say, $\mathbb{R}$ are usually not compact, so the naive formulation fails, whereas covering by maps with local sections (compact neighbourhoods of each point) works.

*Remark* 23.24 (A derived fly in the ointment). As with the Huber theory (Section 10), the presentation starts from classical objects – Banach rings – and when one works locally on the spectrum of such an object one may a priori encounter derived objects. It is slightly more acute here: the basic building blocks are Banach rings, but the structure sheaf is overconvergent, so it is essentially never a sheaf of Banach rings, and there is a mismatch between the global object one starts with and the local values. In practice this does not obstruct anything – the neighbourhood bases one computes land in degree 0 and fit back into the framework – and the definitions can be modified to accommodate the general case. For instance, one may enlarge the class of affine models by passing to arbitrary inverse limits of the $\mathrm{Spec}^{\mathrm{Berk}}(R, |\cdot|)$ (filtered colimits of analytic rings, over inverse limits of compact Hausdorff spaces); the overconvergent objects then count as affine, replacing Banach rings by condensed rings with suitable norm data. We did not try to give the best possible formulation, only to connect with the classical picture.

# 24 Outlook: Directions for Analytic Geometry

This is the final lecture of the course. With Clausen's treatment of Berkovich spaces (Section 23) the programme announced in Section 1 is complete: the traditional frameworks of analytic geometry – complex-analytic, rigid, adic, Berkovich – all sit inside analytic stacks, carrying a genuine derived structure sheaf and a full six-functor formalism. Scholze devotes the last hour to an *outlook*: a list of directions in which the machinery might be pushed, ordered from the most developed to the most speculative. The register is deliberately informal, and much of what follows is flagged by Scholze himself as conjecture, hope, or a computation he has not carried out; the notes preserve those hedges.

The personal motivation, restated from the first lecture: for some five or six years Scholze wanted to do $p$-adic geometry not only $p$-adically but also with real (archimedean) coefficients and, ideally, in a single framework spanning both – and it was always clear that this required a genuinely new language. That language now exists, and the sensible next question is simply to use it.

Throughout, "ring" means commutative ring and everything is (light) condensed.

## 24.1 Direction 1: quasi-coherent sheaves and six functors, in all flavours of analytic geometry

This first item is not a speculation but something that works. One now has a general theory of quasi-coherent sheaves – with no finiteness conditions imposed on the modules – in *all* flavours of analytic geometry, and all those flavours are unified into the single notion of an analytic stack. Better still, this is a full six-functor formalism

$$f^*, \quad f_*, \quad f_!, \quad f^!, \quad \otimes, \quad \mathrm{RHom},$$

which lets one play. Two immediate payoffs Scholze singled out.

*Remark* 24.1 (A structure sheaf for free). As Clausen emphasised in Section 23, even producing a well-behaved *structure sheaf* is a nontrivial application. For Berkovich spaces – even those of finite type over $\mathbb{Z}$ – one classically had to work hard to define anything; for adic spaces there were the non-sheafiness issues that had to be resolved by passing to a derived object by hand (Section 10). In the present framework an analytic stack simply comes with its structure sheaf, period.

*Remark* 24.2 (GAGA and descent for vector bundles). One can reprove the various GAGA theorems and also prove genuinely new ones (Sections 15, 19 and 21). There are also results on descent for vector bundles. Scholze recalled Drinfeld's work on (infinite-dimensional) vector bundles [Dri04], where for the assignment $R \mapsto \{\text{vector bundles on } R(\!(t)\!)\}$ – something that arises naturally when thinking about loop groups – one proves descent results; a difficult paper, recently revisited by Mathew using his descendability ideas [Mat16]. A further variant of such Drinfeld-type statements, put to Scholze by a colleague, can be proved very simply using the general RHom-formulas of the six-functor formalism.

Related to Riemann–Roch and to the Atiyah–Singer index theorem (which Scholze is optimistic could in principle be proved with this technology, though this has not been figured out): all of these are closely tied to $K$-theory, which is the next item.

## 24.2 Direction 2: $K$-theory of analytic spaces

In algebraic geometry, $K$-theory was defined first by Quillen and then for general schemes by Thomason, using the well-behaved stable $\infty$-category of perfect complexes. In analytic geometry the obstruction was the absence of a good enough category of modules to which one could apply categorical $K$-theoretic techniques. With the present technology one can do it – not with perfect complexes exactly, but with a variant of *nuclear* modules (Definition 10.24). This uses Efimov's $K$-theory of dualizable $\infty$-categories; the analytic side has been developed in particular by Andreychev [And23], who has worked out the $K$-theory and THH of Tate adic spaces using Efimov's results on continuous $K$-theory.

*Remark* 24.3 (Localizing motives and refined trace theories). Efimov's strong results on the category of localizing motives (proving it is rigid, hence dualizable) allow one to define refined variants of cyclic and topological cyclic homology which do *not* take values in some complete category: usually, taking $S^1$-fixed points produces a completed power-series object, whereas here one can instead land in nuclear modules in our sense, tying these trace theories back into analytic geometry. Clausen mentioned a joint project bearing on this circle of ideas that would have used nuclear $K$-theory, though he immediately added that they had since found a way to avoid it.

## 24.3 Direction 3: $p$-adic cohomology theories

A great deal of work, much of it already done, concerns $p$-adic coefficient systems and cohomology theories, where the geometric language of analytic stacks is directly relevant. Among the incarnations Scholze listed:

- the *analytic de Rham stack* (Rodríguez Camargo [Cam24]), which gives a six-functor formalism for what are essentially $\widehat{\mathcal{D}}$-modules – modules over a completion of the ring of differential operators to differential operators of infinite order, in the sense of Ardakov–Wadsley – avoiding the usual functional-analysis difficulties;

- the *analytic Hodge–Tate stack* (Anschütz–Heuer–Le Bras);

- *geometric Sen theory* (Pan, Rodríguez Camargo), with applications to global automorphic forms, closely related to geometric Langlands;

- *analytic prismatization*;

- $p$-adic local Langlands for locally analytic representations, phrased in terms of geometric Langlands on the Fargues–Fontaine curve.

Scholze noted Pilloni's recent talk on joint work with Boxer, Calegari and Gee proving modularity of genus-two curves, whose key technical part uses this kind of technology, and that one expects a local geometric language for $p$-adic/locally-analytic representations – a sheaf-theoretic version of the Fargues–Scholze geometrization of local Langlands [FS24], and eventually a global one.

### 24.4 Direction 4: the real analogue of the $p$-adic coefficient theories

A focus of the 2023 Hausdorff trimester in Bonn was the realisation that, now that the $p$-adic story is well understood and phrased in the correct geometric language, one can more or less translate all of its ingredients to the *real* world one-to-one. There is a real analogue of virtually everything:

- a real *analytic de Rham stack*, which in this case is isomorphic to the *Betti stack* – the statement of a Riemann–Hilbert correspondence;

- a real *analytic Hodge stack* (where the relevant gerbe is trivial, whereas $p$-adically it can be a nontrivial gerbe);

- an analytic "prismatization"-type stack whose vector bundles encode, in the $p$-adic world, variations of Hodge structure; in the real world one gets *variations of twistor structure*, generalising variations of Hodge structure (Simpson; Mochizuki's variations of mixed twistor structure), carrying a $U(1)$-action;

- and, synthesising these, a *real local Langlands* for locally analytic representations of $G(\mathbb{R})$, phrased as geometric Langlands on the *twistor*-$\mathbb{P}^1$ (the real analogue of the Fargues–Fontaine curve). Here the classical Harish-Chandra $(\mathfrak{g}, K)$-modules are replaced by a genuinely geometric object: the derived category of the classifying stack of the real-locally-analytic group, together with its forgetful functor to $\mathbb{C}_{\mathrm{gas}}$-modules,

$$D\big(\operatorname{Spec}\mathbb{C}_{\mathrm{gas}} \,/\, G(\mathbb{R})^{R\text{-la}}\big) \;\longrightarrow\; D(\mathbb{C}_{\mathrm{gas}}).$$

*Remark* 24.4 (Representations of $G(\mathbb{R})$ without Harish-Chandra modules). Usually, representations of $G(\mathbb{R})$ force functional-analysis issues, and one replaces them by the more algebraic Harish-Chandra $(\mathfrak{g}, K)$-modules – passing to $K$-finite vectors for a maximal compact $K$ – at the cost of collapsing many representations (with different function spaces but the same Harish-Chandra module) to one. In the present world one can encode representations of the real group directly and need not do this. Concretely, $G(\mathbb{R})$ is a group object in real-analytic manifolds, so its real-analytic incarnation is a group object in analytic stacks; one forms the classifying stack $BG(\mathbb{R})$ (over $\mathbb{C}_{\mathrm{gas}}$, say), whose quasi-coherent sheaves are representations of the group. To realise them as representations one pulls back along the projection $\pi$ from a point: for a canonical object attached to a Harish-Chandra module, $\pi^*$ should produce the minimal globalisation and $\pi^!$ the maximal globalisation (this has not yet been checked). This is closely related to the existence of enough analytic vectors in representations.

### 24.5 Direction 5: metrics and the extended Berkovich line

Here we enter the speculative realm, though with a strong belief. Classically, in questions of functoriality and in geometry, one often has to put *metrics* on things – to prove a Hodge decomposition, say, one does $L^2$-analysis. We cannot yet translate anything involving metrics into our world. Scholze's guess for where they should enter is the *extended* Berkovich line of $\mathbb{Z}$ from Section 22 (Theorem 22.18, Figure 22.1): the Berkovich tree with its central trivial-norm point, one branch per prime $p$ ending at $\mathbb{F}_p$, and the archimedean branch – but where the classical Berkovich space stops the archimedean branch at $|\cdot|_\infty$ (because of the triangle inequality), our space of norms imposes no triangle inequality, so the archimedean branch is prolonged past $|\cdot|_\infty$ and then one-point compactified, adding a point at infinity. The intuition is that *this point at infinity must be related to metrics.*

*Remark* 24.5 (Evidence from the $p$-adic side). Working 2-adically, extending over the endpoint $\mathbb{F}_2$ of the 2-branch means precisely putting a $\mathbb{Z}_2$-module over $\mathbb{Z}_2$ above the generic $\mathbb{Q}_2$-vector space – i.e. a 2-adic *integral* structure, a $p$-adic metric. So by analogy, extending over the archimedean point at infinity ought to mean putting some kind of metric on real objects.

The analogy between $p$ and $\infty$ can be made sharper. Over the open part of a $p$-branch one has a $p$-adic locally analytic space

$$\mathbb{Q}_p^{\mathrm{la}} \;=\; \bigcup_n p^{-n}\,\mathbb{Z}_p^{\mathrm{la}}, \qquad \mathbb{Z}_p^{\mathrm{la}} \;=\; \mathrm{AnSpec}\; C^{\mathrm{la}}(\mathbb{Z}_p, \mathbb{Q}_p),$$

where $C^{\mathrm{la}}(\mathbb{Z}_p, \mathbb{Q}_p)$ are the locally analytic functions – those locally developable into a power-series expansion. Since it has $\mathbb{Q}_p$-coefficients, this space, which lives over the open $p$-adic ray, has a canonical extension over the endpoint $\mathbb{F}_p$; this is closely related to a theory of locally analytic representations of $G(\mathbb{Q}_p)$. The canonical extension is somewhat subtle to write down but its existence matters for the $p$-adic story.

The real analogue: over the open archimedean ray one has the real-analytic $\mathbb{R}$, i.e. $\mathbb{R}$ as a real-analytic manifold with functions the real-analytic functions (again, locally power-series expansions), each fibre over a point of the ray being a copy of $\mathbb{R}^{\mathrm{an}}$. Scholze conjectures this extends canonically over the closed point at infinity, and likewise that the real-analytic group $G(\mathbb{R})^{\mathrm{an}}$ extends canonically over that point. (One uses the gaseous structure at all points; the liquid structure is not needed here.) At every point of the ray there is a rescaling action of the norm, so a whole family of copies of the reals, mapping to the space of norms as an open subspace.

*Remark* 24.6 (Geometry over the point at infinity). More broadly, one can look at analytic stacks over the space of norms that map to the closed point at infinity, and ask what geometry lives there. It is a very peculiar geometry – still about real/complex manifolds, but where one can localise much more, zooming in infinitesimally. For instance, on the Riemann sphere the "bounded" part is not quite the closed unit disc but some minimally overconvergent neighbourhood of it: essentially the closed unit disc, but any real number bigger than 1 is an unbounded function on it. Scholze expects that whatever geometry lives over this point is exactly what is needed to prove statements (like the Hodge decomposition for complex projective manifolds) that classically require putting metrics on things; this remains to be understood.

## 24.6 Direction 6: sheaves of higher categories, and quantum Chern–Simons

Our formalism removes the classical obstruction of having to do functional analysis and fancy category theory at the same time; once that is gone, one can go completely fancy. To an affine analytic space one first associates its derived category $D(A)$, then the module category $\mathrm{Pr}^L_{D(A)} = \mathrm{Mod}_{D(A)}(\mathrm{Pr}^L)$; but one can iterate, using Stefanich's presentable $(\infty, n)$-categories $n\mathrm{Pr}^L$ for any $n$,

$$A \longmapsto D(A), \quad \mathrm{Pr}^L_{D(A)}, \quad 2\mathrm{Pr}^L_{D(A)}, \quad \ldots, \quad n\mathrm{Pr}^L_X,$$

defining $n\mathrm{Pr}^L_X$ for any stack $X$, again with a six-functor formalism to play with. The existence of this is not speculative; what follows is.

The speculation is a construction of *quantum Chern–Simons theory* via the cobordism hypothesis. Let $G$ be a compact, simple, simply connected Lie group, so that the first interesting cohomology is $H^3(G, \mathbb{Z}) \cong \mathbb{Z}$; fix a level

$$\ell \in H^3(G, \mathbb{Z}).$$

Regarding $\ell$ as a map $G \to B^3\mathbb{Z}$ of groups, it deloops to a map $BG \to B^4\mathbb{Z}$, i.e. a class in $H^4(BG, \mathbb{Z}) = \mathbb{Z}$. Now restrict to the real-analytic incarnation $BG^{\mathrm{an}}$ (using that any real-analytic manifold $M$ maps to its condensed incarnation $M^{\mathrm{cond}}$). Using the exponential sequence

$$0 \longrightarrow 2\pi i\,\mathbb{Z} \longrightarrow \mathbb{G}_a \xrightarrow{\exp} \mathbb{G}_m \longrightarrow 0, \tag{24.1}$$

the composite $BG^{\mathrm{an}} \to B^4\mathbb{Z} \to B^4\mathbb{G}_a$ vanishes – this is coherent cohomology, and it is here that compactness of $G$ is crucial – so the level lifts to a map

$$\alpha \colon BG^{\mathrm{an}} \longrightarrow B^3\mathbb{G}_m,$$

as in the lifting triangle

$$\begin{tikzcd} & B^3\mathbb{G}_m \arrow[d] & \\ BG^{\mathrm{an}} \arrow[ur, dashed, "\alpha"] \arrow[r, "\ell"'] & B^4\mathbb{Z} \arrow[r] & B^4\mathbb{G}_a \end{tikzcd}$$

with the bottom composite zero.

A map to $B\mathbb{G}_m$ is a line bundle; a map to $B^2\mathbb{G}_m$ gives a twist of the category of modules; a map to $B^3\mathbb{G}_m$ gives a twist of the category of *categories*, i.e. $\alpha$ specifies an invertible object $L \in 2\mathrm{Pr}^L_{BG^{\mathrm{an}}}$. One then pushes forward along the projection $\pi \colon BG^{\mathrm{an}} \to *$ (with $* = \operatorname{Spec} \mathbb{C}_{\mathrm{gas}}$), forming – morally the space of conformal blocks –

$$\pi_!\, \alpha^* L \in 2\mathrm{Pr}^L_{\mathbb{C}_{\mathrm{gas}}}. \tag{24.2}$$

(For $G$ a finite group this is literally an integral over the space of $G$-bundles, the situation of Freed–Hopkins–Lurie–Teleman [FHLT09].)

*Remark* 24.7 (Is it 3-dualizable?). A three-dimensional topological field theory should, by the cobordism hypothesis, be a 3-dualizable object of a 3-category, and $2\mathrm{Pr}^L$ furnishes one. So: is (24.2) 3-dualizable? An expert would say immediately there is no chance, because a holomorphicity constraint (the space of conformal blocks) is missing. But one is not far off: Scholze believes it is 2-dualizable, and 3-dualizability would require, for $\pi \colon BG^{\mathrm{an}} \to *$ and its iterated diagonals, that

$$\pi, \quad \Delta_\pi, \quad \Delta_{\Delta_\pi} \quad \text{each be proper and smooth,}$$

six conditions in all. Only one fails: the *smoothness of* $\pi$ (of $G^{\mathrm{an}}$ itself). The obstruction is exactly coherent cohomology – for the condensed-set classifying stack, coherent cohomology is singular cohomology, so had one used $B^4\mathbb{G}_a$ over the complex numbers ($BG_{\mathbb{C}}$) the lift would have failed on $H^4(BG;\mathbb{C})$; it is the compact/real-analytic structure that makes the relevant cohomology vanish. Scholze's hope is that tweaking the construction – e.g. replacing $\mathbb{R}^{\mathrm{an}}$ by the canonical extension over the point at infinity of Direction 5, or incorporating the connection / Deligne-cohomology (weight-two) structure that the map to $B^3\mathbb{G}_m$ really carries – supplies the missing holomorphicity constraint. The point is that one can now just *play* with these objects naively, where classically one would drown in functional analysis.

*Remark* 24.8 (The exponential sequence as a sequence of stacks)*.* Asked in what sense (24.1) is exact: all of $2\pi i\,\mathbb{Z}$, $\mathbb{G}_a^{\mathrm{an}}$, $\mathbb{G}_m^{\mathrm{an}}$ are analytic stacks with a commutative group structure – sheaves of abelian-group objects in the higher (anima) sense – and (24.1) is an exact sequence of these, connected and also surjective on the right. It is not stable, but passing to the $B$-groups puts it in a stable context.

### 24.7 Direction 7: virtual fundamental classes

The last direction is treated in more detail; after the very fancy Direction 6 it is comparatively concrete. It concerns a theory of *virtual fundamental classes.* Such classes arise, for instance, in symplectic geometry (Gromov–Witten invariants). In algebraic situations there has been a known solution for some twenty or thirty years, but the genuinely analytic context is much more subtle, and it is exactly the kind of setting – objects that are simultaneously analytic, derived and stacky – where our theory should be useful. Scholze reports that it can indeed do something.

*Question* 24.9 (Signed intersection count)*.* Consider a fibre product of oriented compact smooth manifolds

$$\begin{tikzcd} X \arrow[r] \arrow[d] & M_1 \arrow[d] \\ M_2 \arrow[r] & M \end{tikzcd} \tag{24.3}$$

(one may assume $M_1, M_2 \hookrightarrow M$ are closed immersions), of *expected dimension* 0:

$$\dim M_1 + \dim M_2 \;=\; \dim M.$$

If the intersection is transverse, $X$ is a finite oriented set of points, so one has a well-defined signed count

$$\#^{\mathrm{sgn}}(X) \;\in\; \mathbb{Z},$$

invariant under perturbations (as long as the intersection stays transverse). Can one define $\#^{\mathrm{sgn}}(X)$ in general – without transversality or perturbation – purely intrinsically on $X$?

Each connected component ought to carry a well-defined local contribution (as in Fulton's intersection theory), but even the local numbers must be defined. The difficulty is that for smooth manifolds the intersection, as a topological space, has essentially no structure: two curves in a surface can be tangent to infinite order, agree along an interval and then separate, accumulate infinitely many components, and so on. (For smooth projective varieties the intersection cannot be so wild, and there the problem is classical.)

**Theorem 24.10** (Virtual fundamental classes exist)**.** *Yes: provided the fibre product* (24.3) *is computed in our category – so that $X$ is a* derived *smooth manifold – and, crucially, worked over the* liquid *real numbers* $\mathbb{R}^{\text{liq}}$ *(for some choice of the liquid parameter, which does not affect the outcome) rather than the gaseous reals. One can define the class of analytic stacks that are locally such an intersection, and construct a virtual fundamental class on them.*

The liquid structure – irrelevant for essentially everything else in the course – is genuinely used here (Remark 24.14 below). Symplectic experts at MPI (Nate Botman, Katrin Wehrheim) pointed out that even the simplest case is surprising; the following explains why.

### The surprising local computation

Take $M_1, M_2$ two curves in a surface meeting locally at a single point, and analyse a neighbourhood via the vanishing of a function.

- Transverse $\Rightarrow$ a point, counting $\pm 1$.
- Quadratic tangency $\Rightarrow 0$; cubic $\Rightarrow \pm 1$; and in general, if the function vanishes to *finite* order one simply remembers the vanishing order, which decides whether the graph crosses the line.

The subtle case is infinite-order tangency. Consider the two functions

$$e^{-1/x^2} \qquad \text{and} \qquad \operatorname{sgn}(x)\, e^{-1/x^2},$$

both vanishing to infinite order at 0; the first stays positive (does not cross the axis), the second crosses.

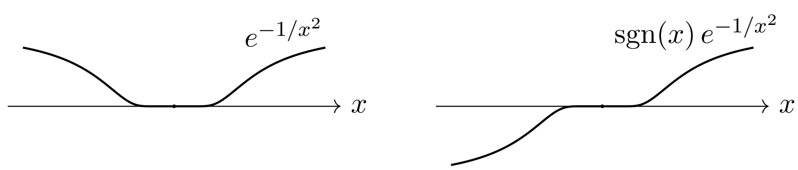



Figure 24.1: Two functions with the same vanishing locus $\{0\}$ set-theoretically: $e^{-1/x^2}$ touches the axis without crossing, $\operatorname{sgn}(x)\, e^{-1/x^2}$ crosses. As liquid analytic rings their vanishing loci differ, and that difference detects the crossing.

The claim is that the *vanishing locus* of such a function distinguishes these two cases – in particular, determines whether the graph crosses the line – in every category. Classically it is unclear what structure the intersection point should carry: it must remember more than all the derivatives of the function (which it cannot double), yet much less than the full germ. In our world it knows exactly the right amount.

**Definition 24.11** (Vanishing locus)**.** For $f \in C^\infty(\mathbb{R}, \mathbb{R})$ with an isolated zero at 0, the *vanishing locus* is

$$\operatorname{AnSpec} C^\infty(\mathbb{R}, \mathbb{R})/(f).$$

If $f$ vanishes to order $n$ this is the truncated polynomial $\mathbb{R}$-algebra $\mathbb{R}[T]/T^n$; if $f$ vanishes to infinite order it is a non-Hausdorff $\mathbb{R}$-algebra. In the latter case $f$ is still a non-zero-divisor, so $C^\infty(\mathbb{R}, \mathbb{R})/(f)$ is concentrated in degree 0 – only non-separated, which is exactly what the condensed/liquid framework handles.

**Proposition 24.12** (The vanishing locus determines $f$ up to a unit)**.** *Let $f, g \in C^\infty(\mathbb{R}, \mathbb{R})$ be as in Definition 24.11, and suppose there is an isomorphism of condensed $\mathbb{R}$-algebras*

$$C^\infty(\mathbb{R}, \mathbb{R})/(f) \;\cong\; C^\infty(\mathbb{R}, \mathbb{R})/(g). \tag{24.4}$$

*Then $f = g \cdot u$ for a unit $u \in C^\infty(\mathbb{R}, \mathbb{R})^\times$. In particular $f$ crosses the line if and only if $g$ does, since an invertible smooth function $\mathbb{R} \to \mathbb{R}$ is everywhere positive or everywhere negative and multiplying by it cannot change the crossing.*

*Remark* 24.13 (What the isomorphism must respect)*.* The isomorphism (24.4) should be one of condensed $\mathbb{R}$-algebras carrying the coordinate; only the isolated zero at 0 is used. The statement is genuinely surprising: functions of visibly different decay rate at 0, such as $e^{-1/x^2}$ and $e^{-1/(cx)^2}$, do have isomorphic vanishing loci (after a reparametrisation), even though the isomorphism of abstract condensed algebras looks as if it should see different coordinates. It is nonetheless true.

*Proof of Proposition 24.12.* The engine is the liquid tensor computation (Remark 24.14): for $f, g$ as above one shows that $C^\infty(\mathbb{R}, \mathbb{R})$ is an *idempotent* algebra over $\mathbb{R}[T]$ ($T$ the coordinate). Both $A := C^\infty(\mathbb{R}, \mathbb{R})/(f)$ and $B := C^\infty(\mathbb{R}, \mathbb{R})/(g)$ carry a unique evaluation map to $\mathbb{R}$ (maps $C^\infty(\mathbb{R}, \mathbb{R}) \to \mathbb{R}$ are classified by real numbers, and only the one at 0 descends to the quotient); these commute with the isomorphism. Now given the map $\mathbb{R}[T] \to B$, one checks that it extends (necessarily uniquely, by idempotency of $C^\infty(\mathbb{R}, \mathbb{R})$ over $\mathbb{R}[T]$) to $C^\infty(\mathbb{R}, \mathbb{R}) \to B$: existence is seen by classifying maps $\mathbb{R}[T] \to C^\infty(\mathbb{R}, \mathbb{R})$, which are given by smooth functions $\mathbb{R} \to \mathbb{R}$ and along which any further smooth function may be composed. Hence the isomorphism upgrades to one respecting the $C^\infty(\mathbb{R}, \mathbb{R})$-structure, the two presentations of the ideal agree, and $f = g \cdot u$ up to a reparametrisation. □

*Remark* 24.14 (Liquid, not gaseous)*.* Working over $\mathbb{R}^{\mathrm{liq}}$ is essential. The liquid tensor product computes the expected thing on nuclear Fréchet spaces (spelled out in the complex-geometry notes [CS20]), so that

$$C^\infty(M_1) \;\otimes^{\mathbb{L}}_{\mathbb{R}^{\mathrm{liq}}}\; C^\infty(M_2) \;\cong\; C^\infty(M_1 \times M_2), \tag{24.5}$$

whereas over $\mathbb{R}_{\mathrm{gas}}$ one would get a strictly smaller ring. The liquid building blocks are essentially the $\alpha$-locally-convex ones with $\alpha$ slightly below 1, so forming these tensor products allows all (nearly) locally-convex combinations; with nuclear transition maps the precise summability parameter does not matter. From (24.5) one deduces the idempotence of $C^\infty(\mathbb{R}, \mathbb{R})$ over $\mathbb{R}[T]$ used above.

*Remark* 24.15 (A $C^\infty$-ring structure for free)*.* Existing theories of derived (or "$C^\infty$-") manifolds build in, by definition, the datum that any function can be composed with an arbitrary smooth function $\mathbb{R} \to \mathbb{R}$. The observation above shows that our funny derived vanishing loci *acquire* this structure canonically – whenever one has a function, it extends uniquely to compose with smooth functions – so it need not be imposed at the outset. This is precisely what makes the objects comparable.

### The general theorem and construction

**Theorem 24.16** (Virtual fundamental class)**.** *Let $S$ be a locally compact Hausdorff, finite-dimensional space equipped with a sheaf $\mathcal{O}_S$ of animated liquid $\mathbb{R}$-algebras, such that $(S, \mathcal{O}_S)$ is locally the (smooth, geometric) intersection of smooth manifolds. Then one can define an* expected dimension *$d \in H^0(S, \mathbb{Z})$, an* orientation sheaf *$\omega_S$ (a $\mathbb{Z}$-local system), and a virtual fundamental class*

$$\mathrm{vfc}(S, \mathcal{O}_S) \;\in\; H^0\big(S,\; \pi^!\mathbb{Z} \otimes \omega_S[-d]\big), \qquad \pi\colon S \to *.$$

*In particular, if $d = 0$, $S$ is compact and an orientation is chosen, one obtains the signed count*

$$\#^{\mathrm{sgn}}(S, \mathcal{O}_S) \;=\; \int_S \mathrm{vfc}(S, \mathcal{O}_S) \;\in\; \mathbb{Z},$$

*the integral being the trace $\pi_! \pi^! \mathbb{Z} \to \mathbb{Z}$ available for compact $S$.*

The construction is the direct analogue of the algebro-geometric one (Behrend–Fantechi [BF96], and later many others including Khan), built on the *intrinsic normal cone*.

*Remark* 24.17 (The intrinsic normal cone). Over $S$ there is an intrinsic normal cone $\mathfrak{C}_S \to S$ whose fibres are a geometric incarnation of the cotangent complex $L_{\mathcal{O}_S/\mathbb{R}}$. For a derived intersection of smooth things this cotangent complex is perfect of amplitude $[0, 1]$, so the cone needs only 1-truncated (group-like) directions: tangent directions give stacky directions, and the obstruction (non-smooth) part gives vector-bundle directions. Its rank difference is the expected dimension $d$, and the same geometry produces the orientation sheaf $\omega_S$.

Concretely, for a compact Hausdorff (or profinite) test object $T$,

$$\mathrm{Hom}(T, S) = \mathrm{Maps}\big(\mathrm{AnSpec}\, C(T, \mathbb{R}), \mathrm{AnSpec}(S, \mathcal{O}_S)\big),$$

which is just maps $T \to$ (underlying set), while maps into the intrinsic normal cone are

$$\mathrm{Hom}(T, \mathfrak{C}_S) = \mathrm{Maps}\big(\mathrm{AnSpec}\big(C(T, \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{R}[\epsilon]\big), \mathrm{AnSpec}(S, \mathcal{O}_S)\big), \qquad |\epsilon| = 1,\ \epsilon^2 = 0$$

– i.e. one takes the source and mods out 0 in the derived sense, adjoining $\epsilon$ in homological degree 1. Deformation theory then reads off the fibres as the geometric incarnation of the dual of the cotangent complex.

To extract a class one uses a deformation-to-the-cone picture. Over $\mathbb{R}$ one writes a cone $\mathrm{Cone}(S)$ which everywhere except at 0 is just a point (the tautological section), but which degenerates, over 0, to the sticky intrinsic normal cone $\mathfrak{C}_S$: on $T$-points, a map corresponds to $g \in C(T, \mathbb{R})$ together with a lift over $C(T, \mathbb{R}) \otimes_{\mathbb{R}} \big(C^\infty(\mathbb{R}, \mathbb{R})/(g)\big)$, so that $g = 0$ recovers $\mathfrak{C}_S$, $g$ invertible gives a point, and $g$ somewhere-zero glues the two. One then forms

$$F \;:=\; \pi_{\mathrm{cone}\,*}\, \pi_{\mathrm{cone}}^{!}\, \mathbb{Z}$$

and, with $j$ the open inclusion of the cone away from 0 and $i$ the closed inclusion of the fibre over 0, the excision triangle

$$j_! j^* F \;\longrightarrow\; F \;\longrightarrow\; i_* i^* F$$

produces a boundary map. Generically $F$ has a canonical section (the point); degenerating that section over the special fibre, and accounting for the fibre bundle by which the cone differs from $S$ (which supplies both the dimension shift and the orientation sheaf), the boundary map lands the tautological class of $H^0(\mathbb{R} \smallsetminus \{0\}, \mathbb{Z})$ – taken to be 0 on one component and 1 on the other, with a shift by $\pm 1$ because $\mathbb{Z}$ is taken on the real line rather than pulled back from a point – into the desired $\mathrm{vfc}(S, \mathcal{O}_S)$.

*Remark* 24.18 (The cone is only shriekable, not smooth). The map $\pi_{\mathrm{cone}}$ is *not* cohomologically smooth – the base $S$ is completely nasty in general, and the cone is some fibrewise vector-bundle-like object over it, with no reason to be smooth – but it is shriekable, which is all one needs to degenerate the tautological fundamental class and read off the virtual class. The whole construction needs only that the relevant map be shriekable; the transversal-manifold case guarantees this and it is stable under fibre products.

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